MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3130 TERTIARY CRUSHING

CIVIL DESIGN CALCULATION

AS BUILT

MD-216-3130-EG-CV-CAL-1007

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2008	Building Code Requirement of Structural Concrete

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The document is prepared for civil design Calculation of teriary cone crusher support structure and its foundation as well as foundation and pedestal for the steel super structure of tertiary crusher building. This building have 2 tertiary cone crusher, a surge bin of 195 m3 capacity, vibrating feeder and the feed conveyor from screen building.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for design of Concrete structure supporting the tertiary cone crusher at RL 919.000m level, Columns and foundations.
This structure as well as foundation has been kept isolated from the steel structure. Separate staad model has been used for the analysis and design.
Single staad file(TC support structure) has been generated for two identical tertiary support structure.
Isolated footing design have been performed using MS-Excel and for combined footing separate FEM staad-pro model has been generated.
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
Design of Reinforced Concrete members followed Strength Design as per SBC 304:2018 & ACI 318:2008.

4. CONCRETE FOUNDATIONS

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1.5m depth below FGL (ie. RL 912.50m) refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5. LOAD CALCULATONS

5.1 Dead Load

Selfweight of structure is applied through Staad Pro.
All Equipment load has been considered as dead load

For Crusher & other equipment load, Refer Attachment 1 - Mechanical General Arrangement drawings.

Dead load on floor @ 919.50m

Weight of RCC Slab	=	24.00	KN/m^3
Slab Thickness	=	0.40	m
Total dead load from Slab	=	9.60	KN/m^2

Chute Material load = 6 x 18 = 108 KN
= 27 KN/support

5.2 Live load

Live load on floor @ 919.000m = 5.00 KN/m^2

5.3 SUPERIMPOSED DEAD LOAD

For Crusher & other equipment load, Refer Attachment 1 - Mechanical General Arrangement drawings.

Dynamic Load Case I (Reversible) :	SDL1_R	=	Dynamic Load of Crusher at start/stop Vertical Load - Downward & Horizontal load in + Z direction
Dynamic Load Case II (Reversible) :	SDL2_R	=	Dynamic Load of Crusher at start/stop Vertical Load - Downward & Horizontal load in + X direction

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5.4 Wind Load

Wind Load Parameters (As per design Criteria) :

Basic wind speed, V_b	=	43	m/s	= 155	Km/h
Exposure category	=	C	(For Process Plant Area)		
Building Category	=	II	(Ref. Table 1.6-1 : SBC 301-2007)		
Wind directionablity factor, K_d	=	0.85	(Ref. Table 6.4-1 : SBC 301-2007)		
Topography Factor, K_{zt}	=	1	(As The project is not located in isolated hills or escarpments)		
Importance Factor, I	=	1	(Ref. Table 6.5-1 : SBC 301-2007)		
Velocity Pressure Co-efficient	=	Kz	(Ref. Table 7.2-2 : SBC 301-2007)		

Wind Velocity Pressure, qz = $0.0473x10^{-3}K_z K_{zt} K_d V^2 I$ (KN/m^2) (Ref. Eq. 7.2-12 : SBC 301-2007)

Level	Height	kz	Pressure	Unit
EL +923.0	10m	1.0	0.966	KN/m^2
EL +933.0	20m	1.16	1.120	KN/m^2
EL +938.2	25.2m	1.21	1.169	KN/m^2

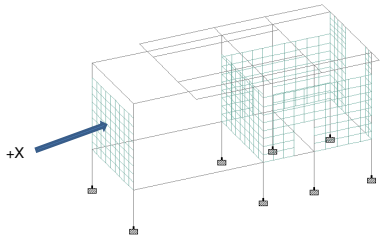
Average Pressure on wall, qz = **1.085** KN/m^2

Gust Effect Factor, G = 0.85 (For Rigid Structure)
(Ref. Section 7.2.7.1 : SBC 301-2007)

Wind Load X-Direction (From Axis 3130/3 to 3130/6)

1) Wind Load on Building Structure

Wind load on Grid 3130/5 Designed as free standing wall as per SBC 301



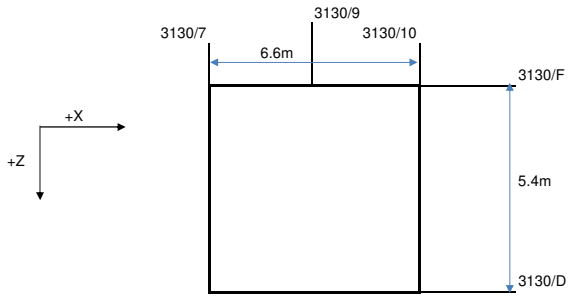
Area for Frame work at Axis 3130 / 5

Member	W (m)	L (m)	A (m^2)	Nos	Total Area (m^2)
Column	0.6	6.00	3.60	2	7.20
Tie beam	0.5	5.40	2.70	1	2.70
Beam	0.6	7.40	4.44	1	4.44
Wall	5.4	4.95	26.73	1	26.73
		Total Effective Projected Area =		A_f =	**41.07**

Height of Building frame	=	6	m
Width of frame	=	5.4	m
Frontal Area	=	32.4	m^2
M/N (Larger Dimension/Smaller Dimension)	=	1.09	
Force Co-efficient, C_f	=	1.20	(Ref. Figure 7.2-16 : SBC 301-2007)
Design Wind Load	F =	$q_z.G.C_f.A_f$	(Ref. Eq. 7.2-22 : SBC 301-2007)
Total Wind load	=	45.5	KN
Wind load per unit frontal area	=	1.40	KN/m2

Wind load in column	Influence wid	Force	Unit
At Row 3130/D	2.70	3.79	KN/m
At Row 3120/F	2.70	3.79	KN/m

Wind Load on Grid 3130/7 & 3130/10



Height of Building Frame "h"	=	6	m
Width of Frame "D"	=	5.4	m
Frontal Area	=	32.4	m^2
h/D	=	1.1	
Force Coefficient "Cf"	=	1.30	(Ref. Figure 7.2-15 : SBC 301-2007)
Design Wind Load	F =	$q_z.G.C_f.A_f$	(Ref. Eq. 7.2-22 : SBC 301-2007)
Total Wind Load	=	38.90	KN
Wind load per unit frontal area	=	1.20	KN/m2

Wind load in column	Influence wid	Force	Unit
At Row 3130/D	2.70	3.24	KN/m
At Row 3120/F	2.70	3.24	KN/m

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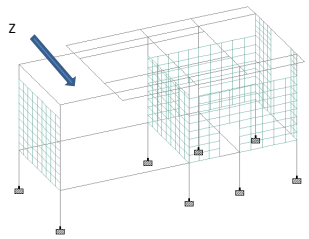
2) Wind Load on crusher

a) Tertiary Crusher Bin

Total Height of Crusher	=	4.5	m	Horizontal Load on beam	= 4.4 KN
Width of Crusher	=	2.6	m	Vertical Load on beam	= ± 3.2 KN
Area of crusher	=	11.7	m^2		
h/d	=	1.73			
Force Co-efficient, Cf	=	1.4	(Ref. Table 7.2-15 : SBC 301-2007)		
Wind pressue at Bin	=	1.09	KN/m^2		
Total Wind Load on crusher	=	17.8	KN		

Wind Load Z-Direction (From Axis 3130/F to 3130/D)

Wind load on Grid 3130/D between 3130/5 & 3130/7



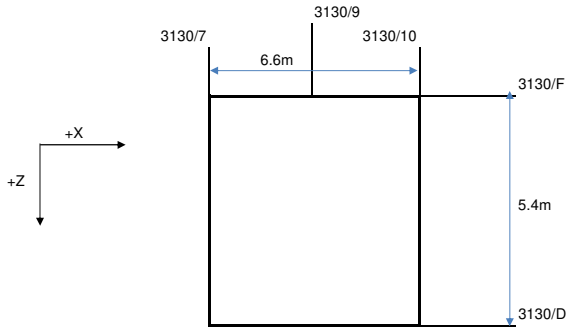
Area for Frame work at Axis 3130 / D between 3130/5 & 3130/7

Member	W (m)	L (m)	A (m^2)	Nos	Total Area (m^2)
Column	0.8	6.00	4.80	2	9.60
Tie beam	0.55	5.40	2.97	1	2.97
Beam	1.1	7.40	8.14	1	8.14
		Total Effective Projected Area =		A_f =	20.71

Height of Building frame	=	6	m
Width of frame	=	5.4	m
Frontal Area	=	32.4	m^2
Solidity Ratio	=	0.64	
Force Co-efficient, C_f	=	1.60	(Ref. Figure 7.2-16 : SBC 301-2007)
Design Wind Load	F =	$q_z.G.C_f.A_f$	(Ref. Eq. 7.2-22 : SBC 301-2007)
Total Wind load	=	30.6	KN
Wind load per unit frontal area	=	0.94	KN/m2

Wind load in column	Influence wid	Force	Unit
At Row 3130/5	2.70	2.55	KN/m
At Row 3120/7	2.70	2.55	KN/m

Wind Load on Grid 3130/D between 3130/7 & 3130/10



Height of Building Frame "h"	=	6	m
Width of Frame "D"	=	6.6	m
Frontal Area	=	39.6	m^2
h/D	=	0.9	
Force Coefficient "Cf"	=	1.30	(Ref. Figure 7.2-15 : SBC 301-2007)
Design Wind Load	F =	$q_z.G.C_f.A_f$	(Ref. Eq. 7.2-22 : SBC 301-2007)
Total Wind Load	=	47.48	KN
Wind load per unit frontal area	=	1.20	KN/m2

Wind load in column	Influence wid	Force	Unit
At Row 3130/7	1.55	1.86	KN/m
At Row 3120/9	3.30	3.96	KN/m
At Row 3120/10	1.75	2.10	KN/m

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5.5 Seismic load

Seismic Design Category		=	A	
Site class		=	B	
Spectral response acceleration (1-sec)	S1	=	0.02	
Spectral response acceleration (.02-sec	Ss	=	0.1	
Importance factor	I	=	1	(From Table 9.5, SBC 301:2007)
Response Reduction factor along X	Rx	=	2.5	(From Table 10.2, SBC 301:2007)
Response Reduction factor along Z	Rz	=	2.5	(From Table 10.2, SBC 301:2007)
Height of building	h	=	6.5	m
Time period along X	Tx	=	0.237	(eqn. 10.9.3.2-1, SBC 301:2007)
Time period along Z	Tz	=	0.237	(eqn. 10.9.3.2-1, SBC 301:2007)
Factor Fa	Fa	=	1	
Factor Fv	Fv	=	1	
SM1	Fa x S1	=	0.02	
SMS	Fa x Ss	=	0.1	
Design spectral acceleration (1-sec)	SD1 = 2/3 x SM1	=	0.0133	
Design spectral acceleration (0.2-sec)	SDS = 2/3 x SMS	=	0.0667	

The above parameters are used in the staad input to find the final seismic load of the structure as per IBC-2003

5.6 Load combination

5.6.1 Load Combination for Base Pressure

Load cases	Reference
LOAD 301 (DL + SDL1_R)	(Eq. 2.4.1-1 of SBC 301)
LOAD 302 (DL + SDL2_R)	
LOAD 303 (DL - SDL1_R)	
LOAD 304 (DL - SDL2_R)	
LOAD 305 (DL + SDL2_R) + LL	(Eq. 2.4.1-2 of SBC 301)
LOAD 306 (DL + SDL1_R) + LL	
LOAD 307 (DL - SDL2_R) + LL	
LOAD 308 (DL - SDL1_R) + LL	
LOAD 309 (DL+ SDL1_R) + 0.75 LL + 0.75 WL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 310 (DL+ SDL1_R) + 0.75 LL + 0.75 WL(-X)	
LOAD 311 (DL+ SDL1_R) + 0.75 LL + 0.75 WL(+Z)	
LOAD 312 (DL+ SDL1_R) + 0.75 LL + 0.75 WL(-Z)	
LOAD 313 (DL+ SDL2_R) + 0.75 LL + 0.75 WL(+X)	
LOAD 314 (DL+ SDL2_R) + 0.75 LL + 0.75 WL(-X)	
LOAD 315 (DL+ SDL2_R) + 0.75 LL + 0.75 WL(+Z)	
LOAD 316 (DL+ SDL2_R) + 0.75 LL + 0.75 WL(-Z)	
LOAD 317 (DL- SDL1_R) + 0.75 LL + 0.75 WL(+X)	
LOAD 318 (DL- SDL1_R) + 0.75 LL + 0.75 WL(-X)	
LOAD 319 (DL- SDL1_R) + 0.75 LL + 0.75 WL(+Z)	
LOAD 320 (DL- SDL1_R) + 0.75 LL + 0.75 WL(-Z)	
LOAD 321 (DL- SDL2_R) + 0.75 LL + 0.75 WL(+X)	
LOAD 322 (DL- SDL2_R) + 0.75 LL + 0.75 WL(-X)	
LOAD 323 (DL- SDL2_R) + 0.75 LL + 0.75 WL(+Z)	
LOAD 324 (DL- SDL2_R) + 0.75 LL + 0.75 WL(-Z)	
LOAD 325 (DL+ SDL1_R) + 0.75 LL + 0.525 EL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 326 (DL+ SDL1_R) + 0.75 LL + 0.525 EL(-X)	
LOAD 327 (DL+ SDL1_R) + 0.75 LL + 0.525 EL(+Z)	
LOAD 328 (DL+ SDL1_R) + 0.75 LL + 0.525 EL(-Z)	
LOAD 329 (DL+ SDL2_R) + 0.75 LL + 0.525 EL(+X)	
LOAD 330 (DL+ SDL2_R) + 0.75 LL + 0.525 EL(-X)	
LOAD 331 (DL+ SDL2_R) + 0.75 LL + 0.525 EL(+Z)	
LOAD 332 (DL+ SDL2_R) + 0.75 LL + 0.525 EL(-Z)	
LOAD 333 (DL- SDL1_R) + 0.75 LL + 0.525 EL(+X)	
LOAD 334 (DL- SDL1_R) + 0.75 LL + 0.525 EL(-X)	
LOAD 335 (DL- SDL1_R) + 0.75 LL + 0.525 EL(+Z)	
LOAD 336 (DL- SDL1_R) + 0.75 LL + 0.525 EL(-Z)	
LOAD 337 (DL- SDL2_R) + 0.75 LL + 0.525 EL(+X)	
LOAD 338 (DL- SDL2_R) + 0.75 LL + 0.525 EL(-X)	
LOAD 339 (DL- SDL2_R) + 0.75 LL + 0.525 EL(+Z)	
LOAD 340 (DL- SDL2_R) + 0.75 LL + 0.525 EL(-Z)	

Load cases	Reference
LOAD 341 0.6 (DL + SDL1_R) + WL(+X)	(Eq. 2.4.1-7 of SBC 301)
LOAD 342 0.6 (DL + SDL1_R) + WL(-X)	
LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)	
LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)	
LOAD 345 0.6 (DL + SDL2_R) + WL(+X)	
LOAD 346 0.6 (DL + SDL2_R) + WL(-X)	
LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)	
LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)	
LOAD 349 0.6 (DL - SDL1_R) + WL(+X)	
LOAD 350 0.6 (DL - SDL1_R) + WL(-X)	
LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)	
LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)	
LOAD 353 0.6 (DL - SDL2_R) + WL(+X)	
LOAD 354 0.6 (DL - SDL2_R) + WL(-X)	
LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)	
LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)	
LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)	
LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)	
LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)	
LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)	
LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)	
LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)	
LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)	
LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)	
LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)	
LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)	
LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)	
LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)	
LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)	
LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)	
LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)	

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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

5.6.2 Load Combination for concrete design

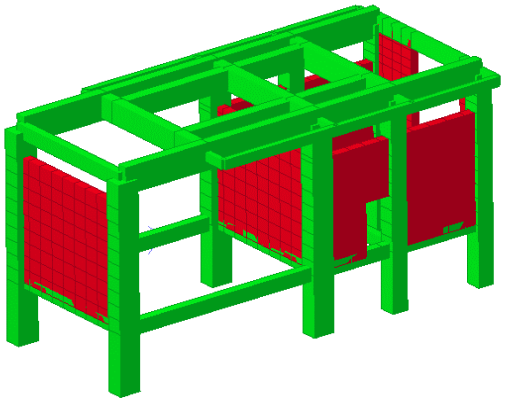
Load cases	Reference
LOAD 401 1.4 (DL+SDL1_R) LOAD 402 1.4 (DL+SDL2_R) LOAD 403 1.4 (DL-SDL1_R) LOAD 404 1.4 (DL-SDL2_R)	(Eq. 2.3.2-1 of SBC 301)
LOAD 405 1.4 (DL + SDL1_R) + 1.7 LL LOAD 406 1.4 (DL + SDL2_R) + 1.7 LL LOAD 407 1.4 (DL - SDL1_R) + 1.7 LL LOAD 408 1.4 (DL - SDL2_R) + 1.7 LL	(Eq. 2.3.2-2 of SBC 301)
LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X) LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X) LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z) LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z) LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X) LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X) LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z) LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z) LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X) LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X) LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z) LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z) LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X) LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X) LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z) LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)	(Eq. 2.3.2-4 of SBC 301)
Load cases	**Reference**
LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X) LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X) LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z) LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X) LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X) LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X) LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z) LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X) LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X) LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X) LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z) LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X) LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X) LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X) LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z) LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)	(Eq. 2.3.2-5 of SBC 301)
LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X) LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X) LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z) LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z) LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X) LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X) LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z) LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z) LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X) LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X) LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z) LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z) LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X) LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X) LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z) LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X) LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X) LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z) LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z) LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X) LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X) LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z) LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z) LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X) LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X) LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z) LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z) LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X) LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X) LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z) LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)	(Eq. 2.3.2-7 of SBC 301)

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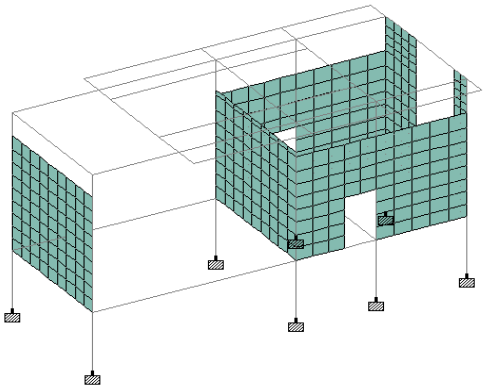
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

6. GEOMETRY

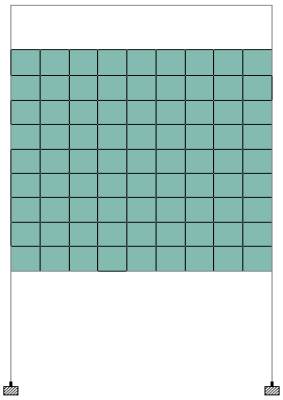
6.1 Geometry of Secondary Crusher Support Structure



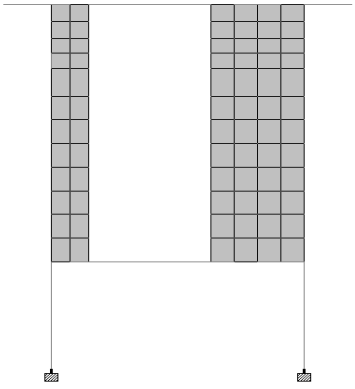
Isometric View of Structure



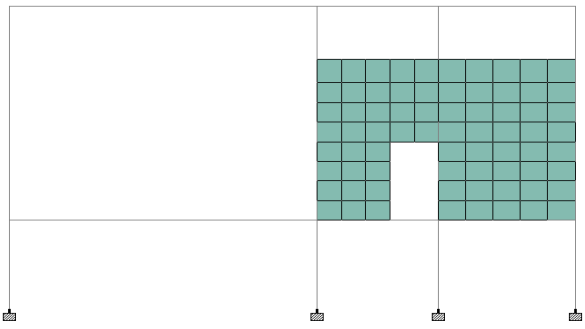
3D view of staad model



Elevation Along Axis - 3130/5



Elevation Along Axis - 3130/10



Elevation Along Row - 3130/D and 3130/J

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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7. DESIGN OF TERTIARY CRUSHER SUPPORT STRUCTURE

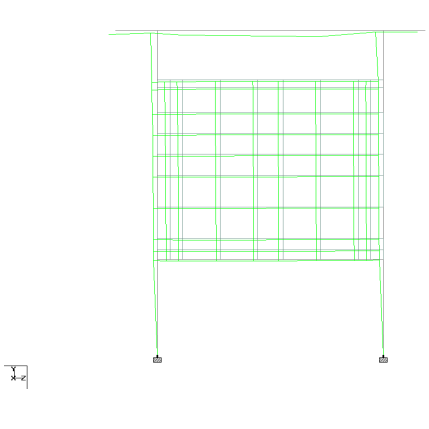
7.1 Analysis Results

7.1.1 Deflection & bending moment diagrams

Allowable deflection H/400 = 21.25 mm *As per Ma'aden Engineering Manual MD-101-SMEM-EG-ST-CRT-001 Clause & 7.8.1*

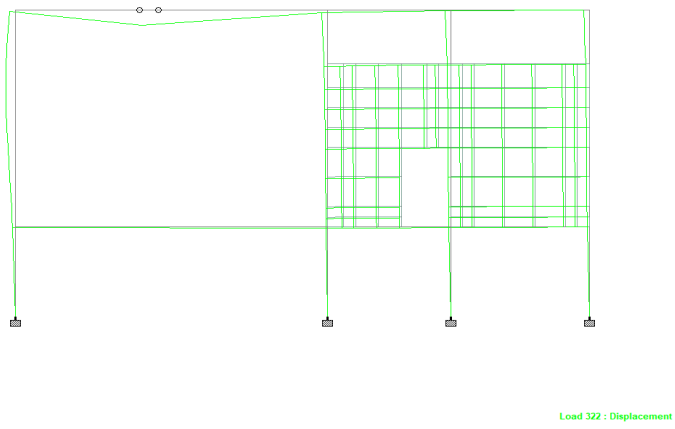
Max Deflection Diagram Along Z

Load case -320 Deflection = 1.6 mm

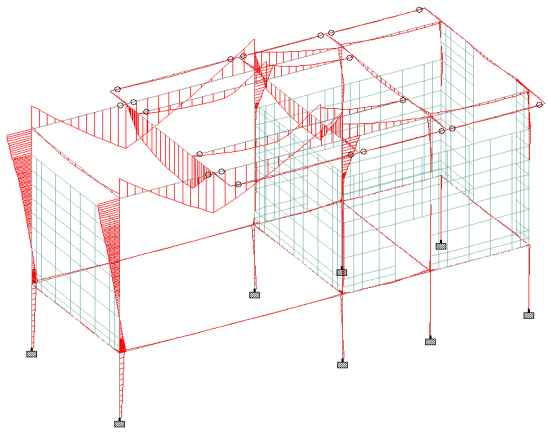


Max Deflection Diagram Along X

Load case - 322 Deflection = 1.2 mm



Bending moment diagram Load Case- 308



7.1.2 Seismic Base Shear

Design base shear along X	=	0.022 x 4898	=	107.76 KN
Design base shear along Z	=	0.022 x 4899	=	107.76 KN

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Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

7.1.3 Frequency Check Of Crusher Support Beam

Dynamic analysis of crusher foundation has been done in Ansys and all results have been reported in document number *MD-216-3130-EG-CV-CAL-1008*. Refer the same for dynamic response.

7.2 Design of column

The column of tertiary crusher support structure is design using staad (TC support structure.STD) and the reinforcements are provided accordingly.

The parameters for design are defined in the staad model.

Column Reinforcement

Member	Width(mm)	depth(mm)	Bar dia 1	Number of bars	Bar dia 2	Number of bars	Total longitudinal reinforcement	Staad Member number	Ast required as per staad (mm2)	Ast provided	link
Column C1 -Grid 3130/5-3130/D	600	800	25	16	25	4	16-25# + 4-25#	4,5	4800	9817.5	10# @ 250 c/c
Column C1 -Grid 3130/5-3130/F	600	800	25	16	25	4	16-25# + 4-25#	1,2	4800	9817.5	10# @ 250 c/c
Column C1 -Grid 3130/7-3130/D	600	800	25	16	25	4	16-25# + 4-25#	13,14	6950	9817.5	10# @ 250 c/c
Column C1 -Grid 3130/7-3130/F	600	800	25	16	25	4	16-25# + 4-25#	10,11	5942	9817.5	10# @ 250 c/c
Column C2 -Grid 3130/9-3130/D	600	600	25	8	16	8	8-25# + 8-16#	22,23	3600	5535.5	10# @ 250 c/c
Column C2 -Grid 3130/9-3130/F	600	600	25	8	16	8	8-25# + 8-16#	19,20	3600	5535.5	10# @ 250 c/c
Column C2 -Grid 3130/10-3130/D	600	600	25	8	16	8	8-25# + 8-16#	31,32	3600	5535.5	10# @ 250 c/c
Column C2 -Grid 3130/10-3130/F	600	600	25	8	16	8	8-25# + 8-16#	28,29	3600	5535.5	10# @ 250 c/c

7.2 Design of Slab

7.2.1 Design of Slab 1- 400mm thick

Slab is designed as simply supported slab along shorter span

Span of slab	=	3.4	m	max span considered	Clear cover	=	40	mm	
Thickness of slab	=	0.4	m		Bar Dia	=	12	mm	
Grade of steel	=	420	N/mm2		Unit weight of concrete	=	24	KN/m3	
Grade of concrete	=	35	N/mm2		Live load on slab	=	5	KN/m2	
Mimimum thickness of one way slab	=	span/20		*Table 7.3.1.1 SBC 304*	β_1	=	0.8		
	=	0.17	m	Hence slab thickness okay					

Load calculation on slab

Dead load from slab	=	0.4 x 24	=	9.6	KN/m2		
Live load from slab	=	5	KN/m2				
Load combination	=	1.4DL + 1.7LL	=	1.4 x 9.6 + 1.7 x 5	=	21.9	KN/m2

Calculation of bending moment and shear force

BM at midspan	=	$wl^2/8$	=	31.7	KNm	
SF at support	=	wl/2	=	37.3	KN	

Calculation of flexural reinforcement

					Effective Depth Along X	=	354	mm
	M	=	31.70	KNm				
	M_{lim}	=	$\emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$		For tension controlled section	$\left(\frac{c}{d}\right)_{max}$ = $\frac{0.003}{0.003 + 0.005}$	=	0.375
		=	855.61	KNm				
$M_{lim} > M$	Tension controlled section				Min reinforcement	A_{stmin} =	$\frac{0.0018 \times 420}{f_y} bd$	*Table 7.6.1.1 SBC 304*
Therefore	Φ	=	0.9	*Table 21.2.2 SBC 304*	greater of or			
						A_{stmin} =	0.0014bd	
	A_{st}	=	$\frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$			A_{stmin} =	637.2	mm^2
						A_{req} =	237.8	mm^2
		=	237.8	mm^2		Required Spacing =	177.5	mm
	Provide	***12mm dia @***	***150***	***c/c***	***Top & bottom both directions***	A_{prov} =	754.0	mm^2

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Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Shear Check

Lesser of the following

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — Table 22.5.5.1 SBC 304

Reinforcement ratio	ρ_w	=	0.0021				λ	=	1	Normal concrete
Effective Depth	d	=	354	mm			Mu	=	31.70	KNm
Shear Force	V_u	=	37.30	KN			φ	=	0.75	for shear
Allowable shear strength	φV_c	=	255.32	KN	O.K		φV_c	=	255.32	KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

φV_c = 260.93 KN — Table 22.5.5.1 SBC 304

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

φV_c = 455.51 KN

7.2.2 Design of Slab 2- 200mm thick

Slab is designed as simply supported slab along shorter span

Span of slab	=	3.5	m	max span considered			Clear cover	=	40	mm
Thickness of slab	=	0.2	m				Bar Dia	=	12	mm
Grade of steel	=	420	N/mm2				Unit weight of concrete	=	24	KN/m3
Grade of concrete	=	35	N/mm2				Live load on slab	=	5	KN/m2
Mimimum thickness of one way slab	=	span/20		Table 7.3.1.1 SBC 304			β_1	=	0.8	
	=	0.175	m	Hence slab thickness okay						

Load calculation on slab

Dead load from slab	=	0.2 x 24	=	4.8	KN/m2		
Live load from slab	=	5	KN/m2				
Load combination	=	1.4DL + 1.7LL	=	1.4 x 4.8 + 1.7 x 5	=	15.2	KN/m2

Calculation of bending moment and shear force

BM at midspan	=	wl2/8	=	23.3	KNm		
SF at support	=	wl/2	=	26.6	KN		

Calculation of flexural reinforcement

Effective Depth Along X = 154 mm

M = 23.31 KNm

$M_{lim} = \emptyset\ 0.85\ f'_c\beta_1\left(\frac{c}{d}\right)_{max}\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)bd^2$ = 161.92 KNm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003+0.005} = 0.375$

$M_{lim} > M$ Tension controlled section

Therefore Φ = 0.9 — Table 21.2.2 SBC 304

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1-\sqrt{1-\frac{2.35M_u}{\emptyset f'_c bd^2}}\right)$ = 407.5 mm^2

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y}bd$ — Table 7.6.1.1 SBC 304

greater of or

A_{stmin} = 0.0014bd

A_{stmin} = 277.2 mm^2

A_{req} = 407.5 mm^2

Required Spacing = 277.6 mm

Provide 12mm dia @ 150 c/c Top & bottom both directions

A_{prov} = 754.0 mm^2

Shear Check

Lesser of the following

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — Table 22.5.5.1 SBC 304

Reinforcement ratio	ρ_w	=	0.0049				λ	=	1	Normal concrete
Effective Depth	d	=	154	mm			Mu	=	23.31	KNm
Shear Force	V_u	=	26.64	KN			φ	=	0.75	for shear
Allowable shear strength	φV_c	=	111.02	KN	O.K		φV_c	=	111.02	KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

φV_c = 118.94 KN — Table 22.5.5.1 SBC 304

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

φV_c = 198.16 KN

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Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4 Design of Beam

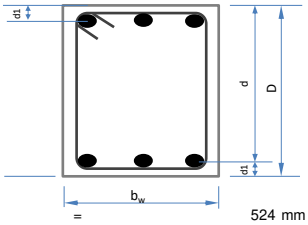
7.4.1 Beam reinforcement summary

Beam mark	Width	Depth	Top reinforcement	Bottom reinforcement	Stirrup
FB1	300	600	2no.- D32 + 1no.- D25	3no.- D25	2L-10mm @ 200c/c
FB2	400	900	4no.- D32	2no.- D32 + 2no.- D25	4L-12mm @ 125c/c
FB3	350	1100	4no.- D32	4no.- D32 + 4no.- D16	4L-10mm @ 200c/c
B1	400	900	4no.- D20	4no.- D25 + 4no.- D25	4L-12mm @ 150c/c
B2	400	800	2no.- D32 + 2no.- D25	4no.- D20	4L-10mm @ 150c/c
TB1	300	500	2no.- D25 + 1no.- D16	3no.- D20	2L-10mm @ 200c/c
TB2	300	550	2no.- D25 + 1no.- D16	2no.- D25 + 1no.- D16	2L-10mm @ 200c/c

7.4.2 Beam marked FB1

Staad beam number - 9,36, 509, 527

Width of beam	=	**300**	mm	
Depth of beam	=	**600**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	524	mm	
Dia of longitudinal bar	=	**32**	mm	
Dia of stirrup bar	=	**10**	mm	



d	=	524	mm
d'	=	76	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	407	Staad beam number -	9	=	**114**	KNm
B.M at span	Mz	L/C -	406	Staad beam number -	9	=	**69**	KNm
Shear Force	Fy	L/C -	407	Staad beam number -	9	=	**102**	KN
Torsion	Mx					=		

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 524 x 524 (1 - 0.8/2 x 0.375)
= 562.4 KNm

max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 114.4 KNm

Mu_2 = Mu - Mu_{lim} = 114.4 - 562.5
= 0.00 KNm

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d-d')}$

= 0.85 x 35x 300 x 524 x (1-√(1-(2.35 x 114.4 x 10^6)/(0.9 x 35 x 300 x 524^2)))/420
= 592.6 mm^2

= 0 x 10^6/(0.9 x 420 x (524 - 76))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0018
Asc = 0.0000 mm^2

= 553.6 mm^2 or 524 mm^2 = 553.6 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 592.7 + 0.00 + 0
= 592.7 mm^2

Top layer1	**2**	*nos*	**32**	*mm*	*dia*	+	**1**	*nos*	**25**	*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 2099.369291 mm^2 Safe

ρ_w = 0.01335477

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Span Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 524 x 524 (1 - 0.8/2 x 0.375)

= 562.4 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 69.3 KNm

Mu_2 = Mu - Mu_{lim} = 69.3 - 562.5

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 524 x (1-√(1-(2.35 x 69.3 x 10^6)/(0.9 x 35 x 300 x 524^2)))/420

= 355.1 mm^2

= 0 x 10^6/(0.9 x 420 x (524 - 76))

= 0 mm^2

e_{sc} = 0.0018

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm^2

= 553.6 mm^2 or 524 mm^2 = 553.57604 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 553.6 + 0 + 0

= 553.6 mm^2

bottom layer1	**3**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1472.621556 mm^2 Safe

ρ_w = 0.009367822

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 165.4 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 184.5 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 269.7 KN

fV_c = 0.75 x 165.5 = 124.08 KN > 101.8 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((101.8-124.1) x 10^3)/(0.75 x 420 x 524) = 0 mm^2/mm

No. of legs = **2** Spacing provided = **200** mm

Provide 2L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement		s =	0 mm	*if Vu > φVc*
ii) Min Asv	= $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s =	599.5455737 mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv	= $0.65\frac{b_w s}{f_{yt}}$	s =	338.3253627 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s =	262 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s =	0 mm	

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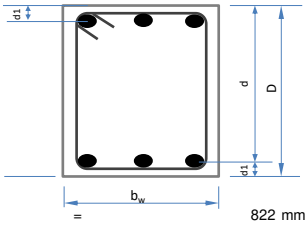
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REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4.3 Beam marked FB2 Staad beam number - 18,59,62,27,64,66, 515, 521, 544, 547, 549, 551

Width of beam	=	**400**	mm	
Depth of beam	=	**900**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	822	mm	
Dia of longitudinal bar	=	**32**	mm	
Dia of stirrup bar	=	**12**	mm	



d	=	822	mm
d'	=	78	mm
h	=	124	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	405	Staad beam number -	62	=	**688**	KNm
B.M at span	Mz	L/C -	406	Staad beam number -	62	=	**500**	KNm
Shear Force	Fy	L/C -	406	Staad beam number -	18	=	**968**	KN
Torsion	Mx					=		

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 822 x 822 (1 - 0.8/2 x 0.375)
= 1845.3 KNm
Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 687.6 KNm

Mu_2 = Mu - Mu_{lim} = 687.6 - 1845.4
= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 400 x 822 x (1-√(1-(2.35 x 687.6 x 10^6)/(0.9 x 35 x 400 x 822^2)))/420
= 2326.374738 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (822 - 78))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0022
Asc = 0.0000 mm^2

= 1157.9 mm^2 or 1096.00 mm^2 = 1157.8613 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 2326.4 + 0.00 + 0
= 2326.4 mm^2

Top layer1	**4**	*nos*	**32**	*mm*	*dia*	+		*nos*		*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 3217.0 mm^2 Safe

ρ_w = 0.00978404

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Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Span Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 822 x 822 (1 - 0.8/2 x 0.375)

= 1845.3 KNm

Singly reinforced beam

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 500.4 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 400 x 822 x (1-√(1-(2.35 x 500.4 x 10^6)/(0.9 x 35 x 400 x 822^2)))/420

= 1668.2 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_wd$ or $\frac{1.4}{f_y}b_wd$

= 1157.861329 mm^2 or 1096 mm^2 = 1157.8613 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1668.3 + 0 + 0

= 1668.3 mm^2

Mu_2 = Mu - Mu_{lim} = 500.4 - 1845.4

= 0 KNm

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (822 - 78))

= 0 mm^2

e_{sc} = 0.00224088

Asc = 0.0000 mm^2

bottom layer1	**2**	*nos*	**32**	*mm*	*dia*	+	**2**	*nos*	**25**	*mm* *dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 2590.2 mm^2 Safe

ρ_w = 0.0079

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

$V_c = \left(0.16\lambda\sqrt{f_c'}b_wd+17\rho_w\left(\frac{V_ud}{M_u}\right)\right)b_wd$ = 374.5 KN

λ = 1 Normal concrete

$V_c = \left(0.16\lambda\sqrt{f_c'}b_wd+17\rho_w\right)b_wd$ = 365.9 KN

$V_c = 0.29\lambda\sqrt{f_c'}b_wd$ = 564.1 KN

fV_c = 0.75 x 366 = 274.4 KN < 968 KN *Shear Reinforcement required*

Vs = $V_U-\emptyset V_C = \frac{\emptyset\ A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U-\emptyset V_C}{\emptyset\ f_y d}$ = ((968-274.5) x 10^3)/(0.75 x 420 x 822) = 2.679 mm^2/mm

No. of legs = **4** Spacing provided = **125** mm

Provide 4L -12 mm stirrups @ 125 C/C

Spacing of stirrup is minimum of the following

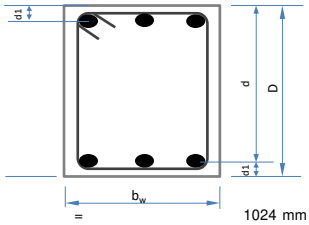
i) Shear Reinforcement		s =	168.9 mm	*if Vu > φVc*
ii) Min Asv or Min Asv	= $0.062\sqrt{f_c'}\frac{b_ws}{f_t}$ = $0.65\frac{b_ws}{f_{yt}}$	s =	1295.0 mm	*Corresponding to table 9.6.3.3 of SBC 304*
		s =	730.8 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s =	411 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s =	0 mm	

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Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4.4 Beam marked FB3 Staad beam number - 49,55,50,51,52,56,53,54, 534-541



Width of beam	=	**350**	mm		
Depth of beam	=	**1100**	mm		
Grade of steel	=	420	MPa		
Grade of concrete	=	35	MPa		
Cover	=	50	mm		
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*	
Effective Depth	=	1024	mm		
Dia of longitudinal bar	=	**32**	mm		
Dia of stirrup bar	=	**10**	mm		

d	=	1024	mm
d'	=	76	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	406	Staad beam number -	55	=	**848**	KNm
B.M at span	Mz	L/C -	405	Staad beam number -	49	=	**1154**	KNm
Shear Force	Fy	L/C -	405	Staad beam number -	49	=	**574**	KN
Torsion	Mx					=		

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 1024 x 1024 (1 - 0.8/2 x 0.375)

= 2505.7 KNm

Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 847.5 KNm

Mu_2 = Mu - Mu_{lim} = 847.5 - 2505.8

= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 350 x 1024 x (1-√(1-(2.35 x 847.5 x 10^6)/(0.9 x 35 x 350 x 1024^2)))/420

= 2290.068991 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (1024 - 76))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0024

Asc = 0.0000 mm^2

= 1262.09702 mm^2 or 1194.666667 mm^2 = 1262.097 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 2290.1 + 0.00 + 0

= 2290.1 mm^2

Top layer1	**4**	*nos*	**32**	*mm*	*dia*	+		*nos*		*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 3216.990877 mm^2 Safe

ρ_w = 0.00897598

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Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
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Span Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 350 x 1024 x 1024 (1 - 0.8/2 x 0.375)

= 2505.7 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 1154.4 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 350 x 1024 x (1-√(1-(2.35 x 1154.4 x 10^6)/(0.9 x 35 x 350 x 1024^2)))/420

= 3177.519881 mm^2

Mu_2 = Mu - Mu_{lim} = 1154.4 - 2505.8

= 0 KNm

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (1024 - 76))

= 0 mm^2

e_{sc} = 0.00240625

Asc = 0.0000 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_wd$ or $\frac{1.4}{f_y}b_wd$

= 1262.09702 mm^2 or 1194.666667 mm^2 = 1262.097 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 3177.6 + 0 + 0

= 3177.6 mm^2

bottom layer1	**4**	*nos*	**32**	*mm*	*dia*	+		*nos*		*mm* *dia*
bottom layer2	**4**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 4021.238597 mm^2 Safe

ρ_w = 0.011219974

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc *Correspo*

$V_c = \left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_ud}{M_u}\right)\right)b_wd$ = 377.181 KN

λ = 1 Normal concrete

$V_c = \left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_wd$ = 393.941 KN

$V_c = 0.29\lambda\sqrt{f_c'}b_wd$ = 614.894 KN

fV_c = 0.75 x 377.2 = 282.8854766 KN < 574 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset A_v f_y d}{s}$

$\frac{A_v}{S} = \frac{V_U - \emptyset V_C}{\emptyset f_y d}$ = ((574-282.9) x 10^3)/(0.75 x 420 x 1024) = 0.902512783 mm^2/mm

No. of legs = **4** Spacing provided = **200** mm

Provide 4L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	348.0939784	mm	*if Vu > φVc*
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_ws}{f_t}$ or Min Asv = $0.65\frac{b_ws}{f_{yt}}$	s =	1027.792412	mm	*Corresponding to table 9.6.3.3 of SBC 304*
	s =	579.986336	mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2	s =	512	mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion	s =	0	mm	

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Project Country	**KINGDOM OF SAUDI ARABIA**

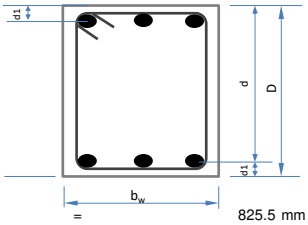
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REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
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7.4.5 Beam marked B1

Staad beam number - 57,58,61



Width of beam	=	**400**	mm		
Depth of beam	=	**900**	mm		
Grade of steel	=	420	MPa		
Grade of concrete	=	35	MPa		
Cover	=	50	mm		
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*	
Effective Depth	=	825.5	mm		
Dia of longitudinal bar	=	**25**	mm		
Dia of stirrup bar	=	**12**	mm		

d	=	825.5	mm
d'	=	74.5	mm
h	=	124	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	408	Staad beam number -	61	=	**33**	KNm
B.M at span	Mz	L/C -	405	Staad beam number -	58	=	**846**	KNm
Shear Force	Fy	L/C -	405	Staad beam number -	57	=	**785**	KN
Torsion	Mx					=		

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 825.5 x 825.5 (1 - 0.8/2 x 0.375)
= 1861.1 KNm

Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 32.9 KNm

Mu_2 = Mu - Mu_{lim} = 32.9 - 1861.1
= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 400 x 825.5 x (1-√(1-(2.35 x 32.9 x 10^6)/(0.9 x 35 x 400 x 825.5^2)))/420
= 105.541869 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (825.5 - 74.5))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0023
Asc = 0.0000 mm^2

= 1162.8 mm^2 or 1100.7 mm^2 = 1162.7914 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 1162.8 + 0.00 + 0
= 1162.8 mm^2

Top layer1	**4**	*nos*		**20**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*			*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1256.64 mm^2 Safe

ρ_w = 0.0038

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Project Country	**KINGDOM OF SAUDI ARABIA**

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Span Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 825.5 x 825.5 (1 - 0.8/2 x 0.375)

= 1861.1 KNm

Singly reinforced beam

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 845.7 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 400 x 825.5 x (1-√(1-(2.35 x 845.7 x 10^6)/(0.9 x 35 x 400 x 825.5^2)))/420

= 2884.748372 mm²

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_wd$ or $\frac{1.4}{f_y}b_wd$

= 1162.8 mm² or 1100.67 mm² = 1162.8 mm² *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 2884.8 + 0 + 0

= 2884.8 mm²

Mu_2 = Mu - Mu_{lim} = 845.7 - 1861.1

= 0 KNm

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (825.5 - 74.5))

= 0 mm²

e_{sc} = 0.00227801

Asc = 0.0000 mm²

bottom layer1	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*
bottom layer2	**4**	*nos*	**25**	*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 3927.0 mm² Safe

ρ_w = 0.011892764

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc *Correspor*

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_ud}{M_u}\right)\right)b_wd$ = 733.1 KN λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_wd$ = 333.9 KN

V_c = $0.29\lambda\sqrt{f_c'}b_wd$ = 566.5 KN

fV_c = 0.75 x 334 = 250.440868 KN < 784.6 KN *Shear Reinforcement required*

Vs = $V_U-\emptyset V_C$ = $\frac{\emptyset\ A_v f_y d}{s}$

$\frac{A_v}{S}$ = $\frac{V_U-\emptyset V_C}{\emptyset\ f_y d}$ = ((784.6-250.5) x 10^3)/(0.75 x 420 x 825.5) = 2.054201425 mm²/mm

No. of legs = **4** Spacing provided = **150** mm

Provide 4L -12 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement s = 220.2263792 mm *if Vu > φVc*

ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_ws}{f_t}$ s = 1295.018439 mm *Corresponding to table 9.6.3.3 of SBC 304*

or

Min Asv = $0.65\frac{b_ws}{f_{yt}}$ s = 730.7827834 mm *Corresponding to table 9.6.3.3 of SBC 304*

iii) effective depth/2 s = 412.75 mm *Corresponding to table 9.7.6.2.2 of SBC 304*

iv) Combined shear & torsion s = 0 mm

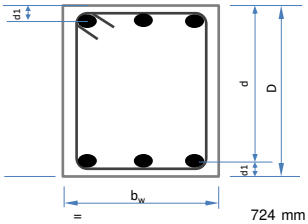
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Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4.6 Beam marked B2

Staad beam number - 60,63,65,67, 545, 548, 550, 552



Width of beam	=	**400**	mm		
Depth of beam	=	**800**	mm		
Grade of steel	=	420	MPa		
Grade of concrete	=	35	MPa		
Cover	=	50	mm		
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*	
Effective Depth	=	724	mm		
Dia of longitudinal bar	=	**32**	mm		
Dia of stirrup bar	=	**10**	mm		

d	=	724	mm
d'	=	76	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	406	Staad beam number -	60	=	**497**	KNm
B.M at span	Mz	L/C -	405	Staad beam number -	63	=	**211**	KNm
Shear Force	Fy	L/C -	406	Staad beam number -	63	=	**529**	KN
Torsion	Mx					=		

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 724 x 724 (1 - 0.8/2 x 0.375)

= 1431.6 KNm

Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 496.9 KNm

Mu_2 = Mu - Mu_{lim} = 496.9 - 1431.6

= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 400 x 724 x (1-√(1-(2.35 x 496.9 x 10^6)/(0.9 x 35 x 400 x 724^2)))/420

= 1901.541997 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (724 - 76))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0022

Asc = 0.0000 mm^2

= 1019.819467 mm^2 or 965.333333 mm^2 = 1019.8195 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1901.6 + 0.00 + 0

= 1901.6 mm^2

Top layer1	**2**	*nos*	**32**	*mm*	*dia*	+	**2**	*nos*	**25**	*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 2590.243143 mm^2 Safe

ρ_w = 0.00894421

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Approved	AG	AG	AG	AG	

Span Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 400 x 724 x 724 (1 - 0.8/2 x 0.375)

= 1431.6 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 210.8 KNm

Mu_2 = Mu - Mu_{lim} = 210.8 - 1431.6

= 0 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 400 x 724 x (1-√(1-(2.35 x 210.8 x 10^6)/(0.9 x 35 x 400 x 724^2)))/420

= 784.2954634 mm²

= 0 x 10^6/(0.9 x 420 x (724 - 76))

= 0 mm²

e_{sc} = 0.00216022

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

Asc = 0.0000 mm²

= 1019.819467 mm² or 965.333333 mm² = 1019.8195 mm² *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 1019.9 + 0 + 0

= 1019.9 mm²

bottom layer1	**4**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm*	*dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1256.637061 mm² Safe

ρ_w = 0.004339216

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc *Correspor*

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 308.093 KN λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 318.162 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 496.856 KN

fV_c = 0.75 x 308.1 = 231.07002 KN < 529.4 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((529.4-231.1) x 10^3)/(0.75 x 420 x 724) = 1.308120583 mm²/mm

No. of legs = **4** Spacing provided = **150** mm

Provide 4L -10 mm stirrups @ 150 C/C

Spacing of stirrup is minimum of the following

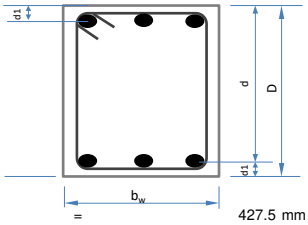
i) Shear Reinforcement		s	=	240.1607845 mm	*if Vu > φVc*
ii) Min Asv or Min Asv	= $0.062\sqrt{f_c'}\frac{b_w S}{f_t}$ = $0.65\frac{b_w S}{f_{yt}}$	s	=	899.3183605 mm	*Corresponding to table 9.6.3.3 of SBC 304*
		s	=	507.488044 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s	=	362 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s	=	0 mm	

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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4.7 Beam marked TB1 Staad beam number - 7,16,25,34, 306-321, 508, 514, 520, 526, 624-639



Width of beam	=	**300**	mm		
Depth of beam	=	**500**	mm		
Grade of steel	=	420	MPa		
Grade of concrete	=	35	MPa		
Cover	=	50	mm		
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*	
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*	
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*	
Effective Depth	=	427.5	mm		
Dia of longitudinal bar	=	**25**	mm		
Dia of stirrup bar	=	**10**	mm		

d	=	427.5	mm
d'	=	72.5	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	411	Staad beam number -	321	=	**71**	KNm
B.M at span	Mz	L/C -	420	Staad beam number -	321	=	**29**	KNm
Shear Force	Fy	L/C -	405	Staad beam number -	321	=	**143**	KN
Torsion	Mx							

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)
= 374.3 KNm
Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 70.8 KNm

Mu_2 = Mu - Mu_{lim} = 70.8 - 374.4
= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 70.8 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420
= 448.6641389 mm^2

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0016
Asc = 0.0000 mm^2

= 451.6293049 mm^2 or 427.5 mm^2 = 451.6293 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 451.7 + 0.00 + 0
= 451.7 mm^2

Top layer1	**2**	*nos*	**25**	*mm*	*dia*	+	**1**	*nos*	**16**	*mm*	*dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm*	*dia*

Ast provided = 1182.809634 mm^2 Safe

ρ_w = 0.00922269

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
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Span Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 427.5 x 427.5 (1 - 0.8/2 x 0.375)

= 374.3 KNm

Singly reinforced beam

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 29.3 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 300 x 427.5 x (1-√(1-(2.35 x 29.3 x 10^6)/(0.9 x 35 x 300 x 427.5^2)))/420

= 182.9327089 mm²

Mu_2 = Mu - Mu_{lim} = 29.3 - 374.4

= 0 KNm

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (427.5 - 72.5))

= 0 mm²

e_{sc} = 0.00164327

Asc = 0.0000 mm²

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

= 451.6293049 mm² or 427.5 mm² = 451.6293 mm²

clause 9.6.1.2 of SBC 304

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 451.7 + 0 + 0

= 451.7 mm²

bottom layer1	**3**	*nos*	**20**	*mm*	*dia*	+		*nos*		*mm* *dia*
bottom layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 942.4777961 mm² Safe

ρ_w = 0.007348755

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc *Correspo*

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ = 138.77 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ = 141.5 KN

V_c = $0.29\lambda\sqrt{f_c'}b_w d$ = 220.0 KN

fV_c = 0.75 x 138.8 = 104.079161 KN < 143.1 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U - \emptyset V_C}{\emptyset\ f_y d}$ = ((143.1-104.1) x 10^3)/(0.75 x 420 x 427.5) = 0.289767668 mm²/mm

No. of legs = **2** Spacing provided = **200** mm

Provide 2L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s =	542.0881977 mm	*if Vu > φVc*
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$ or Min Asv = $0.65\frac{b_w s}{f_{yt}}$	s =	599.5455737 mm	*Corresponding to table 9.6.3.3 of SBC 304*
	s =	338.3253627 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2	s =	213.75 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion	s =	0 mm	

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Project Country	**KINGDOM OF SAUDI ARABIA**

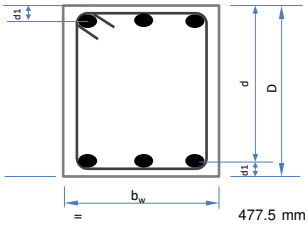
Ma'aden Revision No:	A01	C01	C02	D01
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023
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Approved	AG	AG	AG	AG

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

7.4.8 Beam marked TB2

Staad beam number - 37,38,39,40,41,42, 82-97, 528-533, 560-575



Width of beam	=	**300**	mm	
Depth of beam	=	**550**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	477.5	mm	
Dia of longitudinal bar	=	**25**	mm	
Dia of stirrup bar	=	**10**	mm	

d	=	477.5 mm
d'	=	72.5 mm
h	=	120 mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	406	Staad beam number -	85	=	**82**	KNm
B.M at span	Mz	L/C -	406	Staad beam number -	83	=	**43**	KNm
Shear Force	Fy	L/C -	408	Staad beam number -	38	=	**107**	KN
Torsion	Mx					=		

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 477.5 x 477.5 (1 - 0.8/2 x 0.375)
= 467.0 KNm
Singly reinforced beam

max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 81.5 KNm

Mu_2 = Mu - Mu_{lim} = 81.5 - 467.1
= 0.00 KNm

A_{st1} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\ f_s'(d-d')}$

= 0.85 x 35x 300 x 477.5 x (1-√(1-(2.35 x 81.5 x 10^6)/(0.9 x 35 x 300 x 477.5^2)))/420
= 461.47 mm^2

= 0 x 10^6/(0.9 x 420 x (477.5 - 72.5))
= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_w d$ or $\frac{1.4}{f_y}b_w d$

e_{sc} = 0.0018
Asc = 0.0000 mm^2

= 504.4514458 mm^2 or 477.5 mm^2 = 504.45145 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}
= 504.5 + 0.00 + 0
= 504.5 mm^2

Top layer1	**2**	*nos*	**25**	*mm*	*dia*	+	**1**	*nos*	**16**	*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 1182.809634 mm^2 Safe

ρ_w = 0.00825696

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Approved	AG	AG	AG	AG	

Span Moment

Mu lim = $\emptyset\ 0.85\ f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 477.5 x 477.5 (1 - 0.8/2 x 0.375)

= 467.0 KNm

Singly reinforced beam

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.375

b_1 = 0.80

Mu = 42.9 KNm

A_{st} = $\frac{0.85\ f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\ M_u}{\emptyset\ f_c'bd^2}}\right)$

= 0.85 x 35x 300 x 477.5 x (1-√(1-(2.35 x 42.9 x 10^6)/(0.9 x 35 x 300 x 477.5^2)))/420

= 240.2262814 mm²

Mu_2 = Mu - Mu_{lim} = 42.9 - 467.1

= 0 KNm

A_{st2} = $\frac{M_u - M_{u\ lim}}{\emptyset\ f_s'(d-d')}$

= 0 x 10^6/(0.9 x 420 x (477.5 - 72.5))

= 0 mm²

e_{sc} = 0.00178534

Asc = 0.0000 mm²

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y}b_wd$ or $\frac{1.4}{f_y}b_wd$

= 504.4514458 mm² or 477.5 mm² = 504.45145 mm²

clause 9.6.1.2 of SBC 304

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 504.5 + 0 + 0

= 504.5 mm²

bottom layer1	2	nos	25	mm	dia	+	1	nos	16	mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 1182.809634 mm² Safe

ρ_w = 0.008256961

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Correspo

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_ud}{M_u}\right)\right)b_wd$ = 148.214 KN

λ = 1 Normal concrete

V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_wd$ = 155.704 KN

V_c = $0.29\lambda\sqrt{f_c'}b_wd$ = 135.597 KN

fV_c = 0.75 x 135.6 = 101.697411 KN < 107.1 KN *Shear Reinforcement required*

Vs = $V_U - \emptyset V_C$ = $\frac{\emptyset\ A_V f_y d}{s}$

$\frac{A_V}{S}$ = $\frac{V_U-\emptyset V_C}{\emptyset\ f_y d}$ = ((107.1-101.7) x 10^3)/(0.75 x 420 x 477.5) = 0.035918481 mm²/mm

No. of legs = 2 Spacing provided = 200 mm

Provide 2L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement	s	=	4373.225932 mm	*if Vu > φVc*
ii) Min Asv = $0.062\sqrt{f_c'}\frac{b_ws}{f_t}$ or Min Asv = $0.65\frac{b_ws}{f_{yt}}$	s	=	599.5455737 mm	*Corresponding to table 9.6.3.3 of SBC 304*
	s	=	338.3253627 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2	s	=	238.75 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion	s	=	0 mm	

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Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
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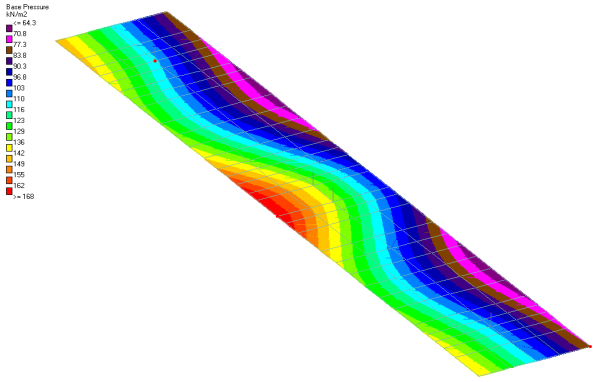
Outotec / L&T Revision No.:	0	1	2	3	
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REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report

7.5.1 Design of combined foundation for column at 3130/J_3130/5 , 3130/G_3130/5 , 3130/F_3130/5 & 3130/D_3130/5 and foundation mkd. as F1A

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **FDN F-1A GR-5 R2.STD**

Staad Support Node for column at **3130/J_3130/5 425 3130/F_3130/5 1**
3130/G_3130/5 422 3130/D_3130/5 5



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	0.5	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	14.5	m	
Width of footing	Lx	=	2.2	m	
Thickness of footing	D	=	0.5	m	
Depth of pedestal	Bx	=	0.8	m	
Width of pedestal	Bz	=	0.6	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	27	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	168.5	KN/m^2	Load Case 322
Minimum Base Pressure		=	30.7	KN/m^2	Load Case 352

2) Design of foundation Raft :

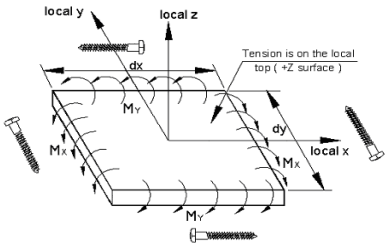
Slab Thickness =	500 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	trength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	ength Reduction Factor of Shear Reinforcement , Φt =	0.75

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :
(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	195.900	kN-m	Corresponding Mxy	1.700	kN-m	*(Plate No. - 152 , L/C No. - 405)*
Min. Mx	-188.500	kN-m	Corresponding Mxy	-8.800	kN-m	*(Plate No. - 161 , L/C No. - 405)*
Max. My	1.600	kN-m	Corresponding Mxy	16.100	kN-m	*(Plate No. - 89 , L/C No. - 454)*
Min. My	-33.000	kN-m	Corresponding Mxy	0.400	kN-m	*(Plate No. - 161 , L/C No. - 405)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	195.9 + 1.7 =	197.600	kN-m
My1 =	My + abs(Mxy) =	1.6 + 16.1 =	17.700	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	195.9 + 1.806 =	197.706	kN-m
My2 =	My+abs(Mxy^2/Mx) =	1.6 + 1.323 =	2.923	kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 197.6 kN-m
Myd = 17.7 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 197.6 kN-m
Φ*Kn = Mu/bd^2 deff. = 412.5 mm
Mu/bd^2 = 197.6X10^6/(1000X412.5^2) = 1.161 N/mm^2 < 6.9 N/mm^2
Pt = 0.0031 Reqd. Ast = 031X1000X412.5/100 = 12.79 cm^2
Provided **25** mm @ **200** C/C (Provide Ast = 24.54)cm^2

Total Provide Ast = 24.54 cm^2 **O.K.**
Total Provide Pt = 0.0059

ALONG LOCAL Y - DIRECTION :

Mu = 17.7 kN-m
Φ*Kn = Mu/bd^2 deff. = 392 mm
Mu/bd^2 = 17.7X10^6/(1000X392^2) = 0.104 N/mm^2 < 6.9 N/mm^2
Pt = 0.0003 Reqd. Ast = 003X1000X412.5/100 = 1.18 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 **O.K.**
Total Provide Pt = 0.0026

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -188.5 - 8.8 = -197.3 kN-m
My1 = My - abs(Mxy) = -33 - 0.4 = -33.4 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -188.5 - 2.347 = -190.847 kN-m
My2 = My-abs(Mxy^2/Mx) = -33 - 0.001 = -32.999 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -197.3 kN-m
Myd = -33.4 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 197.3 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 197.3X10^6/(1000X412.5^2) = 1.16 N/mm^2 deff. = 412.5 mm < 6.9 N/mm^2
Pt = 0.0031 Reqd. Ast = 031X1000X412.5/100 = 12.79 cm^2
Provided **25** mm @ **200** C/C (Provide Ast = 24.54)cm^2

Total Provide Ast = 24.54 cm^2 **O.K.**
Total Provide Pt = 0.0059

ALONG LOCAL Y - DIRECTION :

Mu = 33.4 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 33.4X10^6/(1000X392^2) = 0.217 N/mm^2 deff. = 392 mm < 6.9 N/mm^2
Pt = 0.0006 Reqd. Ast = .0006X1000X392/100 = 2.35 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 **O.K.**
Total Provide Pt = 0.0026

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.5 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 85 , L/C No. - 406)*
Vu = 206.25 kN

Provided Pt = 0.0059
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0059X206.25X10^3X412.5/197.3/10^6)X1000X412.5/10^3
= 408.302 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0059)X1000X412.5/10^3
= 431.835 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X412.5/10^3
= 707.711 kN
HENCE, Vc = 408.302 kN
Φt*Vc = 0.75X408.302 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 306.227 kN **> 206.25KN , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.3 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 33 , L/C No. - 408)*
Vu = 117.6 kN

Pt provided = 0.0026
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0026X117.6X10^3X392/33.4/10^3)X1000X392/10^3
= 394.971 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0026)X1000X392/10^3
= 388.383 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X392/10^3
= 672.54 kN
HENCE, Vc = 388.383 kN
Φt*Vc = 0.75X388.383 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 291.287 kN **> 117.6KN , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

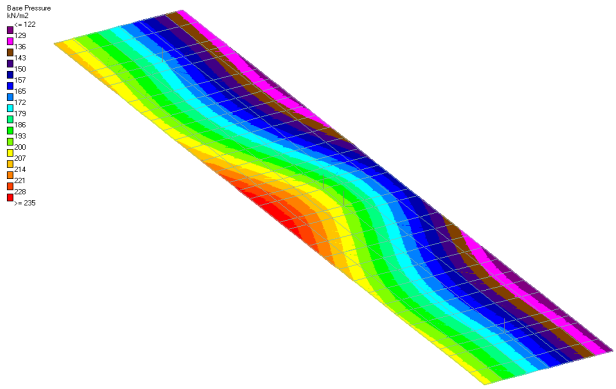
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report

7.5.2 Design of combined foundation for column at 3130/J_3130/7 , 3130/G_3130/7 , 3130/F_3130/7 & 3130/D_3130/7 and foundation mkd. as F1

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **FDN F-1 GR-7 R2.STD**

Staad Support Node for column at **3130/J_3130/7 431 3130/F_3130/7 9**
3130/G_3130/7 428 3130/D_3130/7 13



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	0.5	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	14.5	m	
Width of footing	Lx	=	2.5	m	
Thickness of footing	D	=	0.7	m	
Depth of pedestal	Bx	=	0.8	m	
Width of pedestal	Bz	=	0.6	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	23.4	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	235.3	KN/m^2	Load Case 322
Minimum Base Pressure		=	42.8	KN/m^2	Load Case 352

2) Design of foundation Raft :

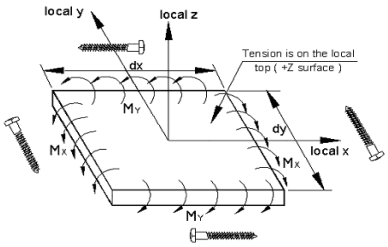
Slab Thickness =	700 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	ength Reduction Factor of Shear Reinforcement , Φt =	0.75

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :
(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	358.200	kN-m	Corresponding Mxy	4.300	kN-m	*(Plate No. - 166 , L/C No. - 405)*
Min. Mx	-354.400	kN-m	Corresponding Mxy	24.300	kN-m	*(Plate No. - 268 , L/C No. - 405)*
Max. My	0.600	kN-m	Corresponding Mxy	-2.500	kN-m	*(Plate No. - 224 , L/C No. - 452)*
Min. My	-134.700	kN-m	Corresponding Mxy	1.500	kN-m	*(Plate No. - 76 , L/C No. - 405)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	358.2 + 4.3 =	362.500 kN-m
My1 =	My + abs(Mxy) =	0.6 + 2.5 =	3.100 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	358.2 + 30.817 =	389.017 kN-m
My2 =	My+abs(Mxy^2/Mx) =	0.6 + 0.017 =	0.617 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 362.5 kN-m
Myd = 3.1 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 362.5 kN-m
Φ*Kn = Mu/bd^2 deff. = 612.5 mm
Mu/bd^2 = 362.5X10^6/(1000X612.5^2) = 0.966 N/mm^2 < 6.9 N/mm^2
Pt = 0.0026 Reqd. Ast = 026X1000X612.5/100 = 15.93 cm^2
Provided 25 mm @ 200 C/C (Provide Ast = 24.54)cm^2

Total Provide Ast = 24.54 cm^2 O.K.
Total Provide Pt = 0.004

ALONG LOCAL Y - DIRECTION :

Mu = 3.1 kN-m
Φ*Kn = Mu/bd^2 deff. = 592 mm
Mu/bd^2 = 3.1X10^6/(1000X592^2) = 0.008 N/mm^2 < 6.9 N/mm^2
Pt = 0.0000 Reqd. Ast = 0X1000X612.5/100 = 0 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0017

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

FOR BOTTOM FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-354.4 - 24.3 =	-378.7	kN-m
My1 =	My - abs(Mxy) =	-134.7 - 1.5 =	-136.2	kN-m
Mx2 =	Mx-abs(Mxy^2/My) =	-354.4 - 4.384 =	-358.784	kN-m
My2 =	My-abs(Mxy^2/Mx) =	-134.7 - 0.006 =	-134.694	kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -378.7 kN-m
Myd = -136.2 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 378.7 kN-m
Φ*Kn = Mu/bd^2 deff. = 612.5 mm
Mu/bd^2 = 378.7X10^6/(1000X612.5^2) = 1.009 N/mm^2 < 6.9 N/mm^2
Pt = 0.0027 Reqd. Ast = 0027X1000X612.5/100 = 16.54 cm^2
Provided 25 mm @ 200 C/C (Provide Ast = 24.54)cm^2

Total Provide Ast = 24.54 cm^2 **O.K.**
Total Provide Pt = 0.004

ALONG LOCAL Y - DIRECTION :

Mu = 136.2 kN-m
Φ*Kn = Mu/bd^2 deff. = 592 mm
Mu/bd^2 = 136.2X10^6/(1000X592^2) = 0.389 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X592/100 = 5.92 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 **O.K.**
Total Provide Pt = 0.0017

SHEAR CHECK

(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION : **(Staad Group : SHEAR X)**

SQx = 0.6 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 162 , L/C No. - 405)*
Vu = 367.50 kN

Provided Pt = 0.004
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.004X367.5X10^3X612.5/378.7/10^6)X1000X612.5/10^3
= 604.532 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.004)X1000X612.5/10^3
= 621.426 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X612.5/10^3
= 1050.844 kN
HENCE, Vc = 604.532 kN
Φt*Vc = 0.75X604.532 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 453.399 kN **> 367.5KN , Hence O.K.**

ALONG LOCAL Y - DIRECTION : **(Staad Group : SHEAR Y)**

SQy = -0.2 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 250 , L/C No. - 408)*
Vu = 118.4 kN

Pt provided = 0.0017
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0017X118.4X10^3X592/136.2/10^3)X1000X592/10^3
= 569.176 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0017)X1000X592/10^3
= 577.48 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X592/10^3
= 1015.673 kN
HENCE, Vc = 569.176 kN
Φt*Vc = 0.75X569.176 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 426.882 kN **> 118.4KN , Hence O.K.**

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

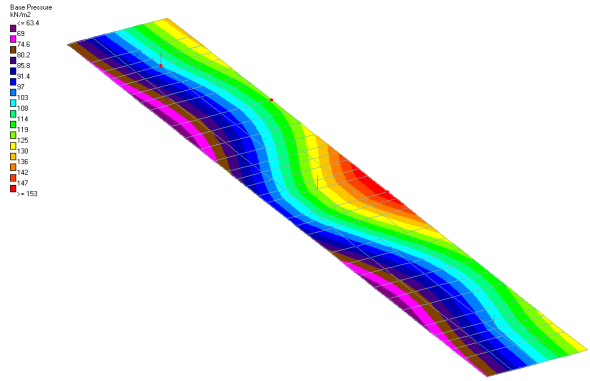
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report

7.5.3 Design of combined foundation for column at 3130/J_3130/9 , 3130/G_3130/9 , 3130/F_3130/9 & 3130/D_3130/9 and foundation mkd. as F1B

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **FDN F-1B GR-9.STD**

Staad Support Node for column at **3130/J_3130/9 17 3130/F_3130/9 17**
3130/G_3130/9 21 3130/D_3130/9 21



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	0.5	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	14.5	m	
Width of footing	Lx	=	2.0	m	
Thickness of footing	D	=	0.5	m	
Depth of pedestal	Bx	=	0.6	m	
Width of pedestal	Bz	=	0.6	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	27	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	152.8	KN/m^2	Load Case 313
Minimum Base Pressure		=	29.4	KN/m^2	Load Case 354

2) Design of foundation Raft :

Slab Thickness =	500 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	trength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	ength Reduction Factor of Shear Reinforcement , Φt =	0.75

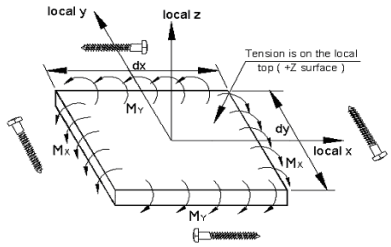
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	139.800	kN-m	Corresponding Mxy	-0.300	kN-m	*(Plate No. - 193 , L/C No. - 405)*
Min. Mx	-141.500	kN-m	Corresponding Mxy	3.400	kN-m	*(Plate No. - 189 , L/C No. - 406)*
Max. My	1.900	kN-m	Corresponding Mxy	11.600	kN-m	*(Plate No. - 179 , L/C No. - 406)*
Min. My	-21.500	kN-m	Corresponding Mxy	-0.900	kN-m	*(Plate No. - 175 , L/C No. - 406)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	139.8 + 0.3 =	140.100 kN-m
My1 =	My + abs(Mxy) =	1.9 + 11.6 =	13.500 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	139.8 + 0.047 =	139.847 kN-m
My2 =	My+abs(Mxy^2/Mx) =	1.9 + 0.963 =	2.863 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 140.1 kN-m
Myd = 13.5 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 140.1 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 140.1X10^6/(1000X417^2) = 0.806 N/mm^2 deff. = 417 mm < 6.9 N/mm^2
Pt = 0.0022 Reqd. Ast = .0022X1000X417/100 = 9.17 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 **O.K.**
Total Provide Pt = 0.0032

ALONG LOCAL Y - DIRECTION :

Mu = 13.5 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 13.5X10^6/(1000X403^2) = 0.078 N/mm^2 deff. = 403 mm < 6.9 N/mm^2
Pt = 0.0002 Reqd. Ast = .0002X1000X417/100 = 0.81 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54)cm^2

Total Provide Ast = 7.54 cm^2 **O.K.**
Total Provide Pt = 0.0019

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -141.5 - 3.4 = -144.9 kN-m
My1 = My - abs(Mxy) = -21.5 - 0.9 = -22.4 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -141.5 - 0.538 = -142.038 kN-m
My2 = My-abs(Mxy^2/Mx) = -21.5 - 0.006 = -21.494 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -144.9 kN-m
Myd = -22.4 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 144.9 kN-m
Φ*Kn = Mu/bd^2 deff. = 417 mm
Mu/bd^2 = 144.9X10^6/(1000X417^2) = 0.833 N/mm^2 < 6.9 N/mm^2
Pt = 0.0022 Reqd. Ast = .0022X1000X417/100 = 9.17 cm^2
Provided 16 mm @ 150 C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 **O.K.**
Total Provide Pt = 0.0032

ALONG LOCAL Y - DIRECTION :

Mu = 22.4 kN-m
Φ*Kn = Mu/bd^2 deff. = 403 mm
Mu/bd^2 = 22.4X10^6/(1000X403^2) = 0.138 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = .0004X1000X403/100 = 1.61 cm^2
Provided 12 mm @ 150 C/C (Provide Ast = 7.54)cm^2

Total Provide Ast = 7.54 cm^2 **O.K.**
Total Provide Pt = 0.0019

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.5 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 189 , L/C No. - 406)*
Vu = 208.50 kN

Provided Pt = 0.0032
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X208.5X10^3X417/144.9/10^6)X1000X417/10^3
= 408.332 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X417/10^3
= 417.406 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X417/10^3
= 715.432 kN
HENCE, Vc = 408.332 kN
Φt*Vc = 0.75X408.332 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 306.249 kN **> 208.5KN , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.3 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 42 , L/C No. - 406)*
Vu = 120.9 kN

Pt provided = 0.0019
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0019X120.9X10^3X403/22.4/10^3)X1000X403/10^3
= 409.782 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0019)X1000X403/10^3
= 394.486 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X403/10^3
= 691.412 kN
HENCE, Vc = 394.486 kN
Φt*Vc = 0.75X394.486 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 295.865 kN **> 120.9KN , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

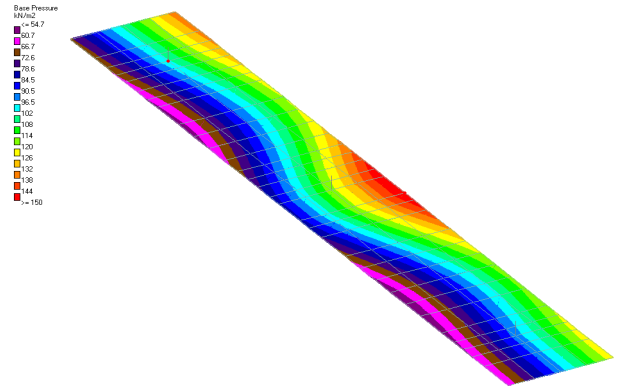
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report

7.5.4 Design of combined foundation for column at 3130/J_3130/10 , 3130/G_3130/10 , 3130/F_3130/10 & 3130/D_3130/10 and foundation mkd. as F1B

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **FDN F1-B GR-10 R.STD**

Staad Support Node for column at **3130/J_3130/10 25 3130/F_3130/10 25**
3130/G_3130/1(29 3130/D_3130/10 29



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	0.5	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	14.5	m	
Width of footing	Lx	=	2.0	m	
Thickness of footing	D	=	0.5	m	
Depth of pedestal	Bx	=	0.6	m	
Width of pedestal	Bz	=	0.6	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	27	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	150.2	KN/m^2	Load Case 313
Minimum Base Pressure		=	3.9	KN/m^2	Load Case 354

2) Design of foundation Raft :

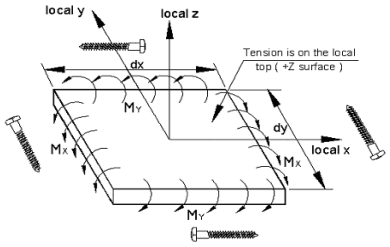
Slab Thickness =	500	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	trength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	ength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :
(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	121.500	kN-m	Corresponding Mxy	-0.300	kN-m	*(Plate No. - 232 , L/C No. - 406)*
Min. Mx	-126.200	kN-m	Corresponding Mxy	0.800	kN-m	*(Plate No. - 189 , L/C No. - 406)*
Max. My	8.500	kN-m	Corresponding Mxy	0.200	kN-m	*(Plate No. - 79 , L/C No. - 454)*
Min. My	-21.200	kN-m	Corresponding Mxy	-1.100	kN-m	*(Plate No. - 175 , L/C No. - 406)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	121.5 + 0.3 =	121.800 kN-m
My1 =	My + abs(Mxy) =	8.5 + 0.2 =	8.700 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	121.5 + 0.011 =	121.511 kN-m
My2 =	My+abs(Mxy^2/Mx) =	8.5 + 0 =	8.5 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 121.8 kN-m
Myd = 8.7 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 121.8 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 121.8X10^6/(1000X417^2) = 0.7 N/mm^2 deff. = 417 mm < 6.9 N/mm^2
Pt = 0.0019 Reqd. Ast = .0019X1000X417/100 = 7.92 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 **O.K.**
Total Provide Pt = 0.0032

ALONG LOCAL Y - DIRECTION :

Mu = 8.7 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 8.7X10^6/(1000X403^2) = 0.05 N/mm^2 deff. = 403 mm < 6.9 N/mm^2
Pt = 0.0001 Reqd. Ast = .0001X1000X417/100 = 0.4 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54)cm^2

Total Provide Ast = 7.54 cm^2 **O.K.**
Total Provide Pt = 0.0019

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

FOR BOTTOM FACE REINFORCEMRENT :

Mx1 = Mx - abs(Mxy) = -126.2 - 0.8 = -127 kN-m
My1 = My - abs(Mxy) = -21.2 - 1.1 = -22.3 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -126.2 - 0.03 = -126.23 kN-m
My2 = My-abs(Mxy^2/Mx) = -21.2 - 0.01 = -21.19 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -127 kN-m
Myd = -22.3 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 127 kN-m
Φ*Kn = Mu/bd^2 deff. = 417 mm
Mu/bd^2 = 127X10^6/(1000X417^2) = 0.73 N/mm^2 < 6.9 N/mm^2
Pt = 0.0020 Reqd. Ast = 0.002X1000X417/100 = 8.34 cm^2
Provided 16 mm @ 150 C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 **O.K.**
Total Provide Pt = 0.0032

ALONG LOCAL Y - DIRECTION :

Mu = 22.3 kN-m
Φ*Kn = Mu/bd^2 deff. = 403 mm
Mu/bd^2 = 22.3X10^6/(1000X403^2) = 0.137 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = .0004X1000X403/100 = 1.61 cm^2
Provided 12 mm @ 150 C/C (Provide Ast = 7.54)cm^2

Total Provide Ast = 7.54 cm^2 **O.K.**
Total Provide Pt = 0.0019

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.4 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 96 , L/C No. - 406)*
Vu = 166.80 kN

Provided Pt = 0.0032
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X166.8X10^3X417/127/10^6)X1000X417/10^3
= 407.145 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X417/10^3
= 417.406 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X417/10^3
= 715.432 kN
HENCE, Vc = 407.145 kN
Φt*Vc = 0.75X407.145 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 305.359 kN **> 166.8KN , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.2 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 42 , L/C No. - 406)*
Vu = 80.6 kN

Pt provided = 0.0019
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0019X80.6X10^3X403/22.3/10^3)X1000X403/10^3
= 400.429 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0019)X1000X403/10^3
= 394.486 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X403/10^3
= 691.412 kN
HENCE, Vc = 394.486 kN
Φt*Vc = 0.75X394.486 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 295.865 kN **> 80.6KN , Hence O.K.**

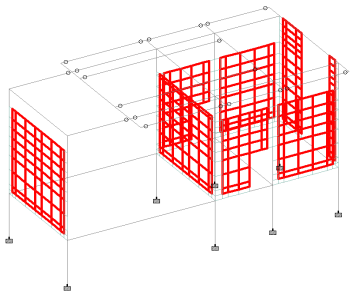
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

7.6 DESIGN OF 300MM WALL :

(STAAD File : TC Support Structure R2)

Wall Thickness =	300	mm
Width =	1000	mm
Clear Cover =	40	mm
Compressive Strength of Concrete , fc' =	45	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	



STAAD Group - M

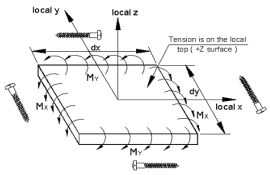
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : M)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	2.3	Mton-m	Corresponding Mxy	-1.3	Mton-m	*(Plate No. - 413 , L/C No. - 401)*
Min. Mx	-2.6	Mton-m	Corresponding Mxy	0.9	Mton-m	*(Plate No. - 419 , L/C No. - 405)*
Max. My	2.4	Mton-m	Corresponding Mxy	-0.8	Mton-m	*(Plate No. - 402 , L/C No. - 405)*
Min. My	-0.9	Mton-m	Corresponding Mxy	0	Mton-m	*(Plate No. - 559 , L/C No. - 408)*

REINFORCEMENT FOR OUTER FACE (TOP +Z):

Mx1 =	Mx + abs(Mxy) =	2.3 + 1.3 =	3.6	Mton-m
My1 =	My + abs(Mxy) =	2.4 + 0.8 =	3.2	Mton-m
Mx2 =	Mx+abs(Mxy^2/My) =	2.3 + 0.704 =	3.004	Mton-m
My2 =	My+abs(Mxy^2/Mx) =	2.4 + 0.278 =	2.678	Mton-m



DESIGN MOMENT CALCULATION FOR OUTER FACE REINFORCEMENT :

Mxd =	3.6 Mton-m
Myd =	3.2 Mton-m

ALONG X - DIRECTION :

Mu =	3.6	Mton-m							
Φ*Kn =	Mu/bd^2					deff. =	254	mm	
Mu/bd^2 =		3.6X10^7/(1000X254^2) =	0.558	N/mm^2	<	8.5	N/mm^2		
Pt =	0.0015	Reqd. Ast =	0.0015X1000X254/100 =	3.81	cm^2				
Provided	12	mm @	200	C/C	(Provide Ast =	5.65	cm2)		
Provided	0	mm @	200	C/C	(Provide Ast =	0	cm2)		
					Total Provide Ast =	5.65	cm2		O.K.
					Total Provide Pt =	0.0022			

ALONG Y - DIRECTION :

Mu =	3.2	Mton-m							
Φ*Kn =	Mu/bd^2					deff. =	242	mm	
Mu/bd^2 =		3.2X10^7/(1000X242^2) =	0.496	N/mm^2	<	8.5	N/mm^2		
Pt =	0.0013	Reqd. Ast =	0.0013X1000X254/100 =	3.15	cm^2				
Provided	12	mm @	200	C/C	(Provide Ast =	5.65	cm2)		
Provided	0	mm @	200	C/C	(Provide Ast =	0	cm2)		
					Total Provide Ast =	5.65	cm2		O.K.
					Total Provide Pt =	0.0023			

REINFORCEMENT FOR INNER FACE (BOTTOM +Z):

Mx1 =	Mx - abs(Mxy) =	-2.6 - 0.9 =	-3.5	Mton-m
My1 =	My - abs(Mxy) =	-0.9 - 0 =	-0.9	Mton-m
Mx2 =	Mx-abs(Mxy^2/My) =	-2.6 - 0.9 =	-3.5	Mton-m
My2 =	My-abs(Mxy^2/Mx) =	-0.9 - 0 =	-0.9	Mton-m

DESIGN MOMENT CALCULATION FOR INNER FACE REINFORCEMENT :

Mxd =	-3.5 Mton-m
Myd =	-0.9 Mton-m

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	

ALONG X - DIRECTION :

Mu = 3.5 Mton-m

Φ*Kn = Mu/bd^2 deff. = 254 mm

Mu/bd^2 = 3.5X10^7/(1000X254^2) = 0.543 N/mm^2 < 8.5 N/mm^2

Pt = 0.0014 Reqd. Ast = 0.0014X1000X254/100 = 3.56 cm^2

Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2)

Provided **0** mm @ **200** C/C (Provide Ast = 0 cm2)

Total Provide Ast = 5.65 cm2) **O.K.**

Total Provide Pt = 0.0022

ALONG Y - DIRECTION :

Mu = 0.9 Mton-m

Φ*Kn = Mu/bd^2 deff. = 242 mm

Mu/bd^2 = 0.9X10^7/(1000X242^2) = 0.154 N/mm^2 < 8.5 N/mm^2

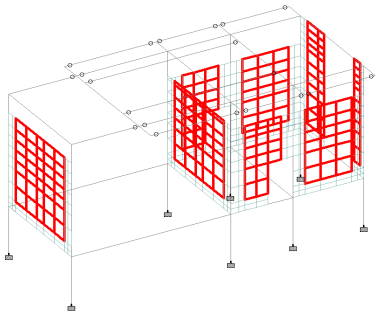
Pt = 0.0004 Reqd. Ast = 0.0004X1000X242/100 = 0.97 cm^2

Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2)

Provided **0** mm @ **200** C/C (Provide Ast = 0 cm2)

Total Provide Ast = 5.65 cm2) **O.K.**

Total Provide Pt = 0.0023



STAAD Group - SF

SHEAR CHECK

(Staad Group : SF)

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = -0.1 N/mm^2 *(At d-distance from face of the Column)* *(Plate No. - 418 , L/C No. - 405)*

Vu = 3 Mton

Provided Pt = 0.0022

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X45^0.5+17X0.0022X3X10^4X254/3.5/10^7)X1000X254/10^4

= 27.469 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X45^0.5+17X0.0022)X1000X254/10^4

= 28.212 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X6.71X1000X254/10^4

= 49.413 Mton

HENCE, Vc = 27.469 Mton

Φ*Vc = 0.75X27.469 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 20.602 Mton **> 3Mton , Hence O.K.**

ALONG Y - DIRECTION :

SQy = 0 N/mm^2 *(At d-distance from face of the Column)* *(Plate No. - 566 , L/C No. - 408)*

Vu = 0 Mton

Pt provided = 0.0023

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X45^0.5+17X0.0023X0X10^4X242/0.9/10^7)X1000X242/10^4

= 25.974 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X45^0.5+17X0.0023)X1000X242/10^4

= 26.92 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X6.71X1000X242/10^4

= 47.078 Mton

HENCE, Vc = 25.974 Mton

Φ*Vc = 0.75X25.974 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 19.481 Mton **> 0Mton , Hence O.K.**

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Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

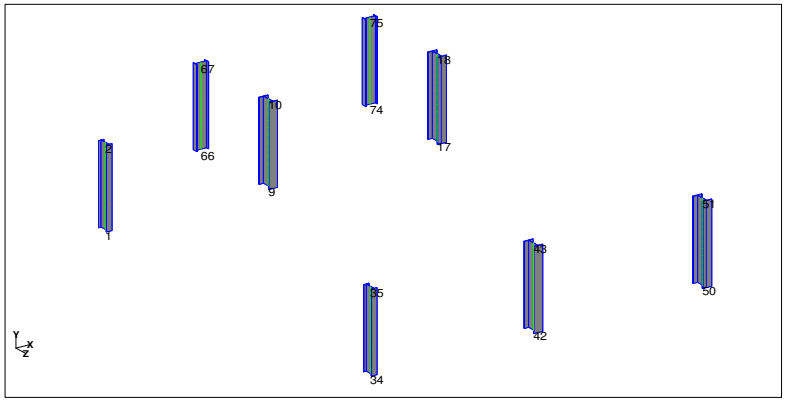
REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.0 DESIGN OF TERTIARY CRUSHER BUILDING FOUNDATION

8.1 Basic load data

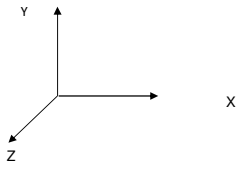
Basic Load cases for column foundation of Crusher house

LOAD 1 DL	Dead Load
LOAD 2 LL	Live Load
LOAD 3 SDL1	Superimposed Dead Load1 - Chute & Bunker empty condition
LOAD 4 SDL2	Superimposed Dead Load2 - Chute & Bunker filled condition
LOAD 5 WL(+X)	Wind Load in (+) X Direction
LOAD 6 WL(-X)	Wind Load in (-) X Direction
LOAD 7 WL(+Z)	Wind Load in (+) Z Direction
LOAD 8 WL(-Z)	Wind Load in (-) Z Direction
LOAD 9 TL1	Temperature Load 1 for (+) 25^0
LOAD 10 TL2	Temperature Load 2 for (-) 25^0
LOAD 11 EL(+X)	Earthquake Load in (+X) Direction
LOAD 12 EL (+Z)	Earthquake Load in (+Z) Direction



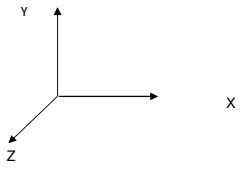
Staad support node - 50 Column support at grid 3130/C-3130/6

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DEAD LOAD	0.0	-260.5	1.6	0	0	0
2 LIVE LOAD	0.0	-266.1	5.0	0	0	0
3 SDL1- BUNKER & CHUTE EMPTY	1.3	-206.4	2.7	0	0	0
4 SDL2- BUNKER & CHUTE FULL	0.4	-2136.2	2.9	0	0	0
5 WIND LOAD +X	0.4	-291.4	0.1	0	0	0
6 WIND LOAD -X	-0.4	293.4	-0.1	0	0	0
7 WIND LOAD +Z	0.1	-57.0	7.1	0	0	0
8 WIND LOAD -Z	-0.1	56.5	-7.0	0	0	0
9 TEMPERATURE LOAD	-6.6	-2.2	-4.2	0	0	0
10 TEMPERATURE LOAD	6.6	2.2	4.2	0	0	0
11 SEISMIC +X	-0.3	-58.0	0.0	0	0	0
12 SEISMIC +Z	0.0	-25.1	0.7	0	0	0



Staad support node - 42 Column support at grid 3130/C-3130/4

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DEAD LOAD	1.4	-265.8	6.5	0	0	0
2 LIVE LOAD	0.0	-308.4	0.9	0	0	0
3 SDL1- BUNKER & CHUTE EMPTY	-44.0	-212.7	-6.9	0	0	0
4 SDL2- BUNKER & CHUTE FULL	-33.0	-1089.0	-5.4	0	0	0
5 WIND LOAD +X	103.9	91.1	-0.3	0	0	0
6 WIND LOAD -X	-103.9	-95.1	0.3	0	0	0
7 WIND LOAD +Z	-0.4	-176.7	89.0	0	0	0
8 WIND LOAD -Z	-0.2	140.4	-77.8	0	0	0
9 TEMPERATURE LOAD	2.7	9.0	-10.8	0	0	0
10 TEMPERATURE LOAD	-2.7	-9.0	10.8	0	0	0
11 SEISMIC +X	13.0	21.9	-0.1	0	0	0
12 SEISMIC +Z	-0.2	-25.1	8.6	0	0	0



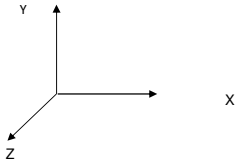
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT		
Prepared By	RPK	RPK	RPK	RPK		
Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG		

Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
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Approved	AG	AG	AG	AG	

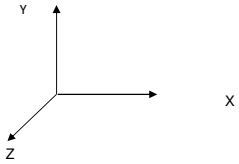
Staad support node - 34 Column support at grid 3130/C-3130/3

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DEAD LOAD	-1.5	-134.4	5.8	0	0	0
2 LIVE LOAD	-0.1	-139.7	0.0	0	0	0
3 SDL1- BUNKER & CHUTE EMPTY	-41.5	-309.9	1.8	0	0	0
4 SDL2- BUNKER & CHUTE FULL	-31.0	-276.9	1.8	0	0	0
5 WIND LOAD +X	97.2	199.4	0.0	0	0	0
6 WIND LOAD -X	-97.2	-197.5	0.0	0	0	0
7 WIND LOAD +Z	-0.5	-115.0	46.0	0	0	0
8 WIND LOAD -Z	-0.1	112.8	-46.0	0	0	0
9 TEMPERATURE LOAD	4.8	0.1	-1.3	0	0	0
10 TEMPERATURE LOAD	-4.8	-0.1	1.3	0	0	0
11 SEISMIC +X	12.1	35.7	0.0	0	0	0
12 SEISMIC +Z	-0.2	-2.9	1.2	0	0	0



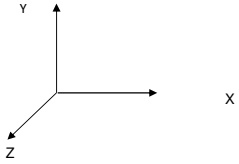
Staad support node - 17 Column support at grid 3130/K-3130/6

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DEAD LOAD	-0.1	-263.7	3.0	0	0	0
2 LIVE LOAD	-0.2	-340.7	2.1	0	0	0
3 SDL1- BUNKER & CHUTE EMPTY	0.0	-97.0	-3.7	0	0	0
4 SDL2- BUNKER & CHUTE FULL	-0.4	-1607.0	15.3	0	0	0
5 WIND LOAD +X	2.4	-276.2	3.6	0	0	0
6 WIND LOAD -X	-2.4	277.8	-3.6	0	0	0
7 WIND LOAD +Z	-0.1	-293.1	58.4	0	0	0
8 WIND LOAD -Z	0.1	282.2	-57.1	0	0	0
9 TEMPERATURE LOAD	-6.6	-8.5	-36.2	0	0	0
10 TEMPERATURE LOAD	6.6	8.5	36.2	0	0	0
11 SEISMIC +X	-0.1	-48.2	0.5	0	0	0
12 SEISMIC +Z	0.0	-72.0	11.5	0	0	0



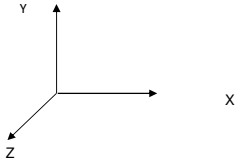
Staad support node - 9 Column support at grid 3130/K-3130/4

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DEAD LOAD	1.5	-246.1	-6.4	0	0	0
2 LIVE LOAD	0.3	-349.4	-0.5	0	0	0
3 SDL1- BUNKER & CHUTE EMPTY	-45.5	-215.9	-9.0	0	0	0
4 SDL2- BUNKER & CHUTE FULL	-34.2	-825.7	-11.2	0	0	0
5 WIND LOAD +X	115.7	59.2	0.9	0	0	0
6 WIND LOAD -X	-115.7	-62.0	-0.9	0	0	0
7 WIND LOAD +Z	0.4	93.9	100.5	0	0	0
8 WIND LOAD -Z	0.2	-78.3	-88.1	0	0	0
9 TEMPERATURE LOAD	1.7	-24.2	8.2	0	0	0
10 TEMPERATURE LOAD	-1.7	24.2	-8.2	0	0	0
11 SEISMIC +X	12.9	14.3	0.2	0	0	0
12 SEISMIC +Z	0.2	8.6	9.5	0	0	0



Staad support node - 1 Column support at grid 3130/K-3130/3

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DEAD LOAD	-1.4	-131.1	-5.8	0	0	0
2 LIVE LOAD	0.0	-130.8	0.0	0	0	0
3 SDL1- BUNKER & CHUTE EMPTY	-43.2	-299.2	1.7	0	0	0
4 SDL2- BUNKER & CHUTE FULL	-32.7	-266.7	1.7	0	0	0
5 WIND LOAD +X	108.0	221.6	0.0	0	0	0
6 WIND LOAD -X	-108.0	-219.7	0.0	0	0	0
7 WIND LOAD +Z	0.5	114.8	46.5	0	0	0
8 WIND LOAD -Z	0.1	-112.9	-46.4	0	0	0
9 TEMPERATURE LOAD	4.0	-1.9	1.3	0	0	0
10 TEMPERATURE LOAD	-4.0	1.9	-1.3	0	0	0
11 SEISMIC +X	12.1	35.7	0.0	0	0	0
12 SEISMIC +Z	0.2	3.0	1.2	0	0	0



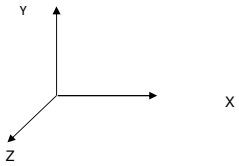
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT		
Prepared By	RPK	RPK	RPK	RPK		
Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG		

Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK	
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Approved	AG	AG	AG	AG	

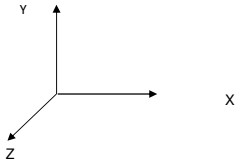
Staad support node - 74 Column support at grid 3130/L-3130/6

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DEAD LOAD	0.8	-66.8	-4.7	0	0	0
2 LIVE LOAD	0.1	-81.5	-7.4	0	0	0
3 SDL1- BUNKER & CHUTE EMPTY	0.0	-29.5	-6.6	0	0	0
4 SDL2- BUNKER & CHUTE FULL	0.2	-317.8	-25.6	0	0	0
5 WIND LOAD +X	12.0	-75.8	-4.4	0	0	0
6 WIND LOAD -X	-12.0	75.9	4.4	0	0	0
7 WIND LOAD +Z	-0.1	349.9	78.7	0	0	0
8 WIND LOAD -Z	0.1	-338.8	-76.7	0	0	0
9 TEMPERATURE LOAD	-2.7	8.9	42.4	0	0	0
10 TEMPERATURE LOAD	2.7	-8.9	-42.4	0	0	0
11 SEISMIC +X	0.4	-8.6	-0.6	0	0	0
12 SEISMIC +Z	0.0	97.2	17.7	0	0	0



Staad support node - 66 Column support at grid 3130/L-3130/4

Load case	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kN-m	Moment-Y kN-m	Moment-Z kN-m
1 DEAD LOAD	-0.8	-85.0	0.0	0	0	0
2 LIVE LOAD	-0.1	-137.1	0.0	0	0	0
3 SDL1- BUNKER & CHUTE EMPTY	0.0	-109.3	1.1	0	0	0
4 SDL2- BUNKER & CHUTE FULL	-0.2	-277.5	1.1	0	0	0
5 WIND LOAD +X	12.0	72.1	0.0	0	0	0
6 WIND LOAD -X	-12.0	-72.8	0.0	0	0	0
7 WIND LOAD +Z	0.1	83.1	1.8	0	0	0
8 WIND LOAD -Z	-0.1	-61.9	-1.8	0	0	0
9 TEMPERATURE LOAD	2.7	18.6	0.6	0	0	0
10 TEMPERATURE LOAD	-2.7	-18.6	-0.6	0	0	0
11 SEISMIC +X	0.4	7.2	0.0	0	0	0
12 SEISMIC +Z	0.0	16.3	0.0	0	0	0



Global Project ID	900707
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Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

8.2 Load Combination

8.2.1 Load Combination for Base pressure calculation

Load Cases	Reference
LOAD 301 (DL + SDL1) LOAD 302 (DL + SDL2)	(Eq. 2.4.1-1 of SBC 301)
LOAD 303 (DL + SDL1 + TL1) + LL LOAD 304 (DL + SDL2 + TL1) + LL LOAD 305 (DL + SDL1 + TL2) + LL LOAD 306 (DL + SDL2 + TL2) + LL	(Eq. 2.4.1-2 of SBC 301)
LOAD 307 (DL+ SDL1) + 0.75 LL + 0.75 WL(+X) LOAD 308 (DL+ SDL1) + 0.75 LL + 0.75 WL(-X) LOAD 309 (DL+ SDL1) + 0.75 LL + 0.75 WL(+Z) LOAD 310 (DL+ SDL1) + 0.75 LL + 0.75 WL(-Z) LOAD 311 (DL+ SDL2) + 0.75 LL + 0.75 WL(+X) LOAD 312 (DL+ SDL2) + 0.75 LL + 0.75 WL(-X) LOAD 313 (DL+ SDL2) + 0.75 LL + 0.75 WL(+Z) LOAD 314 (DL+ SDL2) + 0.75 LL + 0.75 WL(-Z)	(Eq. 2.4.1-6 of SBC 301)
LOAD 315 (DL+ SDL1) + 0.75 LL + 0.525 EL(+X) LOAD 316 (DL+ SDL1) + 0.75 LL + 0.525 EL(-X) LOAD 317 (DL+ SDL1) + 0.75 LL + 0.525 EL(+Z) LOAD 318 (DL+ SDL1) + 0.75 LL + 0.525 EL(-Z) LOAD 319 (DL+ SDL2) + 0.75 LL + 0.525 EL(+X) LOAD 320 (DL+ SDL2) + 0.75 LL + 0.525 EL(-X) LOAD 321 (DL+ SDL2) + 0.75 LL + 0.525 EL(+Z) LOAD 322 (DL+ SDL2) + 0.75 LL + 0.525 EL(-Z)	(Eq. 2.4.1-6 of SBC 301)

Load Cases	Reference
LOAD 323 0.6 (DL+SDL1) + 1.0 WL(+X) LOAD 324 0.6 (DL+SDL1) + 1.0 WL(-X) LOAD 325 0.6 (DL+SDL1) + 1.0 WL(+Z) LOAD 326 0.6 (DL+SDL1) + 1.0 WL(-Z) LOAD 327 0.6 (DL+SDL2) + 1.0 WL(+X) LOAD 328 0.6 (DL+SDL2) + 1.0 WL(-X) LOAD 329 0.6 (DL+SDL2) + 1.0 WL(+Z) LOAD 330 0.6 (DL+SDL2) + 1.0 WL(-Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD 331 0.6 (DL+SDL1) + 0.7 EL(+X) LOAD 332 0.6 (DL+SDL1) + 0.7 EL(-X) LOAD 333 0.6 (DL+SDL1) + 0.7 EL(+Z) LOAD 334 0.6 (DL+SDL1) + 0.7 EL(-Z) LOAD 335 0.6 (DL+SDL2) + 0.7 EL(+X) LOAD 336 0.6 (DL+SDL2) + 0.7 EL(-X) LOAD 337 0.6 (DL+SDL2) + 0.7 EL(+Z) LOAD 338 0.6 (DL+SDL2) + 0.7 EL(-Z)	(Eq. 2.4.1-8 of SBC 301)

8.2.2 Load Combination for concrete design

Load Case	Reference
LOAD 401 1.4 (DL + SDL1) LOAD 402 1.4 (DL + SDL2)	(Eq. 2.3.2-1 of SBC 301)
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL	(Eq. 2.3.2-2 of SBC 301)
LOAD 407 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(+X) LOAD 408 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(-X) LOAD 409 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(+Z) LOAD 410 1.2 (DL+ SDL1)+ 0.5 LL + 1.6 WL(-Z) LOAD 411 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(+X) LOAD 412 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(-X) LOAD 413 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(+Z) LOAD 414 1.2 (DL+ SDL2)+ 0.5 LL + 1.6 WL(-Z)	(Eq. 2.3.2-4 of SBC 301)
LOAD 415 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(+X) LOAD 416 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(-X) LOAD 417 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(+Z) LOAD 418 1.2 (DL+ SDL1)+ 0.5 LL + 1.0 EL(-Z) LOAD 419 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(+X) LOAD 420 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(-X) LOAD 421 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(+Z) LOAD 422 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(-Z)	(Eq. 2.3.2-5 of SBC 301)

LOAD CASE	Reference
LOAD 423 0.9 (DL+ SDL1) + 1.6 WL(+X) LOAD 424 0.9 (DL+ SDL1) + 1.6 WL(-X) LOAD 425 0.9 (DL+ SDL1) + 1.6 WL(+Z) LOAD 426 0.9 (DL+ SDL1) + 1.6 WL(-Z) LOAD 427 0.9 (DL+ SDL2) + 1.6 WL(+X) LOAD 428 0.9 (DL+ SDL2) + 1.6 WL(-X) LOAD 429 0.9 (DL+ SDL2) + 1.6 WL(+Z) LOAD 430 0.9 (DL+ SDL2) + 1.6 WL(-Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD 431 0.9 (DL+ SDL1) + 1.0 EL(+X) LOAD 432 0.9 (DL+ SDL1) + 1.0 EL(-X) LOAD 433 0.9 (DL+ SDL1) + 1.0 EL(+Z) LOAD 434 0.9 (DL+ SDL1) + 1.0 EL(-Z) LOAD 435 0.9 (DL+ SDL2) + 1.0 EL(+X) LOAD 436 0.9 (DL+ SDL2) + 1.0 EL(-X) LOAD 437 0.9 (DL+ SDL2) + 1.0 EL(+Z) LOAD 438 0.9 (DL+ SDL2) + 1.0 EL(-Z)	(Eq. 2.3.2-7 of SBC 301)

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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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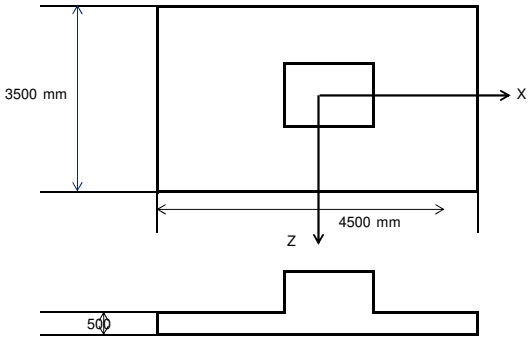
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T F	0	1	2		3	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY		SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG		AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.3 Design of foundation

8.3.1 Design of foundation F3 at Grid 3130/C-3130/3

Finished Ground level	F.G.L	=	0.000	m	Weight of soil	=	407.97 KN	*inputs
Bottom of foundation	B.O.F	=	-2.000	m	Weight of foundation	=	227.40 KN	
Top of footing (staad support)	T.O.F	=	1.000	m	Uplift	=	0.00 KN	
Ground Water Level	G.W.L	=	-10.000	m	Surcharge	=	0 KN	
Surcharge	Ws	=	0.000	KN/m^2	Pressure on top for tension	=	40.3 KN/m^2	
Length of footing	Lz	=	3.50	m				
Width of footing	Lx	=	4.50	m				
Thickness of footing	D	=	0.50	m				
Depth of pedestal	Bx	=	0.80	m				
Width of pedestal	Bz	=	0.80	m				
Height of Load point from bottom of footing	H	=	3.000	m				
Section Modulus of Pedestal	Zxx	=	9.19	m^3				
Section Modulus of Pedestal	Zzz	=	11.81	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	536	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	50	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	16	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

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Project Country	**KINGDOM OF SAUDI ARABIA**

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CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	-43.0	442.93	7.6	22.92	0.0	129.0	1078.3	81.88	55.05	1	0.15
302	-32.5	410.5	7.6	22.92	0	97.6	1045.9	77.16	55.65	1	0.14
303	-38.3	582.36	6.4	19.17	0	115.0	1217.7	89.13	65.50	1	0.17
304	-27.9	549.93	6.4	19.17	0	83.6	1185.3	84.42	66.10	1	0.16
305	-48.0	582.67	9.0	26.85	0	143.9	1218.0	92.44	62.23	1	0.17
306	-37.5	550.24	9.0	26.85	0.0	112.5	1185.6	87.72	62.83	1	0.16
307	29.8	398.09	7.7	22.95	0	-89.3	1033.5	75.68	55.56	1	0.14
308	-116.0	695.71	7.7	23.01	0	347.9	1331.1	116.47	52.55	1	0.22
309	-43.5	633.86	42.2	126.57	0	130.4	1269.2	105.40	55.77	1	0.20
310	-43.2	463.02	-26.8	-80.46	0	129.5	1098.4	89.46	50.02	1	0.17
311	40.2	365.66	7.7	22.95	0.0	-120.7	1001.0	76.28	50.84	1	0.14
312	-105.5	663.28	7.7	23.01	0	316.5	1298.7	111.75	53.15	1	0.21
313	-33.0	601.43	42.2	126.54	0	98.9	1236.8	100.68	56.38	1	0.19
314	-32.7	430.59	-26.8	-80.49	0.0	98.0	1066.0	84.74	50.62	1	0.16
315	-43.1	547.62	7.7	22.98	0	129.3	1183.0	88.56	61.66	1	0.17
316	-43.1	547.62	7.7	22.98	0	129.3	1183.0	88.56	61.66	1	0.17
317	-43.1	547.62	7.7	22.98	0	129.3	1183.0	88.56	61.66	1	0.17
318	-43.1	547.62	7.7	22.98	0	129.3	1183.0	88.56	61.66	1	0.17
319	-32.6	515.19	7.7	22.98	0.0	97.9	1150.6	83.84	62.26	1	0.16
320	-32.6	515.19	7.7	22.98	0	97.9	1150.6	83.84	62.26	1	0.16
321	-32.6	515.19	7.7	22.98	0	97.9	1150.6	83.84	62.26	1	0.16
322	-32.6	515.19	7.7	22.98	0	97.9	1150.6	83.84	62.26	1	0.16
323	71.4	66.39	4.6	13.71	0	-214.1	701.8	64.17	24.94	1	0.12
324	-123.0	463.21	4.6	13.77	0.0	368.9	1098.6	102.48	37.02	1	0.19
325	-26.3	380.74	50.6	151.86	0	78.8	1016.1	87.71	41.32	1	0.16
326	-25.9	152.96	-41.4	-124.20	0	77.6	788.3	70.14	29.97	1	0.13
327	77.7	46.93	4.6	13.71	0	-233.0	682.3	64.53	22.11	1	0.12
328	-116.7	443.76	4.6	13.77	0.0	350.0	1079.1	99.65	37.38	1	0.19
329	-20.0	361.28	50.6	151.83	0	59.9	996.7	84.88	41.68	1	0.16
330	-19.6	133.5	-41.4	-124.20	0	58.7	768.9	67.31	30.33	1	0.13
331	-25.8	265.76	4.6	13.74	0	77.4	901.1	65.26	49.17	1	0.12
332	-25.8	265.76	4.6	13.74	0	77.4	901.1	65.26	49.17	1	0.12
333	-25.8	265.76	4.6	13.74	0.0	77.4	901.1	65.26	49.17	1	0.12
334	-25.8	265.76	4.6	13.74	0	77.4	901.1	65.26	49.17	1	0.12
335	-19.5	246.3	4.6	13.74	0	58.6	881.7	62.43	49.53	1	0.12
336	-19.5	246.3	4.6	13.74	0	58.6	881.7	62.43	49.53	1	0.12
337	-19.5	246.3	4.6	13.74	0	58.6	881.7	62.43	49.53	1	0.12
338	-19.5	246.3	4.6	13.74	0.0	58.6	881.7	62.43	49.53	1	0.12

Max P_{max} = 111.75 KN/m^2

Max P_{min} = 22.11 KN/m^2

Resultant F.O.S = 0.21

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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	-60.1	622.1	10.7	32.10	0.0	180.33	58.26	20.74
402	-45.5	575.9	10.7	32.07	0	136.44	51.61	21.53
403	-53.6	859.4	9.0	26.88	0	160.80	71.10	38.02
404	-67.1	859.7	12.5	37.62	0	201.30	75.72	33.45
405	-39.0	813.2	9.0	26.88	0	116.91	64.45	38.81
406	-52.5	813.6	12.5	37.62	0	157.41	69.08	34.23
407	103.9	284.0	9.2	27.51	0	-311.61	47.41	-11.34
408	-207.1	919.1	9.2	27.60	0	621.15	113.94	2.77
409	-52.3	787.0	82.8	248.49	0	156.96	90.30	9.64
410	-51.7	422.6	-64.4	-193.17	0	155.04	60.98	-7.32
411	116.4	244.4	9.2	27.48	0	-349.23	48.07	-17.04
412	-194.5	879.5	9.2	27.60	0	583.56	108.25	3.44
413	-39.8	747.5	82.8	248.49	0	119.34	84.61	10.31
414	-39.1	383.0	-64.4	-193.17	0	117.42	55.28	-6.65
415	-51.6	603.1	9.2	27.54	0	154.77	54.39	22.19
416	-51.6	603.1	9.2	27.54	0	154.77	54.39	22.19
417	-51.6	603.1	9.2	27.54	0	154.77	54.39	22.19
418	-51.6	603.1	9.2	27.54	0	154.77	54.39	22.19
419	-39.1	563.5	9.2	27.54	0	117.18	48.69	22.86
420	-39.1	563.5	9.2	27.54	0	117.18	48.69	22.86
421	-39.1	563.5	9.2	27.54	0	117.18	48.69	22.86
422	-39.1	563.5	9.2	27.54	0	117.18	48.69	22.86
423	116.8	80.9	6.9	20.58	0	-350.46	37.04	-26.77
424	-194.1	716.0	6.9	20.67	0	582.30	97.00	-6.09
425	-39.4	583.9	80.5	241.59	0	118.11	73.37	0.78
426	-38.7	219.5	-66.7	-200.10	0	116.19	45.55	-17.68
427	126.2	51.2	6.9	20.58	0	-378.66	37.54	-31.05
428	-184.7	686.3	6.9	20.67	0	554.10	92.73	-5.59
429	-30.0	554.2	80.5	241.56	0	89.91	69.09	1.28
430	-29.3	189.8	-66.7	-200.10	0	87.99	41.28	-17.18
431	-38.6	399.9	6.9	20.64	0	115.92	37.45	13.33
432	-38.6	399.9	6.9	20.64	0	115.92	37.45	13.33
433	-38.6	399.9	6.9	20.64	0	115.92	37.45	13.33
434	-38.6	399.9	6.9	20.64	0	115.92	37.45	13.33
435	-29.2	370.2	6.9	20.61	0	87.72	33.18	13.84
436	-29.2	370.2	6.9	20.61	0	87.72	33.18	13.84
437	-29.2	370.2	6.9	20.61	0	87.72	33.18	13.84
438	-29.2	370.2	6.9	20.61	0	87.72	33.18	13.84

Max Net factored B.P = 113.94 KN/m^2

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.85 m
Moment about Z — Muz = 194.99 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 1333.87 KNm

As $M_{uzlim} > M_{uz}$ — Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 1188.1 mm^2

Effective Depth Along X = 442 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
$A_{stmin} = 0.0014bd$

A_{stmin} = 795.6 mm^2
A_{req} = 1188.1 mm^2
Required Spacing = **169.2** mm

Length of cantilever along Z = 1.35 m
Moment about X — Mux = 103.83 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 1239.05 KNm

As $M_{uxlim} > M_{ux}$ — Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 651.0 mm^2

Effective Depth Along X = 426 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
$A_{stmin} = 0.0014bd$

A_{stmin} = 766.8 mm^2
A_{req} = 651.0 mm^2
Required Spacing = **262.2** mm

Provide Bottom Reinforcement = **16 mm dia @ 150 c/c**
Ast prov = 1340.4 mm^2

Top Reinforcement

Length of cantilever along X = 1.85 m
Moment about Z — Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 1345.97 KNm

As $M_{uzlim} > M_{uz}$ — Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 0.0 mm^2

Effective Depth Along X = 444 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 399.6 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **283.0** mm

Length of cantilever along Z = 1.35 m
Moment about X — Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 1274.20 KNm

As $M_{uxlim} > M_{ux}$ — Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 0.0 mm^2

Effective Depth Along Z = 432 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

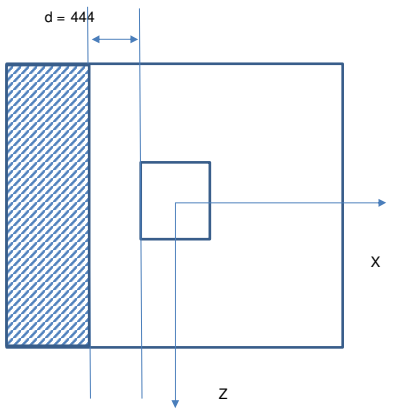
A_{stmin} = 388.8 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **290.9** mm

Provide top Reinforcement = **12 mm dia @ 150 c/c**
Ast prov = 754.0 mm^2

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Project Country	KINGDOM OF SAUDI ARABIA

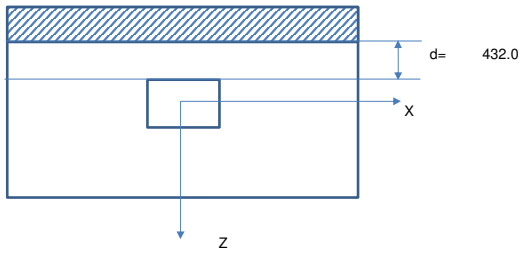
Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T R	0	1	2	3		
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023		
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

ONE WAY SHEAR CKECK



Along X Direction	Cantilever	L	=	1.406	m	Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}b_w d + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0030						*Table 22.5.5.1 SBC 304*
Effective Depth Along X		d	=	444	mm			λ	=	1 Normal concrete
Shear Force at d distance		V_u	=	160.20	KN			Mu	=	112.62 KNm
Allowable shear strength		φVc	=	326.00	KN	O.K		φ	=	0.75 for shear
								φVc	=	326.00 KN
							ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}b_w d + 17\rho_w\right)b_w d$
								φVc	=	332.298995 KN *Table 22.5.5.1 SBC 304*
							iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
									=	571.316 KN

ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.918	m	Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0031						*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	432	mm			λ	=	1 Normal concrete
Shear Force at d distance		V_u	=	104.60	KN			Mu	=	48.01 KNm
Allowable shear strength		φVc	=	322.77	KN	O.K		φ	=	0.75 for shear
								φVc	=	322.77 KN
							ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
								φVc	=	323.77984 KN
							iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
								φVc	=	555.87 KN

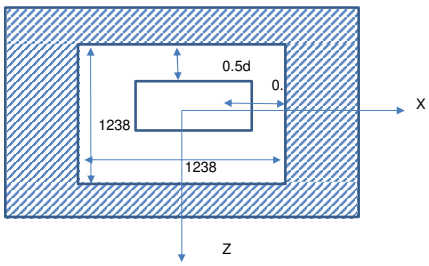
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	15.75	m2
Punching area	=	1.53	m2
λ	=	1	
β	=	1	
αs	=	40	
b0	=	4952.0	mm

Allowable shear stress least of *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.72 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	4.95	m	
Depth at section	d	=	438	mm	
Shear Force at section	Vu	=	1620.0	KN	
Allowable shear strength	φVc	=	3175.88	KN	O.K

Vc	=	$v_c b_0 d$	
	=	4234.51	KN
φ	=	0.75	for shear
φV_n	=	φVc	
φVc	=	3175.9	KN

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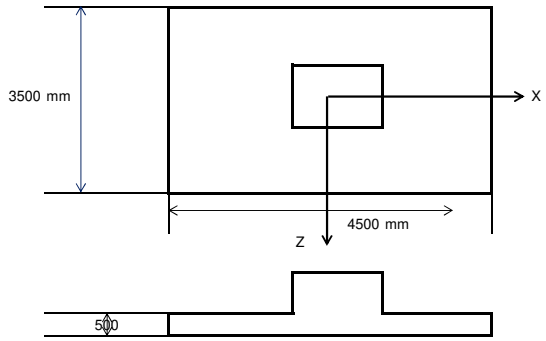
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REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.3.2 Design of foundation F3 at Grid 3130/K-3130/3

Finished Ground level	F.G.L	=	0.000	m	Weight of soil	=	407.97 KN
Bottom of foundation	B.O.F	=	-2.000	m	Weight of foundation	=	227.40 KN
Top of footing (staad support)	T.O.F	=	1.000	m	Uplift	=	0.00 KN
Ground Water Level	G.W.L	=	-10.000	m	Surcharge	=	0 KN
Surcharge	Ws	=	0.000	KN/m^2	Pressure on top for tension	=	40.3 KN/m^2
Length of footing	Lz	=	3.50	m			
Width of footing	Lx	=	4.50	m			
Thickness of footing	D	=	0.50	m			
Depth of pedestal	Bx	=	0.80	m			
Width of pedestal	Bz	=	0.80	m			
Height of Load point from bottom of footing	H	=	3.000	m			
Section Modulus of Pedestal	Zxx	=	9.19	m^3			
Section Modulus of Pedestal	Zzz	=	11.81	m^3			
Allowable Net Bearing Capacity		=	500	KN/m^2			
Gross Bearing Capacity		=	536	KN/m^2			
Grade of concrete		=	35	N/mm^2			
Grade of steel		=	420	N/mm^2			
Clear cover		=	50	mm			
Top Reinfocement		=	12	mm			
Bottom Reinforcement		=	16	mm			
Co-efficient of friction between soil & concrete		=	0.3				
Unit wt of soil		=	18	KN/m^3			
Unit wt of concrete		=	24	KN/m^3			

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

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CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	-44.5	427.93	-4.1	-12.42	0.0	133.6	1063.3	80.17	54.85	1	0.15
302	-34.1	396.28	-4.1	-12.39	0	102.2	1031.7	75.50	55.50	1	0.14
303	-40.6	560.41	-2.9	-8.67	0	121.7	1195.8	87.16	64.68	1	0.16
304	-30.1	528.76	-2.9	-8.67	0	90.3	1164.1	82.50	65.33	1	0.15
305	-48.5	556.73	-5.5	-16.35	0	145.4	1192.1	89.77	61.60	1	0.17
306	-38.0	525.07	-5.5	-16.35	0.0	114.0	1160.4	85.11	62.25	1	0.16
307	36.5	359.78	-4.2	-12.45	0	-109.5	995.2	73.81	52.56	1	0.14
308	-125.5	690.54	-4.2	-12.51	0	376.6	1325.9	117.42	50.95	1	0.22
309	-44.2	439.82	30.7	92.07	0	132.5	1075.2	89.50	47.03	1	0.17
310	-44.5	610.58	-39.0	-116.94	0	133.4	1246.0	103.13	55.08	1	0.19
311	47.0	328.12	-4.2	-12.45	0.0	-140.9	963.5	74.46	47.89	1	0.14
312	-115.1	658.89	-4.2	-12.51	0	345.2	1294.3	112.76	51.59	1	0.21
313	-33.7	408.16	30.7	92.07	0	101.1	1043.5	84.83	47.68	1	0.16
314	-34.0	578.93	-39.0	-116.91	0.0	102.0	1214.3	98.46	55.74	1	0.18
315	-44.5	525.91	-4.2	-12.48	0	133.5	1161.3	86.39	61.07	1	0.16
316	-44.5	525.91	-4.2	-12.48	0	133.5	1161.3	86.39	61.07	1	0.16
317	-44.5	525.91	-4.2	-12.48	0	133.5	1161.3	86.39	61.07	1	0.16
318	-44.5	525.91	-4.2	-12.48	0	133.5	1161.3	86.39	61.07	1	0.16
319	-34.0	494.25	-4.2	-12.48	0.0	102.1	1129.6	81.73	61.72	1	0.15
320	-34.0	494.25	-4.2	-12.48	0	102.1	1129.6	81.73	61.72	1	0.15
321	-34.0	494.25	-4.2	-12.48	0	102.1	1129.6	81.73	61.72	1	0.15
322	-34.0	494.25	-4.2	-12.48	0	102.1	1129.6	81.73	61.72	1	0.15
323	81.3	35.25	-2.5	-7.41	0	-243.9	670.6	64.03	21.13	1	0.12
324	-134.7	476.27	-2.5	-7.47	0.0	404.2	1111.6	105.61	35.55	1	0.20
325	-26.3	141.97	44.0	131.97	0	78.8	777.3	70.39	28.32	1	0.13
326	-26.7	369.66	-48.9	-146.70	0	80.0	1005.0	86.55	41.07	1	0.16
327	87.6	16.25	-2.5	-7.41	0	-262.7	651.6	64.42	18.33	1	0.12
328	-128.5	457.28	-2.5	-7.47	0.0	385.4	1092.7	102.81	35.94	1	0.19
329	-20.0	122.98	44.0	131.97	0	59.9	758.4	67.59	28.71	1	0.13
330	-20.4	350.66	-48.9	-146.70	0	61.2	986.0	83.75	41.46	1	0.16
331	-26.7	256.76	-2.5	-7.44	0	80.2	892.1	64.24	49.05	1	0.12
332	-26.7	256.76	-2.5	-7.44	0	80.2	892.1	64.24	49.05	1	0.12
333	-26.7	256.76	-2.5	-7.44	0.0	80.2	892.1	64.24	49.05	1	0.12
334	-26.7	256.76	-2.5	-7.44	0	80.2	892.1	64.24	49.05	1	0.12
335	-20.4	237.77	-2.5	-7.44	0	61.3	873.1	61.44	49.44	1	0.11
336	-20.4	237.77	-2.5	-7.44	0	61.3	873.1	61.44	49.44	1	0.11
337	-20.4	237.77	-2.5	-7.44	0	61.3	873.1	61.44	49.44	1	0.11
338	-20.4	237.77	-2.5	-7.44	0.0	61.3	873.1	61.44	49.44	1	0.11

Max P_{max} = 112.76 KN/m^2

Max P_{min} = 18.33 KN/m^2

Resultant F.O.S = 0.21

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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	-62.5	602.4	-5.8	-17.37	0.0	187.35	56.00	20.50
402	-47.8	557.0	-5.8	-17.37	0	143.28	49.39	21.34
403	-56.9	827.4	-4.1	-12.18	0	170.61	68.30	36.76
404	-67.9	822.1	-7.6	-22.92	0	203.79	71.94	32.45
405	-42.2	782.0	-4.1	-12.15	0	126.54	61.68	37.62
406	-53.2	776.7	-7.6	-22.89	0	159.72	65.33	33.30
407	119.3	227.1	-5.0	-14.88	0	-357.93	46.34	-17.50
408	-226.3	933.2	-5.0	-15.00	0	678.99	118.36	0.14
409	-52.8	398.1	69.4	208.11	0	158.25	61.32	-10.78
410	-53.4	762.4	-79.3	-237.75	0	160.32	87.86	8.96
411	131.9	188.2	-5.0	-14.88	0	-395.70	47.07	-23.17
412	-213.7	894.3	-5.0	-15.00	0	641.22	112.70	0.86
413	-40.2	359.1	69.4	208.11	0	120.48	55.65	-10.05
414	-40.9	723.5	-79.3	-237.75	0	122.55	82.19	9.68
415	-53.5	581.7	-5.0	-14.94	0	160.53	52.15	21.72
416	-53.5	581.7	-5.0	-14.94	0	160.53	52.15	21.72
417	-53.5	581.7	-5.0	-14.94	0	160.53	52.15	21.72
418	-53.5	581.7	-5.0	-14.94	0	160.53	52.15	21.72
419	-40.9	542.8	-5.0	-14.94	0	122.76	46.48	22.45
420	-40.9	542.8	-5.0	-14.94	0	122.76	46.48	22.45
421	-40.9	542.8	-5.0	-14.94	0	122.76	46.48	22.45
422	-40.9	542.8	-5.0	-14.94	0	122.76	46.48	22.45
423	132.7	32.7	-3.7	-11.13	0	-398.04	36.98	-32.83
424	-213.0	738.7	-3.7	-11.22	0	638.88	102.21	-8.40
425	-39.4	203.6	70.6	211.89	0	118.17	45.99	-20.14
426	-40.1	567.9	-78.0	-234.00	0	120.21	71.70	0.41
427	142.1	3.5	-3.7	-11.10	0	-426.36	37.52	-37.08
428	-203.5	709.5	-3.7	-11.22	0	610.56	97.96	-7.86
429	-29.9	174.4	70.6	211.89	0	89.82	41.74	-19.59
430	-30.6	538.7	-78.0	-233.97	0	91.89	67.45	0.96
431	-40.2	387.3	-3.7	-11.16	0	120.45	36.00	13.18
432	-40.2	387.3	-3.7	-11.16	0	120.45	36.00	13.18
433	-40.2	387.3	-3.7	-11.16	0	120.45	36.00	13.18
434	-40.2	387.3	-3.7	-11.16	0	120.45	36.00	13.18
435	-30.7	358.1	-3.7	-11.16	0	92.10	31.75	13.72
436	-30.7	358.1	-3.7	-11.16	0	92.10	31.75	13.72
437	-30.7	358.1	-3.7	-11.16	0	92.10	31.75	13.72
438	-30.7	358.1	-3.7	-11.16	0	92.10	31.75	13.72

Max Net factored B.P = 118.36 KN/m^2

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Ma'aden Revision No:	A01	C01	C02	D01		
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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.85 m
Moment about Z — Muz = 202.55 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1333.87 KNm

As $M_{uzlim} > M_{uz}$ — Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 1235.2 mm^2

Effective Depth Along X = 442 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
$A_{stmin} = 0.0014bd$

A_{stmin} = 795.6 mm^2
A_{req} = 1235.2 mm^2

Required Spacing = **162.8** mm

Length of cantilever along Z = 1.35 m
Moment about X — Mux = 107.86 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1239.05 KNm

As $M_{uxlim} > M_{ux}$ — Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 676.6 mm^2

Effective Depth Along X = 426 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
$A_{stmin} = 0.0014bd$

A_{stmin} = 766.8 mm^2
A_{req} = 676.6 mm^2

Required Spacing = **262.2** mm

Provide Bottom Reinforcement = **16 mm dia @ 150 c/c** — Ast prov = 1340.4 mm^2

Top Reinforcement

Length of cantilever along X = 1.85 m
Moment about Z — Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1345.97 KNm

As $M_{uzlim} > M_{uz}$ — Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 444 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 399.6 mm^2
A_{req} = 0.0 mm^2

Required Spacing = **283.0** mm

Length of cantilever along Z = 1.35 m
Moment about X — Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1274.20 KNm

As $M_{uxlim} > M_{ux}$ — Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 432 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 388.8 mm^2
A_{req} = 0.0 mm^2

Required Spacing = **290.9** mm

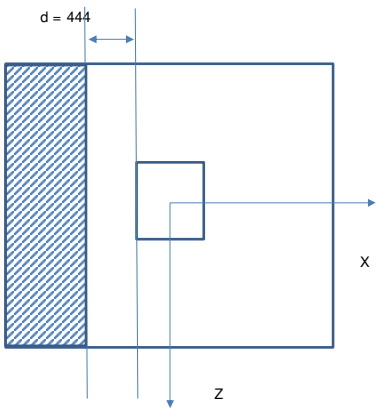
Provide top Reinforcement = **12 mm dia @ 150 c/c** — Ast prov = 754.0 mm^2

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ONE WAY SHEAR CKECK



Along X Direction Cantilever
Reinforcement ratio
Effective Depth Along X
Shear Force at d distance
Allowable shear strength

L	=	1.406	m		
ρ_w	=	0.0030			
d	=	444	mm		
V_u	=	166.42	KN		
φVc	=	326.00	KN	O.K	

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ	=	1	Normal concrete
Mu	=	116.99	KNm
φ	=	0.75	for shear
φVc	=	326.0026	KN

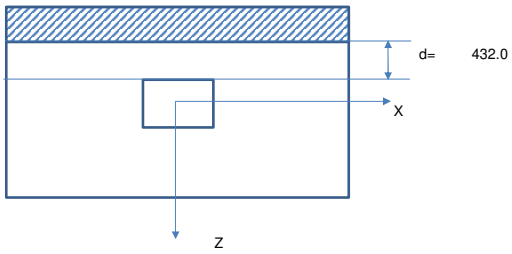
ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

φVc = 332.298995 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

= 571.315825 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever
Reinforcement ratio
Effective Depth Along Z
Shear Force at d distance
Allowable shear strength

L	=	0.918	m		
ρ_w	=	0.0031			
d	=	432	mm		
V_u	=	108.66	KN		
φVc	=	322.77	KN	O.K	

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ	=	1	Normal concrete
Mu	=	49.87	KNm
φ	=	0.75	for shear
φVc	=	322.77	KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

φVc = 323.77984 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

φVc = 555.87 KN

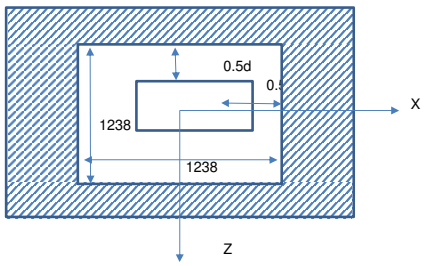
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TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	15.75	m2	
Punching area		=	1.53	m2	
	λ	=	1		
	β	=	1		
	αs	=	40		
	b0	=	4952.0	mm	

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.72 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	4.95	m	
Depth at section	d	=	438	mm	
Shear Force at section	Vu	=	1682.8	KN	
Allowable shear strength	φVc	=	3175.88	KN	O.K

Vc	=	$v_c b_0 d$	
	=	4234.51	KN
φ	=	0.75	for shear
φV_n	=	φVc	
φVc	=	3175.9	KN

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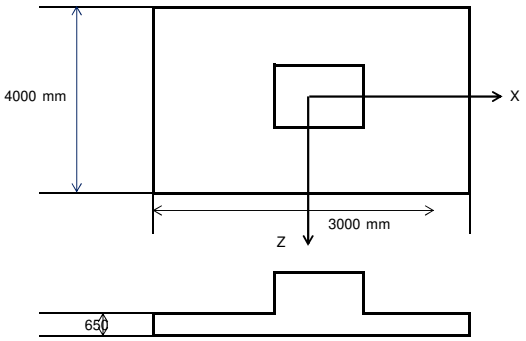
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REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.3.3 Design of foundation F2 at Grid 3130/C-3130/4

Finished Ground level	F.G.L	=	0.000	m	Weight of soil	=	276.05	KN
Bottom of foundation	B.O.F	=	-2.000	m	Weight of foundation	=	223.30	KN
Top of footing (staad support)	T.O.F	=	1.000	m	Uplift	=	0.00	KN
Ground Water Level	G.W.L	=	-10.000	m	Surcharge	=	0	KN
Surcharge	Ws	=	0.000	KN/m^2	Pressure on top for tension	=	41.6	KN/m^2
Length of footing	Lz	=	4.00	m				
Width of footing	Lx	=	3.00	m				
Thickness of footing	D	=	0.65	m				
Depth of pedestal	Bx	=	0.80	m				
Width of pedestal	Bz	=	0.80	m				
Height of Load point from bottom of footing	H	=	3.000	m				
Section Modulus of Pedestal	Zxx	=	8.00	m^3				
Section Modulus of Pedestal	Zzz	=	6.00	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	536	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	50	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	25	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

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Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG		

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	-42.7	482.1	-0.3	-0.84	0.0	128.0	981.4	103.23	60.35	1	0.19
302	-31.6	1357.11	1.2	3.60	0	94.7	1856.5	170.94	138.46	1	0.32
303	-40.1	781.8	-10.2	-30.66	0	120.2	1281.1	130.62	82.90	1	0.24
304	-29.0	1656.81	-8.7	-26.22	0	86.9	2156.2	197.44	161.92	1	0.37
305	-45.3	799.73	11.4	34.11	0	136.0	1299.1	135.19	81.32	1	0.25
306	-34.3	1674.74	12.9	38.55	0.0	102.8	2174.1	203.12	159.23	1	0.38
307	35.3	645.2	0.1	0.39	0	-105.8	1144.5	113.05	77.70	1	0.21
308	-120.6	785.05	0.6	1.83	0	361.9	1284.4	167.58	46.49	1	0.31
309	-43.0	846.1	67.1	201.27	0	128.9	1345.4	158.76	65.48	1	0.30
310	-42.8	608.3	-58.0	-174.00	0	128.5	1107.6	135.47	49.14	1	0.25
311	46.3	1520.21	1.6	4.83	0.0	-139.0	2019.6	192.07	144.52	1	0.36
312	-109.5	1660.06	2.1	6.27	0	328.6	2159.4	235.50	124.40	1	0.44
313	-31.9	1721.11	68.6	205.68	0	95.6	2220.5	226.68	143.39	1	0.42
314	-31.7	1483.31	-56.5	-169.56	0.0	95.2	1982.7	202.29	128.16	1	0.38
315	-42.7	713.6	0.4	1.08	0	128.1	1212.9	122.56	79.60	1	0.23
316	-42.7	713.6	0.4	1.08	0	128.1	1212.9	122.56	79.60	1	0.23
317	-42.7	713.6	0.4	1.08	0	128.1	1212.9	122.56	79.60	1	0.23
318	-42.7	713.6	0.4	1.08	0	128.1	1212.9	122.56	79.60	1	0.23
319	-31.6	1588.61	1.8	5.52	0.0	94.8	2088.0	190.49	157.51	1	0.36
320	-31.6	1588.61	1.8	5.52	0	94.8	2088.0	190.49	157.51	1	0.36
321	-31.6	1588.61	1.8	5.52	0	94.8	2088.0	190.49	157.51	1	0.36
322	-31.6	1588.61	1.8	5.52	0	94.8	2088.0	190.49	157.51	1	0.36
323	78.3	198.06	-0.5	-1.44	0	-234.9	697.4	97.45	18.78	1	0.18
324	-129.5	384.53	0.2	0.48	0.0	388.6	883.9	138.48	8.84	1	0.26
325	-26.0	465.93	88.8	266.40	0	77.9	965.3	126.72	34.15	1	0.24
326	-25.8	148.86	-78.0	-233.94	0	77.4	648.2	96.15	11.88	1	0.18
327	85.0	723.07	0.4	1.23	0	-254.9	1222.4	144.51	59.23	1	0.27
328	-122.9	909.53	1.1	3.15	0.0	368.6	1408.9	179.23	55.58	1	0.33
329	-19.3	990.94	89.7	269.04	0	57.9	1490.3	167.48	80.91	1	0.31
330	-19.1	673.87	-77.1	-231.30	0	57.4	1173.2	136.25	59.29	1	0.25
331	-25.6	289.26	-0.2	-0.51	0	76.8	788.6	78.58	52.85	1	0.15
332	-25.6	289.26	-0.2	-0.51	0	76.8	788.6	78.58	52.85	1	0.15
333	-25.6	289.26	-0.2	-0.51	0.0	76.8	788.6	78.58	52.85	1	0.15
334	-25.6	289.26	-0.2	-0.51	0	76.8	788.6	78.58	52.85	1	0.15
335	-19.0	814.27	0.7	2.16	0	56.9	1313.6	119.21	99.72	1	0.22
336	-19.0	814.27	0.7	2.16	0	56.9	1313.6	119.21	99.72	1	0.22
337	-19.0	814.27	0.7	2.16	0	56.9	1313.6	119.21	99.72	1	0.22
338	-19.0	814.27	0.7	2.16	0.0	56.9	1313.6	119.21	99.72	1	0.22

Max P_{max} = 235.50 KN/m^2

Max P_{min} = 8.84 KN/m^2

Resultant F.O.S = 0.44

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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	-59.6	669.9	-0.6	-1.86	0.0	178.89	85.87	25.78
402	-44.2	1896.7	1.5	4.59	0	132.45	180.71	135.41
403	-56.0	1181.6	-14.3	-42.90	0	167.91	131.82	65.12
404	-63.4	1206.9	15.9	47.79	0	190.14	138.24	62.91
405	-40.5	2408.5	-12.1	-36.42	0	121.44	225.50	175.91
406	-47.9	2433.7	18.1	54.24	0	143.70	233.54	172.08
407	115.1	582.7	-0.6	-1.80	0	-345.39	106.35	-9.23
408	-217.4	880.5	0.4	1.20	0	652.17	182.22	-35.47
409	-51.7	1011.1	142.2	426.69	0	155.16	163.46	5.06
410	-51.4	503.7	-124.6	-373.86	0	154.29	114.43	-30.47
411	128.4	1634.3	1.3	3.75	0	-385.23	200.86	71.52
412	-204.1	1932.1	2.3	6.75	0	612.36	263.91	58.11
413	-38.4	2062.7	144.1	432.24	0	115.32	245.14	98.64
414	-38.2	1555.3	-122.8	-368.31	0	114.45	194.72	64.50
415	-51.1	728.4	-0.1	-0.33	0	153.39	86.31	35.09
416	-51.1	728.4	-0.1	-0.33	0	153.39	86.31	35.09
417	-51.1	728.4	-0.1	-0.33	0	153.39	86.31	35.09
418	-51.1	728.4	-0.1	-0.33	0	153.39	86.31	35.09
419	-37.9	1780.0	1.7	5.22	0	113.55	167.91	128.76
420	-37.9	1780.0	1.7	5.22	0	113.55	167.91	128.76
421	-37.9	1780.0	1.7	5.22	0	113.55	167.91	128.76
422	-37.9	1780.0	1.7	5.22	0	113.55	167.91	128.76
423	127.9	284.9	-0.9	-2.67	0	-383.76	88.04	-40.55
424	-204.6	582.8	0.1	0.33	0	613.80	150.91	-53.78
425	-38.9	713.4	142.0	425.85	0	116.79	132.14	-13.25
426	-38.6	206.0	-124.9	-374.73	0	115.92	83.33	-49.00
427	137.9	1073.6	0.5	1.50	0	-413.64	158.60	20.34
428	-194.7	1371.5	1.5	4.50	0	583.95	212.18	16.40
429	-29.0	1502.0	143.3	429.99	0	86.91	193.40	56.94
430	-28.7	994.7	-123.5	-370.56	0	86.04	143.55	22.23
431	-38.3	430.6	-0.4	-1.20	0	115.02	55.21	16.57
432	-38.3	430.6	-0.4	-1.20	0	115.02	55.21	16.57
433	-38.3	430.6	-0.4	-1.20	0	115.02	55.21	16.57
434	-38.3	430.6	-0.4	-1.20	0	115.02	55.21	16.57
435	-28.4	1219.3	1.0	2.97	0	85.14	116.17	87.05
436	-28.4	1219.3	1.0	2.97	0	85.14	116.17	87.05
437	-28.4	1219.3	1.0	2.97	0	85.14	116.17	87.05
438	-28.4	1219.3	1.0	2.97	0	85.14	116.17	87.05

Max Net factored B.P = 263.91 KN/m^2

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Project Country	**KINGDOM OF SAUDI ARABIA**

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.1 m
Moment about Z: Muz = 159.67 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 2356.60 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 724.4 mm^2

Effective Depth Along X = 587.5 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
$A_{stmin} = 0.0014bd$

A_{stmin} = 1057.5 mm^2
A_{req} = 724.4 mm^2
Required Spacing = **464.2** mm

Length of cantilever along Z = 1.6 m
Moment about X: Mux = 337.81 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 2160.30 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 1619.7 mm^2

Effective Depth Along X = 562.5 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
$A_{stmin} = 0.0014bd$

A_{stmin} = 1012.5 mm^2
A_{req} = 1619.7 mm^2
Required Spacing = **303.1** mm

Provide Bottom Reinforcement = **25 mm dia @ 200 c/c**
Ast prov = 2454.4 mm^2

Top Reinforcement

Length of cantilever along X = 1.1 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 2409.03 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 0.0 mm^2

Effective Depth Along X = 594 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 534.6 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **211.6** mm

Length of cantilever along Z = 1.6 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 2312.68 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 0.0 mm^2

Effective Depth Along Z = 582 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

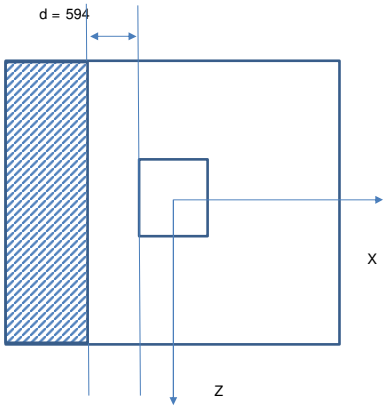
A_{stmin} = 523.8 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **215.9** mm

Provide top Reinforcement = **12 mm dia @ 200 c/c**
Ast prov = 565.5 mm^2

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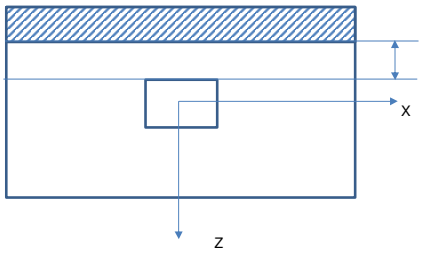
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ONE WAY SHEAR CKECK



Along X Direction Cantilever	L	=	0.506	m		Least of	i)	V_c	=	$\left(0.16\,\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio	ρ_w	=	0.0041							*Table 22.5.5.1 SBC 304*
Effective Depth Along X	d	=	594	mm				λ	=	1 Normal concrete
Shear Force at d distance	V_u	=	133.54	KN				Mu	=	33.79 KNm
Allowable shear strength	φVc	=	452.99	KN	O.K			φ	=	0.75 for shear
								φVc	=	495.169177 KN
							ii)	V_c	=	$\left(0.16\,\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	=	452.991375 KN *Table 22.5.5.1 SBC 304*
							iii)	V_c	=	$0.29\,\lambda\sqrt{f_c'}\,b_w d$
									=	764.327928 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever	L	=	1.018	m		Least of	i)	V_c	=	$\left(0.16\,\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio	ρ_w	=	0.0042							*Table 22.5.5.1 SBC 304*
Effective Depth Along Z	d	=	582	mm				λ	=	1 Normal concrete
Shear Force at d distance	V_u	=	268.66	KN				Mu	=	136.75 KNm
Allowable shear strength	φVc	=	444.47	KN	O.K			φ	=	0.75 for shear
								φVc	=	448.96 KN
							ii)	V_c	=	$\left(0.16\,\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	=	444.47222 KN
							iii)	V_c	=	$0.29\,\lambda\sqrt{f_c'}\,b_w d$
								φVc	=	748.89 KN

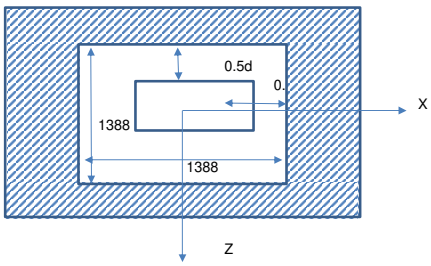
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TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	12	m2
Punching area	=	1.93	m2
λ	=	1	
β	=	1	
αs	=	40	
b0	=	5552.0	mm

Allowable shear stress least of *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 3.06 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	5.55	m	
Depth at section	d	=	588	mm	
Shear Force at section	Vu	=	2658.5	KN	
Allowable shear strength	φVc	=	4780.09	KN	O.K

Vc	=	$v_c b_0 d$	
	=	6373.45	KN
φ	=	0.75	for shear
φV_n	=	φVc	
φVc	=	4780.1	KN

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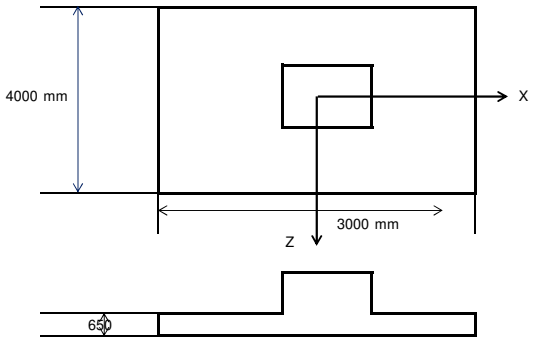
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REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.3.4 Design of foundation F2 at Grid 3130/C-3130/6

Finished Ground level	F.G.L	=	0.000	m	Weight of soil	=	276.05	KN
Bottom of foundation	B.O.F	=	-2.000	m	Weight of foundation	=	223.30	KN
Top of footing (staad support)	T.O.F	=	1.000	m	Uplift	=	0.00	KN
Ground Water Level	G.W.L	=	-10.000	m	Surcharge	=	0	KN
Surcharge	Ws	=	0.000	KN/m^2	Pressure on top for tension	=	41.6	KN/m^2
Length of footing	Lz	=	4.00	m				
Width of footing	Lx	=	3.00	m				
Thickness of footing	D	=	0.65	m				
Depth of pedestal	Bx	=	0.80	m				
Width of pedestal	Bz	=	0.80	m				
Height of Load point from bottom of footing	H	=	3.000	m				
Section Modulus of Pedestal	Zxx	=	8.00	m^3				
Section Modulus of Pedestal	Zzz	=	6.00	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	536	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	50	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	25	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

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CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
401	1.9	653.73	6.0	17.91	0.0	-5.7	1153.1	99.27	92.91	1	0.19
402	0.6	3355.47	6.2	18.57	0	-1.9	3854.8	323.88	318.59	1	0.60
403	-7.3	1109.19	8.6	25.68	0	21.9	1608.5	140.90	127.19	1	0.26
404	11.2	1103.15	20.3	60.96	0	-33.5	1602.5	146.74	120.35	1	0.27
405	-8.5	3810.93	8.8	26.34	0	25.6	4310.3	366.75	351.63	1	0.68
406	9.9	3804.89	20.6	61.65	0.0	-29.7	4304.2	371.35	346.02	1	0.69
407	2.3	1159.69	7.8	23.31	0	-7.0	1659.0	142.33	134.17	1	0.27
408	0.9	223.99	7.4	22.32	0	-2.8	723.3	63.53	57.03	1	0.12
409	1.7	784.65	18.9	56.73	0	-5.2	1284.0	114.95	99.05	1	0.21
410	1.5	603.01	-3.6	-10.68	0	-4.6	1102.4	93.96	89.77	1	0.18
411	1.3	3475.46	8.0	23.91	0.0	-3.8	3974.8	334.85	327.61	1	0.62
412	-0.2	2539.77	7.6	22.89	0	0.5	3039.1	256.20	250.32	1	0.48
413	0.7	3100.43	19.1	57.33	0	-2.0	3599.8	307.47	292.49	1	0.57
414	0.5	2918.79	-3.4	-10.11	0.0	-1.4	3418.1	286.33	283.36	1	0.53
415	1.6	693.41	7.6	22.83	0	-4.9	1192.8	103.06	95.73	1	0.19
416	1.6	693.41	7.6	22.83	0	-4.9	1192.8	103.06	95.73	1	0.19
417	1.6	693.41	7.6	22.83	0	-4.9	1192.8	103.06	95.73	1	0.19
418	1.6	693.41	7.6	22.83	0	-4.9	1192.8	103.06	95.73	1	0.19
419	0.6	3009.19	7.8	23.40	0.0	-1.7	3508.5	295.58	289.17	1	0.55
420	0.6	3009.19	7.8	23.40	0	-1.7	3508.5	295.58	289.17	1	0.55
421	0.6	3009.19	7.8	23.40	0	-1.7	3508.5	295.58	289.17	1	0.55
422	0.6	3009.19	7.8	23.40	0	-1.7	3508.5	295.58	289.17	1	0.55
423	1.9	886.53	4.0	12.00	0	-5.7	1385.9	117.94	113.03	1	0.22
424	0.5	-49.16	3.7	11.01	0.0	-1.5	450.2	39.14	35.89	1	0.07
425	1.3	511.5	15.1	45.42	0	-3.9	1010.8	90.57	77.90	1	0.17
426	1.1	329.86	-7.3	-21.99	0	-3.3	829.2	72.40	65.80	1	0.14
427	1.1	2623.36	4.2	12.45	0	-3.3	3122.7	262.34	258.11	1	0.49
428	-0.3	1687.67	3.8	11.43	0.0	0.9	2187.0	183.83	180.67	1	0.34
429	0.5	2248.33	15.3	45.87	0	-1.5	2747.7	234.96	222.99	1	0.44
430	0.3	2066.69	-7.2	-21.57	0	-0.9	2566.0	216.68	210.99	1	0.40
431	1.2	420.25	3.8	11.52	0	-3.6	919.6	78.68	74.59	1	0.15
432	1.2	420.25	3.8	11.52	0	-3.6	919.6	78.68	74.59	1	0.15
433	1.2	420.25	3.8	11.52	0.0	-3.6	919.6	78.68	74.59	1	0.15
434	1.2	420.25	3.8	11.52	0	-3.6	919.6	78.68	74.59	1	0.15
435	0.4	2157.09	4.0	11.94	0	-1.2	2656.4	223.07	219.67	1	0.42
436	0.4	2157.09	4.0	11.94	0	-1.2	2656.4	223.07	219.67	1	0.42
437	0.4	2157.09	4.0	11.94	0	-1.2	2656.4	223.07	219.67	1	0.42
438	0.4	2157.09	4.0	11.94	0.0	-1.2	2656.4	223.07	219.67	1	0.42

Max P_{max} = 334.85 KN/m^2

Max P_{min} = 35.89 KN/m^2

Resultant F.O.S = 0.62

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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	1.9	650.7	6.0	17.88	0.0	-5.64	57.40	51.05
402	0.6	3353.3	6.2	18.57	0	-1.89	282.08	276.80
403	-7.3	1105.9	8.5	25.62	0	21.90	99.01	85.31
404	11.2	1099.9	20.3	60.93	0	-33.45	104.85	78.47
405	-8.6	3808.5	8.8	26.31	0	25.65	324.94	309.81
406	9.9	3802.6	20.5	61.62	0	-29.70	329.53	304.23
407	2.3	1157.1	7.8	23.28	0	-6.96	100.50	92.36
408	0.9	221.2	7.4	22.29	0	-2.73	21.67	15.19
409	1.7	782.0	18.9	56.70	0	-5.16	73.11	57.22
410	1.5	600.3	-3.6	-10.71	0	-4.53	52.12	47.93
411	1.3	3473.7	8.0	23.88	0	-3.75	293.08	285.86
412	-0.2	2537.7	7.6	22.89	0	0.45	214.41	208.54
413	0.7	3098.5	19.1	57.30	0	-1.95	265.70	250.72
414	0.5	2916.8	-3.4	-10.11	0	-1.35	244.56	241.58
415	1.6	690.7	7.6	22.80	0	-4.86	61.22	53.90
416	1.6	690.7	7.6	22.80	0	-4.86	61.22	53.90
417	1.6	690.7	7.6	22.80	0	-4.86	61.22	53.90
418	1.6	690.7	7.6	22.80	0	-4.86	61.22	53.90
419	0.6	3007.3	7.8	23.37	0	-1.68	253.81	247.40
420	0.6	3007.3	7.8	23.37	0	-1.68	253.81	247.40
421	0.6	3007.3	7.8	23.37	0	-1.68	253.81	247.40
422	0.6	3007.3	7.8	23.37	0	-1.68	253.81	247.40
423	1.6	709.8	3.9	11.79	0	-4.92	61.44	56.86
424	0.8	124.8	3.7	11.16	0	-2.31	12.18	8.62
425	1.3	475.3	10.9	32.70	0	-3.81	44.33	34.89
426	1.1	361.8	-3.2	-9.45	0	-3.42	31.90	28.40
427	0.8	2447.2	4.1	12.24	0	-2.52	205.88	201.98
428	0.0	1862.2	3.9	11.61	0	0.12	156.65	153.71
429	0.5	2212.7	11.0	33.12	0	-1.38	188.76	180.02
430	0.3	2099.2	-3.0	-9.00	0	-1.02	176.23	173.64
431	1.2	418.3	3.8	11.49	0	-3.63	36.90	32.82
432	1.2	418.3	3.8	11.49	0	-3.63	36.90	32.82
433	1.2	418.3	3.8	11.49	0	-3.63	36.90	32.82
434	1.2	418.3	3.8	11.49	0	-3.63	36.90	32.82
435	0.4	2155.7	4.0	11.94	0	-1.23	181.34	177.94
436	0.4	2155.7	4.0	11.94	0	-1.23	181.34	177.94
437	0.4	2155.7	4.0	11.94	0	-1.23	181.34	177.94
438	0.4	2155.7	4.0	11.94	0	-1.23	181.34	177.94

Max Net factored B.P = 329.53 KN/m^2

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.1 m
Moment about Z: Muz = 199.37 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 2356.60 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 906.5 mm^2

Effective Depth Along X = 587.5 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
$A_{stmin} = 0.0014bd$

A_{stmin} = 1057.5 mm^2
A_{req} = 906.5 mm^2
Required Spacing = **464.2** mm

Length of cantilever along Z = 1.6 m
Moment about X: Mux = 421.80 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 2160.30 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 2033.2 mm^2

Effective Depth Along X = 562.5 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
$A_{stmin} = 0.0014bd$

A_{stmin} = 1012.5 mm^2
A_{req} = 2033.2 mm^2
Required Spacing = **241.4** mm

Provide Bottom Reinforcement = **25 mm dia @ 200 c/c**
Ast prov = 2454.4 mm^2

Top Reinforcement

Length of cantilever along X = 1.1 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 2409.03 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 0.0 mm^2

Effective Depth Along X = 594 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 534.6 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **211.6** mm

Length of cantilever along Z = 1.6 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$
= 2312.68 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$
= 0.0 mm^2

Effective Depth Along Z = 582 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 523.8 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **215.9** mm

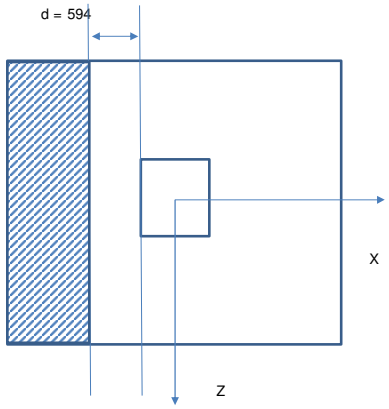
Provide top Reinforcement = **12 mm dia @ 200 c/c**
Ast prov = 565.5 mm^2

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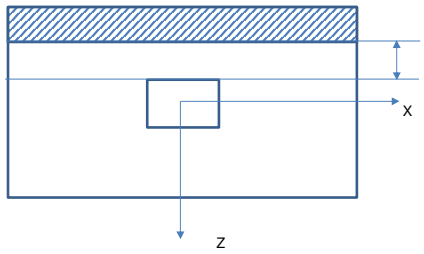
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ONE WAY SHEAR CKECK



Along X Direction	Cantilever	L	=	0.506	m	Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0041						*Table 22.5.5.1 SBC 304*
Effective Depth Along X		d	=	594	mm			λ	=	1 Normal concrete
Shear Force at d distance		V_u	=	166.74	KN			Mu	=	42.19 KNm
Allowable shear strength		φVc	=	452.99	KN	O.K		φ	=	0.75 for shear
								φVc	=	495.2 KN
							ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	=	453.0 KN *Table 22.5.5.1 SBC 304*
							iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
									=	764.3 KN

ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	1.018	m	Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0042						*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	582	mm			λ	=	1 Normal concrete
Shear Force at d distance		V_u	=	335.46	KN			Mu	=	170.75 KNm
Allowable shear strength		φVc	=	444.47	KN	O.K		φ	=	0.75 for shear
								φVc	=	449.0 KN
							ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	=	444.5 KN
							iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
								φVc	=	748.9 KN

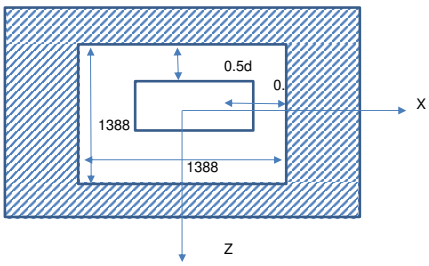
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TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	12	m2
Punching area	=	1.93	m2
λ	=	1	
β	=	1	
αs	=	40	
b0	=	5552.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 3.06 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	5.55	m		
Depth at section	d	=	588	mm		
Shear Force at section	Vu	=	3319.5	KN		
Allowable shear strength	φVc	=	4780.09	KN	O.K	

Vc	=	$v_c b_0 d$	
	=	6373.45	KN
φ	=	0.75	for shear
φV_n	=	φVc	
φVc	=	4780.1	KN

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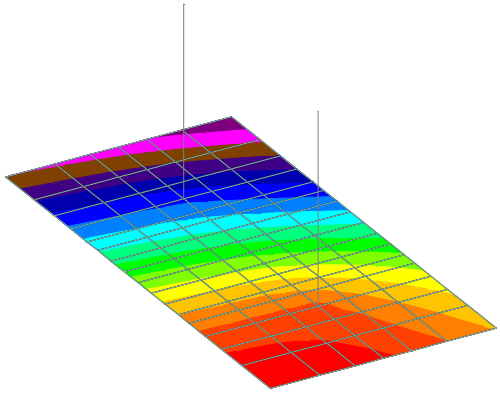
REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report

8.3.5 Design of combined footing for column at 3130/L_3130/4 & 3130/K_3130/4 and foundation mkd. as F4

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **FDN F-4 GR-4.STD**

Staad Supoort Node for column at **3130/L_3130/4** **66**
3130/K_3130/4 **9**





Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	0.5	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	6.5	m	
Width of footing	Lx	=	3.2	m	
Thickness of footing	D	=	0.5	m	
Depth of pedestal	Bx	=	0.8	m	
Width of pedestal	Bz	=	0.8	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	27	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	215.2	KN/m^2	Load Case 312
Minimum Base Pressure		=	4.6	KN/m^2	Load Case 323

2) Design of foundation Raft :

Slab Thickness =	500	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	rength Reduction Factor of Shear Reinforcement , Φt =	0.75	

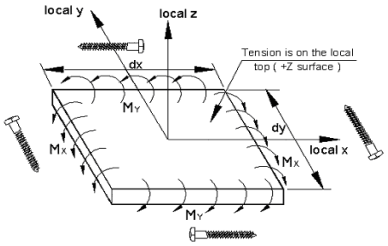
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : RAFT_MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	105.710	kN-m	Corresponding Mxy	-15.350	kN-m	*(Plate No. - 86 , L/C No. - 425)*
Min. Mx	-247.380	kN-m	Corresponding Mxy	17.570	kN-m	*(Plate No. - 76 , L/C No. - 414)*
Max. My	66.820	kN-m	Corresponding Mxy	-31.430	kN-m	*(Plate No. - 96 , L/C No. - 424)*
Min. My	-161.730	kN-m	Corresponding Mxy	-39.060	kN-m	*(Plate No. - 76 , L/C No. - 405)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	105.71 + 15.35 =	121.060 kN-m
My1 =	My + abs(Mxy) =	66.82 + 31.43 =	98.250 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	105.71 + 3.526 =	109.236 kN-m
My2 =	My+abs(Mxy^2/Mx) =	66.82 + 9.345 =	76.165 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 121.06 kN-m
Myd = 98.25 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 121.06 kN-m
Φ*Kn = Mu/bd^2 deff. = 419 mm
Mu/bd^2 = 121.06X10^6/(1000X419^2) = 0.69 N/mm^2 < 6.9 N/mm^2
Pt = 0.0018 Reqd. Ast =).0018X1000X419/100 = 7.54 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54)cm^2

Total Provide Ast = 7.54 cm^2 **O.K.**
Total Provide Pt = 0.0018

ALONG LOCAL Y - DIRECTION :

Mu = 98.25 kN-m
Φ*Kn = Mu/bd^2 deff. = 407 mm
Mu/bd^2 = 98.25X10^6/(1000X407^2) = 0.56 N/mm^2 < 6.9 N/mm^2
Pt = 0.0015 Reqd. Ast =).0015X1000X419/100 = 6.11 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54)cm^2

Total Provide Ast = 7.54 cm^2 **O.K.**
Total Provide Pt = 0.0019

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -247.38 - 17.57 = -264.95 kN-m
My1 = My - abs(Mxy) = -161.73 - 39.06 = -200.79 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -247.38 - 1.909 = -249.289 kN-m
My2 = My-abs(Mxy^2/Mx) = -161.73 - 6.167 = -155.563 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -264.95 kN-m
Myd = -200.79 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 264.95 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 264.95X10^6/(1000X415^2) = 1.538 N/mm^2 < 6.9 N/mm^2 deff. = 415 mm
Pt = 0.0042 Reqd. Ast =).0042X1000X415/100 = 17.43 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 **O.K.**
Total Provide Pt = 0.005

ALONG LOCAL Y - DIRECTION :

Mu = 200.79 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 200.79X10^6/(1000X395^2) = 1.287 N/mm^2 < 6.9 N/mm^2 deff. = 395 mm
Pt = 0.0035 Reqd. Ast =).0035X1000X395/100 = 13.83 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 **O.K.**
Total Provide Pt = 0.0053

SHEAR CHECK

(Staad Group : RAFT_SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.44 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 74 , L/C No. - 414)*
Vu = 182.60 kN

Provided Pt = 0.005
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.005X182.6X10^3X415/264.95/10^6)X1000X415/10^3
= 402.917 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.005)X1000X415/10^3
= 428.103 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X415/10^3
= 712 kN
HENCE, Vc = 402.917 kN
Φt*Vc = 0.75X402.917 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 302.188 kN **> 182.6T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.28 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 30 , L/C No. - 412)*
Vu = 110.6 kN

Pt provided = 0.0053
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0053X110.6X10^3X395/200.79/10^3)X1000X395/10^3
= 381.64 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0053)X1000X395/10^3
= 409.486 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X395/10^3
= 677.687 kN
HENCE, Vc = 381.64 kN
Φt*Vc = 0.75X381.64 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 286.23 kN **> 110.6T , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	01/27/2019	11-03-2020	28-04-2022	01-11-2023			Date	01/27/2019	11-03-2020	28-04-2022	01-11-2023	
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Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

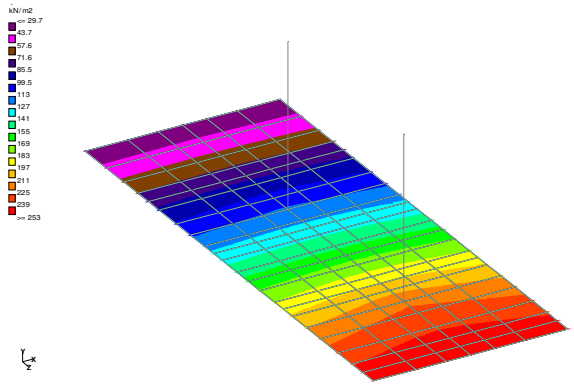
REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report

8.3.6 Design of combined footing for column at 3130/L_3130/6 & 3130/K_3130/6 and foundation mkd. as F5

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **FDN F-5 GR-5.STD**

Staad Supoort Node for column at **3130/L_3130/6** **74**
3130/K_3130/6 **17**

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +313.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	0.5	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	8.7	m	
Width of footing	Lx	=	3.0	m	
Thickness of footing	D	=	0.6	m	
Depth of pedestal	Bx	=	0.8	m	
Width of pedestal	Bz	=	0.8	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	25.2	KN/m^2	staad input



1) Check for Gross Base Pressure :

Maximum Base Pressure	=	223.3	KN/m^2	Load Case 311
Minimum Base Pressure	=	-13.7	KN/m^2	Load Case 329

2) Design of foundation Raft :

Slab Thickness =	600	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	rength Reduction Factor of Shear Reinforcement , Φt =	0.75	

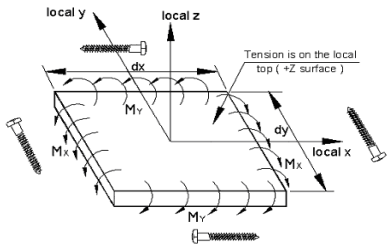
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : RAFT_MOMENT)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	139.940	kN-m	Corresponding Mxy	-2.440	kN-m	*(Plate No. - 155 , L/C No. - 425)*
Min. Mx	-467.350	kN-m	Corresponding Mxy	50.560	kN-m	*(Plate No. - 76 , L/C No. - 405)*
Max. My	24.880	kN-m	Corresponding Mxy	-4.130	kN-m	*(Plate No. - 155 , L/C No. - 425)*
Min. My	-129.260	kN-m	Corresponding Mxy	73.970	kN-m	*(Plate No. - 86 , L/C No. - 405)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	139.94 + 2.44 =	142.380 kN-m
My1 =	My + abs(Mxy) =	24.88 + 4.13 =	29.010 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	139.94 + 0.239 =	140.179 kN-m
My2 =	My+abs(Mxy^2/Mx) =	24.88 + 0.122 =	25.002 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 142.38 kN-m
Myd = 29.01 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 142.38 kN-m
Φ*Kn = Mu/bd^2 deff. = 517 mm
Mu/bd^2 = 142.38X10^6/(1000X517^2) = 0.533 N/mm^2 < 6.9 N/mm^2
Pt = 0.0014 Reqd. Ast =).0014X1000X517/100 = 7.24 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4 cm2) at Horizontal

Total Provide Ast = 13.4 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.0026

ALONG LOCAL Y - DIRECTION :

Mu = 29.01 kN-m
Φ*Kn = Mu/bd^2 deff. = 501 mm
Mu/bd^2 = 29.01X10^6/(1000X501^2) = 0.109 N/mm^2 < 6.9 N/mm^2
Pt = 0.0003 Reqd. Ast =).0003X1000X517/100 = 1.5 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 **O.K.**
Total Provide Pt = 0.0027

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	01/27/2019	11-03-2020	28-04-2022	01-11-2023			Date	01/27/2019	11-03-2020	28-04-2022	01-11-2023	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

FOR BOTTOM FACE REINFORCEMENT :

Mx1 = Mx - abs(Mxy) = -467.35 - 50.56 = -517.91 kN-m
My1 = My - abs(Mxy) = -129.26 - 73.97 = -203.23 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -467.35 - 19.777 = -487.127 kN-m
My2 = My-abs(Mxy^2/Mx) = -129.26 - 11.708 = -117.552 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -517.91 kN-m
Myd = -203.23 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 517.91 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 517.91X10^6/(1000X512.5^2) = 1.972 N/mm^2 < deff. = 512.5 mm 6.9 N/mm^2
Pt = 0.0054 Reqd. Ast =)054X1000X512.5/100 = 27.68 cm^2
Provided 25 mm @ 150 C/C (Provide Ast = 32.72)cm^2

Total Provide Ast = 32.72 cm^2 **O.K.**
Total Provide Pt = 0.0064

ALONG LOCAL Y - DIRECTION :

Mu = 203.23 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 203.23X10^6/(1000X487.5^2) = 0.855 N/mm^2 < deff. = 487.5 mm 6.9 N/mm^2
Pt = 0.0023 Reqd. Ast =)023X1000X487.5/100 = 11.21 cm^2
Provided 25 mm @ 150 C/C (Provide Ast = 32.72)cm^2

Total Provide Ast = 32.72 cm^2 **O.K.**
Total Provide Pt = 0.0067

SHEAR CHECK

(Staad Group : RAFT_SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.72 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 169 , L/C No. - 406)*
Vu = 369.00 kN

Provided Pt = 0.0064
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0064X369X10^3X512.5/517.91/10^6)X1000X512.5/10^3
= 505.479 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0064)X1000X512.5/10^3
= 540.879 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X512.5/10^3
= 879.277 kN
HENCE, Vc = 505.479 kN
Φt*Vc = 0.75X505.479 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 379.109 kN **> 369T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.26 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 45 , L/C No. - 406)*
Vu = 126.75 kN

Pt provided = 0.0067
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0067X126.75X10^3X487.5/203.23/10^3)X1000X487.5/10^3
= 478.337 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0067)X1000X487.5/10^3
= 516.98 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X487.5/10^3
= 836.386 kN
HENCE, Vc = 478.337 kN
Φt*Vc = 0.75X478.337 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 358.753 kN **> 126.75T , Hence O.K.**

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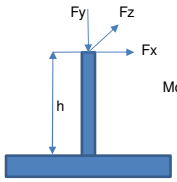
Ma'aden Revision No:	A01	C01	C02	D01		
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Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG		

Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
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Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4 Design of pedestal for building foundation

8.4.1 Design of pedestal P1 at grid 3130/C-3130/6



Moment at pedestal Bottom,
Mx = Fz *h
Mz = Fx *h

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Thickness of footing	=	0.5 m
Height of column - h	=	2.500 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	1.9	653.7	6.0	14.9	0.0	4.7
402	0.6	3355.5	6.2	15.5	0.0	1.6
403	-7.3	1109.2	8.6	21.4	0.0	-18.2
404	11.2	1103.2	20.3	50.8	0.0	27.9
405	-8.5	3810.9	8.8	22.0	0.0	-21.4
406	9.9	3804.9	20.6	51.4	0.0	24.8
407	2.3	1159.7	7.8	19.4	0.0	5.8
408	0.9	224.0	7.4	18.6	0.0	2.3
409	1.7	784.7	18.9	47.3	0.0	4.3
410	1.5	603.0	-3.6	-8.9	0.0	3.8
411	1.3	3475.5	8.0	19.9	0.0	3.2
412	-0.2	2539.8	7.6	19.1	0.0	-0.4
413	0.7	3100.4	19.1	47.8	0.0	1.6
414	0.5	2918.8	-3.4	-8.4	0.0	1.1
415	1.6	693.4	7.6	19.0	0.0	4.1
416	1.6	693.4	7.6	19.0	0.0	4.1
417	1.6	693.4	7.6	19.0	0.0	4.1
418	1.6	693.4	7.6	19.0	0.0	4.1
419	0.6	3009.2	7.8	19.5	0.0	1.4
420	0.6	3009.2	7.8	19.5	0.0	1.4
421	0.6	3009.2	7.8	19.5	0.0	1.4
422	0.6	3009.2	7.8	19.5	0.0	1.4
423	1.9	886.5	4.0	10.0	0.0	4.8
424	0.5	-49.2	3.7	9.2	0.0	1.3
425	1.3	511.5	15.1	37.9	0.0	3.3
426	1.1	329.9	-7.3	-18.3	0.0	2.8
427	1.1	2623.4	4.2	10.4	0.0	2.8
428	-0.3	1687.7	3.8	9.5	0.0	-0.8
429	0.5	2248.3	15.3	38.2	0.0	1.3
430	0.3	2066.7	-7.2	-18.0	0.0	0.8
431	1.2	420.3	3.8	9.6	0.0	3.0
432	1.2	420.3	3.8	9.6	0.0	3.0
433	1.2	420.3	3.8	9.6	0.0	3.0
434	1.2	420.3	3.8	9.6	0.0	3.0
435	0.4	2157.1	4.0	10.0	0.0	1.0
436	0.4	2157.1	4.0	10.0	0.0	1.0
437	0.4	2157.1	4.0	10.0	0.0	1.0
438	0.4	2157.1	4.0	10.0	0.0	1.0

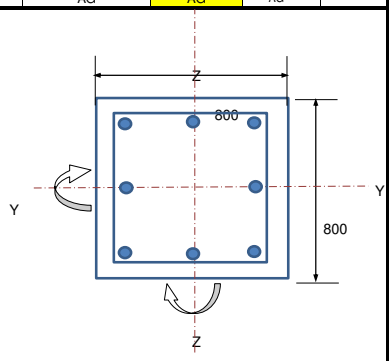
Summary of load in pedestal

	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	405	3811	22	-21
Fy min	424	-49	9	1
Mx max	406	3805	51	25
Mx min	426	330	-18	25
Mz max	404	1103	51	28
Mz min	405	3811	22	-21

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Dimension of column along Z	Dz	=	800	mm
Dimenson of column along Y	Dy	=	800	mm
Unsupported length of column along Z	Lz	=	2500	mm
Unsupported length of column along Y	Ly	=	2500	mm
Grade of steel	f_y	=	420	MPa
Grade of concrete	f_c'	=	35	MPa
Effective length factor along Z	Kz	=	2	
Effective length factor along Y	Ky	=	2	
Effective Depth in Z dir		=	655	mm
Effective Depth in Y dir		=	655	mm
Cover		=	50	mm
Effective cover		=	72.5	mm
Dia of longitudinal bar		=	25	mm
Dia of link		=	10	mm
Percentage of steel reinforcement		=	1	%



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	1103	KN	Critical Load Case No.
B.M about Z	Mz	=	M_{uz}	=	28	KNm	404 1.4 (DL + SDL1 + TL2) + 1.7 LL
B.M about Y	Mx	=	M_{uy}	=	51	KNm	

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \leq 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2500 / 240 r_z = 240

= 20.8 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2500 / 240 r_z = 240

= 20.8 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u = 1103 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\emptyset}$ = 1696.92 KN γ_z = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.076

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagra in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.09 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.09x640000x35x800/10^6 = 1612.8 KNm

Max moment capacity M_{uz} = ØMnz = 1048.32 KNm > 28 KNm *safe*

Utility ratio = 0.0267094

Moment Capacity about Y axis

P_u = 1103 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\emptyset}$ = 1696.92 KN γ_y = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.076

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagra in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.09 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.09x640000x35x800/10^6 = 1612.8 KNm

Max moment capacity M_{uy} = ØMny = 1048.32 KNm > 51 KNm *safe*

Utility ratio = ***0.049***

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Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023		
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT		
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Approved	AG	AG	AG	AG		

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Approved	AG	AG	AG	AG	

Combined moment capacity check

M_{noz} = 1612.8 KNm $M_{nz} = M_{uz}/\varphi$ = 43.0769231 KNm

M_{noy} = 1612.8 KNm $M_{ny} = M_{uy}/\varphi$ = 78.4615385 KNm

Assume β = 0.65 log β = -0.187087

log 0.5 = -0.30103

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta} < 1$ *Equation 3-15 SP-17M(09)*

= 0.00 + 0.01 = 0.0107 < 1 *Safe*

Longitudinal reinforcement: 16 nos 20 mm — Required Ast = 6400 mm^2

Spread equally on all four side: 4 nos 25 mm

Total Ast = 6990.0437 mm^2 O.K

Minimim Reinforcement = 0.01Ag = 6400 mm^2

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist.
a) 16db = 400 mm
b) 48db = 480 mm
c) Smaller column dimension = 800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 49 KN M_{uz} = 1 KNm M_{uy} = 9 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\emptyset}$ = 54.44 KN A_g = 800 x 800 = 640000 mm^2

f_c' = 35 N/mm^2

$K_n = \frac{P_n}{A_g f_c'}$ = 0.002 ρ_s = 0.01

γ_z = 0.81875

γ_y = 0.81875

From the tension curve corresponding to the values of K_n γ_y f_y f_c' ρ_s as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05

M_{nz} = 896 KNm M_{ny} = 896 KNm

$\emptyset M_{nz}$ = 806.4 KNm >1 *safe* $\emptyset M_{ny}$ = 806.4 KNm >9 *safe*

Utility ratio = ***0.00*** *Utility ratio =* ***0.01***

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

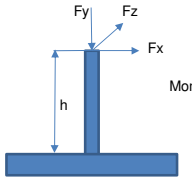
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REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.2 Design of pedestal P1 at grid 3130/C-3130/4

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Thickness of footing	=	0.5 m
Height of column - h	=	2.500 m



Moment at pedestal Bottom,
Mx = Fz *h
Mz = Fx *h

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	-59.6	669.9	-0.6	-1.6	0.0	-149.1
402	-44.2	1896.7	1.5	3.8	0.0	-110.4
403	-56.0	1181.6	-14.3	-35.8	0.0	-139.9
404	-63.4	1206.9	15.9	39.8	0.0	-158.5
405	-40.5	2408.5	-12.1	-30.4	0.0	-101.2
406	-47.9	2433.7	18.1	45.2	0.0	-119.8
407	115.1	582.7	-0.6	-1.5	0.0	287.8
408	-217.4	880.5	0.4	1.0	0.0	-543.5
409	-51.7	1011.1	142.2	355.6	0.0	-129.3
410	-51.4	503.7	-124.6	-311.6	0.0	-128.6
411	128.4	1634.3	1.3	3.1	0.0	321.0
412	-204.1	1932.1	2.3	5.6	0.0	-510.3
413	-38.4	2062.7	144.1	360.2	0.0	-96.1
414	-38.2	1555.3	-122.8	-306.9	0.0	-95.4
415	-51.1	728.4	-0.1	-0.3	0.0	-127.8
416	-51.1	728.4	-0.1	-0.3	0.0	-127.8
417	-51.1	728.4	-0.1	-0.3	0.0	-127.8
418	-51.1	728.4	-0.1	-0.3	0.0	-127.8
419	-37.9	1780.0	1.7	4.4	0.0	-94.6
420	-37.9	1780.0	1.7	4.4	0.0	-94.6
421	-37.9	1780.0	1.7	4.4	0.0	-94.6
422	-37.9	1780.0	1.7	4.4	0.0	-94.6
423	65.6	339.6	-0.7	-1.8	0.0	164.0
424	-142.3	525.7	-0.1	-0.2	0.0	-355.6
425	-38.7	607.3	88.6	221.4	0.0	-96.8
426	-38.5	290.2	-78.2	-195.6	0.0	-96.3
427	75.5	1128.3	0.7	1.7	0.0	188.8
428	-132.3	1314.4	1.3	3.3	0.0	-330.8
429	-28.8	1396.0	90.0	224.9	0.0	-71.9
430	-28.6	1078.9	-76.8	-192.1	0.0	-71.4
431	-38.3	430.6	-0.4	-1.0	0.0	-95.9
432	-38.3	430.6	-0.4	-1.0	0.0	-95.9
433	-38.3	430.6	-0.4	-1.0	0.0	-95.9
434	-38.3	430.6	-0.4	-1.0	0.0	-95.9
435	-28.4	1219.3	1.0	2.5	0.0	-71.0
436	-28.4	1219.3	1.0	2.5	0.0	-71.0
437	-28.4	1219.3	1.0	2.5	0.0	-71.0
438	-28.4	1219.3	1.0	2.5	0.0	-71.0

Summary of load in pedestal

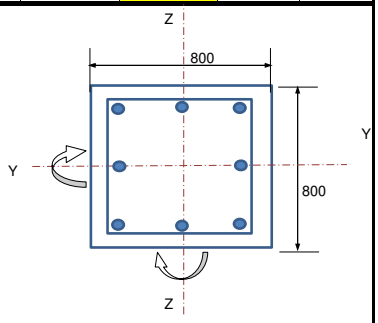
	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	406	2434	45	-120
Fy min	426	290	-196	-96
Mx max	413	2063	360	-96
Mx min	410	504	-312	-96
Mz max	411	1634	3	321
Mz min	408	881	1	-543

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023		
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Dimension of column along Z	Dz	=	800 mm
Dimenson of column along Y	Dy	=	800 mm
Unsupported length of column along Z	Lz	=	2500 mm
Unsupported length of column along Y	Ly	=	2500 mm
Grade of steel	f_y	=	420 MPa
Grade of concrete	f_c'	=	35 MPa
Effective length factor along Z	Kz	=	2
Effective length factor along Y	Ky	=	2
Effective Depth in Z dir		=	655 mm
Effective Depth in Y dir		=	655 mm
Cover		=	50 mm
Effective cover		=	72.5 mm
Dia of longitudinal bar		=	25 mm
Dia of link		=	10 mm
Percentage of steel reinforcement		=	1 %



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	504 KN	Critical Load Case No.	
B.M about Z	Mz	=	M_{uz}	=	96 KNm	410	1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
B.M about Y	Mx	=	M_{uy}	=	312 KNm		

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \leq 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2500 / 240 r_z = 240

= 20.8 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2500 / 240 r_z = 240

= 20.8 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u = 504 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = 775.38 KN γ_z = 0.81875

$K_n = \frac{P_n}{A_g f_c'}$ = 0.035

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagram in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.06 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.06x640000x35x800/10^6 = 1075.2 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 698.88 KNm > 96 KNm *safe*

Utility ratio = ***0.137***

Moment Capacity about Y axis

P_u = 504 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = 775.38 KN γ_y = 0.82

$K_n = \frac{P_n}{A_g f_c'}$ = 0.035

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagram in SP-17M(09)

#DIV/0!

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.06 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.06x640000x35x800/10^6 = 1075.2 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 698.88 KNm > 312 KNm *safe*

Utility ratio = ***0.45***

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
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Combined moment capacity check

M_{noz} = 1075.2 KNm M_{nz} = M_{uz}/φ = 147.6923077 KNm

M_{noy} = 1075.2 KNm M_{ny} = M_{uy}/φ = 480 KNm

Assume β = 0.65 log β = -0.1870866

log 0.5 = -0.30103

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta}$ < 1 *Equation 3-15 SP-17M(09)*

= 0.04 + 0.27 = **0.31** < 1 *Safe*

Longitudinal reinforcement: 16 nos 20 mm Required Ast = 6400 mm^2

Spread equally on all four side: 4 nos 25 mm

Total Ast = 6990.0437 mm^2 O.K

Minimim Reinforcement = 0.01Ag = 6400 mm^2

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist. a) 16db = 400 mm

b) 48db = 480 mm

c) Smaller column dimension = 800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 287 KN M_{uz} = 1 KNm M_{uy} = 192 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\emptyset}$ = 318.89 KN A_g = 800 x 800 = 640000 mm^2

f_c' = 35 N/mm^2

$K_n = \frac{P_n}{A_g f_c'}$ = 0.014 ρ_s = 0.01

γ_z = 0.81875

γ_y = 0.81875

From the tension curve corresponding to the values of K_n γ_y f_y f_c' ρ_s as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05

M_{nz} = 896 KNm M_{ny} = 896 KNm

$\emptyset M_{nz}$ = 806.4 KNm >1 $\emptyset M_{ny}$ = 806.4 KNm >192

safe *safe*

Utility ratio = ***0.00*** *Utility ratio* = ***0.24***

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
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Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.3 Design of pedestal P1 at grid 3130/C-3130/3

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Height of column - h	=	0.5 m
Height of column	=	2.500 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	-60.1	622.1	10.7	26.8	0.0	-150.3
402	-45.5	575.9	10.7	26.7	0.0	-113.7
403	-53.6	859.4	9.0	22.4	0.0	-134.0
404	-67.1	859.7	12.5	31.4	0.0	-167.8
405	-39.0	813.2	9.0	22.4	0.0	-97.4
406	-52.5	813.6	12.5	31.4	0.0	-131.2
407	103.9	284.0	9.2	22.9	0.0	259.7
408	-207.1	919.1	9.2	23.0	0.0	-517.6
409	-52.3	787.0	82.8	207.1	0.0	-130.8
410	-51.7	422.6	-64.4	-161.0	0.0	-129.2
411	116.4	244.4	9.2	22.9	0.0	291.0
412	-194.5	879.5	9.2	23.0	0.0	-486.3
413	-39.8	747.5	82.8	207.1	0.0	-99.5
414	-39.1	383.0	-64.4	-161.0	0.0	-97.9
415	-51.6	603.1	9.2	23.0	0.0	-129.0
416	-51.6	603.1	9.2	23.0	0.0	-129.0
417	-51.6	603.1	9.2	23.0	0.0	-129.0
418	-51.6	603.1	9.2	23.0	0.0	-129.0
419	-39.1	563.5	9.2	23.0	0.0	-97.7
420	-39.1	563.5	9.2	23.0	0.0	-97.7
421	-39.1	563.5	9.2	23.0	0.0	-97.7
422	-39.1	563.5	9.2	23.0	0.0	-97.7
423	116.8	80.9	6.9	17.2	0.0	292.1
424	-194.1	716.0	6.9	17.2	0.0	-485.3
425	-39.4	583.9	80.5	201.3	0.0	-98.4
426	-38.7	219.5	-66.7	-166.8	0.0	-96.8
427	126.2	51.2	6.9	17.2	0.0	315.6
428	-184.7	686.3	6.9	17.2	0.0	-461.8
429	-30.0	554.2	80.5	201.3	0.0	-74.9
430	-29.3	189.8	-66.7	-166.8	0.0	-73.3
431	-38.6	399.9	6.9	17.2	0.0	-96.6
432	-38.6	399.9	6.9	17.2	0.0	-96.6
433	-38.6	399.9	6.9	17.2	0.0	-96.6
434	-38.6	399.9	6.9	17.2	0.0	-96.6
435	-29.2	370.2	6.9	17.2	0.0	-73.1
436	-29.2	370.2	6.9	17.2	0.0	-73.1
437	-29.2	370.2	6.9	17.2	0.0	-73.1
438	-29.2	370.2	6.9	17.2	0.0	-73.1

Summary of load in pedestal

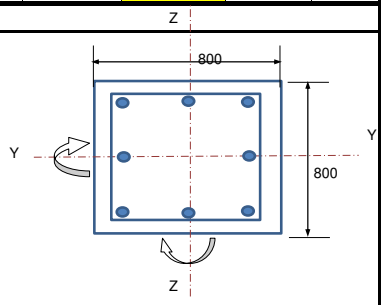
	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	408	919	23	-518
Fy min	427	51	17	316
Mx max	409	787	207	-131
Mx min	426	219	-167	-131
Mz max	427	51	17	316
Mz min	408	919	23	-518

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023		
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Prepared By	RPK	RPK	RPK	RPK		
Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG		

Outotec / L&T Revision No.:	0	1	2	3	
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Dimension of column along Z	D_z	=	800 mm
Dimenson of column along Y	D_y	=	800 mm
Unsupported length of column along Z	L_z	=	2500 mm
Unsupported length of column along Y	L_y	=	2500 mm
Grade of steel	f_y	=	420 MPa
Grade of concrete	f_c'	=	35 MPa
Effective length factor along Z	K_z	=	2
Effective length factor along Y	K_y	=	2
Effective Depth in Z dir		=	655 mm
Effective Depth in Y dir		=	655 mm
Cover		=	50 mm
Effective cover		=	72.5 mm
Dia of longitudinal bar		=	25 mm
Dia of link		=	10 mm
Percentage of steel reinforcement		=	1 %



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	919 KN	Load Case No.	
B.M about Z	Mz	=	M_{uz}	=	518 KNm	408	1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
B.M about Y	Mx	=	M_{uy}	=	23 KNm		

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \le 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2500 / 240 r_z = 240

= 20.8 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2500 / 240 r_z = 240

= 20.8 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u	=	919 KN			A_g	=	800 x 800 =	640000 mm^2
Assume Strength reduction factor φ	=	0.65 *Table 21.2.2*	*SBC 304*		f_c'	=	35 N/mm^2	
					ρ_s	=	0.01	
$P_n = \frac{P_u}{\phi}$	=	1413.85 KN			γ_z	=	0.82	
$K_n = \frac{P_n}{A_g f_c'}$	=	0.063						

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagrar in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.07 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.07x640000x35x800/10^6 = 1254.4 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 815.36 KNm > 518 KNm *safe*

Utility ratio = ***0.635***

Moment Capacity about Y axis

P_u	=	919 KN			A_g	=	800 x 800 =	640000 mm^2
Assume Strength reduction factor φ	=	0.65 *Table 21.2.2*	*SBC 304*		f_c'	=	35 N/mm^2	
					ρ_s	=	0.01	
$P_n = \frac{P_u}{\phi}$	=	1413.85 KN			γ_y	=	0.82	
$K_n = \frac{P_n}{A_g f_c'}$	=	0.063						

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagrar in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.07 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.07x640000x35x800/10^6 = 1254.4 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 815.36 KNm > 23 KNm *safe*

Utility ratio = ***0.03***

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Combined moment capacity check

M_{noz} = 1254.4 KNm $M_{nz} = M_{uz}/\varphi$ = 796.9230769 KNm

M_{noy} = 1254.4 KNm $M_{ny} = M_{uy}/\varphi$ = 35.38461538 KNm

Assume β = 0.65 log β = -0.1870866

log 0.5 = -0.30103

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta} < 1$ *Equation 3-15 SP-17M(09)*

= 0.48 + 0.00 = **0.49** < 1 *Safe*

Longitudinal reinforcement		16	nos	20	mm	Required Ast = 6400 mm^2
Spread equally on all four side		4	nos	25	mm	
		Total Ast	=	6990.0437	mm^2	O.K
Minimim Reinforcement	=	0.01Ag	=	6400	mm^2	

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist.	a)	16db	=	400 mm
	b)	48db	=	480 mm
	c)	Smaller column dimension	=	800 mm

Provide lateral ties 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 0 KN M_{uz} = 0 KNm M_{uy} = 0 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\emptyset}$ = 0.00 KN A_g = 800 x 800 = 640000 mm^2

$K_n = \frac{P_n}{A_g f_c'}$ = 0.000 f_c' = 35 N/mm^2

ρ_s = 0.01

γ_z = 0.82

γ_y = 0.82

From the tension curve corresponding to the values of K_n γ_y f_y f_c' ρ_s as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$	=	0.05		$\frac{M_{ny}}{A_g f_c' D_y}$	=	0.05	
M_{nz}	=	896 KNm		M_{ny}	=	896 KNm	
$\emptyset M_{nz}$	=	806.4 KNm	>0	$\emptyset M_{ny}$	=	806.4 KNm	>0
			safe				*safe*
		Utility ratio =	***0.00***			*Utility ratio =*	***0.00***

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
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Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.4 Design of pedestal P1 at grid 3130/K-3130/6

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Height of column - h	=	0.5 m
Height of column	=	2.500 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	-0.1	505.0	-1.0	-2.4	0.0	-0.2
402	-0.6	2619.0	25.6	63.9	0.0	-1.5
403	-9.5	1096.1	-48.1	-120.2	0.0	-23.7
404	8.8	1072.5	53.3	133.2	0.0	22.1
405	-10.0	3210.1	-21.6	-53.9	0.0	-25.1
406	8.3	3186.4	79.8	199.5	0.0	20.8
407	3.7	1045.2	6.0	15.0	0.0	9.3
408	-4.0	158.8	-5.6	-14.0	0.0	-10.0
409	-0.3	1072.2	93.7	234.2	0.0	-0.8
410	0.0	151.8	-91.1	-227.7	0.0	0.1
411	3.2	2857.1	28.8	71.9	0.0	8.1
412	-4.4	1970.7	17.1	42.8	0.0	-11.1
413	-0.8	2884.1	116.4	291.0	0.0	-1.9
414	-0.4	1963.8	-68.4	-170.9	0.0	-1.1
415	-0.1	603.3	0.2	0.6	0.0	-0.4
416	-0.1	603.3	0.2	0.6	0.0	-0.4
417	-0.1	603.3	0.2	0.6	0.0	-0.4
418	-0.1	603.3	0.2	0.6	0.0	-0.4
419	-0.6	2415.2	23.0	57.4	0.0	-1.5
420	-0.6	2415.2	23.0	57.4	0.0	-1.5
421	-0.6	2415.2	23.0	57.4	0.0	-1.5
422	-0.6	2415.2	23.0	57.4	0.0	-1.5
423	3.8	766.6	5.2	13.0	0.0	9.5
424	-3.9	-119.8	-6.4	-16.1	0.0	-9.7
425	-0.2	793.6	92.8	232.1	0.0	-0.5
426	0.1	-126.8	-91.9	-229.8	0.0	0.3
427	3.5	2125.5	22.2	55.6	0.0	8.6
428	-4.2	1239.1	10.6	26.5	0.0	-10.6
429	-0.6	2152.6	109.9	274.7	0.0	-1.4
430	-0.2	1232.2	-74.9	-187.2	0.0	-0.6
431	-0.1	324.7	-0.6	-1.5	0.0	-0.1
432	-0.1	324.7	-0.6	-1.5	0.0	-0.1
433	-0.1	324.7	-0.6	-1.5	0.0	-0.1
434	-0.1	324.7	-0.6	-1.5	0.0	-0.1
435	-0.4	1683.6	16.4	41.1	0.0	-1.0
436	-0.4	1683.6	16.4	41.1	0.0	-1.0
437	-0.4	1683.6	16.4	41.1	0.0	-1.0
438	-0.4	1683.6	16.4	41.1	0.0	-1.0

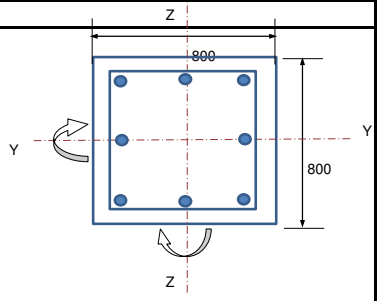
Summary of load in pedestal

	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	405	3210	-54	-25
Fy min	426	-127	-230	0
Mx max	413	2884	291	-2
Mx min	426	-127	-230	-2
Mz max	404	1072	133	22
Mz min	405	3210	-54	-25

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Dimension of column along Z	Dz	=	800	mm
Dimenson of column along Y	Dy	=	800	mm
Unsupported length of column along Z	Lz	=	2500	mm
Unsupported length of column along Y	Ly	=	2500	mm
Grade of steel	f_y	=	420	MPa
Grade of concrete	f_c'	=	35	MPa
Effective length factor along Z	Kz	=	2	
Effective length factor along Y	Ky	=	2	
Effective Depth in Z dir		=	655	mm
Effective Depth in Y dir		=	655	mm
Cover		=	50	mm
Effective cover		=	72.5	mm
Dia of longitudinal bar		=	25	mm
Dia of link		=	10	mm
Percentage of steel reinforcement		=	1	%



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	2884	KN	Load Case No.
B.M about Z	Mz	=	M_{uz}	=	2	KNm	410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
B.M about Y	Mx	=	M_{uy}	=	291	KNm	

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \leq 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z} = 2 \times 2500 / 240 = 20.8$ $r_z = 240$ *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y} = 2 \times 2500 / 240 = 20.8$ $r_z = 240$ *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u	=	2884	KN			A_g	=	800 x 800 =	640000	mm^2
Assume Strength reduction factor φ	=	0.65	*Table 21.2.2*	*SBC 304*		f_c'	=	35	N/mm^2	
						ρ_s	=	0.01		
$P_n = \frac{P_u}{\phi}$	=	4436.92	KN			γ_z	=	0.82		
$K_n = \frac{P_n}{A_g f_c'}$	=	0.198								

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagra in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.1 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.1x640000x35x800/10^6 = 1792 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 1164.8 KNm > 2 KNm *safe*

Utility ratio = ***0.002***

Moment Capacity about Y axis

P_u	=	2884	KN			A_g	=	800 x 800	640000	mm^2
Assume Strength reduction factor φ	=	0.65	*Table 21.2.2*	*SBC 304*		f_c'	=	35	N/mm^2	
						ρ_s	=	0.01		
$P_n = \frac{P_u}{\phi}$	=	4436.92	KN			γ_y	=	0.82		
$K_n = \frac{P_n}{A_g f_c'}$	=	0.198								

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagra in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.1 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.1x640000x35x800/10^6 = 1792 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 1164.8 KNm > 291 KNm *safe*

Utility ratio = ***0.25***

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Combined moment capacity check

M_{noz} = 1792 KNm $M_{nz} = M_{uz}/\varphi$ = 3.08 KNm

M_{noy} = 1792 KNm $M_{ny} = M_{uy}/\varphi$ = 447.69 KNm

Assume β = 0.65 log β = -0.19 log 0.5 = -0.30

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta} < 1$ *Equation 3-15 SP-17M(09)*

= 0.00 + 0.11 = **0.11** < 1 *Safe*

Longitudinal reinforcement: 16 nos 20 mm Required Ast = 6400 mm^2

Spread equally on all four side: 4 nos 25 mm

Total Ast = 6990.0437 mm^2 O.K

Minimim Reinforcement = 0.01Ag = 6400 mm^2

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist.
a) 16db = 400 mm
b) 48db = 480 mm
c) Smaller column dimension = 800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 127 KN M_{uz} = 0 KNm M_{uy} = 130 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\emptyset}$ = 141.11 KN A_g = 800 x 800 = 640000 mm^2

$K_n = \frac{P_n}{A_g f_c'}$ = 0.006 f_c' = 35 N/mm^2

ρ_s = 0.01

γ_z = 0.82

γ_y = 0.82

From the tension curve corresponding to the values of K_n γ_y f_y f_c' ρ_s as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05

M_{nz} = 896 KNm M_{ny} = 896 KNm

$\emptyset M_{nz}$ = 806.4 KNm >0 *safe* $\emptyset M_{ny}$ = 806.4 KNm >130 *safe*

Utility ratio = ***0.00*** *Utility ratio* = ***0.16***

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023		
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT		
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Checked	SBY	SBY	SBY	SBY		
Approved	AG	AG	AG	AG		

Outotec / L&T Revision No.:	0	1	2	3	
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Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.4 Design of pedestal P1 at grid 3130/K-3130/4

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Height of column - h	=	0.5 m
Height of column	=	2.500 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	-61.5	646.9	-21.4	-53.6	0.0	-153.9
402	-45.8	1500.6	-24.7	-61.6	0.0	-114.4
403	-58.7	1274.7	-10.8	-27.1	0.0	-146.7
404	-63.5	1206.9	-33.9	-84.6	0.0	-158.8
405	-42.9	2128.4	-14.0	-35.1	0.0	-107.3
406	-47.7	2060.6	-37.1	-92.7	0.0	-119.3
407	132.5	634.4	-17.2	-42.9	0.0	331.2
408	-237.7	828.4	-20.1	-50.3	0.0	-594.2
409	-52.0	578.9	142.2	355.5	0.0	-130.0
410	-52.4	854.5	-159.6	-399.1	0.0	-130.9
411	146.0	1366.2	-19.9	-49.8	0.0	365.0
412	-224.2	1560.2	-22.9	-57.2	0.0	-560.4
413	-38.5	1310.6	139.4	348.6	0.0	-96.2
414	-38.8	1586.2	-162.4	-406.0	0.0	-97.1
415	-52.6	729.1	-18.6	-46.6	0.0	-131.6
416	-52.6	729.1	-18.6	-46.6	0.0	-131.6
417	-52.6	729.1	-18.6	-46.6	0.0	-131.6
418	-52.6	729.1	-18.6	-46.6	0.0	-131.6
419	-39.1	1460.9	-21.4	-53.5	0.0	-97.7
420	-39.1	1460.9	-21.4	-53.5	0.0	-97.7
421	-39.1	1460.9	-21.4	-53.5	0.0	-97.7
422	-39.1	1460.9	-21.4	-53.5	0.0	-97.7
423	145.5	321.1	-12.3	-30.7	0.0	363.8
424	-224.6	515.1	-15.3	-38.2	0.0	-561.6
425	-38.9	265.5	147.0	367.6	0.0	-97.4
426	-39.3	541.2	-154.8	-387.0	0.0	-98.2
427	155.7	869.9	-14.4	-35.9	0.0	389.2
428	-214.5	1063.9	-17.3	-43.4	0.0	-536.2
429	-28.8	814.4	145.0	362.5	0.0	-72.0
430	-29.2	1090.0	-156.8	-392.1	0.0	-72.9
431	-39.6	415.8	-13.8	-34.5	0.0	-98.9
432	-39.6	415.8	-13.8	-34.5	0.0	-98.9
433	-39.6	415.8	-13.8	-34.5	0.0	-98.9
434	-39.6	415.8	-13.8	-34.5	0.0	-98.9
435	-29.4	964.7	-15.9	-39.6	0.0	-73.6
436	-29.4	964.7	-15.9	-39.6	0.0	-73.6
437	-29.4	964.7	-15.9	-39.6	0.0	-73.6
438	-29.4	964.7	-15.9	-39.6	0.0	-73.6

Summary of load in pedestal

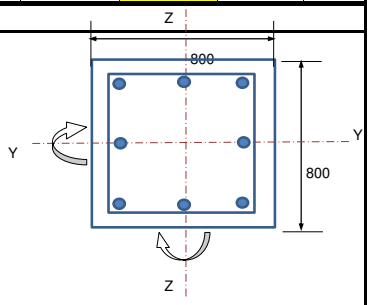
	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	405	2128	-35	-107
Fy min	425	266	368	-97
Mx max	425	266	368	-97
Mx min	414	1586	-406	-97
Mz max	427	870	-36	389
Mz min	408	828	-50	-594

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023		
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Outotec / L&T Revision No.:	0	1	2	3	
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Dimension of column along Z	Dz	=	800 mm
Dimenson of column along Y	Dy	=	800 mm
Unsupported length of column along Z	Lz	=	2500 mm
Unsupported length of column along Y	Ly	=	2500 mm
Grade of steel	f_y	=	420 MPa
Grade of concrete	f_c'	=	35 MPa
Effective length factor along Z	Kz	=	2
Effective length factor along Y	Ky	=	2
Effective Depth in Z dir		=	655 mm
Effective Depth in Y dir		=	655 mm
Cover		=	50 mm
Effective cover		=	72.5 mm
Dia of longitudinal bar		=	25 mm
Dia of link		=	10 mm
Percentage of steel reinforcement		=	1 %



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	266 KN	Load Case No.	
B.M about Z	Mz	=	M_{uz}	=	97 KNm	425	0.9 (DL + SDL1) + 1.6 WL(+Z)
B.M about Y	Mx	=	M_{uy}	=	268 KNm		

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \leq 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2500 / 240 r_z = 240

= 20.8 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2500 / 240 r_z = 240

= 20.8 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u = 266 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = 409.23 KN γ_z = 0.82

$K_n = \frac{P_n}{A_g f_c'}$ = 0.018

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagrar in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.07 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.07x640000x35x800/10^6 = 1254.4 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 815.36 KNm > 97 KNm *safe*

Utility ratio = ***0.119***

Moment Capacity about Y axis

P_u = 266 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\phi}$ = 409.23 KN γ_y = 0.82

$K_n = \frac{P_n}{A_g f_c'}$ = 0.018

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagrar in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.07 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.07x640000x35x800/10^6 = 1254.4 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 815.36 KNm > 268 KNm *safe*

Utility ratio = ***0.33***

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Combined moment capacity check

M_{noz} = 1254.4 KNm M_{nz} = M_{uz}/φ = 149.23 KNm

M_{noy} = 1254.4 KNm M_{ny} = M_{uy}/φ = 412.31 KNm

Assume β = 0.65 log β = -0.19 log 0.5 = -0.30

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta}$ < 1 *Equation 3-15 SP-17M(09)*

= 0.03 + 0.17 = **0.20** < 1 *Safe*

Longitudinal reinforcement		16	nos	20	mm	Required Ast =	6400 mm²
Spread equally on all four side		4	nos	25	mm		
		Total Ast	=	6990.0437	mm²	O.K	
Minimim Reinforcement	=	0.01Ag	=	6400	mm²		

Transverse Reinforcement

Minimum diameter of tie		=	10 mm		
Maximum c/c dist.	a)	16db	=	400	mm
	b)	48db	=	480	mm
	c)	Smaller column dimension	=	800	mm
Provide lateral ties		10 mm	@	250	c/c

Check for Tension

Tension Forces P_u = 0 KN M_{uz} = 0 KNm M_{uy} = 0 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\emptyset}$ = 0.00 KN A_g = 800 x 800 = 640000 mm²

f_c' = 35 N/mm²

$K_n = \frac{P_n}{A_g f_c'}$ = 0.000 ρ_s = 0.01

γ_z = 0.82

γ_y = 0.82

From the tension curve corresponding to the values of K_n γ_y f_y f_c' ρ_s as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$	=	0.05		$\frac{M_{ny}}{A_g f_c' D_y}$	=	0.05	
M_{nz}	=	896 KNm		M_{ny}	=	896 KNm	
$\emptyset M_{nz}$	=	806.4 KNm	>0	$\emptyset M_{ny}$	=	806.4 KNm	>0
			safe				*safe*
		Utility ratio =	***0.00***			*Utility ratio =*	***0.00***

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023		
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Outotec / L&T Revision No.:	0	1	2	3	
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Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.5 Design of pedestal P1 at grid 3130/K-3130/3

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Height of column - h	=	0.5 m
Height of column	=	2.500 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	-62.5	602.4	-5.8	-14.5	0.0	-156.1
402	-47.8	557.0	-5.8	-14.5	0.0	-119.4
403	-56.9	827.4	-4.1	-10.2	0.0	-142.2
404	-67.9	822.1	-7.6	-19.1	0.0	-169.8
405	-42.2	782.0	-4.1	-10.1	0.0	-105.5
406	-53.2	776.7	-7.6	-19.1	0.0	-133.1
407	119.3	227.1	-5.0	-12.4	0.0	298.3
408	-226.3	933.2	-5.0	-12.5	0.0	-565.8
409	-52.8	398.1	69.4	173.4	0.0	-131.9
410	-53.4	762.4	-79.3	-198.1	0.0	-133.6
411	131.9	188.2	-5.0	-12.4	0.0	329.8
412	-213.7	894.3	-5.0	-12.5	0.0	-534.4
413	-40.2	359.1	69.4	173.4	0.0	-100.4
414	-40.9	723.5	-79.3	-198.1	0.0	-102.1
415	-53.5	581.7	-5.0	-12.5	0.0	-133.8
416	-53.5	581.7	-5.0	-12.5	0.0	-133.8
417	-53.5	581.7	-5.0	-12.5	0.0	-133.8
418	-53.5	581.7	-5.0	-12.5	0.0	-133.8
419	-40.9	542.8	-5.0	-12.5	0.0	-102.3
420	-40.9	542.8	-5.0	-12.5	0.0	-102.3
421	-40.9	542.8	-5.0	-12.5	0.0	-102.3
422	-40.9	542.8	-5.0	-12.5	0.0	-102.3
423	132.7	32.7	-3.7	-9.3	0.0	331.7
424	-213.0	738.7	-3.7	-9.4	0.0	-532.4
425	-39.4	203.6	70.6	176.6	0.0	-98.5
426	-40.1	567.9	-78.0	-195.0	0.0	-100.2
427	142.1	3.5	-3.7	-9.3	0.0	355.3
428	-203.5	709.5	-3.7	-9.4	0.0	-508.8
429	-29.9	174.4	70.6	176.6	0.0	-74.9
430	-30.6	538.7	-78.0	-195.0	0.0	-76.6
431	-40.2	387.3	-3.7	-9.3	0.0	-100.4
432	-40.2	387.3	-3.7	-9.3	0.0	-100.4
433	-40.2	387.3	-3.7	-9.3	0.0	-100.4
434	-40.2	387.3	-3.7	-9.3	0.0	-100.4
435	-30.7	358.1	-3.7	-9.3	0.0	-76.8
436	-30.7	358.1	-3.7	-9.3	0.0	-76.8
437	-30.7	358.1	-3.7	-9.3	0.0	-76.8
438	-30.7	358.1	-3.7	-9.3	0.0	-76.8

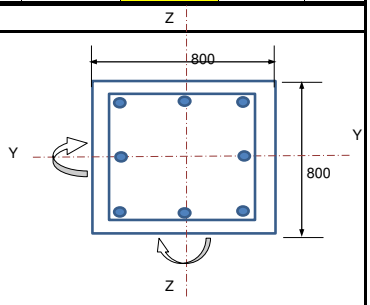
Summary of load in pedestal

	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	408	933	-13	-566
Fy min	427	3	-9	355
Mx max	425	204	177	-98
Mx min	410	762	-198	-98
Mz max	427	3	-9	355
Mz min	408	933	-13	-566

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Dimension of column along Z	D_z	=	800 mm
Dimenson of column along Y	D_y	=	800 mm
Unsupported length of column along Z	L_z	=	2500 mm
Unsupported length of column along Y	L_y	=	2500 mm
Grade of steel	f_y	=	420 MPa
Grade of concrete	f_c'	=	35 MPa
Effective length factor along Z	K_z	=	2
Effective length factor along Y	K_y	=	2
Effective Depth in Z dir		=	655 mm
Effective Depth in Y dir		=	655 mm
Cover		=	50 mm
Effective cover		=	72.5 mm
Dia of longitudinal bar		=	25 mm
Dia of link		=	10 mm
Percentage of steel reinforcement		=	1 %



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	204 KN	Load Case No.	
B.M about Z	Mz	=	M_{uz}	=	98 KNm	425	0.9 (DL + SDL1) + 1.6 WL(+Z)
B.M about Y	Mx	=	M_{uy}	=	177 KNm		

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \leq 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2500 / 240 r_z = 240

= 20.8 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2500 / 240 r_z = 240

= 20.8 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u	=	204 KN			A_g	=	800 x 800 =	640000 mm^2
Assume Strength reduction factor φ	=	0.65 *Table 21.2.2*	*SBC 304*		f_c'	=	35 N/mm^2	
					ρ_s	=	0.01	
$P_n = \frac{P_u}{\phi}$	=	313.85 KN			γ_z	=	0.82	
$K_n = \frac{P_n}{A_g f_c'}$	=	0.014						

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagrar in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.05x640000x35x800/10^6 = 896 KNm

Max moment capacity M_{uz} = ϕM_{nz} = 582.4 KNm > 98 KNm *safe*

Utility ratio = ***0.168***

Moment Capacity about Y axis

P_u	=	204 KN			A_g	=	800 x 800 =	640000 mm^2
Assume Strength reduction factor φ	=	0.65 *Table 21.2.2*	*SBC 304*		f_c'	=	35 N/mm^2	
					ρ_s	=	0.01	
$P_n = \frac{P_u}{\phi}$	=	313.85 KN			γ_y	=	0.82	
$K_n = \frac{P_n}{A_g f_c'}$	=	0.014						

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagrar in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.05x640000x35x800/10^6 = 896 KNm

Max moment capacity M_{uy} = ϕM_{ny} = 582.4 KNm > 177 KNm *safe*

Utility ratio = ***0.30***

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Combined moment capacity check

M_{noz} = 896 KNm M_{nz} = M_{uz}/φ = 150.77 KNm

M_{noy} = 896 KNm M_{ny} = M_{uy}/φ = 272.31 KNm

Assume β = 0.65 log β = -0.19 log 0.5 = -0.30

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta}$ < 1 *Equation 3-15 SP-17M(09)*

= 0.06 + 0.15 = **0.20** < 1 *Safe*

Longitudinal reinforcement	16	nos	20	mm	Required Ast =	6400 mm²
Spread equally on all four side	4	nos	25	mm		
	Total Ast =	6990.0437	mm²	O.K		
Minimim Reinforcement =	0.01Ag =	6400	mm²			

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist.	a)	16db	=	400 mm
	b)	48db	=	480 mm
	c)	Smaller column dimension	=	800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 0 KN M_{uz} = 0 KNm M_{uy} = 0 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\emptyset}$ = 0.00 KN A_g = 800 x 800 = 640000 mm²

$K_n = \frac{P_n}{A_g f_c'}$ = 0.000 f_c' = 35 N/mm² ρ_s = 0.01 γ_z = 0.82 γ_y = 0.82

From the tension curve corresponding to the values of K_n γ_y f_y f_c' ρ_s as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05

M_{nz} = 896 KNm M_{ny} = 896 KNm

$\emptyset M_{nz}$ = 806.4 KNm >0 *safe* $\emptyset M_{ny}$ = 806.4 KNm >0 *safe*

Utility ratio = ***0.00*** *Utility ratio* = ***0.00***

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01		
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Outotec / L&T Revision No.:	0	1	2	3	
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Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.6 Design of pedestal P1 at grid 3130/L-3130/6

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Height of column - h	=	0.5 m
Height of column	=	2.500 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	1.1	134.7	-15.8	-39.5	0.0	2.8
402	1.3	538.4	-42.3	-105.8	0.0	3.3
403	-2.5	260.7	31.0	77.4	0.0	-6.3
404	4.9	285.7	-87.7	-219.2	0.0	12.3
405	-2.3	664.4	4.5	11.1	0.0	-5.7
406	5.2	689.4	-114.2	-285.5	0.0	12.9
407	20.2	277.5	-24.2	-60.6	0.0	50.6
408	-18.3	34.7	-10.3	-25.7	0.0	-45.7
409	0.8	-403.7	108.7	271.7	0.0	2.0
410	1.2	698.3	-139.9	-349.8	0.0	2.9
411	20.4	623.4	-47.0	-117.4	0.0	51.1
412	-18.1	380.7	-33.0	-82.5	0.0	-45.2
413	1.0	-57.7	85.9	214.8	0.0	2.4
414	1.4	1044.2	-162.6	-406.6	0.0	3.4
415	1.0	156.2	-17.2	-43.1	0.0	2.5
416	1.0	156.2	-17.2	-43.1	0.0	2.5
417	1.0	156.2	-17.2	-43.1	0.0	2.5
418	1.0	156.2	-17.2	-43.1	0.0	2.5
419	1.2	502.2	-40.0	-99.9	0.0	2.9
420	1.2	502.2	-40.0	-99.9	0.0	2.9
421	1.2	502.2	-40.0	-99.9	0.0	2.9
422	1.2	502.2	-40.0	-99.9	0.0	2.9
423	20.0	207.9	-17.2	-42.9	0.0	49.9
424	-18.5	-34.9	-3.2	-8.0	0.0	-46.4
425	0.5	-473.3	115.7	289.4	0.0	1.3
426	0.9	628.7	-132.8	-332.1	0.0	2.2
427	20.1	467.4	-34.2	-85.5	0.0	50.3
428	-18.4	224.6	-20.2	-50.6	0.0	-46.0
429	0.7	-213.8	98.7	246.7	0.0	1.7
430	1.0	888.1	-149.9	-374.7	0.0	2.6
431	0.7	86.6	-10.2	-25.4	0.0	1.8
432	0.7	86.6	-10.2	-25.4	0.0	1.8
433	0.7	86.6	-10.2	-25.4	0.0	1.8
434	0.7	86.6	-10.2	-25.4	0.0	1.8
435	0.9	346.1	-27.2	-68.0	0.0	2.1
436	0.9	346.1	-27.2	-68.0	0.0	2.1
437	0.9	346.1	-27.2	-68.0	0.0	2.1
438	0.9	346.1	-27.2	-68.0	0.0	2.1
439	0.7	421.1	-92.4	-231.1	0.0	1.7
440	0.6	72.3	-8.8	-22.0	0.0	1.4
441	0.6	72.3	-8.8	-22.0	0.0	1.4
442	0.6	72.3	-8.8	-22.0	0.0	1.4
443	0.6	72.3	-8.8	-22.0	0.0	1.4

Summary of load in pedestal

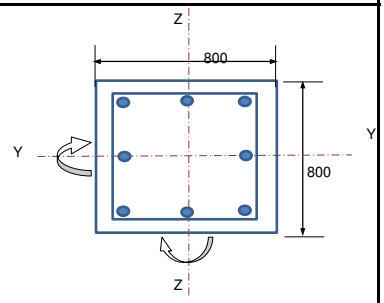
	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	414	1044	-407	3
Fy min	425	-473	289	1
Mx max	425	-473	289	1
Mx min	414	1044	-407	1
Mz max	411	623	-117	51
Mz min	424	-35	-8	-46

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Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
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Outotec / L&T Revision No.:	0	1	2	3	
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Dimension of column along Z	Dz	=	800 mm
Dimenson of column along Y	Dy	=	800 mm
Unsupported length of column along Z	Lz	=	2500 mm
Unsupported length of column along Y	Ly	=	2500 mm
Grade of steel	f_y	=	420 MPa
Grade of concrete	f_c'	=	35 MPa
Effective length factor along Z	Kz	=	2
Effective length factor along Y	Ky	=	2
Effective Depth in Z dir		=	655 mm
Effective Depth in Y dir		=	655 mm
Cover		=	50 mm
Effective cover		=	72.5 mm
Dia of longitudinal bar		=	25 mm
Dia of link		=	10 mm
Percentage of steel reinforcement		=	1 %



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	1044 KN	Load Case No.	
B.M about Z	Mz	=	M_{uz}	=	2 KNm	414	1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
B.M about Y	Mx	=	M_{uy}	=	407 KNm		

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \le 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2500 / 240 = 20.8 r_z = 240 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2500 / 240 = 20.8 r_z = 240 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u = 1044 KN
Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304*
$P_n = \frac{P_u}{\phi}$ = 1606.15 KN
$K_n = \frac{P_n}{A_g f_c'}$ = 0.072

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagrai in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.08 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.08x640000x35x800/10^6 = 1433.6 KNm

Max moment capacity $M_{uz} = \phi M_{nz}$ = 931.84 KNm > 2 KNm *safe*
Utility ratio = ***0.002***

Moment Capacity about Y axis

P_u = 1044 KN
Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304*
$P_n = \frac{P_u}{\phi}$ = 1606.15 KN
$K_n = \frac{P_n}{A_g f_c'}$ = 0.072

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagrai in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.08 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.08x640000x35x800/10^6 = 1433.6 KNm

Max moment capacity $M_{uy} = \phi M_{ny}$ = 931.84 KNm > 407 KNm *safe*
Utility ratio = ***0.44***

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Combined moment capacity check

M_{noz} = 1433.6 KNm M_{nz} = M_{uz}/φ = 3.08 KNm

M_{noy} = 1433.6 KNm M_{ny} = M_{uy}/φ = 626.15 KNm

Assume β = 0.65 log β = -0.19 log 0.5 = -0.30

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta}$ < 1 *Equation 3-15 SP-17M(09)*

= 0.00 + 0.26 = **0.26** < 1 *Safe*

Longitudinal reinforcement		16	nos	20	mm	Required Ast = 6400 mm²
Spread equally on all four side		4	nos	25	mm	
		Total Ast	=	6990.0437	mm²	O.K
Minimim Reinforcement	=	0.01Ag	=	6400	mm²	

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist.
- a) 16db = 400 mm
- b) 48db = 480 mm
- c) Smaller column dimension = 800 mm

Provide lateral ties: 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 473 KN M_{uz} = 1 KNm M_{uy} = 289 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\phi}$ = 525.56 KN A_g = 800 x 800 = 640000 mm²

$K_n = \frac{P_n}{A_g f_c'}$ = 0.023 f_c' = 35 N/mm² ρ_s = 0.01 γ_z = 0.82 γ_y = 0.82

From the tension curve corresponding to the values of K_n γ_y f_y f_c' ρ_s as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.04 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.04

M_{nz} = 716.8 KNm M_{ny} = 716.8 KNm

$\emptyset M_{nz}$ = 645.12 KNm >1 $\emptyset M_{ny}$ = 645.12 KNm >289

safe *safe*

Utility ratio = ***0.00*** *Utility ratio =* ***0.45***

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Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01		
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Approved	AG	AG	AG	AG	

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria

8.4.6 Design of pedestal P1 at grid 3130/L-3130/4

Underside of base plate	=	0.500 m
Bottom of foundation	=	-2.500 m
Height of column - h	=	0.5 m
Height of column	=	2.500 m

Calculation of load in pedestal for all load combinations

L/C	Fx	Fy	Fz	Mx	My	Mz
401	-1.1	272.0	1.5	3.8	0.0	-2.8
402	-1.3	507.5	1.5	3.7	0.0	-3.3
403	2.5	478.9	2.3	5.9	0.0	6.3
404	-4.9	531.1	0.7	1.7	0.0	-12.3
405	2.3	714.4	2.3	5.8	0.0	5.7
406	-5.2	766.6	0.6	1.6	0.0	-12.9
407	18.2	186.3	1.3	3.3	0.0	45.5
408	-20.2	418.2	1.3	3.2	0.0	-50.4
409	-0.8	168.7	4.2	10.5	0.0	-2.0
410	-1.2	400.6	-1.6	-3.9	0.0	-2.9
411	18.0	388.2	1.3	3.2	0.0	45.0
412	-20.3	620.0	1.2	3.1	0.0	-50.9
413	-1.0	370.6	4.2	10.4	0.0	-2.4
414	-1.4	602.5	-1.6	-4.0	0.0	-3.4
415	-1.0	301.6	1.3	3.2	0.0	-2.5
416	-1.0	301.6	1.3	3.2	0.0	-2.5
417	-1.0	301.6	1.3	3.2	0.0	-2.5
418	-1.0	301.6	1.3	3.2	0.0	-2.5
419	-1.2	503.5	1.3	3.2	0.0	-2.9
420	-1.2	503.5	1.3	3.2	0.0	-2.9
421	-1.2	503.5	1.3	3.2	0.0	-2.9
422	-1.2	503.5	1.3	3.2	0.0	-2.9
423	18.5	59.5	1.0	2.5	0.0	46.2
424	-19.9	291.3	1.0	2.4	0.0	-49.7
425	-0.5	41.9	3.9	9.7	0.0	-1.3
426	-0.9	273.8	-1.9	-4.7	0.0	-2.2
427	18.3	210.9	1.0	2.4	0.0	45.8
428	-20.0	442.8	0.9	2.3	0.0	-50.1
429	-0.7	193.3	3.8	9.6	0.0	-1.7
430	-1.0	425.2	-1.9	-4.8	0.0	-2.6
431	-0.7	174.8	1.0	2.4	0.0	-1.8
432	-0.7	174.8	1.0	2.4	0.0	-1.8
433	-0.7	174.8	1.0	2.4	0.0	-1.8
434	-0.7	174.8	1.0	2.4	0.0	-1.8
435	-0.9	326.2	0.9	2.4	0.0	-2.1
436	-0.9	326.2	0.9	2.4	0.0	-2.1
437	-0.9	326.2	0.9	2.4	0.0	-2.1
438	-0.9	326.2	0.9	2.4	0.0	-2.1
439	-0.7	118.9	-1.6	-3.9	0.0	-1.7
440	-0.6	77.2	0.0	0.0	0.0	-1.4
441	-0.6	77.2	0.0	0.0	0.0	-1.4
442	-0.6	77.2	0.0	0.0	0.0	-1.4
443	-0.6	77.2	0.0	0.0	0.0	-1.4

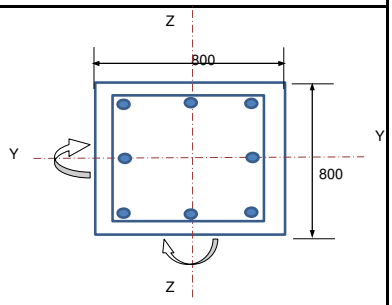
Summary of load in pedestal

	L/C	Fy(KN)	Mx(KNm)	Mz(KNm)
FY max	406	767	2	-13
Fy min	425	42	10	-1
Mx max	409	169	10	-2
Mx min	430	425	-5	-2
Mz max	423	60	2	46
Mz min	412	620	3	-51

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Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Dimension of column along Z	Dz	=	800 mm
Dimenson of column along Y	Dy	=	800 mm
Unsupported length of column along Z	Lz	=	2500 mm
Unsupported length of column along Y	Ly	=	2500 mm
Grade of steel	f_y	=	420 MPa
Grade of concrete	f_c'	=	35 MPa
Effective length factor along Z	Kz	=	2
Effective length factor along Y	Ky	=	2
Effective Depth in Z dir		=	655 mm
Effective Depth in Y dir		=	655 mm
Cover		=	50 mm
Effective cover		=	72.5 mm
Dia of longitudinal bar		=	25 mm
Dia of link		=	10 mm
Percentage of steel reinforcement		=	1 %



ANALYSIS RESULTS

Analysis result as obtained from staad analysis

Axial load	Fy	=	P_u	=	60 KN	Load Case No.	
B.M about Z	Mz	=	M_{uz}	=	46 KNm	423	0.9 (DL + SDL1) + 1.6 WL(+X)
B.M about Y	Mx	=	M_{uy}	=	3 KNm		

CHECK FOR SLENDERNESS

Allowable slenderness for sway column $\frac{kl}{r} \le 22$ where r = 0.3h *Clause 6.2.5 SBC 304*

Slenderness check along Z axis $\frac{k_z l_z}{r_z}$ = 2 x 2500 / 240 r_z = 240

= 20.8 *Non-slender along Z*

Slenderness check along Y axis $\frac{k_y l_y}{r_y}$ = 2 x 2500 / 240 r_z = 240

= 20.8 *Non-slender along Y*

DESIGN FOR CAPACITY OF COLUMN

Moment Capacity about Z axis

P_u = 60 KN A_g = 800 x 800 = 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\emptyset}$ = 92.31 KN γ_z = 0.82

$K_n = \frac{P_n}{A_g f_c'}$ = 0.004

Corresponding to the values of K_n ρ_s γ_z f_y f_c' the value of $\frac{M_{nz}}{A_g f_c' D_z}$ is interpolated from Interaction diagran in SP-17M(09)

We get $\frac{M_{nz}}{A_g f_c' D_z}$ = 0.05 *As per graph* *3.3.3* *SP-17M(09)*

M_{nz} = 0.05x640000x35x800/10^6 = 896 KNm

Max moment capacity M_{uz} = $\emptyset M_{nz}$ = 582.4 KNm > 46 KNm *safe*

Utility ratio = ***0.079***

Moment Capacity about Y axis

P_u = 60 KN A_g = 800 x 800 640000 mm^2

Assume Strength reduction factor φ = 0.65 *Table 21.2.2 SBC 304* f_c' = 35 N/mm^2

ρ_s = 0.01

$P_n = \frac{P_u}{\emptyset}$ = 92.31 KN γ_y = 0.82

$K_n = \frac{P_n}{A_g f_c'}$ = 0.004

Corresponding to the values of K_n ρ_s γ_y f_y f_c' the value of $\frac{M_{ny}}{A_g f_c' D_y}$ is interpolated from Interaction diagran in SP-17M(09)

We get $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.05 *As per graph* *3.3.2* *SP-17M(09)*

M_{ny} = 0.05x640000x35x800/10^6 = 896 KNm

Max moment capacity M_{uy} = $\emptyset M_{ny}$ = 582.4 KNm > 3 KNm *safe*

Utility ratio = ***0.01***

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER CLIENTS COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

Combined moment capacity check

M_{noz} = 896 KNm M_{nz} = M_{uz}/φ = 70.77 KNm

M_{noy} = 896 KNm M_{ny} = M_{uy}/φ = 4.62 KNm

Assume β = 0.65 log β = -0.19

log 0.5 = -0.30

Check $\left(\frac{M_{nz}}{M_{noz}}\right)^{\log 0.5/\log \beta} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\log 0.5/\log \beta}$ < 1 *Equation 3-15 SP-17M(09)*

= 0.02 + 0.00 = **0.02** < 1 *Safe*

Longitudinal reinforcement		16	nos	20	mm	Required Ast =	6400 mm²
Spread equally on all four side		4	nos	25	mm		
		Total Ast	=	6990.0437	mm²	O.K	
Minimim Reinforcement	=	0.01Ag	=	6400	mm²		

Transverse Reinforcement

Minimum diameter of tie = 10 mm

Maximum c/c dist. a) 16db = 400 mm

b) 48db = 480 mm

c) Smaller column dimension = 800 mm

Provide lateral ties 10 mm @ 250 c/c

Check for Tension

Tension Forces P_u = 0 KN M_{uz} = 0 KNm M_{uy} = 0 KNm

Assume Strength reduction factor φ = 0.9 Tension controlled *Table 21.2.2 SBC 304*

$P_n = \frac{P_u}{\emptyset}$ = 0.00 KN A_g = 800 x 800 = 640000 mm²

f_c' = 35 N/mm²

$K_n = \frac{P_n}{A_g f_c'}$ = 0.000 ρ_s = 0.01

γ_z = 0.82

γ_y = 0.82

From the tension curve corresponding to the values of K_n γ_y f_y f_c' ρ_s as per interaction diagram generated using clause 3.2.2 of SP-17M(09)

We get,

$\frac{M_{nz}}{A_g f_c' D_z}$ = 0.04 $\frac{M_{ny}}{A_g f_c' D_y}$ = 0.04

M_{nz} = 716.8 KNm M_{ny} = 716.8 KNm

$\emptyset M_{nz}$ = 645.12 KNm >0 $\emptyset M_{ny}$ = 645.12 KNm >0

safe *safe*

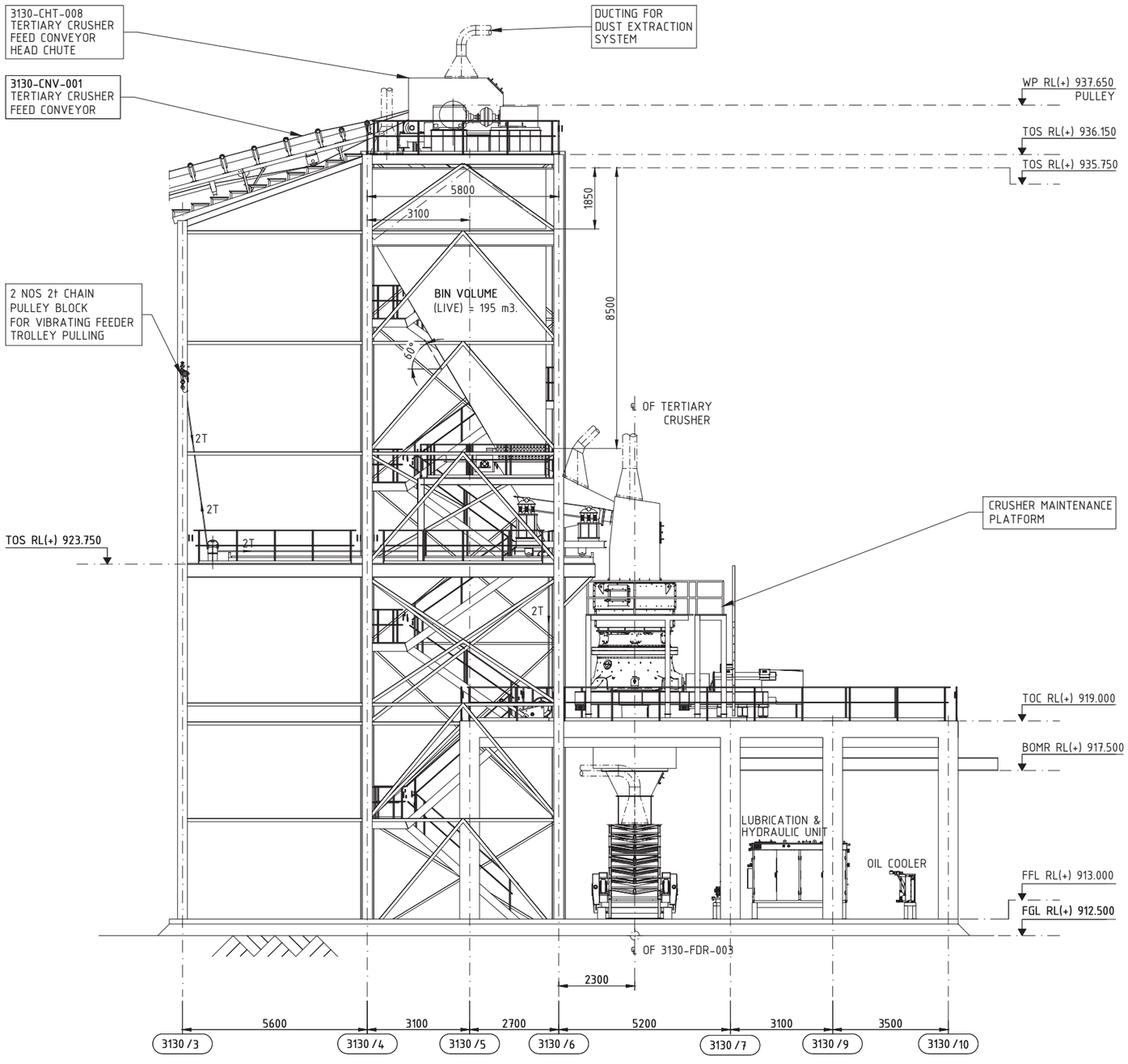
Utility ratio = ***0.00*** *Utility ratio =* ***0.00***

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 1 MECHANICAL GENERAL ARRANGEMENT DRAWING

DRAWING NUMBER MD-216-3130-EG-ME-ELV-1017 REV A01

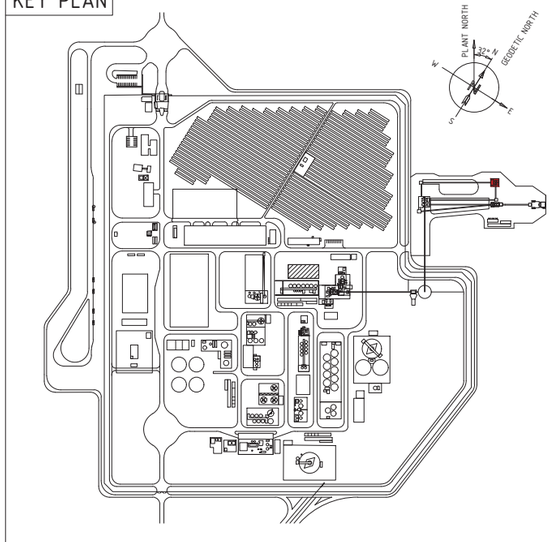


ELEVATION A-A
SCALE 1:75

NOTE:
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3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+) 912.500 m
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CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3100-EG-ME-GAD-1016	DUCT - TERTIARY CRUSHER HOUSE								
	DUST EXTRACTION SYSTEM								
MD-216-3130-EG-ME-LPX-1014	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1013	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1012	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1010	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-ELV-1018	TERTIARY CRUSHER HOUSE-ELEVATION								
MD-216-3130-EG-ME-LPX-1009	TERTIARY CRUSHER HOUSE-PLAN VIEW	A01	06/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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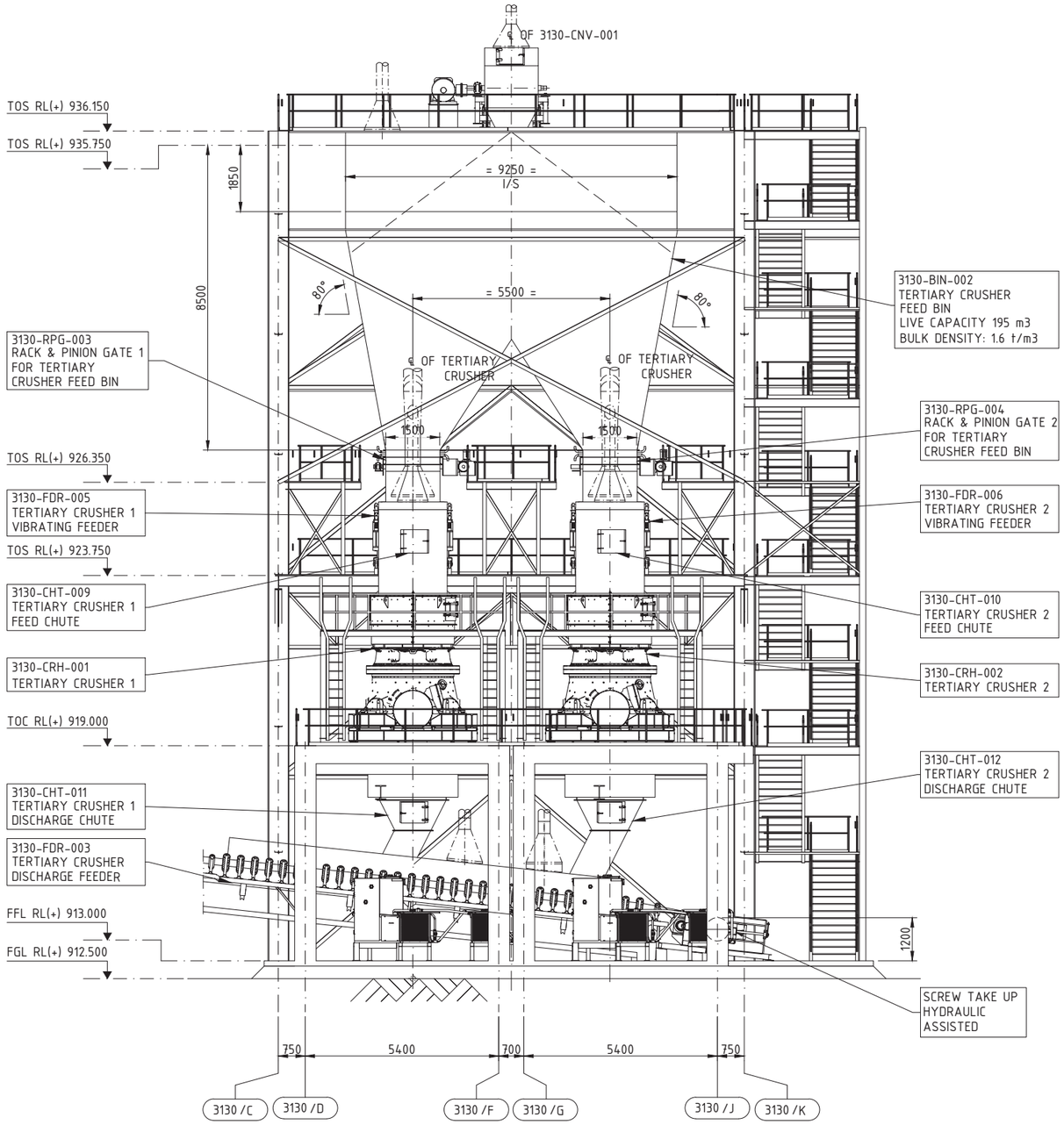
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OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MLD13__31301 — DOCUMENT ID: OU100667060 — REVISION: 0

DRAWN	SKH	03/01/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	04/01/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	04/01/2020	TERTIARY CRUSHING
APPROVED	BGR	06/01/2020	TERTIARY CRUSHER HOUSE-ELEVATION
EMPLOYER			
SCALE	1:75	A1	DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1017 — REV: A01

DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1018
REV: A01

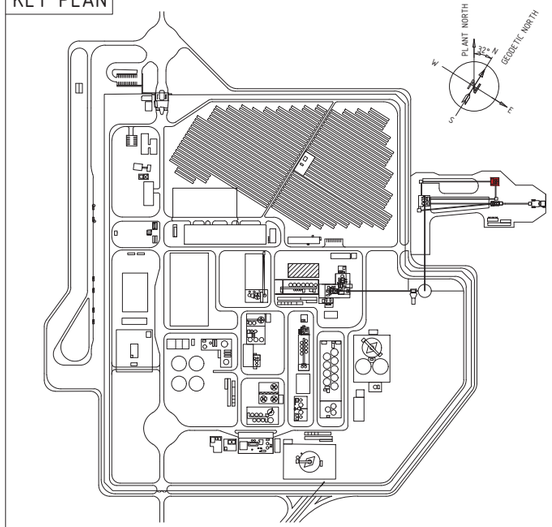


ELEVATION B-B
SCALE 1:75

NOTE:
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CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3100-EG-ME-GAD-1016	DUCT - TERTIARY CRUSHER HOUSE								
	DUST EXTRACTION SYSTEM								
MD-216-3130-EG-ME-LPX-1014	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1013	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1012	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1010	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-ELV-1017	TERTIARY CRUSHER HOUSE-ELEVATION								
MD-216-3130-EG-ME-LPX-1009	TERTIARY CRUSHER HOUSE-PLAN VIEW	A01	06/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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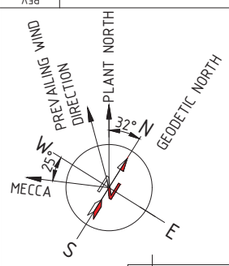
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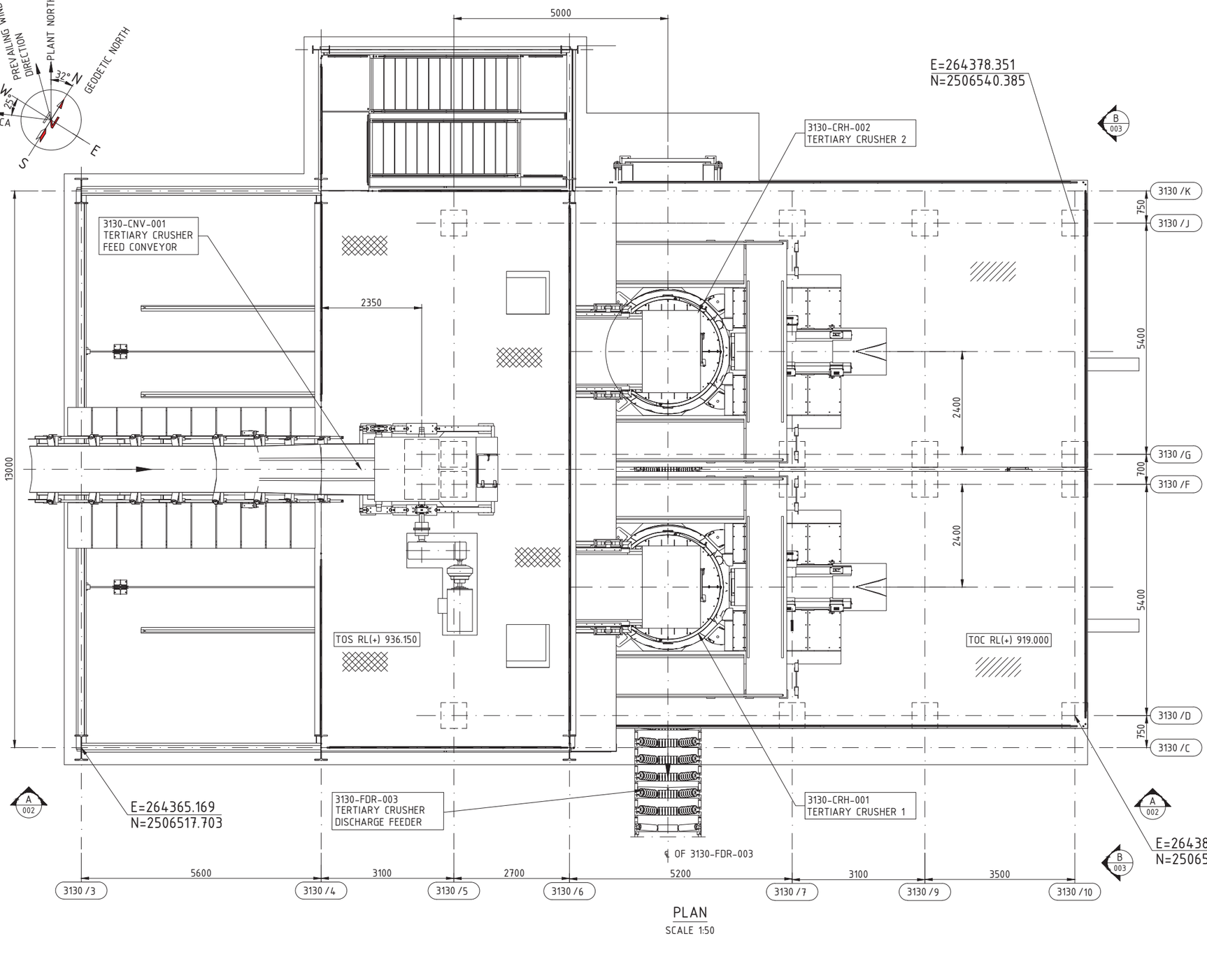
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CHECKED	BRD	04/01/2020
ENGINEER	BD	04/01/2020
APPROVED	BGR	06/01/2020
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
TERTIARY CRUSHER HOUSE-ELEVATION

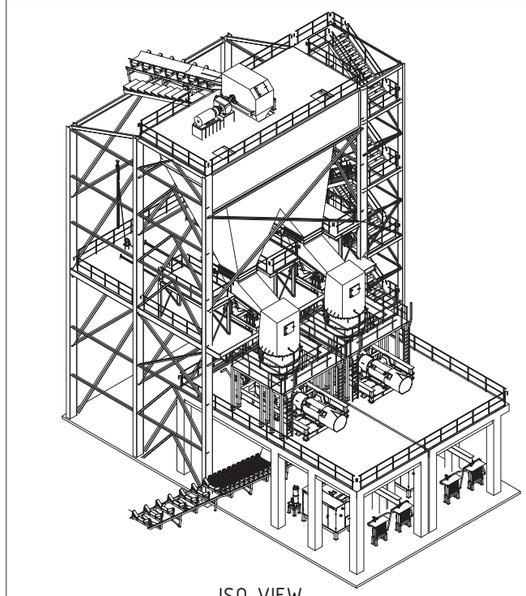
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REV: A01

DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1009
REV: A01





PLAN
SCALE 1:50

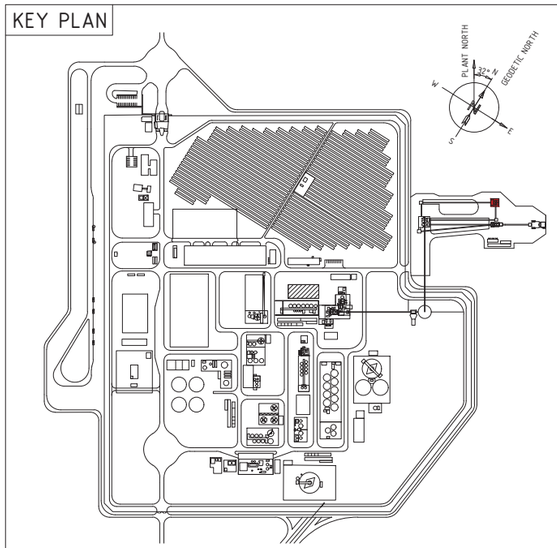


ISO VIEW

NOTE:
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CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1014	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1013	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1012	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1010	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-ELV-1018	TERTIARY CRUSHER HOUSE-ELEVATION								
MD-216-3130-EG-ME-ELV-1017	TERTIARY CRUSHER HOUSE-ELEVATION	A01	06/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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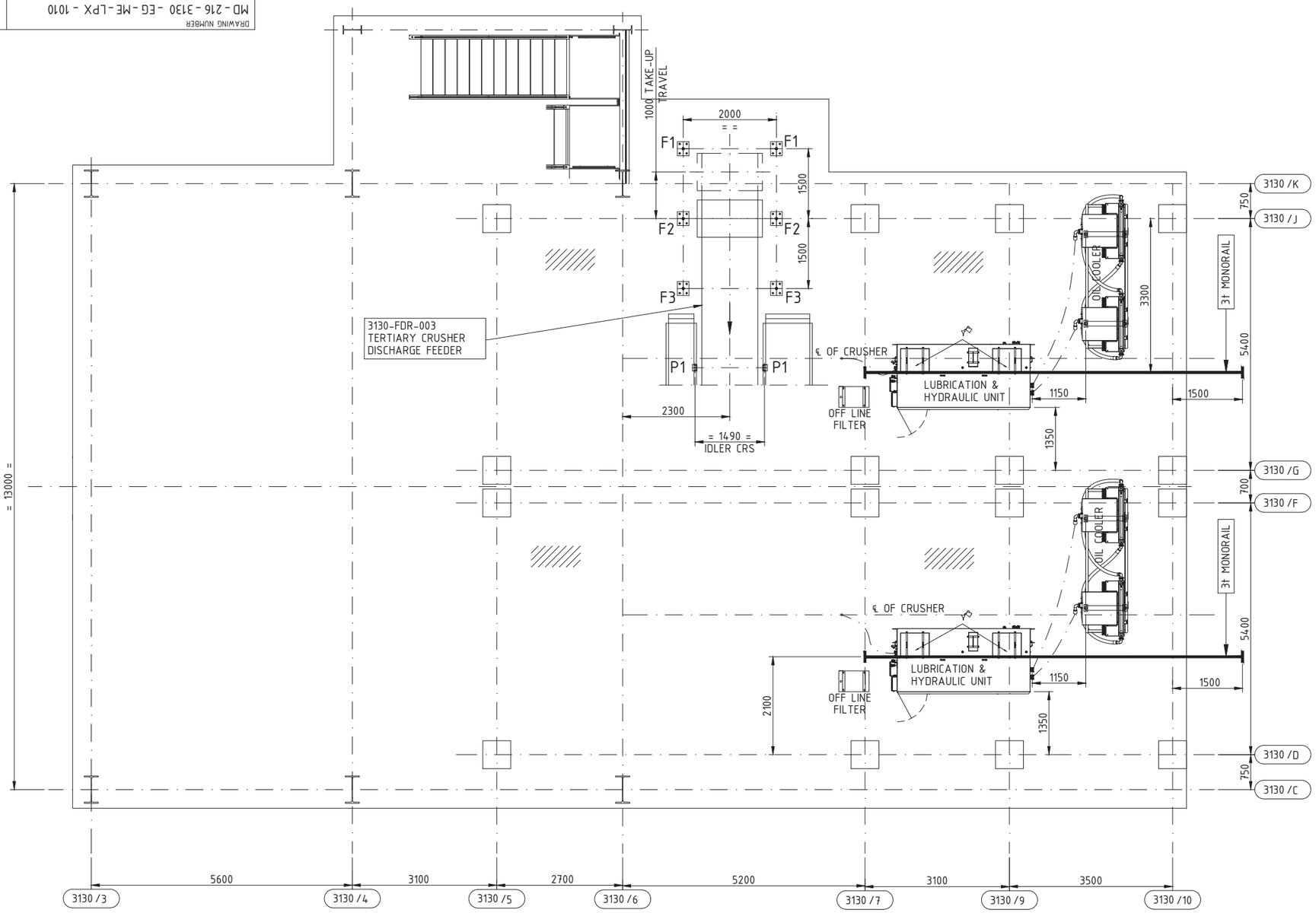
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OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31331	DOCUMENT ID OU100669072 / REVISION 0

DRAWN	SKH	03/01/2020
CHECKED	BRD	04/01/2020
ENGINEER	BD	04/01/2020
APPROVED	BGR	06/01/2020
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
TERTIARY CRUSHER HOUSE-PLAN VIEW

DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1009
REV: A01

DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1010 REV A01



PLAN AT RL(+) 913.000

SCALE 1:50

NOTE:
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CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN

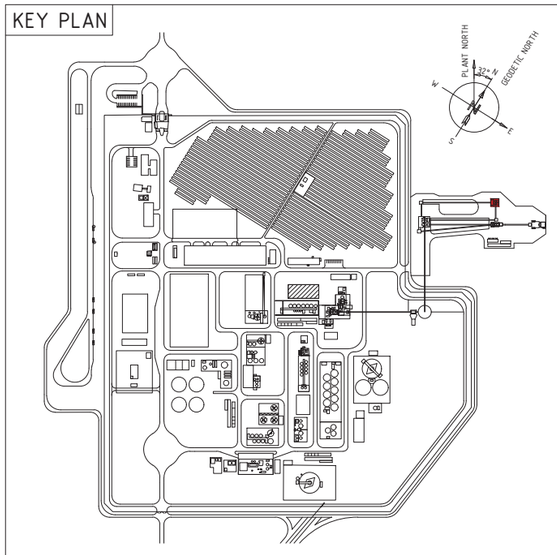


TABLE-1: LOAD FOR SCREW TAKE-UP FRAME

FOR BELT FEEDER : 3130-FDR-003

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN "t")	
			ALONG THE CONV.	ACROSS THE CONV.
F1	(+)913.000	(+)2.0	(+)2.5	(±)0.20
F2		(+)2.0	(+)2.5	(±)0.20
F3		(+)2.0	(+)2.5	(±)0.20

TABLE-2: LOAD FOR CONVEYOR SHORT POST

FOR BELT FEEDER : 3130-FDR-003

	POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")		HORIZONTAL LOAD (IN "t")	
			DEAD LOAD	LIVE LOAD	ALONG	ACROSS
IMPACT ZONE	P1	(+)913.000	(+) 0.98	(+) 0.55	(±) 0.17	(±) 0.09

SIGN CONVENTION

(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)

(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

Note: Wind load to be considered during civil & structural design

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1014	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1013	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1012	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-ELV-1018	TERTIARY CRUSHER HOUSE-ELEVATION								
MD-216-3130-EG-ME-ELV-1017	TERTIARY CRUSHER HOUSE-ELEVATION								
MD-216-3130-EG-ME-LPX-1009	TERTIARY CRUSHER HOUSE-PLAN VIEW	A01	06/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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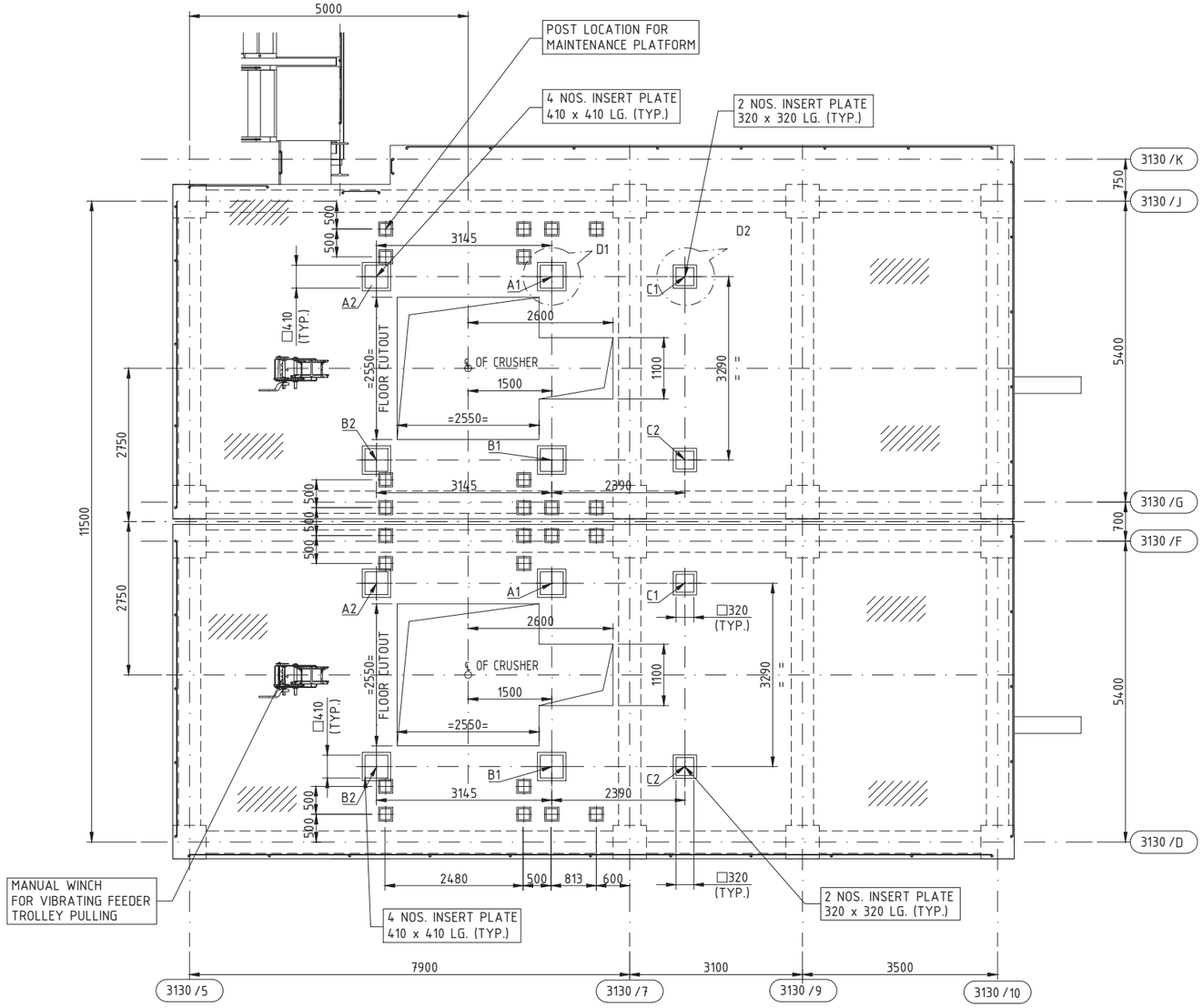
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CHECKED	BRD	04/01/2020
ENGINEER	BD	04/01/2020
APPROVED	BGR	06/01/2020
EMPLOYER		
SCALE	1:50	A1

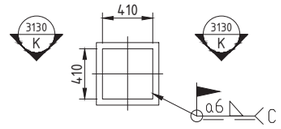
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
TERTIARY CRUSHER HOUSE-PLAN VIEW

DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1010 REV: A01

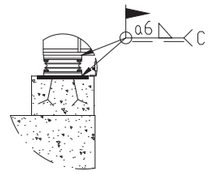
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REV: A01



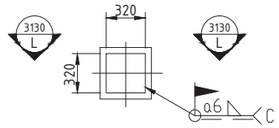
PLAN AT RL(+) 919.000
SCALE 1:50



DETAIL D1
SCALE 1:25



SECTION K-K
SCALE 1:25



DETAIL D2
SCALE 1:25

SECTION L-L
SCALE 1:25

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+) 913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+) 912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN

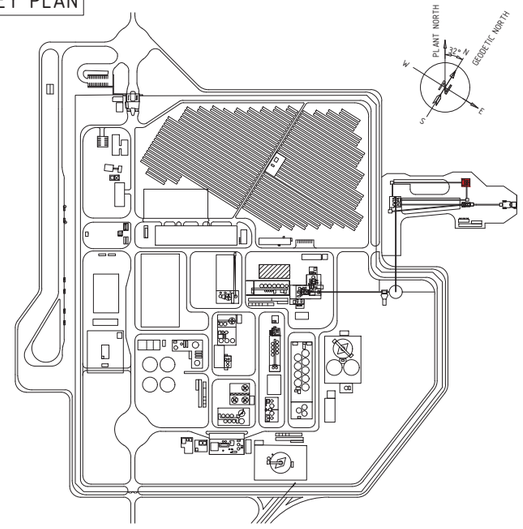


TABLE-3: LOAD FOR TERTIARY CRUSHER

3130-CRH-001 & 3130-CRH-002

	Static Load	Dynamic Load, normal operation f=4,67 Hz		Dynamic Load, start/stop	
Damper position	Vertical	Vertical	Horizontal	Vertical	Horizontal
A [1,2]	200kN	± 13 kN	15 kN	± 65 kN	68 kN
B [1,2]	200kN	± 14 kN	16 kN	± 70 kN	73 kN
C [1,2]	45kN	± 4 kN	2 kN	± 22 kN	12 kN
Vertical load, positive downwards					
Horizontal load, (all horizontal directions)					
Material volume=6.0 m^3, Density=1800 kg/m^3					

Drive unit weights	
Component	kg
Sub-frame	1650
Electric motor	7100
Coupling	240

Note: Wind load to be considered during civil & structural design

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1014	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1013	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1010	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-ELV-1018	TERTIARY CRUSHER HOUSE-ELEVATION								
MD-216-3130-EG-ME-ELV-1017	TERTIARY CRUSHER HOUSE-ELEVATION								
MD-216-3130-EG-ME-LPX-1009	TERTIARY CRUSHER HOUSE-PLAN VIEW	A01	06/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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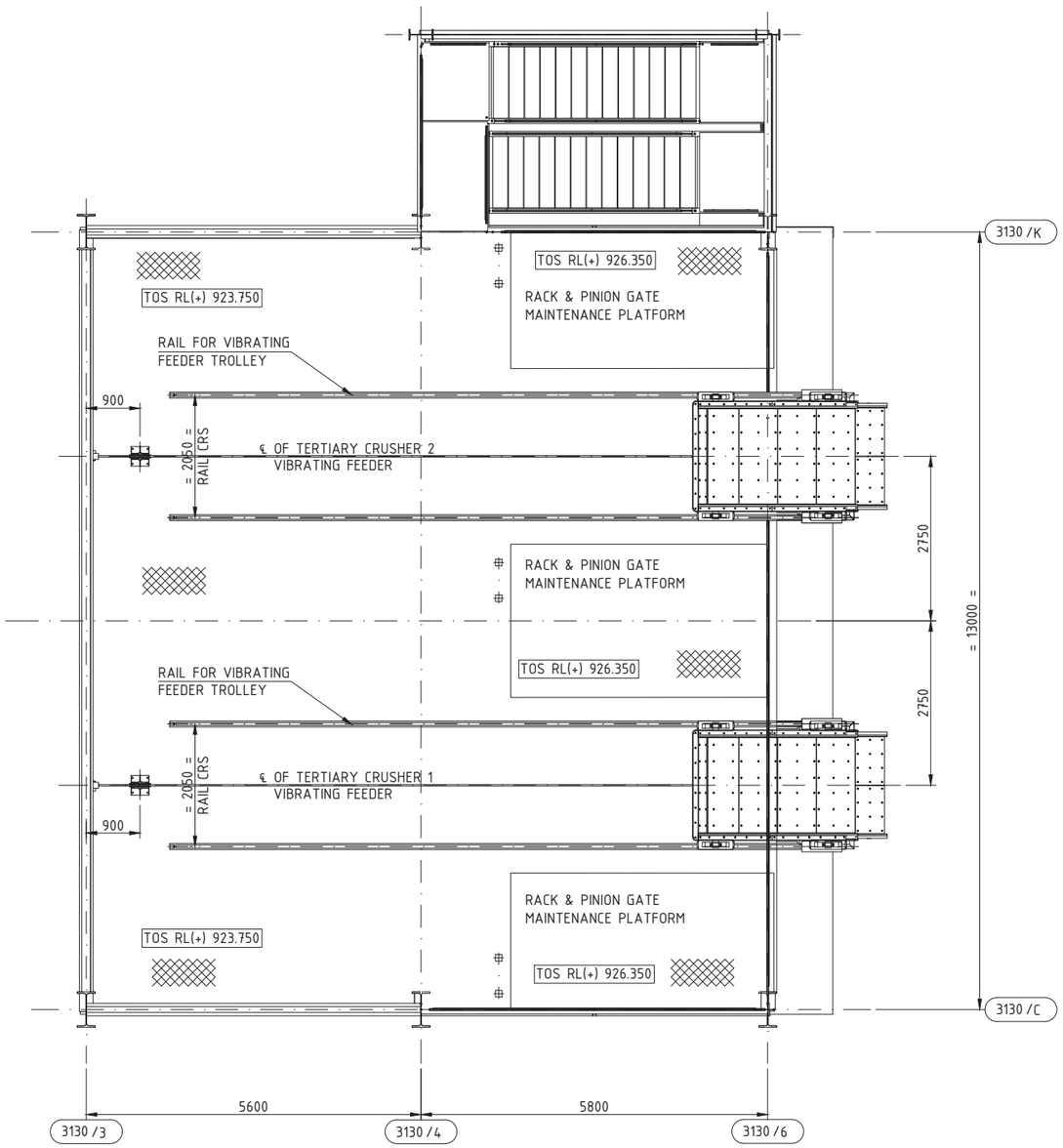
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CHECKED	BRD	04/01/2020
ENGINEER	BD	04/01/2020
APPROVED	BGR	06/01/2020
EMPLOYER		
SCALE	1:50, 1:25	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
TERTIARY CRUSHER HOUSE-PLAN VIEW

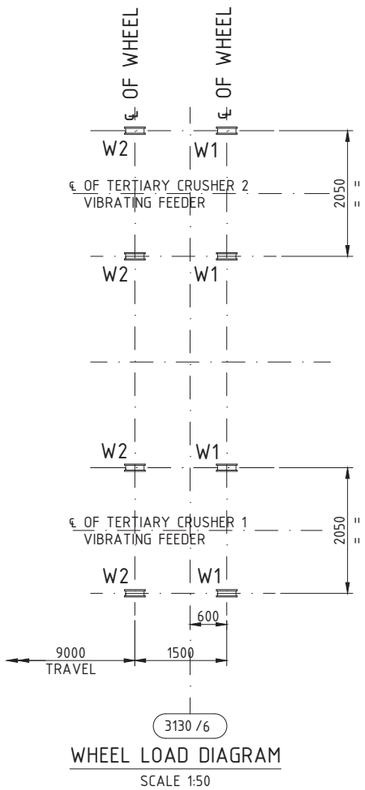
DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1012 — REV: A01

DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1013
REV: A01



PLAN AT RL(+) 923.750

SCALE 1:50



WHEEL LOAD DIAGRAM

SCALE 1:50

TABLE-4: LOAD FOR WHEEL

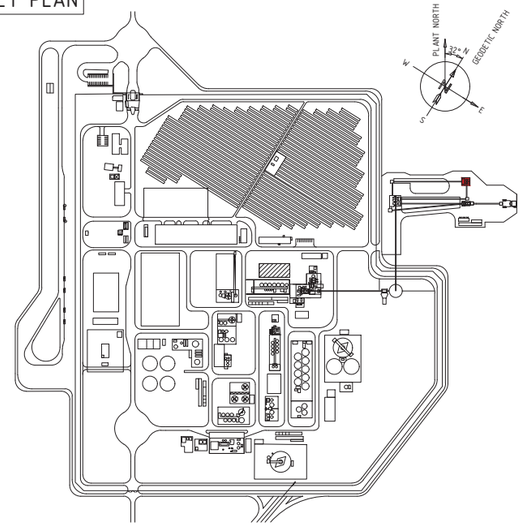
NO.	WHEEL MARKING	LOAD IN t
1	W1	4.0
2	W2	4.0

Note: Wind load to be considered during civil & structural design

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- CONCRETE FLOOR
- CHECKERED PLATE
- GRATING
- BRICK WALL
- CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1014	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1012	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1010	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-ELV-1018	TERTIARY CRUSHER HOUSE-ELEVATION								
MD-216-3130-EG-ME-ELV-1017	TERTIARY CRUSHER HOUSE-ELEVATION								
MD-216-3130-EG-ME-LPX-1009	TERTIARY CRUSHER HOUSE-PLAN VIEW	A01	06/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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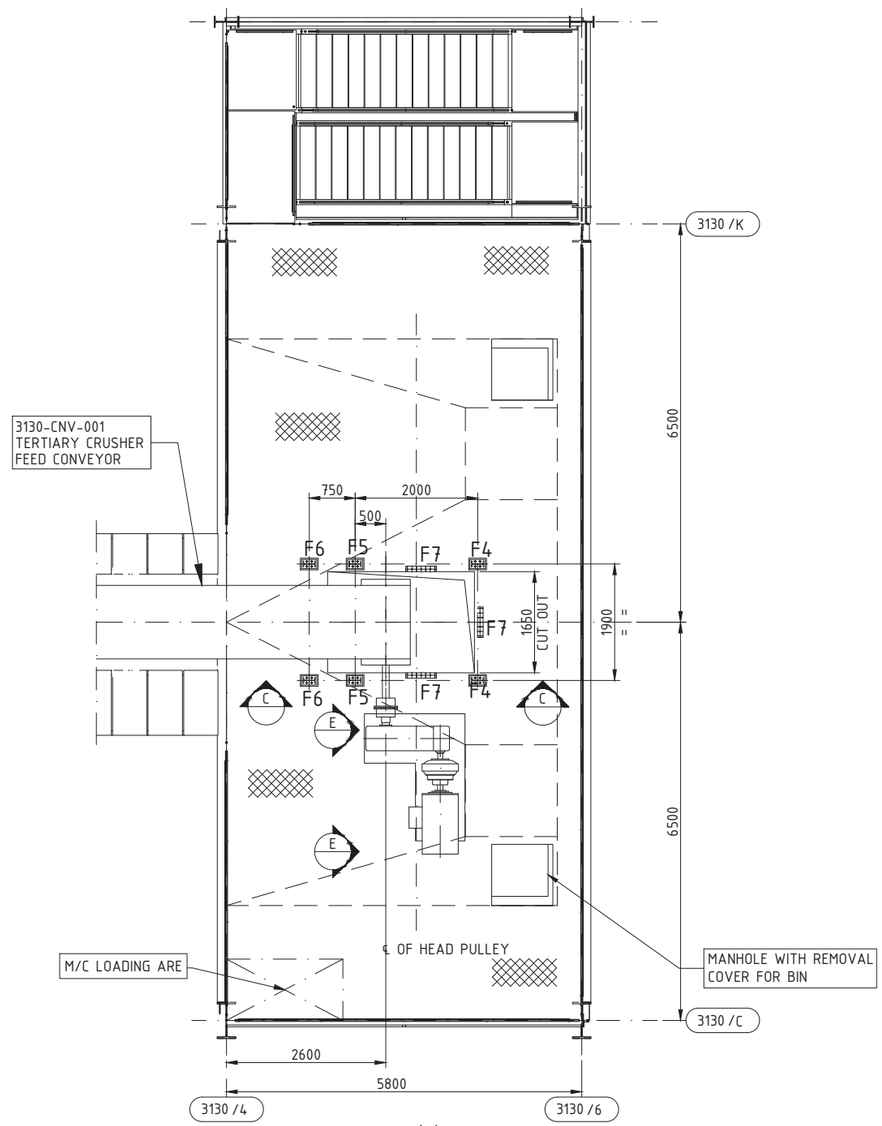
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DRAWN	SKH	03/01/2020
CHECKED	BRD	04/01/2020
ENGINEER	BD	04/01/2020
APPROVED	BGR	06/01/2020
EMPLOYER		
SCALE	1:50	A1

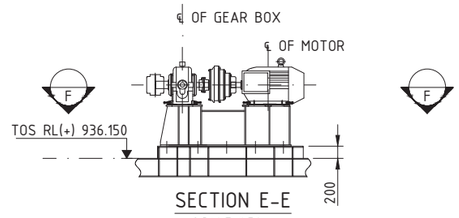
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
TERTIARY CRUSHER HOUSE-PLAN VIEW

DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1013
REV: A01

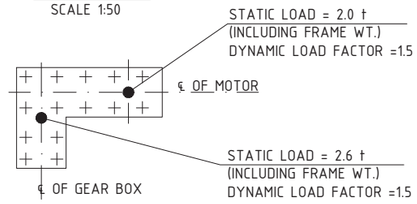
DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1014 | REV: A01



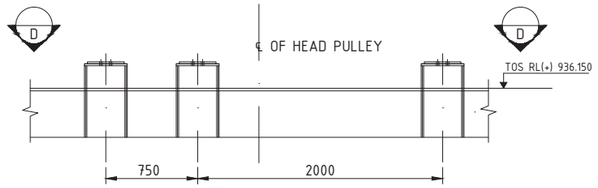
PLAN AT RL(+) 936.150
SCALE 1:50



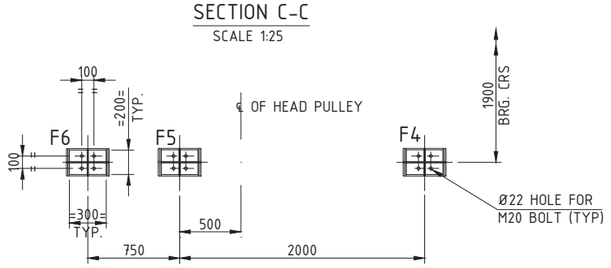
SECTION E-E
SCALE 1:50



VIEW F-F
SCALE 1:50



SECTION C-C
SCALE 1:25



VIEW D-D
SCALE 1:25

NOTE:
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- CONCRETE FLOOR
- CHECKERED PLATE
- GRATING
- BRICK WALL
- CONCRETE WALL

KEY PLAN

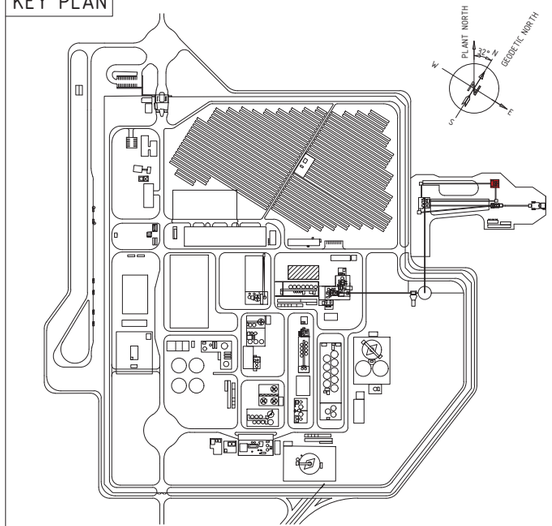


TABLE-5 : FOR CHUTE SADDLE LOAD
FOR CHUTE 3130-CHT-008

POINTS	ELEVATIONS IN "m"	V (IN "t")
F7	(+)936.150	(+)1.5

SIGN CONVENTION
(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)
(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

Note: Wind load to be considered during civil & structural design

TABLE-5 : FOR DRIVE PULLEY FRAME LOAD
FOR CONV. 3130-CNV-001

POINTS	ELEVATIONS IN "m"	LOAD CASE I			LOAD CASE II		
		V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")	V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")
F4	(+)936.150	(+) 1.70	(-) 0.91	(±) 0.17	(-) 1.14	(-) 2.16	(±) 0.12
F5		(+) 4.37	(-) 2.45	(±) 0.45	(+) 3.35	(-) 2.48	(±) 0.35
F6		(+) 4.70	(-) 3.18	(±) 0.45	(+) 5.95	(-) 4.00	(±) 0.60

NOTE:
CASE I : CONVEYOR NORMAL RUNNING + MATERIAL LOAD + DEAD LOAD
CASE II : CONVEYOR STARTING + CHUTE EMPTY + DEAD LOAD

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1013	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1012	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1010	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-ELV-1018	TERTIARY CRUSHER HOUSE-ELEVATION								
MD-216-3130-EG-ME-ELV-1017	TERTIARY CRUSHER HOUSE-ELEVATION								
MD-216-3130-EG-ME-LPX-1009	TERTIARY CRUSHER HOUSE-PLAN VIEW	A01	06/01/2020	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_100_H_G_16129_008		0
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__31356	OU100669128	0

DRAWN	SKH	03/01/2020
CHECKED	BRD	04/01/2020
ENGINEER	BD	04/01/2020
APPROVED	BGR	06/01/2020
EMPLOYER		
SCALE	1:50, 1:25	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
TERTIARY CRUSHER HOUSE-PLAN VIEW

DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1014 | REV: A01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 2 STAADPRO INPUT FILE OF TC SUPPORT STRUCTURE

Job Title:

Client:

Engineer:

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 13-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
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146 TO 148 151 TO 158 161 TO 168 171 TO 178 181 TO 228 231 TO 238 241 TO 248 -
251 TO 258 261 TO 305 338 TO 409 412 TO 492 494 TO 501 503 510 512 -
514 TO 543 545 TO 560 563 565 566 568 569 571 573 574 577 TO 580 582 TO 585 -
587 TO 590 592 594 596 597 THICKNESS 0.3
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+07
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-05
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
19 20 22 23 28 29 31 32 100 TO 103 118 TO 145 564 567 570 576 581 -
586 PRIS YD 0.6 ZD 0.6
7 16 25 34 306 TO 321 504 506 509 511 513 544 PRIS YD 0.5 ZD 0.3
37 TO 42 82 TO 97 PRIS YD 0.55 ZD 0.3
49 TO 56 PRIS YD 1.1 ZD 0.35
18 27 57 TO 59 61 62 64 66 PRIS YD 0.9 ZD 0.4
9 36 505 561 572 591 593 595 PRIS YD 0.6 ZD 0.3
68 TO 81 PRIS YD 0.4 ZD 0.2
MEMBER PROPERTY AMERICAN
60 63 65 67 PRIS YD 0.8 ZD 0.4
MEMBER PROPERTY AMERICAN
1 2 4 5 10 11 13 14 98 99 104 TO 117 322 TO 337 410 411 493 -
502 PRIS YD 0.8 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 5 9 13 17 21 25 29 FIXED
*2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 PINNED
MEMBER RELEASE
60 63 76 TO 81 START MP 0.99
65 67 76 TO 81 END MP 0.99
55 56 START MX
49 52 END MX
*SLAVE ZX MASTER 38 JOINT 4 8 12 16 20 24 28 32 TO 48
DEFINE IBC 2003
SDS 0.066 SD1 0.0133 S1 0.2 IE 1 RX 2.5 RZ 2.5 SCLASS 2 PX 0.237 PZ 0.237
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JOINT WEIGHT
1 WEIGHT 14
2 WEIGHT 129
4 WEIGHT 485
5 WEIGHT 14
6 WEIGHT 128
8 WEIGHT 442
9 WEIGHT 14
10 WEIGHT 94
12 WEIGHT 1257
13 WEIGHT 14
14 WEIGHT 103
16 WEIGHT 1110
17 WEIGHT 11
18 WEIGHT 94
20 WEIGHT 49
21 WEIGHT 11
22 WEIGHT 93
24 WEIGHT 83
25 WEIGHT 11
26 WEIGHT 230
28 WEIGHT 175
29 WEIGHT 11
30 WEIGHT 181
32 WEIGHT 145
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LOAD 11 SEISMIC LOAD (+X)
IBC LOAD X 1
************************************************
LOAD 12 SEISMIC LOAD (+Z)
IBC LOAD Z 1
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
FLOOR LOAD
YRANGE 5.9 6.1 FLOAD -10 XRANGE -0.1 11  ZRANGE -1.05 6.45  GY
FLOOR LOAD
YRANGE 5.9 6.1 FLOAD -5 XRANGE 11 15  ZRANGE -1.05 6.45  GY
JOINT LOAD
35 37 FY -200
MEMBER LOAD
60 63 CON GY -200 3.245
65 67 CON GY -45 0.99
***material load
JOINT LOAD
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35 37 FY -27
MEMBER LOAD
60 63 CON GY -27 3.245
**Dedusting duct support load
9 58 CON GY -10
**Cable tray support load
50 53 55 56 60 63 65 67 CON GY -6 0.3
36 71 75 80 81 505 561 572 591 593 595 UNI GY -2.3
**Chute weight at soffit
JOINT LOAD
35 37 FY -20
MEMBER LOAD
60 63 CON GY -20 3
***Screed concrete load
LOAD 2 LIVE LOAD
FLOOR LOAD
YRANGE 5.9 6.1 FLOAD -5 XRANGE -0.1 14.6  ZRANGE -1.05 6.45  GY
*******************************************
LOAD 3 LOADTYPE None  TITLE SDL1- CRUSHER DYNAMIC LOAD
***Dynamic load at Z direction from crusher start & stop
JOINT LOAD
35 37 FY -70 FZ 73
MEMBER LOAD
60 63 CON GY -70 3.245
60 63 CON GZ 73 3.145
65 67 CON GY -22 0.99
65 67 CON GZ 12 0.99
***Monorail load
59 CON GY -40
505 CON GY -40 0.7
*******************************************
LOAD 4 LOADTYPE None  TITLE SDL-2 CRUSHER DYNAMIC LOAD
***Dynamic load at Z direction from crusher start & stop
JOINT LOAD
35 37 FX 73 FY -70
MEMBER LOAD
60 63 CON GY -70 3.245
60 63 CON GX 73 3.145
65 67 CON GY -22 0.99
65 67 CON GX 12 0.99
***Monorail load
64 CON GY -40
*******************************************
LOAD 5 WIND LOAD (+X)
```

```
MEMBER LOAD
**Wind load on building frame
2 5 322 TO 337 410 411 UNI GX 3.8
11 14 98 99 104 TO 117 493 502 UNI GX 3.24
**Wind load on crusher
JOINT LOAD
35 37 FX 5 FY 3.5
MEMBER LOAD
60 63 CON GY -3.5 3.145
60 63 CON GX 5 3.145
*******************************************
LOAD 6 WIND LOAD (-X)
MEMBER LOAD
**Wind load on building frame
2 5 322 TO 337 410 411 UNI GX -3.8
29 32 102 103 132 TO 145 564 567 570 576 581 586 UNI GX -3.24
**Wind load on crusher
JOINT LOAD
35 37 FX -5 FY -3.5
MEMBER LOAD
60 63 CON GY 3.5 3.145
60 63 CON GX -5 3.145
*******************************************
LOAD 7 WIND LOAD (+Z)
**Wind load on building frame
MEMBER LOAD
2 5 11 14 98 99 104 TO 117 322 TO 337 410 411 493 502 UNI GZ 2.6
11 98 104 TO 110 493 UNI GZ 1.9
20 100 118 TO 124 UNI GZ 4
29 102 132 TO 138 576 581 586 UNI GZ 2.1
**Wind load on crusher
JOINT LOAD
35 FY 3.5 FZ 5
37 FY -3.5 FZ 5
MEMBER LOAD
60 CON GY 3.5 3.145
60 CON GZ 5 3.145
63 CON GY -3.5 3.145
63 CON GZ 5 3.145
*******************************************
LOAD 8 WIND LOAD (-Z)
**Wind load on building frame
MEMBER LOAD
2 5 11 14 98 99 104 TO 117 322 TO 337 410 411 493 502 UNI GZ -2.6
```

```
11 14 98 99 104 TO 117 493 502 UNI GZ -1.9
23 101 125 TO 131 UNI GZ -4
32 103 139 TO 145 564 567 570 UNI GZ -2.1
**Wind load on crusher
JOINT LOAD
35 FY -3.5 FZ -5
37 FY 3.5 FZ -5
MEMBER LOAD
60 CON GY -3.5 3.145
60 CON GZ -5 3.145
63 CON GY 3.5 3.145
63 CON GZ -5 3.145
LOAD 22 MODAL ANALYSIS
SELFWEIGHT X 1
FLOOR LOAD
YRANGE 5.9 6.1 FLOAD 10 XRANGE -0.1 11  ZRANGE -1.05 6.45  GX
FLOOR LOAD
YRANGE 5.9 6.1 FLOAD 10 XRANGE 11 15  ZRANGE -1.05 6.45  GX
**crusher load
JOINT LOAD
35 37 FX 200
MEMBER LOAD
60 63 CON GX 200 3.245
65 67 CON GX 45 0.99
***material load
JOINT LOAD
35 37 FX 27
MEMBER LOAD
60 63 CON GX 27 3.245
**Dedusting duct support load
9 58 CON GX 10
**Cable tray support load
50 53 55 56 60 63 65 67 CON GX 6 0.3
36 71 75 80 81 505 561 572 591 593 595 UNI GX 2.3
**Chute weight at soffit
JOINT LOAD
35 37 FX 20
MEMBER LOAD
60 63 CON GX 20 3
SELFWEIGHT Y 1
FLOOR LOAD
YRANGE 5.9 6.1 FLOAD 10 XRANGE -0.1 11  ZRANGE -1.05 6.45  GY
FLOOR LOAD
YRANGE 5.9 6.1 FLOAD 10 XRANGE 11 15  ZRANGE -1.05 6.45  GY
```

```
**crusher load
JOINT LOAD
35 37 FY 200
MEMBER LOAD
60 63 CON GY 200 3.245
65 67 CON GY 45 0.99
***material load
JOINT LOAD
35 37 FY 27
MEMBER LOAD
60 63 CON GX 27 3.245
**Dedusting duct support load
9 58 CON GY 10
**Cable tray support load
50 53 55 56 60 63 65 67 CON GY 6 0.3
36 71 75 80 81 505 561 572 591 593 595 UNI GY 2.3
**Chute weight at soffit
JOINT LOAD
35 37 FY 20
MEMBER LOAD
60 63 CON GY 20 3
SELFWEIGHT Z 1
FLOOR LOAD
YRANGE 5.9 6.1 FLOAD 10 XRANGE -0.1 11  ZRANGE -1.05 6.45  GZ
FLOOR LOAD
YRANGE 5.9 6.1 FLOAD 10 XRANGE 11 15  ZRANGE -1.05 6.45  GZ
**crusher load
JOINT LOAD
35 37 FZ 200
MEMBER LOAD
60 63 CON GZ 200 3.245
65 67 CON GZ 45 0.99
***material load
JOINT LOAD
35 37 FZ 27
MEMBER LOAD
60 63 CON GZ 27 3.245
**Dedusting duct support load
9 58 CON GZ 10
**Cable tray support load
50 53 55 56 60 63 65 67 CON GZ 6 0.3
36 71 75 80 81 505 561 572 591 593 595 UNI GZ 2.3
**Chute weight at soffit
JOINT LOAD
```

```
35 37 FZ 20
MEMBER LOAD
60 63 CON GZ 20 3
MODAL CALCULATION REQUESTED
************************************************
LOAD 301 DL+SDL1_R
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2_R
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL-SDL1_R
REPEAT LOAD
1 1.0 3 -1.0
LOAD 304 DL-SDL2_R
REPEAT LOAD
1 1.0 4 -1.0
**********************************
LOAD 305 DL + SDL1_R + LL
REPEAT LOAD
1 1.0 3 1.0 2 1.0
LOAD 306 DL + SDL2_R + LL
REPEAT LOAD
1 1.0 4 1.0 2 1.0
LOAD 307 DL - SDL1_R + LL
REPEAT LOAD
1 1.0 3 -1.0 2 1.0
LOAD 308 DL - SDL2_R + LL
REPEAT LOAD
1 1.0 4 -1.0 2 1.0
************************************************
LOAD 309 DL + SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 310 DL + SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 311 DL + SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 312 DL + SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 313 DL + SDL2_R + 0.75 LL + 0.75 WL(+X)
```

```
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 314 DL + SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 315 DL + SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 316 DL + SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 317 DL - SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 5 0.75
LOAD 318 DL - SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 6 0.75
LOAD 319 DL - SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 7 0.75
LOAD 320 DL - SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 8 0.75
LOAD 321 DL - SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 5 0.75
LOAD 322 DL - SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 6 0.75
LOAD 323 DL - SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 7 0.75
LOAD 324 DL - SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 8 0.75
************************************************
LOAD 325 DL + SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 326 DL + SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 327 DL + SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
```

```
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 328 DL + SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 329 DL + SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 330 DL + SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 331 DL + SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 332 DL + SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 333 DL - SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 0.525
LOAD 334 DL - SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 -0.525
LOAD 335 DL - SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 0.525
LOAD 336 DL - SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 -0.525
LOAD 337 DL - SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 0.525
LOAD 338 DL - SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 -0.525
LOAD 339 DL - SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 0.525
LOAD 340 DL - SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 -0.525
************************************************
LOAD 341 0.6 (DL + SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
```

```
LOAD 342 0.6 (DL + SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 345 0.6 (DL + SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 346 0.6 (DL + SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 349 0.6 (DL - SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 -0.6 5 1.0
LOAD 350 0.6 (DL - SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 -0.6 6 1.0
LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 7 1.0
LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 8 1.0
LOAD 353 0.6 (DL - SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 -0.6 5 1.0
LOAD 354 0.6 (DL - SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 -0.6 6 1.0
LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 7 1.0
LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)
REPEAT LOAD
```

```
1 0.6 4 -0.6 8 1.0
************************************************
LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 -0.6 11 0.7
LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 -0.6 11 -0.7
LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 12 0.7
LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 12 -0.7
LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 -0.6 11 0.7
LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 -0.6 11 -0.7
```

```
LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
************************************************
*Load Combination for Concrete design
************************************************
LOAD 401 1.4 (DL+SDL1_R)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2_R)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL-SDL1_R)
REPEAT LOAD
1 1.4 3 -1.4
LOAD 404 1.4 (DL-SDL2_R)
REPEAT LOAD
1 1.4 4 -1.4
************************************************
LOAD 405 1.4 (DL + SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 2 1.7
LOAD 407 1.4 (DL - SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 -1.4 2 1.7
LOAD 408 1.4 (DL - SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 -1.4 2 1.7
************************************************
LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
```

```
LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 5 1.6
LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 6 1.6
LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 7 1.6
LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 8 1.6
LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 5 1.6
LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 6 1.6
LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 7 1.6
LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 8 1.6
************************************************
LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
```

```
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 1.0
LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 -1.0
LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 1.0
LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 -1.0
LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 1.0
LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 -1.0
LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 1.0
LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 -1.0
```

```
**********************************************
LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 -0.9 5 1.6
LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 -0.9 6 1.6
LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 7 1.6
LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 8 1.6
LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 -0.9 5 1.6
LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 -0.9 6 1.6
LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)
```

```
REPEAT LOAD
1 0.9 4 -0.9 7 1.6
LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 8 1.6
***********************************************
LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 -0.9 11 1.0
LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 -0.9 11 -1.0
LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 12 1.0
LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 12 -1.0
LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
```

```
1 0.9 4 -0.9 11 1.0
LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 -0.9 11 -1.0
LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 12 1.0
LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 12 -1.0
**********
**********
*LOAD COMBINATION 1001
*1 1.0 2 0.25 4 1.0
PERFORM ANALYSIS
*LOAD LIST 401 TO 472
*START CONCRETE DESIGN
*CODE ACI
*FC 35000 MEMB 1 2 4 5 7 9 TO 11 13 14 16 18 TO 20 22 23 25 27 TO 29 31 32 -
*34 36 TO 42 49 TO 145 306 TO 337 410 411 493 502 504 TO 506 509 511 513 544 -
*561 564 567 570 572 576 581 586 591 593 595
*FYMAIN 420000 MEMB 1 2 4 5 7 9 TO 11 13 14 16 18 TO 20 22 23 25 27 TO 29 31 -
*32 34 36 TO 42 49 TO 145 306 TO 337 410 411 493 502 504 TO 506 509 511 513 -
*544 561 564 567 570 572 576 581 586 591 593 595
*DESIGN COLUMN 1 2 4 5 10 11 13 14 19 20 22 23 28 29 31 32 98 TO 145 -
*322 TO 337 410 411 493 502 564 567 570 576 581 586
*MAXMAIN 32 ALL
*END CONCRETE DESIGN
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 3 STAADPRO OUTPUT FILE OF TC SUPPORT STRUCTURE

```
**************************************************
*                                                *
*          STAAD.Pro CONNECT Edition             *
*          Version  22.06.00.138                 *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    APR 28, 2022                 *
*          Time=    18:34:34                     *
*                                                *
*  Licensed to: Larsen & Toubro Limited          *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\GET Period\EDRC\EDRC Work\Maaden Gold\Tertiary Crusher Support Final\Wall Design\STAAD\.. .STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 13-JAN-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 -2.5 0; 2 0 0 0; 4 0 6 0; 5 0 -2.5 5.4; 6 0 0 5.4; 8 0 6 5.4; 9 7.9 -2.5 0
     9. 10 7.9 0 0; 12 7.9 6 0; 13 7.9 -2.5 5.4; 14 7.9 0 5.4; 16 7.9 6 5.4
    10. 17 11 -2.5 0; 18 11 0 0; 20 11 6 0; 21 11 -2.5 5.4; 22 11 0 5.4; 24 11 6 5.4
    11. 25 14.5 -2.5 0; 26 14.5 0 0; 28 14.5 6 0; 29 14.5 -2.5 5.4; 30 14.5 0 5.4
    12. 32 14.5 6 5.4; 33 3.35 6 0; 34 3.35 6 5.4; 35 3.35 6 0.755; 36 7.9 6 0.755
    13. 37 3.35 6 4.045; 38 7.9 6 4.045; 39 11 6 0.755; 40 11 6 4.045; 41 7.9 6 -1.
    14. 42 11 6 -1; 43 14.5 6 -1; 44 3.35 6 -1; 45 7.9 6 6.4; 46 11 6 6.4
    15. 47 14.5 6 6.4; 48 3.35 6 6.4; 49 8.3 0 0; 50 8.6 0 0; 51 9.76 0 0
    16. 52 10.38 0 0; 53 11.3 0 0; 54 11.6 0 0; 55 13.9 0 0; 56 14.2 0 0; 57 8.3 0 5.4
    17. 58 8.6 0 5.4; 59 9.76 0 5.4; 60 10.38 0 5.4; 61 11.3 0 5.4; 62 11.6 0 5.4
    18. 63 13.9 0 5.4; 64 14.2 0 5.4; 65 7.9 4.5 0; 66 7.9 4.5 5.4; 67 11 4.5 0
    19. 68 11 4.5 5.4; 69 14.5 4.5 0; 70 14.5 4.5 5.4; 71 7.9 0.249999 0
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    21. 77 7.9 3.85 0; 78 7.9 0.249999 5.4; 79 7.9 0.549999 5.4; 80 7.9 1.375 5.4
    22. 81 7.9 2.2 5.4; 82 7.9 2.75 5.4; 83 7.9 3.3 5.4; 84 7.9 3.85 5.4
    23. 85 11 0.274999 0; 86 11 0.574999 0; 87 11 1.3875 0; 88 11 2.2 0; 89 11 2.75 0
    24. 90 11 3.3 0; 91 11 3.85 0; 92 11 0.274999 5.4; 93 11 0.574999 5.4
    25. 94 11 1.3875 5.4; 95 11 2.2 5.4; 96 11 2.75 5.4; 97 11 3.3 5.4; 98 11 3.85 5.4
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    27. 103 14.5 2.75 0; 104 14.5 3.3 0; 105 14.5 3.85 0; 106 14.5 0.274999 5.4
    28. 107 14.5 0.574999 5.4; 108 14.5 1.3875 5.4; 109 14.5 2.2 5.4
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175. 626 525 55; 650 541 59; 675 555 51
176. ELEMENT INCIDENCES SHELL
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178. 154 63 64 144 143; 155 64 30 106 144; 156 78 137 153 79; 157 137 138 154 153
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204. 294 215 216 232 231; 295 216 104 105 232; 296 77 225 113 65
205. 297 225 226 114 113; 300 228 91 67 116; 301 91 229 117 67; 302 229 230 118 117
206. 304 231 232 120 119; 305 232 105 69 120; 338 2 241 273 259
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289. 620 518 223 239 520; 621 238 519 521 126; 622 519 520 522 521
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307. 678 130 556 557 146; 679 556 131 147 557; 680 146 557 558 162
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310. 687 561 180 196 562; 688 194 560 563 210; 689 560 195 211 563
311. 690 195 562 564 211; 691 562 196 212 564; 692 210 563 565 226
312. 693 563 211 227 565; 694 211 564 566 227; 695 564 212 228 566
313. 696 226 565 567 114; 697 565 227 115 567; 698 227 566 568 115
314. 699 566 228 116 568
315. START GROUP DEFINITION
316. ELEMENT
317. _M 157 162 164 167 172 174 177 182 184 187 192 194 197 202 204 207 212 214 -
318. 217 222 224 237 242 244 247 252 254 257 262 264 267 272 274 277 282 284 287 -
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319. 292 294 297 302 304 348 TO 354 357 TO 363 366 TO 372 375 TO 381 384 TO 390 -
320. 393 TO 399 402 TO 408 413 TO 419 431 TO 437 440 TO 446 449 TO 455 -
321. 458 TO 464 467 TO 473 476 TO 482 485 TO 491 495 TO 501 517 TO 519 -
322. 521 TO 523 525 TO 527 529 TO 531 533 TO 535 537 TO 539 541 TO 543 547 549 -
323. 551 553 555 557 559 563 566 569 573 578 TO 580 583 TO 585 588 TO 590 594 -
324. 596 597 603 TO 623 629 TO 649 653 TO 674 678 TO 699
325. _SF 358 TO 362 367 TO 371 376 TO 380 385 TO 389 394 TO 398 403 TO 407 414 -
326. 415 TO 418 441 TO 445 450 TO 454 459 TO 463 468 TO 472 477 TO 481 486 TO 490 -
327. 496 TO 500 522 523 526 527 530 531 534 535 538 539 542 543 549 551 553 555 -
328. 557 559 563 566 569 573 579 580 584 585 589 590 596 597 606 TO 623 -
329. 632 TO 649 655 TO 661 663 TO 665 667 TO 669 671 TO 673 680 TO 686 -
330. 688 TO 690 692 TO 694 696 TO 698
331. END GROUP DEFINITION
332. ELEMENT PROPERTY
333. 146 147 151 152 154 TO 157 161 162 164 TO 167 171 172 174 TO 177 181 182 184 -
334. 185 TO 187 190 TO 192 194 TO 197 200 TO 202 204 TO 207 210 TO 212 214 TO 217 -
335. 220 TO 222 224 TO 227 231 232 234 TO 237 241 242 244 TO 247 251 252 -
336. 254 TO 257 261 262 264 TO 267 270 TO 272 274 TO 277 280 TO 282 284 TO 287 -
337. 290 TO 292 294 TO 297 300 TO 302 304 305 338 TO 409 412 TO 492 494 TO 501 -
338. 503 510 512 514 TO 543 545 TO 560 563 565 566 568 569 571 573 574 -
339. 577 TO 580 582 TO 585 587 TO 590 592 594 596 597 599 601 TO 623 625 -
340. 627 TO 649 651 TO 674 676 TO 699 THICKNESS 0.3
341. DEFINE MATERIAL START
342. ISOTROPIC CONCRETE
343. E 2.17185E+07
344. POISSON 0.17
345. DENSITY 23.5616
346. ALPHA 1E-05
347. DAMP 0.05
348. TYPE CONCRETE
349. STRENGTH FCU 27579
350. END DEFINE MATERIAL
351. MEMBER PROPERTY AMERICAN
352. 19 20 22 23 28 29 31 32 100 TO 103 118 TO 145 564 567 570 576 581 -
353. 586 PRIS YD 0.6 ZD 0.6
354. 7 16 25 34 306 TO 321 504 506 509 511 513 544 PRIS YD 0.5 ZD 0.3
355. 37 TO 42 82 TO 97 598 600 624 626 650 675 PRIS YD 0.55 ZD 0.3
356. 49 TO 56 PRIS YD 1.1 ZD 0.35
357. 18 27 57 TO 59 61 62 64 66 PRIS YD 0.9 ZD 0.4
358. 9 36 505 561 572 591 593 595 PRIS YD 0.6 ZD 0.3
359. 68 TO 81 PRIS YD 0.4 ZD 0.2
360. MEMBER PROPERTY AMERICAN
361. 60 63 65 67 PRIS YD 0.8 ZD 0.4
362. MEMBER PROPERTY AMERICAN
363. 1 2 4 5 10 11 13 14 98 99 104 TO 117 322 TO 337 410 411 493 -
364. 502 PRIS YD 0.8 ZD 0.6
365. CONSTANTS
366. MATERIAL CONCRETE ALL
367. SUPPORTS
368. 1 5 9 13 17 21 25 29 FIXED
369. *2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 PINNED
370. MEMBER RELEASE
371. 60 63 76 TO 81 START MP 0.99
372. 65 67 76 TO 81 END MP 0.99
373. 55 56 START MX
374. 49 52 END MX
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  375. *SLAVE ZX MASTER 38 JOINT 4 8 12 16 20 24 28 32 TO 48
  376. DEFINE IBC 2003
  377. SDS 0.066 SD1 0.0133 S1 0.2 IE 1 RX 2.5 RZ 2.5 SCLASS 2 PX 0.237 PZ 0.237
  378. JOINT WEIGHT
  379. 1 WEIGHT 14
  380. 2 WEIGHT 129
  381. 4 WEIGHT 485
  382. 5 WEIGHT 14
  383. 6 WEIGHT 128
  384. 8 WEIGHT 442
  385. 9 WEIGHT 14
  386. 10 WEIGHT 94
  387. 12 WEIGHT 1257
  388. 13 WEIGHT 14
  389. 14 WEIGHT 103
  390. 16 WEIGHT 1110
  391. 17 WEIGHT 11
  392. 18 WEIGHT 94
  393. 20 WEIGHT 49
  394. 21 WEIGHT 11
  395. 22 WEIGHT 93
  396. 24 WEIGHT 83
  397. 25 WEIGHT 11
  398. 26 WEIGHT 230
  399. 28 WEIGHT 175
  400. 29 WEIGHT 11
  401. 30 WEIGHT 181
  402. 32 WEIGHT 145
  403. ************************************************
  404. LOAD 11 SEISMIC LOAD (+X)
  405. IBC LOAD X 1
  406. ************************************************
  407. LOAD 12 SEISMIC LOAD (+Z)
  408. IBC LOAD Z 1
  409. LOAD 1 DEAD LOAD
  410. SELFWEIGHT Y -1 LIST ALL
  411. FLOOR LOAD
  412. YRANGE 5.9 6.1 FLOAD -10 XRANGE -0.1 11  ZRANGE -1.05 6.45  GY
**NOTE** about Floor/OneWay Loads/Weights.
  Please note that depending on the shape of the floor you may
  have to break up the FLOOR/ONEWAY LOAD into multiple commands.
  For details please refer to Technical Reference Manual
  Section 5.32.4.2 Note d and/or "5.32.4.3 Note f.

  413. FLOOR LOAD
  414. YRANGE 5.9 6.1 FLOAD -5 XRANGE 11 15  ZRANGE -1.05 6.45  GY
  415. JOINT LOAD
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416. 35 37 FY -200
417. MEMBER LOAD
418. 60 63 CON GY -200 3.245
419. 65 67 CON GY -45 0.99
420. ***MATERIAL LOAD
421. JOINT LOAD
422. 35 37 FY -27
423. MEMBER LOAD
424. 60 63 CON GY -27 3.245
425. **DEDUSTING DUCT SUPPORT LOAD
426. 9 58 CON GY -10
427. **CABLE TRAY SUPPORT LOAD
428. 50 53 55 56 60 63 65 67 CON GY -6 0.3
429. 36 71 75 80 81 505 561 572 591 593 595 UNI GY -2.3
430. **CHUTE WEIGHT AT SOFFIT
431. JOINT LOAD
432. 35 37 FY -20
433. MEMBER LOAD
434. 60 63 CON GY -20 3
435. ***SCREED CONCRETE LOAD
436. LOAD 2 LIVE LOAD
437. FLOOR LOAD
438. YRANGE 5.9 6.1 FLOAD -5 XRANGE -0.1 14.6  ZRANGE -1.05 6.45  GY
439. ************************************************
440. LOAD 3 LOADTYPE NONE  TITLE SDL1- CRUSHER DYNAMIC LOAD
441. ***DYNAMIC LOAD AT Z DIRECTION FROM CRUSHER START & STOP
442. JOINT LOAD
443. 35 37 FY -70 FZ 73
444. MEMBER LOAD
445. 60 63 CON GY -70 3.245
446. 60 63 CON GZ 73 3.145
447. 65 67 CON GY -22 0.99
448. 65 67 CON GZ 12 0.99
449. ***MONORAIL LOAD
450. 59 CON GY -40
451. 505 CON GY -40 0.7
452. ****************************************
453. LOAD 4 LOADTYPE NONE  TITLE SDL-2 CRUSHER DYNAMIC LOAD
454. ***DYNAMIC LOAD AT Z DIRECTION FROM CRUSHER START & STOP
455. JOINT LOAD
456. 35 37 FX 73 FY -70
457. MEMBER LOAD
458. 60 63 CON GY -70 3.245
459. 60 63 CON GX 73 3.145
460. 65 67 CON GY -22 0.99
461. 65 67 CON GX 12 0.99
462. ***MONORAIL LOAD
463. 64 CON GY -40
464. ****************************************
465. LOAD 5 WIND LOAD (+X)
466. MEMBER LOAD
467. **WIND LOAD ON BUILDING FRAME
468. 2 5 322 TO 337 410 411 UNI GX 3.8
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469. 11 14 98 99 104 TO 117 493 502 UNI GX 3.24
470. **WIND LOAD ON CRUSHER
471. JOINT LOAD
472. 35 37 FX 5 FY 3.5
473. MEMBER LOAD
474. 60 63 CON GY -3.5 3.145
475. 60 63 CON GX 5 3.145
476. ************************************************
477. LOAD 6 WIND LOAD (-X)
478. MEMBER LOAD
479. **WIND LOAD ON BUILDING FRAME
480. 2 5 322 TO 337 410 411 UNI GX -3.8
481. 29 32 102 103 132 TO 145 564 567 570 576 581 586 UNI GX -3.24
482. **WIND LOAD ON CRUSHER
483. JOINT LOAD
484. 35 37 FX -5 FY -3.5
485. MEMBER LOAD
486. 60 63 CON GY 3.5 3.145
487. 60 63 CON GX -5 3.145
488. ************************************************
489. LOAD 7 WIND LOAD (+Z)
490. **WIND LOAD ON BUILDING FRAME
491. MEMBER LOAD
492. 2 5 11 14 98 99 104 TO 117 322 TO 337 410 411 493 502 UNI GZ 2.6
493. 11 98 104 TO 110 493 UNI GZ 1.9
494. 20 100 118 TO 124 UNI GZ 4
495. 29 102 132 TO 138 576 581 586 UNI GZ 2.1
496. **WIND LOAD ON CRUSHER
497. JOINT LOAD
498. 35 FY 3.5 FZ 5
499. 37 FY -3.5 FZ 5
500. MEMBER LOAD
501. 60 CON GY 3.5 3.145
502. 60 CON GZ 5 3.145
503. 63 CON GY -3.5 3.145
504. 63 CON GZ 5 3.145
505. ************************************************
506. LOAD 8 WIND LOAD (-Z)
507. **WIND LOAD ON BUILDING FRAME
508. MEMBER LOAD
509. 2 5 11 14 98 99 104 TO 117 322 TO 337 410 411 493 502 UNI GZ -2.6
510. 11 14 98 99 104 TO 117 493 502 UNI GZ -1.9
511. 23 101 125 TO 131 UNI GZ -4
512. 32 103 139 TO 145 564 567 570 UNI GZ -2.1
513. **WIND LOAD ON CRUSHER
514. JOINT LOAD
515. 35 FY -3.5 FZ -5
516. 37 FY 3.5 FZ -5
517. MEMBER LOAD
518. 60 CON GY -3.5 3.145
519. 60 CON GZ -5 3.145
520. 63 CON GY 3.5 3.145
521. 63 CON GZ -5 3.145
522. LOAD 22 MODAL ANALYSIS
523. SELFWEIGHT X 1
524. FLOOR LOAD
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525. YRANGE 5.9 6.1 FLOAD 10 XRANGE -0.1 11  ZRANGE -1.05 6.45  GX
526. FLOOR LOAD
527. YRANGE 5.9 6.1 FLOAD 10 XRANGE 11 15  ZRANGE -1.05 6.45  GX
528. **CRUSHER LOAD
529. JOINT LOAD
530. 35 37 FX 200
531. MEMBER LOAD
532. 60 63 CON GX 200 3.245
533. 65 67 CON GX 45 0.99
534. ***MATERIAL LOAD
535. JOINT LOAD
536. 35 37 FX 27
537. MEMBER LOAD
538. 60 63 CON GX 27 3.245
539. **DEDUSTING DUCT SUPPORT LOAD
540. 9 58 CON GX 10
541. **CABLE TRAY SUPPORT LOAD
542. 50 53 55 56 60 63 65 67 CON GX 6 0.3
543. 36 71 75 80 81 505 561 572 591 593 595 UNI GX 2.3
544. **CHUTE WEIGHT AT SOFFIT
545. JOINT LOAD
546. 35 37 FX 20
547. MEMBER LOAD
548. 60 63 CON GX 20 3
549. SELFWEIGHT Y 1
550. FLOOR LOAD
551. YRANGE 5.9 6.1 FLOAD 10 XRANGE -0.1 11  ZRANGE -1.05 6.45  GY
552. FLOOR LOAD
553. YRANGE 5.9 6.1 FLOAD 10 XRANGE 11 15  ZRANGE -1.05 6.45  GY
554. **CRUSHER LOAD
555. JOINT LOAD
556. 35 37 FY 200
557. MEMBER LOAD
558. 60 63 CON GY 200 3.245
559. 65 67 CON GY 45 0.99
560. ***MATERIAL LOAD
561. JOINT LOAD
562. 35 37 FY 27
563. MEMBER LOAD
564. 60 63 CON GX 27 3.245
565. **DEDUSTING DUCT SUPPORT LOAD
566. 9 58 CON GY 10
567. **CABLE TRAY SUPPORT LOAD
568. 50 53 55 56 60 63 65 67 CON GY 6 0.3
569. 36 71 75 80 81 505 561 572 591 593 595 UNI GY 2.3
570. **CHUTE WEIGHT AT SOFFIT
571. JOINT LOAD
572. 35 37 FY 20
573. MEMBER LOAD
574. 60 63 CON GY 20 3
```

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575. SELFWEIGHT Z 1
576. FLOOR LOAD
577. YRANGE 5.9 6.1 FLOAD 10 XRANGE -0.1 11  ZRANGE -1.05 6.45  GZ
578. FLOOR LOAD
579. YRANGE 5.9 6.1 FLOAD 10 XRANGE 11 15  ZRANGE -1.05 6.45  GZ
580. **CRUSHER LOAD
581. JOINT LOAD
582. 35 37 FZ 200
583. MEMBER LOAD
584. 60 63 CON GZ 200 3.245
585. 65 67 CON GZ 45 0.99
586. ***MATERIAL LOAD
587. JOINT LOAD
588. 35 37 FZ 27
589. MEMBER LOAD
590. 60 63 CON GZ 27 3.245
591. **DEDUSTING DUCT SUPPORT LOAD
592. 9 58 CON GZ 10
593. **CABLE TRAY SUPPORT LOAD
594. 50 53 55 56 60 63 65 67 CON GZ 6 0.3
595. 36 71 75 80 81 505 561 572 591 593 595 UNI GZ 2.3
596. **CHUTE WEIGHT AT SOFFIT
597. JOINT LOAD
598. 35 37 FZ 20
599. MEMBER LOAD
600. 60 63 CON GZ 20 3
601. MODAL CALCULATION REQUESTED
602. ********************************************
603. LOAD 301 DL+SDL1_R
604. REPEAT LOAD
605. 1 1.0 3 1.0
606. LOAD 302 DL+SDL2_R
607. REPEAT LOAD
608. 1 1.0 4 1.0
609. LOAD 303 DL-SDL1_R
610. REPEAT LOAD
611. 1 1.0 3 -1.0
612. LOAD 304 DL-SDL2_R
613. REPEAT LOAD
614. 1 1.0 4 -1.0
615. **********************************
616. LOAD 305 DL + SDL1_R + LL
617. REPEAT LOAD
618. 1 1.0 3 1.0 2 1.0
619. LOAD 306 DL + SDL2_R + LL
620. REPEAT LOAD
621. 1 1.0 4 1.0 2 1.0
622. LOAD 307 DL - SDL1_R + LL
623. REPEAT LOAD
624. 1 1.0 3 -1.0 2 1.0
625. LOAD 308 DL - SDL2_R + LL
626. REPEAT LOAD
627. 1 1.0 4 -1.0 2 1.0
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628. ************************************************
629. LOAD 309 DL + SDL1_R + 0.75 LL + 0.75 WL(+X)
630. REPEAT LOAD
631. 1 1.0 3 1.0 2 0.75 5 0.75
632. LOAD 310 DL + SDL1_R + 0.75 LL + 0.75 WL(-X)
633. REPEAT LOAD
634. 1 1.0 3 1.0 2 0.75 6 0.75
635. LOAD 311 DL + SDL1_R + 0.75 LL + 0.75 WL(+Z)
636. REPEAT LOAD
637. 1 1.0 3 1.0 2 0.75 7 0.75
638. LOAD 312 DL + SDL1_R + 0.75 LL + 0.75 WL(-Z)
639. REPEAT LOAD
640. 1 1.0 3 1.0 2 0.75 8 0.75
641. LOAD 313 DL + SDL2_R + 0.75 LL + 0.75 WL(+X)
642. REPEAT LOAD
643. 1 1.0 4 1.0 2 0.75 5 0.75
644. LOAD 314 DL + SDL2_R + 0.75 LL + 0.75 WL(-X)
645. REPEAT LOAD
646. 1 1.0 4 1.0 2 0.75 6 0.75
647. LOAD 315 DL + SDL2_R + 0.75 LL + 0.75 WL(+Z)
648. REPEAT LOAD
649. 1 1.0 4 1.0 2 0.75 7 0.75
650. LOAD 316 DL + SDL2_R + 0.75 LL + 0.75 WL(-Z)
651. REPEAT LOAD
652. 1 1.0 4 1.0 2 0.75 8 0.75
653. LOAD 317 DL - SDL1_R + 0.75 LL + 0.75 WL(+X)
654. REPEAT LOAD
655. 1 1.0 3 -1.0 2 0.75 5 0.75
656. LOAD 318 DL - SDL1_R + 0.75 LL + 0.75 WL(-X)
657. REPEAT LOAD
658. 1 1.0 3 -1.0 2 0.75 6 0.75
659. LOAD 319 DL - SDL1_R + 0.75 LL + 0.75 WL(+Z)
660. REPEAT LOAD
661. 1 1.0 3 -1.0 2 0.75 7 0.75
662. LOAD 320 DL - SDL1_R + 0.75 LL + 0.75 WL(-Z)
663. REPEAT LOAD
664. 1 1.0 3 -1.0 2 0.75 8 0.75
665. LOAD 321 DL - SDL2_R + 0.75 LL + 0.75 WL(+X)
666. REPEAT LOAD
667. 1 1.0 4 -1.0 2 0.75 5 0.75
668. LOAD 322 DL - SDL2_R + 0.75 LL + 0.75 WL(-X)
669. REPEAT LOAD
670. 1 1.0 4 -1.0 2 0.75 6 0.75
671. LOAD 323 DL - SDL2_R + 0.75 LL + 0.75 WL(+Z)
672. REPEAT LOAD
673. 1 1.0 4 -1.0 2 0.75 7 0.75
674. LOAD 324 DL - SDL2_R + 0.75 LL + 0.75 WL(-Z)
675. REPEAT LOAD
676. 1 1.0 4 -1.0 2 0.75 8 0.75
677. ************************************************
678. LOAD 325 DL + SDL1_R + 0.75 LL + 0.525 EL(+X)
679. REPEAT LOAD
680. 1 1.0 3 1.0 2 0.75 11 0.525
681. LOAD 326 DL + SDL1_R + 0.75 LL + 0.525 EL(-X)
682. REPEAT LOAD
683. 1 1.0 3 1.0 2 0.75 11 -0.525
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684. LOAD 327 DL + SDL1_R + 0.75 LL + 0.525 EL(+Z)
685. REPEAT LOAD
686. 1 1.0 3 1.0 2 0.75 12 0.525
687. LOAD 328 DL + SDL1_R + 0.75 LL + 0.525 EL(-Z)
688. REPEAT LOAD
689. 1 1.0 3 1.0 2 0.75 12 -0.525
690. LOAD 329 DL + SDL2_R + 0.75 LL + 0.525 EL(+X)
691. REPEAT LOAD
692. 1 1.0 4 1.0 2 0.75 11 0.525
693. LOAD 330 DL + SDL2_R + 0.75 LL + 0.525 EL(-X)
694. REPEAT LOAD
695. 1 1.0 4 1.0 2 0.75 11 -0.525
696. LOAD 331 DL + SDL2_R + 0.75 LL + 0.525 EL(+Z)
697. REPEAT LOAD
698. 1 1.0 4 1.0 2 0.75 12 0.525
699. LOAD 332 DL + SDL2_R + 0.75 LL + 0.525 EL(-Z)
700. REPEAT LOAD
701. 1 1.0 4 1.0 2 0.75 12 -0.525
702. LOAD 333 DL - SDL1_R + 0.75 LL + 0.525 EL(+X)
703. REPEAT LOAD
704. 1 1.0 3 -1.0 2 0.75 11 0.525
705. LOAD 334 DL - SDL1_R + 0.75 LL + 0.525 EL(-X)
706. REPEAT LOAD
707. 1 1.0 3 -1.0 2 0.75 11 -0.525
708. LOAD 335 DL - SDL1_R + 0.75 LL + 0.525 EL(+Z)
709. REPEAT LOAD
710. 1 1.0 3 -1.0 2 0.75 12 0.525
711. LOAD 336 DL - SDL1_R + 0.75 LL + 0.525 EL(-Z)
712. REPEAT LOAD
713. 1 1.0 3 -1.0 2 0.75 12 -0.525
714. LOAD 337 DL - SDL2_R + 0.75 LL + 0.525 EL(+X)
715. REPEAT LOAD
716. 1 1.0 4 -1.0 2 0.75 11 0.525
717. LOAD 338 DL - SDL2_R + 0.75 LL + 0.525 EL(-X)
718. REPEAT LOAD
719. 1 1.0 4 -1.0 2 0.75 11 -0.525
720. LOAD 339 DL - SDL2_R + 0.75 LL + 0.525 EL(+Z)
721. REPEAT LOAD
722. 1 1.0 4 -1.0 2 0.75 12 0.525
723. LOAD 340 DL - SDL2_R + 0.75 LL + 0.525 EL(-Z)
724. REPEAT LOAD
725. 1 1.0 4 -1.0 2 0.75 12 -0.525
726. ************************************************
727. LOAD 341 0.6 (DL + SDL1_R) + WL(+X)
728. REPEAT LOAD
729. 1 0.6 3 0.6 5 1.0
730. LOAD 342 0.6 (DL + SDL1_R) + WL(-X)
731. REPEAT LOAD
732. 1 0.6 3 0.6 6 1.0
733. LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)
734. REPEAT LOAD
735. 1 0.6 3 0.6 7 1.0
736. LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)
737. REPEAT LOAD
738. 1 0.6 3 0.6 8 1.0
739. LOAD 345 0.6 (DL + SDL2_R) + WL(+X)
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740. REPEAT LOAD
741. 1 0.6 4 0.6 5 1.0
742. LOAD 346 0.6 (DL + SDL2_R) + WL(-X)
743. REPEAT LOAD
744. 1 0.6 4 0.6 6 1.0
745. LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)
746. REPEAT LOAD
747. 1 0.6 4 0.6 7 1.0
748. LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)
749. REPEAT LOAD
750. 1 0.6 4 0.6 8 1.0
751. LOAD 349 0.6 (DL - SDL1_R) + WL(+X)
752. REPEAT LOAD
753. 1 0.6 3 -0.6 5 1.0
754. LOAD 350 0.6 (DL - SDL1_R) + WL(-X)
755. REPEAT LOAD
756. 1 0.6 3 -0.6 6 1.0
757. LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)
758. REPEAT LOAD
759. 1 0.6 3 -0.6 7 1.0
760. LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)
761. REPEAT LOAD
762. 1 0.6 3 -0.6 8 1.0
763. LOAD 353 0.6 (DL - SDL2_R) + WL(+X)
764. REPEAT LOAD
765. 1 0.6 4 -0.6 5 1.0
766. LOAD 354 0.6 (DL - SDL2_R) + WL(-X)
767. REPEAT LOAD
768. 1 0.6 4 -0.6 6 1.0
769. LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)
770. REPEAT LOAD
771. 1 0.6 4 -0.6 7 1.0
772. LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)
773. REPEAT LOAD
774. 1 0.6 4 -0.6 8 1.0
775. **********
776. LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)
777. REPEAT LOAD
778. 1 0.6 3 0.6 11 0.7
779. LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)
780. REPEAT LOAD
781. 1 0.6 3 0.6 11 -0.7
782. LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
783. REPEAT LOAD
784. 1 0.6 3 0.6 12 0.7
785. LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
786. REPEAT LOAD
787. 1 0.6 3 0.6 12 -0.7
788. LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)
789. REPEAT LOAD
790. 1 0.6 4 0.6 11 0.7
791. LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)
792. REPEAT LOAD
793. 1 0.6 4 0.6 11 -0.7
794. LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
795. REPEAT LOAD

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796. 1 0.6 4 0.6 12 -0.7
797. LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
798. REPEAT LOAD
799. 1 0.6 4 0.6 12 -0.7
800. LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)
801. REPEAT LOAD
802. 1 0.6 3 -0.6 11 0.7
803. LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)
804. REPEAT LOAD
805. 1 0.6 3 -0.6 11 -0.7
806. LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
807. REPEAT LOAD
808. 1 0.6 3 -0.6 12 0.7
809. LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
810. REPEAT LOAD
811. 1 0.6 3 -0.6 12 -0.7
812. LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)
813. REPEAT LOAD
814. 1 0.6 4 -0.6 11 0.7
815. LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)
816. REPEAT LOAD
817. 1 0.6 4 -0.6 11 -0.7
818. LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
819. REPEAT LOAD
820. 1 0.6 4 -0.6 12 -0.7
821. LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
822. REPEAT LOAD
823. 1 0.6 4 -0.6 12 -0.7
824. ************************************************
825. *LOAD COMBINATION FOR CONCRETE DESIGN
826. ************************************************
827. LOAD 401 1.4 (DL+SDL1_R)
828. REPEAT LOAD
829. 1 1.4 3 1.4
830. LOAD 402 1.4 (DL+SDL2_R)
831. REPEAT LOAD
832. 1 1.4 4 1.4
833. LOAD 403 1.4 (DL-SDL1_R)
834. REPEAT LOAD
835. 1 1.4 3 -1.4
836. LOAD 404 1.4 (DL-SDL2_R)
837. REPEAT LOAD
838. 1 1.4 4 -1.4
839. ************************************************
840. LOAD 405 1.4 (DL + SDL1_R ) + 1.7 LL
841. REPEAT LOAD
842. 1 1.4 3 1.4 2 1.7
843. LOAD 406 1.4 (DL + SDL2_R ) + 1.7 LL
844. REPEAT LOAD
845. 1 1.4 4 1.4 2 1.7
846. LOAD 407 1.4 (DL - SDL1_R ) + 1.7 LL
847. REPEAT LOAD
848. 1 1.4 3 -1.4 2 1.7
849. LOAD 408 1.4 (DL - SDL2_R ) + 1.7 LL
850. REPEAT LOAD
851. 1 1.4 4 -1.4 2 1.7
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852. ************************************************
853. LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)
854. REPEAT LOAD
855. 1 1.2 3 1.2 2 0.5 5 1.6
856. LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)
857. REPEAT LOAD
858. 1 1.2 3 1.2 2 0.5 6 1.6
859. LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)
860. REPEAT LOAD
861. 1 1.2 3 1.2 2 0.5 7 1.6
862. LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)
863. REPEAT LOAD
864. 1 1.2 3 1.2 2 0.5 8 1.6
865. LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)
866. REPEAT LOAD
867. 1 1.2 4 1.2 2 0.5 5 1.6
868. LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)
869. REPEAT LOAD
870. 1 1.2 4 1.2 2 0.5 6 1.6
871. LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)
872. REPEAT LOAD
873. 1 1.2 4 1.2 2 0.5 7 1.6
874. LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)
875. REPEAT LOAD
876. 1 1.2 4 1.2 2 0.5 8 1.6
877. LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)
878. REPEAT LOAD
879. 1 1.2 3 -1.2 2 0.5 5 1.6
880. LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)
881. REPEAT LOAD
882. 1 1.2 3 -1.2 2 0.5 6 1.6
883. LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)
884. REPEAT LOAD
885. 1 1.2 3 -1.2 2 0.5 7 1.6
886. LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)
887. REPEAT LOAD
888. 1 1.2 3 -1.2 2 0.5 8 1.6
889. LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)
890. REPEAT LOAD
891. 1 1.2 4 -1.2 2 0.5 5 1.6
892. LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)
893. REPEAT LOAD
894. 1 1.2 4 -1.2 2 0.5 6 1.6
895. LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)
896. REPEAT LOAD
897. 1 1.2 4 -1.2 2 0.5 7 1.6
898. LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)
899. REPEAT LOAD
900. 1 1.2 4 -1.2 2 0.5 8 1.6
901. ************************************************
902. LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)
903. REPEAT LOAD
904. 1 1.2 3 1.2 2 0.5 11 1.0
905. LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
906. REPEAT LOAD
907. 1 1.2 3 1.2 2 0.5 11 -1.0
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908. LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)
909. REPEAT LOAD
910. 1 1.2 3 1.2 2 0.5 12 1.0
911. LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
912. REPEAT LOAD
913. 1 1.2 3 1.2 2 0.5 12 -1.0
914. LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)
915. REPEAT LOAD
916. 1 1.2 4 1.2 2 0.5 11 1.0
917. LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
918. REPEAT LOAD
919. 1 1.2 4 1.2 2 0.5 11 -1.0
920. LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)
921. REPEAT LOAD
922. 1 1.2 4 1.2 2 0.5 12 1.0
923. LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
924. REPEAT LOAD
925. 1 1.2 4 1.2 2 0.5 12 -1.0
926. LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)
927. REPEAT LOAD
928. 1 1.2 3 -1.2 2 0.5 11 1.0
929. LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
930. REPEAT LOAD
931. 1 1.2 3 -1.2 2 0.5 11 -1.0
932. LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)
933. REPEAT LOAD
934. 1 1.2 3 -1.2 2 0.5 12 1.0
935. LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
936. REPEAT LOAD
937. 1 1.2 3 -1.2 2 0.5 12 -1.0
938. LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)
939. REPEAT LOAD
940. 1 1.2 4 -1.2 2 0.5 11 1.0
941. LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
942. REPEAT LOAD
943. 1 1.2 4 -1.2 2 0.5 11 -1.0
944. LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)
945. REPEAT LOAD
946. 1 1.2 4 -1.2 2 0.5 12 1.0
947. LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
948. REPEAT LOAD
949. 1 1.2 4 -1.2 2 0.5 12 -1.0
950. ************************************************
951. LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)
952. REPEAT LOAD
953. 1 0.9 3 0.9 5 1.6
954. LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)
955. REPEAT LOAD
956. 1 0.9 3 0.9 6 1.6
957. LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)
958. REPEAT LOAD
959. 1 0.9 3 0.9 7 1.6
960. LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)
961. REPEAT LOAD
962. 1 0.9 3 0.9 8 1.6
963. LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)
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 964. REPEAT LOAD
 965. 1 0.9 4 0.9 5 1.6
 966. LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)
 967. REPEAT LOAD
 968. 1 0.9 4 0.9 6 1.6
 969. LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)
 970. REPEAT LOAD
 971. 1 0.9 4 0.9 7 1.6
 972. LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)
 973. REPEAT LOAD
 974. 1 0.9 4 0.9 8 1.6
 975. LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)
 976. REPEAT LOAD
 977. 1 0.9 3 -0.9 5 1.6
 978. LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)
 979. REPEAT LOAD
 980. 1 0.9 3 -0.9 6 1.6
 981. LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)
 982. REPEAT LOAD
 983. 1 0.9 3 -0.9 7 1.6
 984. LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)
 985. REPEAT LOAD
 986. 1 0.9 3 -0.9 8 1.6
 987. LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)
 988. REPEAT LOAD
 989. 1 0.9 4 -0.9 5 1.6
 990. LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)
 991. REPEAT LOAD
 992. 1 0.9 4 -0.9 6 1.6
 993. LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)
 994. REPEAT LOAD
 995. 1 0.9 4 -0.9 7 1.6
 996. LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)
 997. REPEAT LOAD
 998. 1 0.9 4 -0.9 8 1.6
 999. ************************************************
1000. LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)
1001. REPEAT LOAD
1002. 1 0.9 3 0.9 11 1.0
1003. LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)
1004. REPEAT LOAD
1005. 1 0.9 3 0.9 11 -1.0
1006. LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)
1007. REPEAT LOAD
1008. 1 0.9 3 0.9 12 1.0
1009. LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)
1010. REPEAT LOAD
1011. 1 0.9 3 0.9 12 -1.0
1012. LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)
1013. REPEAT LOAD
1014. 1 0.9 4 0.9 11 1.0
1015. LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)
1016. REPEAT LOAD
1017. 1 0.9 4 0.9 11 -1.0
1018. LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)
1019. REPEAT LOAD
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1020. 1 0.9 4 0.9 12 1.0
1021. LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)
1022. REPEAT LOAD
1023. 1 0.9 4 0.9 12 -1.0
1024. LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)
1025. REPEAT LOAD
1026. 1 0.9 3 -0.9 11 1.0
1027. LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)
1028. REPEAT LOAD
1029. 1 0.9 3 -0.9 11 -1.0
1030. LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)
1031. REPEAT LOAD
1032. 1 0.9 3 -0.9 12 1.0
1033. LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)
1034. REPEAT LOAD
1035. 1 0.9 3 -0.9 12 -1.0
1036. LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)
1037. REPEAT LOAD
1038. 1 0.9 4 -0.9 11 1.0
1039. LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)
1040. REPEAT LOAD
1041. 1 0.9 4 -0.9 11 -1.0
1042. LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)
1043. REPEAT LOAD
1044. 1 0.9 4 -0.9 12 1.0
1045. LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)
1046. REPEAT LOAD
1047. 1 0.9 4 -0.9 12 -1.0
1048. **********
1049. **********
1050. *LOAD COMBINATION 1001
1051. *1 1.0 2 0.25 4 1.0
1052. PERFORM ANALYSIS


         P R O B L E M   S T A T I S T I C S
         -----------------------------------

  NUMBER OF JOINTS        554  NUMBER OF MEMBERS     187
  NUMBER OF PLATES        434  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS      8

        Using 64-bit analysis engine.

        SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

 TOTAL      PRIMARY LOAD CASES =   155, TOTAL DEGREES OF FREEDOM =    3276
 TOTAL LOAD COMBINATION  CASES =     0  SO FAR.
```

***NOTE: MASSES DEFINED UNDER LOAD# 22 WILL FORM
THE FINAL MASS MATRIX FOR DYNAMIC ANALYSIS.

**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS & CHANGE.

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG X  :                        *
*     CT =  0.016 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.320 T =  0.237 Tuser = 0.237                 *
*  TIME PERIOD USED (T) =  0.237                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.022 X      4898.00   *
*                       =       109.95  KN                 *
************************************************************
```

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************************************************************
*  IBC 2003 SEISMIC LOAD ALONG Z  :                        *
*     CT =  0.016 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.320 T =  0.237 Tuser = 0.237                 *
*  TIME PERIOD USED (T) =  0.237                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.022 X      4898.00   *
*                       =       109.95  KN                 *
************************************************************
```

```
EIGEN METHOD   : SUBSPACE
-------------------------
NUMBER OF MODES REQUESTED              =     6
NUMBER OF EXISTING MASSES IN THE MODEL =  1638
NUMBER OF MODES THAT WILL BE USED      =     6

  *** EIGENSOLUTION: ADVANCED METHOD ***
```

CALCULATED FREQUENCIES FOR LOAD CASE 22

MODE	FREQUENCY(CYCLES/SEC)	PERIOD(SEC)
1	3.427	0.29178
2	4.435	0.22546
3	5.414	0.18472
4	6.645	0.15049
5	7.395	0.13523
6	7.622	0.13120

MODE	MODAL WEIGHT (MODAL MASS TIMES g) IN KN X	Y	Z	GENERALIZED WEIGHT
1	2.006354E-03	7.669999E-04	2.071866E+03	1.065363E+03
2	1.976678E+00	2.433861E-01	1.477572E+03	8.148263E+02
3	4.104152E+03	1.480510E+01	8.421746E-01	2.746966E+03
4	7.984210E-02	6.887311E-01	1.808979E+02	9.774974E+02
5	1.991101E+00	3.974950E+00	2.777843E+01	1.590372E+03
6	4.091005E+01	1.101309E+03	5.464934E-02	7.299382E+02

MASS PARTICIPATION FACTORS

MASS PARTICIPATION FACTORS IN PERCENT

MODE	X	Y	Z	SUMM-X	SUMM-Y	SUMM-Z
1	0.00	0.00	46.70	0.000	0.000	46.703
2	0.04	0.01	33.31	0.044	0.006	80.009
3	91.40	0.34	0.02	91.445	0.343	80.028
4	0.00	0.02	4.08	91.446	0.359	84.106
5	0.04	0.09	0.63	91.491	0.450	84.732
6	0.91	25.13	0.00	92.402	25.581	84.733

1053. LOAD LIST 401 TO 472
1054. START CONCRETE DESIGN

```
CONCRETE DESIGN
  1055. CODE ACI
  1056. FC 35000 MEMB 1 2 4 5 7 9 TO 11 13 14 16 18 TO 20 22 23 25 27 TO 29 31 32 -
  1057. 34 36 TO 42 49 TO 145 306 TO 337 410 411 493 502 504 TO 506 509 511 513 544 -
  1058. 561 564 567 570 572 576 581 586 591 593 595
  1059. FYMAIN 420000 MEMB 1 2 4 5 7 9 TO 11 13 14 16 18 TO 20 22 23 25 27 TO 29 31 -
  1060. 32 34 36 TO 42 49 TO 145 306 TO 337 410 411 493 502 504 TO 506 509 511 513 -
  1061. 544 561 564 567 570 572 576 581 586 591 593 595
  1062. DESIGN COLUMN 1 2 4 5 10 11 13 14 19 20 22 23 28 29 31 32 98 TO 145 -
  1063. 322 TO 337 410 411 493 502 564 567 570 576 581 586
```

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 1
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.63	1.25	1.88	2.50
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 2
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.06	0.12	0.19	0.25
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 4
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.63	1.25	1.88	2.50
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 5
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.06	0.12	0.19	0.25
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 10
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.63	1.25	1.88	2.50
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 11
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.06	0.12	0.19	0.25
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 13
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.63	1.25	1.88	2.50
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 14
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.06	0.12	0.19	0.25
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 19
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.63	1.25	1.88	2.50
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 20
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.07	0.14	0.21	0.27
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 22
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.63	1.25	1.88	2.50
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 23
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.07	0.14	0.21	0.27
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 28
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.63	1.25	1.88	2.50
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 29
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.07	0.14	0.21	0.27
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 31
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.63	1.25	1.88	2.50
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 32
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.07	0.14	0.21	0.27
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 98
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.05	0.10	0.15	0.20
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 99
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.05	0.10	0.15	0.20
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 100
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.37	0.75	1.12	1.50
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 101
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.37	0.75	1.12	1.50
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 102
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.09	0.17	0.26	0.35
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 103
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.09	0.17	0.26	0.35
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 104
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.07	0.15	0.22	0.30
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 105
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.21	0.41	0.62	0.83
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 106					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.21	0.41	0.62	0.83
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 107					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 108					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.14	0.28	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 109
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 110
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.16	0.33	0.49	0.65
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 111
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.07	0.15	0.22	0.30
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 112
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.21	0.41	0.62	0.83
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 113
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.21	0.41	0.62	0.83
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 114
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 115
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.28	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 116
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 117
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.16	0.33	0.49	0.65
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 118
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.07	0.15	0.22	0.30
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 119
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.20	0.41	0.61	0.81
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 120
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.20	0.41	0.61	0.81
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 121
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 122
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.28	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 123
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 124
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.16	0.33	0.49	0.65
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 125
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.07	0.15	0.22	0.30
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 126
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.20	0.41	0.61	0.81
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 127
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.20	0.41	0.61	0.81
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 128
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 129
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.28	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 130
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 131
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.16	0.33	0.49	0.65
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 132
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.07	0.15	0.22	0.30
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 133					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.20	0.41	0.61	0.81
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 134					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.20	0.41	0.61	0.81
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 135					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 136
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.28	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 137
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 138
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.16	0.33	0.49	0.65
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 139
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.07	0.15	0.22	0.30
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 140
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.20	0.41	0.61	0.81
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 141
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.20	0.41	0.61	0.81
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 142
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 143
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.28	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 144
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 145
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.16	0.33	0.49	0.65
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 322
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.15	0.30	0.45	0.60
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 323
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.15	0.30	0.45	0.60
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 324
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.07	0.15	0.22	0.30
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 325
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.21	0.41	0.62	0.83
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 326
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.21	0.41	0.62	0.83
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 327
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 328
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.28	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 329
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0

Units: KNS , METER (Unless Noted Otherwise)

Member : 330					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.14	0.28	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 331					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.07	0.15	0.22	0.30
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 332					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.21	0.41	0.62	0.83
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 333					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.21	0.41	0.62	0.83
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 334					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 335					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.14	0.28	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 336
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.27	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 337
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.14	0.28	0.41	0.55
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 410
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.25	0.50	0.75	1.00
As(Longitudinal)	0.00	0.00	0.00	0.00	0.01
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 411
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.25	0.50	0.75	1.00
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 493
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.33	0.65	0.98	1.30
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 502					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.33	0.65	0.98	1.30
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 564					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.09	0.17	0.26	0.35
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 567					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.10	0.20	0.30	0.40
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 570
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.10	0.20	0.30	0.40
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 576
Type : Column Shape : Rectangular

AREA OF STEEL REQUIRED

Section -	1	2	3	4	5
Location	0.00	0.09	0.17	0.26	0.35
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

STAAD.PRO CONCRETE DESIGN - (ACI-318-14) v2.0
****

Units: KNS , METER (Unless Noted Otherwise)

Member : 581					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.10	0.20	0.30	0.40
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

Member : 586					
Type : Column	Shape : Rectangular				
AREA OF STEEL REQUIRED					
Section -	1	2	3	4	5
Location	0.00	0.10	0.20	0.30	0.40
As(Longitudinal)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Y)	0.00	0.00	0.00	0.00	0.00
As/sv(Trans Z)	0.00	0.00	0.00	0.00	0.00

1064. MAXMAIN 32 ALL
1065. END CONCRETE DESIGN
1066. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 28,2022 TIME= 18:36:58 ****

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     ************************************************************
     *   For technical assistance on STAAD.Pro, please visit    *
     *   http://www.bentley.com/en/support/                     *
     *                                                          *
     *   Details about additional assistance from               *
     *   Bentley and Partners can be found at program menu      *
     *   Help->Technical Support                                *
     *                                                          *
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     *                http://www.bentley.com                    *
     ************************************************************
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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 4 STAADPRO INPUT FILE OF FOUNDATION F1

Job Title:

Client:

Engineer:

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
5 0 -1.7 0; 6 0 -1.7 5.4; 101 0 -2 0; 102 0 -2 5.4; 114 -1.25 -2 -1;
117 -1.25 -2 -0.3; 118 -1.25 -2 0; 119 -0.4 -2 -1; 121 -0.4 -2 -0.3;
122 -0.4 -2 0; 123 0 -2 -1; 124 0 -2 -0.3; 125 0.4 -2 -1; 127 0.4 -2 -0.3;
128 0.4 -2 0; 129 1.25 -2 -1; 131 1.25 -2 -0.3; 132 1.25 -2 0;
136 -1.25 -2 6.1; 137 -1.25 -2 5.4; 139 -1.25 -2 13; 141 -0.4 -2 6.1;
142 -0.4 -2 5.4; 143 -0.4 -2 13; 145 0 -2 6.1; 146 0 -2 13; 147 0.4 -2 6.1;
148 0.4 -2 5.4; 149 0.4 -2 13; 151 1.25 -2 6.1; 152 1.25 -2 5.4;
153 1.25 -2 13; 158 -1.25 -2 0.3; 160 -1.25 -2 1; 162 -1.25 -2 1.43;
164 -1.25 -2 1.76; 166 -1.25 -2 2.18; 168 -1.25 -2 3.94; 170 -1.25 -2 4.4;
172 -1.25 -2 4.73; 174 -1.25 -2 5.1; 175 -0.4 -2 0.3; 176 -0.4 -2 1;
177 -0.4 -2 1.43; 178 -0.4 -2 1.76; 179 -0.4 -2 2.18; 180 -0.4 -2 3.94;
181 -0.4 -2 4.4; 182 -0.4 -2 4.73; 183 -0.4 -2 5.1; 184 0 -2 0.3; 185 0 -2 1;
186 0 -2 1.43; 187 0 -2 1.76; 188 0 -2 2.18; 189 0 -2 3.94; 190 0 -2 4.4;
191 0 -2 4.73; 192 0 -2 5.1; 193 0.4 -2 0.3; 194 0.4 -2 1; 195 0.4 -2 1.43;
196 0.4 -2 1.76; 197 0.4 -2 2.18; 198 0.4 -2 3.94; 199 0.4 -2 4.4;
200 0.4 -2 4.73; 201 0.4 -2 5.1; 202 1.25 -2 0.3; 203 1.25 -2 1;
204 1.25 -2 1.43; 205 1.25 -2 1.76; 206 1.25 -2 2.18; 207 1.25 -2 3.94;
208 1.25 -2 4.4; 209 1.25 -2 4.73; 210 1.25 -2 5.1; 221 -1.25 -2 -1.5;
222 -0.4 -2 -1.5; 223 0 -2 -1.5; 224 0.4 -2 -1.5; 225 1.25 -2 -1.5;
235 -1.25 -2 6.4; 237 -1.25 -2 7.1; 239 -1.25 -2 7.6; 241 -1.25 -2 8.1;
243 -1.25 -2 8.6; 245 -1.25 -2 9.1; 247 -1.25 -2 9.6; 249 -1.25 -2 10;
251 -1.25 -2 10.5; 253 -1.25 -2 11.2; 255 -1.25 -2 11.5; 257 -1.25 -2 11.8;
259 -1.25 -2 12.5; 260 -0.4 -2 6.4; 261 -0.4 -2 7.1; 262 -0.4 -2 7.6;
263 -0.4 -2 8.1; 264 -0.4 -2 8.6; 265 -0.4 -2 9.1; 266 -0.4 -2 9.6;
267 -0.4 -2 10; 268 -0.4 -2 10.5; 269 -0.4 -2 11.2; 270 -0.4 -2 11.5;
271 -0.4 -2 11.8; 272 -0.4 -2 12.5; 273 0 -2 6.4; 274 0 -2 7.1; 275 0 -2 7.6;
276 0 -2 8.1; 277 0 -2 8.6; 278 0 -2 9.1; 279 0 -2 9.6; 280 0 -2 10;
281 0 -2 10.5; 282 0 -2 11.2; 283 0 -2 11.5; 284 0 -2 11.8; 285 0 -2 12.5;
286 0.4 -2 6.4; 287 0.4 -2 7.1; 288 0.4 -2 7.6; 289 0.4 -2 8.1; 290 0.4 -2 8.6;
291 0.4 -2 9.1; 292 0.4 -2 9.6; 293 0.4 -2 10; 294 0.4 -2 10.5;
295 0.4 -2 11.2; 296 0.4 -2 11.5; 297 0.4 -2 11.8; 298 0.4 -2 12.5;
299 1.25 -2 6.4; 300 1.25 -2 7.1; 301 1.25 -2 7.6; 302 1.25 -2 8.1;
303 1.25 -2 8.6; 304 1.25 -2 9.1; 305 1.25 -2 9.6; 306 1.25 -2 10;
307 1.25 -2 10.5; 308 1.25 -2 11.2; 309 1.25 -2 11.5; 310 1.25 -2 11.8;
311 1.25 -2 12.5; 326 -1.25 -2 2.58; 328 -1.25 -2 2.98; 330 -1.25 -2 3.48;
331 -0.4 -2 2.58; 332 -0.4 -2 2.98; 333 -0.4 -2 3.48; 334 0 -2 2.58;
335 0 -2 2.98; 336 0 -2 3.48; 337 0.4 -2 2.58; 338 0.4 -2 2.98;
339 0.4 -2 3.48; 340 1.25 -2 2.58; 341 1.25 -2 2.98; 342 1.25 -2 3.48;
346 0 -1.7 6.1; 347 0 -1.7 11.5; 348 -0.9 -2 -0.3; 349 -0.9 -2 -1;
350 -0.9 -2 0; 351 0.9 -2 -0.3; 352 0.9 -2 -1; 353 0.9 -2 0; 354 -0.9 -2 6.1;
355 -0.9 -2 5.4; 356 0.9 -2 6.1; 357 0.9 -2 5.4; 358 -0.9 -2 0.3;
359 -0.9 -2 1; 360 -0.9 -2 1.43; 361 -0.9 -2 1.76; 362 -0.9 -2 2.18;
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363 -0.9 -2 4.4; 364 -0.9 -2 3.94; 365 -0.9 -2 4.73; 366 -0.9 -2 5.1;
367 0.9 -2 0.3; 368 0.9 -2 1; 369 0.9 -2 1.43; 370 0.9 -2 1.76;
371 0.9 -2 2.18; 372 0.9 -2 4.4; 373 0.9 -2 3.94; 374 0.9 -2 4.73;
375 0.9 -2 5.1; 376 -0.9 -2 -1.5; 377 0.9 -2 -1.5; 378 -0.9 -2 6.4;
379 -0.9 -2 7.1; 380 -0.9 -2 7.6; 381 -0.9 -2 8.1; 382 -0.9 -2 8.6;
383 -0.9 -2 9.1; 384 -0.9 -2 9.6; 385 -0.9 -2 10; 386 -0.9 -2 10.5;
387 -0.9 -2 11.2; 388 -0.9 -2 11.5; 389 -0.9 -2 11.8; 390 -0.9 -2 12.5;
391 -0.9 -2 13; 392 0.9 -2 6.4; 393 0.9 -2 7.1; 394 0.9 -2 7.6; 395 0.9 -2 8.1;
396 0.9 -2 8.6; 397 0.9 -2 9.1; 398 0.9 -2 9.6; 399 0.9 -2 10; 400 0.9 -2 10.5;
401 0.9 -2 11.2; 402 0.9 -2 11.5; 403 0.9 -2 11.8; 404 0.9 -2 12.5;
405 0.9 -2 13; 406 -0.9 -2 2.58; 407 -0.9 -2 2.98; 408 -0.9 -2 3.48;
409 0.9 -2 2.58; 410 0.9 -2 2.98; 411 0.9 -2 3.48;
MEMBER INCIDENCES
1 101 5; 2 102 6; 240 145 346; 241 283 347;
ELEMENT INCIDENCES SHELL
19 119 121 124 123; 20 121 122 101 124; 22 123 124 127 125; 23 124 101 128 127;
36 142 141 145 102; 39 102 145 147 148; 68 122 175 184 101; 69 175 176 185 184;
70 176 177 186 185; 71 177 178 187 186; 72 178 179 188 187; 74 180 181 190 189;
75 181 182 191 190; 76 182 183 192 191; 77 183 142 102 192; 78 101 184 193 128;
79 184 185 194 193; 80 185 186 195 194; 81 186 187 196 195; 82 187 188 197 196;
84 189 190 199 198; 85 190 191 200 199; 86 191 192 201 200; 87 192 102 148 201;
113 222 119 123 223; 115 223 123 125 224; 160 141 260 273 145;
161 260 261 274 273; 162 261 262 275 274; 163 262 263 276 275;
164 263 264 277 276; 165 264 265 278 277; 166 265 266 279 278;
167 266 267 280 279; 168 267 268 281 280; 169 268 269 282 281;
170 269 270 283 282; 171 270 271 284 283; 172 271 272 285 284;
173 272 143 146 285; 174 145 273 286 147; 175 273 274 287 286;
176 274 275 288 287; 177 275 276 289 288; 178 276 277 290 289;
179 277 278 291 290; 180 278 279 292 291; 181 279 280 293 292;
182 280 281 294 293; 183 281 282 295 294; 184 282 283 296 295;
185 283 284 297 296; 186 284 285 298 297; 187 285 146 149 298;
224 179 331 334 188; 225 331 332 335 334; 226 332 333 336 335;
227 333 180 189 336; 228 188 334 337 197; 229 334 335 338 337;
230 335 336 339 338; 231 336 189 198 339; 242 114 117 348 349;
243 349 348 121 119; 244 117 118 350 348; 245 348 350 122 121;
246 125 127 351 352; 247 352 351 131 129; 248 127 128 353 351;
249 351 353 132 131; 250 137 136 354 355; 251 355 354 141 142;
252 148 147 356 357; 253 357 356 151 152; 254 118 158 358 350;
255 350 358 175 122; 256 158 160 359 358; 257 358 359 176 175;
258 160 162 360 359; 259 359 360 177 176; 260 162 164 361 360;
261 360 361 178 177; 262 164 166 362 361; 263 361 362 179 178;
264 168 170 363 364; 265 364 363 181 180; 266 170 172 365 363;
267 363 365 182 181; 268 172 174 366 365; 269 365 366 183 182;
270 174 137 355 366; 271 366 355 142 183; 272 128 193 367 353;
273 353 367 202 132; 274 193 194 368 367; 275 367 368 203 202;
276 194 195 369 368; 277 368 369 204 203; 278 195 196 370 369;
279 369 370 205 204; 280 196 197 371 370; 281 370 371 206 205;
282 198 199 372 373; 283 373 372 208 207; 284 199 200 374 372;
285 372 374 209 208; 286 200 201 375 374; 287 374 375 210 209;

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288 201 148 357 375; 289 375 357 152 210; 290 221 114 349 376;
291 376 349 119 222; 292 224 125 352 377; 293 377 352 129 225;
294 136 235 378 354; 295 354 378 260 141; 296 235 237 379 378;
297 378 379 261 260; 298 237 239 380 379; 299 379 380 262 261;
300 239 241 381 380; 301 380 381 263 262; 302 241 243 382 381;
303 381 382 264 263; 304 243 245 383 382; 305 382 383 265 264;
306 245 247 384 383; 307 383 384 266 265; 308 247 249 385 384;
309 384 385 267 266; 310 249 251 386 385; 311 385 386 268 267;
312 251 253 387 386; 313 386 387 269 268; 314 253 255 388 387;
315 387 388 270 269; 316 255 257 389 388; 317 388 389 271 270;
318 257 259 390 389; 319 389 390 272 271; 320 259 139 391 390;
321 390 391 143 272; 322 147 286 392 356; 323 356 392 299 151;
324 286 287 393 392; 325 392 393 300 299; 326 287 288 394 393;
327 393 394 301 300; 328 288 289 395 394; 329 394 395 302 301;
330 289 290 396 395; 331 395 396 303 302; 332 290 291 397 396;
333 396 397 304 303; 334 291 292 398 397; 335 397 398 305 304;
336 292 293 399 398; 337 398 399 306 305; 338 293 294 400 399;
339 399 400 307 306; 340 294 295 401 400; 341 400 401 308 307;
342 295 296 402 401; 343 401 402 309 308; 344 296 297 403 402;
345 402 403 310 309; 346 297 298 404 403; 347 403 404 311 310;
348 298 149 405 404; 349 404 405 153 311; 350 166 326 406 362;
351 362 406 331 179; 352 326 328 407 406; 353 406 407 332 331;
354 328 330 408 407; 355 407 408 333 332; 356 330 168 364 408;
357 408 364 180 333; 358 197 337 409 371; 359 371 409 340 206;
360 337 338 410 409; 361 409 410 341 340; 362 338 339 411 410;
363 410 411 342 341; 364 339 198 373 411; 365 411 373 207 342;
START GROUP DEFINITION
ELEMENT
_MX 19 22 69 TO 72 74 TO 76 79 TO 82 84 TO 86 113 115 161 TO 169 172 173 175 -
176 TO 183 186 187 224 TO 231 242 243 246 247 256 TO 269 274 TO 287 -
290 TO 293 296 TO 313 318 TO 321 324 TO 341 346 TO 365
_MY 242 TO 365
_SHEARX 70 TO 72 74 80 TO 82 84 113 115 162 TO 168 173 176 TO 182 187 224 -
225 TO 231 258 TO 265 276 TO 283 290 TO 293 298 TO 311 320 321 326 TO 339 -
348 TO 365
_SHEARY 242 244 247 249 250 253 254 256 258 260 262 264 266 268 270 273 275 -
277 279 281 283 285 287 289 290 293 294 296 298 300 302 304 306 308 310 312 -
314 316 318 320 323 325 327 329 331 333 335 337 339 341 343 345 347 349 350 -
352 354 356 359 361 363 365
END GROUP DEFINITION
ELEMENT PROPERTY
19 20 22 23 36 39 68 TO 72 74 TO 82 84 TO 87 113 115 160 TO 187 224 TO 231 -
242 TO 365 THICKNESS 0.7
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+07
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-05
```

```
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 240 241 PRIS YD 0.8 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
101 102 114 117 TO 119 121 TO 125 127 TO 129 131 132 136 137 139 141 TO 143 -
145 TO 149 151 TO 153 158 160 162 164 166 168 170 172 174 TO 210 221 TO 225 -
235 237 239 241 243 245 247 249 251 253 255 257 259 TO 311 326 328 -
330 TO 342 348 TO 411 ELASTIC MAT DIRECT Y SUBGRADE 20000 PRINT
LOAD 1 DEAD  LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
19 20 22 23 36 39 68 TO 72 74 TO 82 84 TO 87 113 115 160 TO 187 224 TO 231 -
242 TO 365 PR GY -28
JOINT LOAD
5   FX  -19.2   FY  -1061.2 FZ  -13.1
5   MX  -11.1   MY  0.1 MZ  18.9
6   FX  -20.8   FY  -998.8  FZ  18.5
6   MX  18.5    MY  -0.7    MZ  22.9
346 FX  -19.2   FY  -1061.2 FZ  -13.1
346 MX  -11.1   MY  0.1 MZ  18.9
347 FX  -20.8   FY  -998.8  FZ  18.5
347 MX  18.5    MY  -0.7    MZ  22.9
LOAD 2 LIVE LOAD
JOINT LOAD
5   FX  -2.9    FY  -113.7  FZ  -1.2
5   MX  -0.8    MY  0.2 MZ  2.7
6   FX  -3.5    FY  -117.4  FZ  2.2
6   MX  2.3 MY  -0.4    MZ  3.7
346 FX  -2.9    FY  -113.7  FZ  -1.2
346 MX  -0.8    MY  0.2 MZ  2.7
347 FX  -3.5    FY  -117.4  FZ  2.2
347 MX  2.3 MY  -0.4    MZ  3.7
LOAD 3 SDL1 CRUSHER DYNAMIC LOAD
JOINT LOAD
5   FX  -8.1    FY  76.8    FZ  93.0
5   MX  135.9   MY  7.0 MZ  11.5
6   FX  1.5 FY  -330.1  FZ  96.5
6   MX  139.0   MY  6.7 MZ  -4.7
346 FX  -8.1    FY  76.8    FZ  93.0
346 MX  135.9   MY  7.0 MZ  11.5
347 FX  1.5 FY  -330.1  FZ  96.5
347 MX  139.0   MY  6.7 MZ  -4.7
LOAD 4 SDL2 CRUSHER DYNAMIC LOAD
JOINT LOAD
```

```
5   FX  58.4    FY  23.6    FZ  -0.5
5   MX  -0.5    MY  0.6 MZ  -93.9
6   FX  49.6    FY  25.9    FZ  0.2
6   MX  0.1 MY  -1.7    MZ  -79.8
346 FX  58.4    FY  23.6    FZ  -0.5
346 MX  -0.5    MY  0.6 MZ  -93.9
347 FX  49.6    FY  25.9    FZ  0.2
347 MX  0.1 MY  -1.7    MZ  -79.8
LOAD 5 WIND LOAD (+X)
JOINT LOAD
5   FX  20  FY  32.4    FZ  0.2
5   MX  0.2 MY  0.0 MZ  -30.7
6   FX  19.4    FY  31.1    FZ  -0.2
6   MX  -0.2    MY  0.0 MZ  -29.8
346 FX  20  FY  32.4    FZ  0.2
346 MX  0.2 MY  0.0 MZ  -30.7
347 FX  19.4    FY  31.1    FZ  -0.2
347 MX  -0.2    MY  0.0 MZ  -29.8
LOAD 6 WIND LOAD (-X)
JOINT LOAD
5   FX  -19.9   FY  -32.2   FZ  -0.2
5   MX  -0.2    MY  0.0 MZ  30.5
6   FX  -19.3   FY  -30.9   FZ  0.2
6   MX  0.2 MY  0.0 MZ  29.5
346 FX  -19.9   FY  -32.2   FZ  -0.2
346 MX  -0.2    MY  0.0 MZ  30.5
347 FX  -19.3   FY  -30.9   FZ  0.2
347 MX  0.2 MY  0.0 MZ  29.5
LOAD 7 WIND LOAD (+Z)
JOINT LOAD
5   FX  0.5 FY  56.6    FZ  30.8
5   MX  44.8    MY  0.1 MZ  -0.8
6   FX  -0.6    FY  -56.2   FZ  30.9
6   MX  44.8    MY  0.9 MZ  0.8
346 FX  0.5 FY  56.6    FZ  30.8
346 MX  44.8    MY  0.1 MZ  -0.8
347 FX  -0.6    FY  -56.2   FZ  30.9
347 MX  44.8    MY  0.9 MZ  0.8
LOAD 8 WIND LOAD (-Z)
JOINT LOAD
5   FX  -0.5    FY  -62.7   FZ  -35.7
5   MX  -51.8   MY  -1.1    MZ  0.8
6   FX  0.5 FY  63.2    FZ  -35.6
6   MX  -51.7   MY  -0.3    MZ  -0.8
346 FX  -0.5    FY  -62.7   FZ  -35.7
346 MX  -51.8   MY  -1.1    MZ  0.8
347 FX  0.5 FY  63.2    FZ  -35.6
347 MX  -51.7   MY  -0.3    MZ  -0.8
LOAD 11 SEISMIC LOAD (+X)
```

```
JOINT LOAD
5   FX  20.6    FY  48.7    FZ  0.4
5   MX  0.4 MY  -0.1    MZ  -32.4
6   FX  19.1    FY  45  FZ  -0.4
6   MX  -0.4    MY  -0.1    MZ  -30.1
346 FX  20.6    FY  48.7    FZ  0.4
346 MX  0.4 MY  -0.1    MZ  -32.4
347 FX  19.1    FY  45  FZ  -0.4
347 MX  -0.4    MY  0.0 MZ  -30.1
LOAD 12 SEISMIC LOAD (+Z)
JOINT LOAD
5   FX  0.4 FY  71.9    FZ  29.7
5   MX  43.1    MY  1.0 MZ  -0.5
6   FX  -0.4    FY  -71.9   FZ  29.7
6   MX  43.1    MY  1.0 MZ  0.5
346 FX  0.4 FY  71.9    FZ  29.7
346 MX  43.1    MY  1.0 MZ  -0.5
347 FX  -0.4    FY  -71.9   FZ  29.7
347 MX  43.1    MY  1.0 MZ  0.5
*****************LOAD COMBINATION FOR BASE PRESSURE CHECK
************************************************
LOAD 301 DL+SDL1_R
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2_R
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL-SDL1_R
REPEAT LOAD
1 1.0 3 -1.0
LOAD 304 DL-SDL2_R
REPEAT LOAD
1 1.0 4 -1.0
**********************************
LOAD 305 DL + SDL1_R + LL
REPEAT LOAD
1 1.0 3 1.0 2 1.0
LOAD 306 DL + SDL2_R + LL
REPEAT LOAD
1 1.0 4 1.0 2 1.0
LOAD 307 DL - SDL1_R + LL
REPEAT LOAD
1 1.0 3 -1.0 2 1.0
LOAD 308 DL - SDL2_R + LL
REPEAT LOAD
1 1.0 4 -1.0 2 1.0
*********************************************
LOAD 309 DL + SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
```

```
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 310 DL + SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 311 DL + SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 312 DL + SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 313 DL + SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 314 DL + SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 315 DL + SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 316 DL + SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 317 DL - SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 5 0.75
LOAD 318 DL - SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 6 0.75
LOAD 319 DL - SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 7 0.75
LOAD 320 DL - SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 8 0.75
LOAD 321 DL - SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 5 0.75
LOAD 322 DL - SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 6 0.75
LOAD 323 DL - SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 7 0.75
LOAD 324 DL - SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 8 0.75
**********************************************
LOAD 325 DL + SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
```

```
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 326 DL + SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 327 DL + SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 328 DL + SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 329 DL + SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 330 DL + SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 331 DL + SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 332 DL + SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 333 DL - SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 0.525
LOAD 334 DL - SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 -0.525
LOAD 335 DL - SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 0.525
LOAD 336 DL - SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 -0.525
LOAD 337 DL - SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 0.525
LOAD 338 DL - SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 -0.525
LOAD 339 DL - SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 0.525
LOAD 340 DL - SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 -0.525
*********************************************
LOAD 341 0.6 (DL + SDL1_R) + WL(+X)
REPEAT LOAD
```

```
1 0.6 3 0.6 5 1.0
LOAD 342 0.6 (DL + SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 345 0.6 (DL + SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 346 0.6 (DL + SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 349 0.6 (DL - SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 -0.6 5 1.0
LOAD 350 0.6 (DL - SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 -0.6 6 1.0
LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 7 1.0
LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 8 1.0
LOAD 353 0.6 (DL - SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 -0.6 5 1.0
LOAD 354 0.6 (DL - SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 -0.6 6 1.0
LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 7 1.0
LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 8 1.0
*********************************************
LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
```

```
1 0.6 3 0.6 11 0.7
LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 -0.6 11 0.7
LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 -0.6 11 -0.7
LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 12 0.7
LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 12 -0.7
LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 -0.6 11 0.7
LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 -0.6 11 -0.7
LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
************************************************
*Load Combination for Concrete design
************************************************
```

```
LOAD 401 1.4 (DL+SDL1_R)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2_R)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL-SDL1_R)
REPEAT LOAD
1 1.4 3 -1.4
LOAD 404 1.4 (DL-SDL2_R)
REPEAT LOAD
1 1.4 4 -1.4
************************************************
LOAD 405 1.4 (DL + SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 2 1.7
LOAD 407 1.4 (DL - SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 -1.4 2 1.7
LOAD 408 1.4 (DL - SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 -1.4 2 1.7
************************************************
LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
```

```
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 5 1.6
LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 6 1.6
LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 7 1.6
LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 8 1.6
LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 5 1.6
LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 6 1.6
LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 7 1.6
LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 8 1.6
************************************************
LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
```

```
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 1.0
LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 -1.0
LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 1.0
LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 -1.0
LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 1.0
LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 -1.0
LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 1.0
LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 -1.0
************************************************
LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
```

```
1 0.9 4 0.9 8 1.6
LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 -0.9 5 1.6
LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 -0.9 6 1.6
LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 7 1.6
LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 8 1.6
LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 -0.9 5 1.6
LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 -0.9 6 1.6
LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 7 1.6
LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 8 1.6
************************************************
LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
```

```
1 0.9 4 0.9 12 -1.0
LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 -0.9 11 1.0
LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 -0.9 11 -1.0
LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 12 1.0
LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 12 -1.0
LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 -0.9 11 1.0
LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 -0.9 11 -1.0
LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 12 1.0
LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 12 -1.0
**********
PERFORM ANALYSIS
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 5 STAADPRO INPUT FILE OF FOUNDATION F1A

Job Title:

Client:

Engineer:

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
5 0 -1.7 0; 6 0 -1.7 5.4; 101 0 -2 0; 102 0 -2 5.4; 114 -1.1 -2 -0.7;
117 -1.1 -2 -0.3; 118 -1.1 -2 0; 119 -0.5 -2 -0.7; 121 -0.5 -2 -0.3;
122 -0.5 -2 0; 123 0 -2 -0.7; 124 0 -2 -0.3; 125 0.5 -2 -0.7; 127 0.5 -2 -0.3;
128 0.5 -2 0; 129 1.1 -2 -0.7; 131 1.1 -2 -0.3; 132 1.1 -2 0; 136 -1.1 -2 6.1;
137 -1.1 -2 5.4; 139 -1.1 -2 13; 141 -0.5 -2 6.1; 142 -0.5 -2 5.4;
143 -0.5 -2 13; 145 0 -2 6.1; 146 0 -2 13; 147 0.5 -2 6.1; 148 0.5 -2 5.4;
149 0.5 -2 13; 151 1.1 -2 6.1; 152 1.1 -2 5.4; 153 1.1 -2 13; 158 -1.1 -2 0.3;
160 -1.1 -2 0.7; 162 -1.1 -2 1.03; 164 -1.1 -2 1.36; 166 -1.1 -2 2.08;
168 -1.1 -2 4.04; 170 -1.1 -2 4.37; 172 -1.1 -2 4.7; 174 -1.1 -2 5.1;
175 -0.5 -2 0.3; 176 -0.5 -2 0.7; 177 -0.5 -2 1.03; 178 -0.5 -2 1.36;
179 -0.5 -2 2.08; 180 -0.5 -2 4.04; 181 -0.5 -2 4.37; 182 -0.5 -2 4.7;
183 -0.5 -2 5.1; 184 0 -2 0.3; 185 0 -2 0.7; 186 0 -2 1.03; 187 0 -2 1.36;
188 0 -2 2.08; 189 0 -2 4.04; 190 0 -2 4.37; 191 0 -2 4.7; 192 0 -2 5.1;
193 0.5 -2 0.3; 194 0.5 -2 0.7; 195 0.5 -2 1.03; 196 0.5 -2 1.36;
197 0.5 -2 2.08; 198 0.5 -2 4.04; 199 0.5 -2 4.37; 200 0.5 -2 4.7;
201 0.5 -2 5.1; 202 1.1 -2 0.3; 203 1.1 -2 0.7; 204 1.1 -2 1.03;
205 1.1 -2 1.36; 206 1.1 -2 2.08; 207 1.1 -2 4.04; 208 1.1 -2 4.37;
209 1.1 -2 4.7; 210 1.1 -2 5.1; 221 -1.1 -2 -1.5; 222 -0.5 -2 -1.5;
223 0 -2 -1.5; 224 0.5 -2 -1.5; 225 1.1 -2 -1.5; 235 -1.1 -2 6.4;
237 -1.1 -2 6.8; 239 -1.1 -2 7.6; 241 -1.1 -2 8.1; 243 -1.1 -2 8.6;
245 -1.1 -2 9.1; 247 -1.1 -2 9.6; 249 -1.1 -2 10.1; 251 -1.1 -2 10.8;
253 -1.1 -2 11.2; 255 -1.1 -2 11.5; 257 -1.1 -2 11.8; 259 -1.1 -2 12.2;
260 -0.5 -2 6.4; 261 -0.5 -2 6.8; 262 -0.5 -2 7.6; 263 -0.5 -2 8.1;
264 -0.5 -2 8.6; 265 -0.5 -2 9.1; 266 -0.5 -2 9.6; 267 -0.5 -2 10.1;
268 -0.5 -2 10.8; 269 -0.5 -2 11.2; 270 -0.5 -2 11.5; 271 -0.5 -2 11.8;
272 -0.5 -2 12.2; 273 0 -2 6.4; 274 0 -2 6.8; 275 0 -2 7.6; 276 0 -2 8.1;
277 0 -2 8.6; 278 0 -2 9.1; 279 0 -2 9.6; 280 0 -2 10.1; 281 0 -2 10.8;
282 0 -2 11.2; 283 0 -2 11.5; 284 0 -2 11.8; 285 0 -2 12.2; 286 0.5 -2 6.4;
287 0.5 -2 6.8; 288 0.5 -2 7.6; 289 0.5 -2 8.1; 290 0.5 -2 8.6; 291 0.5 -2 9.1;
292 0.5 -2 9.6; 293 0.5 -2 10.1; 294 0.5 -2 10.8; 295 0.5 -2 11.2;
296 0.5 -2 11.5; 297 0.5 -2 11.8; 298 0.5 -2 12.2; 299 1.1 -2 6.4;
300 1.1 -2 6.8; 301 1.1 -2 7.6; 302 1.1 -2 8.1; 303 1.1 -2 8.6; 304 1.1 -2 9.1;
305 1.1 -2 9.6; 306 1.1 -2 10.1; 307 1.1 -2 10.8; 308 1.1 -2 11.2;
309 1.1 -2 11.5; 310 1.1 -2 11.8; 311 1.1 -2 12.2; 326 -1.1 -2 2.58;
328 -1.1 -2 3.08; 330 -1.1 -2 3.58; 331 -0.5 -2 2.58; 332 -0.5 -2 3.08;
333 -0.5 -2 3.58; 334 0 -2 2.58; 335 0 -2 3.08; 336 0 -2 3.58; 337 0.5 -2 2.58;
338 0.5 -2 3.08; 339 0.5 -2 3.58; 340 1.1 -2 2.58; 341 1.1 -2 3.08;
342 1.1 -2 3.58; 346 0 -1.7 6.1; 347 0 -1.7 11.5;
MEMBER INCIDENCES
1 101 5; 2 102 6; 240 145 346; 241 283 347;
ELEMENT INCIDENCES SHELL
16 114 117 121 119; 17 117 118 122 121; 19 119 121 124 123; 20 121 122 101 124;
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22 123 124 127 125; 23 124 101 128 127; 25 125 127 131 129; 26 127 128 132 131;
33 137 136 141 142; 36 142 141 145 102; 39 102 145 147 148; 42 148 147 151 152;
58 118 158 175 122; 59 158 160 176 175; 60 160 162 177 176; 61 162 164 178 177;
62 164 166 179 178; 64 168 170 181 180; 65 170 172 182 181; 66 172 174 183 182;
67 174 137 142 183; 68 122 175 184 101; 69 175 176 185 184; 70 176 177 186 185;
71 177 178 187 186; 72 178 179 188 187; 74 180 181 190 189; 75 181 182 191 190;
76 182 183 192 191; 77 183 142 102 192; 78 101 184 193 128; 79 184 185 194 193;
80 185 186 195 194; 81 186 187 196 195; 82 187 188 197 196; 84 189 190 199 198;
85 190 191 200 199; 86 191 192 201 200; 87 192 102 148 201; 88 128 193 202 132;
89 193 194 203 202; 90 194 195 204 203; 91 195 196 205 204; 92 196 197 206 205;
94 198 199 208 207; 95 199 200 209 208; 96 200 201 210 209; 97 201 148 152 210;
111 221 114 119 222; 113 222 119 123 223; 115 223 123 125 224;
117 224 125 129 225; 146 136 235 260 141; 147 235 237 261 260;
148 237 239 262 261; 149 239 241 263 262; 150 241 243 264 263;
151 243 245 265 264; 152 245 247 266 265; 153 247 249 267 266;
154 249 251 268 267; 155 251 253 269 268; 156 253 255 270 269;
157 255 257 271 270; 158 257 259 272 271; 159 259 139 143 272;
160 141 260 273 145; 161 260 261 274 273; 162 261 262 275 274;
163 262 263 276 275; 164 263 264 277 276; 165 264 265 278 277;
166 265 266 279 278; 167 266 267 280 279; 168 267 268 281 280;
169 268 269 282 281; 170 269 270 283 282; 171 270 271 284 283;
172 271 272 285 284; 173 272 143 146 285; 174 145 273 286 147;
175 273 274 287 286; 176 274 275 288 287; 177 275 276 289 288;
178 276 277 290 289; 179 277 278 291 290; 180 278 279 292 291;
181 279 280 293 292; 182 280 281 294 293; 183 281 282 295 294;
184 282 283 296 295; 185 283 284 297 296; 186 284 285 298 297;
187 285 146 149 298; 188 147 286 299 151; 189 286 287 300 299;
190 287 288 301 300; 191 288 289 302 301; 192 289 290 303 302;
193 290 291 304 303; 194 291 292 305 304; 195 292 293 306 305;
196 293 294 307 306; 197 294 295 308 307; 198 295 296 309 308;
199 296 297 310 309; 200 297 298 311 310; 201 298 149 153 311;
220 166 326 331 179; 221 326 328 332 331; 222 328 330 333 332;
223 330 168 180 333; 224 179 331 334 188; 225 331 332 335 334;
226 332 333 336 335; 227 333 180 189 336; 228 188 334 337 197;
229 334 335 338 337; 230 335 336 339 338; 231 336 189 198 339;
232 197 337 340 206; 233 337 338 341 340; 234 338 339 342 341;
235 339 198 207 342;
START GROUP DEFINITION
ELEMENT
_MX 16 19 22 25 59 TO 62 64 TO 66 69 TO 72 74 TO 76 79 TO 82 84 TO 86 -
89 TO 92 94 TO 96 111 113 115 117 147 TO 155 158 159 161 TO 169 172 173 175 -
176 TO 183 186 187 189 TO 197 200 201 220 TO 235
_MY 16 17 25 26 33 42 58 TO 62 64 TO 67 88 TO 92 94 TO 97 111 117 146 TO 159 -
188 TO 201 220 TO 223 232 TO 235
_SHEARX 60 TO 62 64 65 70 TO 72 74 75 80 TO 82 84 85 90 TO 92 94 95 111 113 -
115 117 148 TO 154 159 162 TO 168 173 176 TO 182 187 190 TO 196 201 -
220 TO 235
_SHEARY 16 17 25 26 33 42 58 TO 62 64 TO 67 88 TO 92 94 TO 97 111 117 146 -
147 TO 159 188 TO 201 220 TO 223 232 TO 235
```

```
END GROUP DEFINITION
ELEMENT PROPERTY
16 17 19 20 22 23 25 26 33 36 39 42 58 TO 62 64 TO 72 74 TO 82 84 TO 92 94 -
95 TO 97 111 113 115 117 146 TO 201 220 TO 235 THICKNESS 0.5
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+07
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-05
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 240 241 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
101 102 114 117 TO 119 121 TO 125 127 TO 129 131 132 136 137 139 141 TO 143 -
145 TO 149 151 TO 153 158 160 162 164 166 168 170 172 174 TO 210 221 TO 225 -
235 237 239 241 243 245 247 249 251 253 255 257 259 TO 311 326 328 -
330 TO 342 ELASTIC MAT DIRECT Y SUBGRADE 20000 PRINT
LOAD 1 DEAD  LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
16 17 19 20 22 23 25 26 33 36 39 42 58 TO 62 64 TO 72 74 TO 82 84 TO 92 94 -
95 TO 97 111 113 115 117 146 TO 201 220 TO 235 PR GY -27
JOINT LOAD
5   FX  14.0    FY  -580.5  FZ  -12.0
5   MX  -11.8   MY  0.1 MZ  27.7
6   FX  10.1    FY  -543.4  FZ  11.5
6   MX  10.8    MY  0.7 MZ  28.7
346 FX  14.0    FY  -580.5  FZ  -12.0
346 MX  -11.8   MY  0.1 MZ  27.7
347 FX  10.1    FY  -543.4  FZ  11.5
347 MX  10.8    MY  0.7 MZ  28.7
LOAD 2 LIVE LOAD
JOINT LOAD
5   FX  2.2 FY  -54.2   FZ  -1.2
5   MX  -1.1    MY  -0.1    MZ  2.9
6   FX  1.9 FY  -54.1   FZ  1.0
6   MX  0.9 MY  0.0 MZ  3.6
346 FX  2.2 FY  -54.2   FZ  -1.2
346 MX  -1.1    MY  -0.1    MZ  2.9
347 FX  1.9 FY  -54.1   FZ  1.0
347 MX  0.9 MY  0.0 MZ  3.6
LOAD 3 SDL1 CRUSHER DYNAMIC LOAD
JOINT LOAD
```

5 FX -1.3 FY 84.4 FZ 50.7
5 MX 74.5 MY -6.6 MZ 7.2
6 FX 7.0 FY -173.6 FZ 52.6
6 MX 76.3 MY -6.3 MZ 1.0
346 FX -1.3 FY 84.4 FZ 50.7
346 MX 74.5 MY -6.6 MZ 7.2
347 FX 7.0 FY -173.6 FZ 52.6
347 MX 76.3 MY -6.3 MZ 1.0
LOAD 4 SDL2 CRUSHER DYNAMIC LOAD
JOINT LOAD
5 FX 32.8 FY -41.6 FZ -1.5
5 MX -1.9 MY -1.4 MZ -59.6
6 FX 27.6 FY -29.7 FZ 0.1
6 MX -0.3 MY 0.6 MZ -50.8
346 FX 32.8 FY -41.6 FZ -1.5
346 MX -1.9 MY -1.4 MZ -59.6
347 FX 27.6 FY -29.7 FZ 0.1
347 MX -0.3 MY 0.6 MZ -50.8
LOAD 5 WIND LOAD (+X)
JOINT LOAD
5 FX 9.7 FY 1.6 FZ 0
5 MX -0.1 MY -0.1 MZ -24.0
6 FX 9.5 FY 1.5 FZ -0.1
6 MX -0.1 MY 0.1 MZ -23.4
346 FX 9.7 FY 1.6 FZ 0
346 MX -0.1 MY -0.1 MZ -24
347 FX 9.5 FY 1.5 FZ -0.1
347 MX -0.1 MY 0.1 MZ -23.4
LOAD 6 WIND LOAD (-X)
JOINT LOAD
5 FX -9.6 FY -1.7 FZ 0
5 MX 0.1 MY 0.1 MZ 23.7
6 FX -9.3 FY -1.6 FZ 0.1
6 MX 0.1 MY -0.1 MZ 23.1
346 FX -9.6 FY -1.7 FZ 0
346 MX 0.1 MY 0.1 MZ 23.7
347 FX -9.3 FY -1.6 FZ 0.1
347 MX 0.1 MY -0.1 MZ 23.1
LOAD 7 WIND LOAD (+Z)
JOINT LOAD
5 FX 0 FY 34.1 FZ 19.1
5 MX 27.9 MY -0.7 MZ -0.8
6 FX 0 FY -34.1 FZ 19.1
6 MX 27.9 MY -0.7 MZ 0.8
346 FX 0 FY 34.1 FZ 19.1
346 MX 27.9 MY -0.7 MZ -0.8
347 FX 0 FY -34.1 FZ 19.1
347 MX 27.9 MY -0.7 MZ 0.8
LOAD 8 WIND LOAD (-Z)

```
JOINT LOAD
5   FX  0   FY  -34.7   FZ  -19.3
5   MX  -28.2   MY  0.8 MZ  0.9
6   FX  -0.1    FY  34.7    FZ  -19.3
6   MX  -28.2   MY  0.8 MZ  -0.9
346 FX  0   FY  -34.7   FZ  -19.3
346 MX  -28.2   MY  0.8 MZ  0.9
347 FX  -0.1    FY  34.7    FZ  -19.3
347 MX  -28.2   MY  0.8 MZ  -0.9
LOAD 11 SEISMIC LOAD (+X)
JOINT LOAD
5   FX  10.2    FY  2.5 FZ  -0.1
5   MX  -0.1    MY  -0.1    MZ  -21.7
6   FX  9.4 FY  2.8 FZ  -0.2
6   MX  -0.2    MY  -0.1    MZ  -20.2
346 FX  10.2    FY  2.5 FZ  -0.1
346 MX  -0.1    MY  -0.1    MZ  -21.7
347 FX  9.4 FY  2.8 FZ  -0.2
347 MX  -0.2    MY  -0.1    MZ  -20.2
LOAD 12 SEISMIC LOAD (+Z)
JOINT LOAD
5   FX  -0.1    FY  35.8    FZ  14.5
5   MX  21.1    MY  -0.4    MZ  -0.4
6   FX  0.1 FY  -35.8   FZ  14.5
6   MX  21.1    MY  -0.4    MZ  0.4
346 FX  -0.1    FY  35.8    FZ  14.5
346 MX  21.1    MY  -0.4    MZ  -0.4
347 FX  0.1 FY  -35.8   FZ  14.5
347 MX  21.1    MY  -0.4    MZ  0.4
*****************LOAD COMBINATION FOR BASE PRESSURE CHECK
************************************************
LOAD 301 DL+SDL1_R
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2_R
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL-SDL1_R
REPEAT LOAD
1 1.0 3 -1.0
LOAD 304 DL-SDL2_R
REPEAT LOAD
1 1.0 4 -1.0
*********************************
LOAD 305 DL + SDL1_R + LL
REPEAT LOAD
1 1.0 3 1.0 2 1.0
LOAD 306 DL + SDL2_R + LL
REPEAT LOAD
```

```
1 1.0 4 1.0 2 1.0
LOAD 307 DL - SDL1_R + LL
REPEAT LOAD
1 1.0 3 -1.0 2 1.0
LOAD 308 DL - SDL2_R + LL
REPEAT LOAD
1 1.0 4 -1.0 2 1.0
*********************************************
LOAD 309 DL + SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 310 DL + SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 311 DL + SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 312 DL + SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 313 DL + SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 314 DL + SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 315 DL + SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 316 DL + SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 317 DL - SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 5 0.75
LOAD 318 DL - SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 6 0.75
LOAD 319 DL - SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 7 0.75
LOAD 320 DL - SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 8 0.75
LOAD 321 DL - SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 5 0.75
LOAD 322 DL - SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
```

```
1 1.0 4 -1.0 2 0.75 6 0.75
LOAD 323 DL - SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 7 0.75
LOAD 324 DL - SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 8 0.75
************************************************
LOAD 325 DL + SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 326 DL + SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 327 DL + SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 328 DL + SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 329 DL + SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 330 DL + SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 331 DL + SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 332 DL + SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 333 DL - SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 0.525
LOAD 334 DL - SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 -0.525
LOAD 335 DL - SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 0.525
LOAD 336 DL - SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 -0.525
LOAD 337 DL - SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 0.525
LOAD 338 DL - SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
```

```
1 1.0 4 -1.0 2 0.75 11 -0.525
LOAD 339 DL - SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 0.525
LOAD 340 DL - SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 -0.525
************************************************
LOAD 341 0.6 (DL + SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 342 0.6 (DL + SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 345 0.6 (DL + SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 346 0.6 (DL + SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 349 0.6 (DL - SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 -0.6 5 1.0
LOAD 350 0.6 (DL - SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 -0.6 6 1.0
LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 7 1.0
LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 8 1.0
LOAD 353 0.6 (DL - SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 -0.6 5 1.0
LOAD 354 0.6 (DL - SDL2_R) + WL(-X)
REPEAT LOAD
```

```
1 0.6 4 -0.6 6 1.0
LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 7 1.0
LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 8 1.0
************************************************
LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 -0.6 11 0.7
LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 -0.6 11 -0.7
LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 12 0.7
LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 12 -0.7
LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 -0.6 11 0.7
LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
```

```
1 0.6 4 -0.6 11 -0.7
LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
************************************************
*Load Combination for Concrete design
************************************************
LOAD 401 1.4 (DL+SDL1_R)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2_R)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL-SDL1_R)
REPEAT LOAD
1 1.4 3 -1.4
LOAD 404 1.4 (DL-SDL2_R)
REPEAT LOAD
1 1.4 4 -1.4
************************************************
LOAD 405 1.4 (DL + SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 2 1.7
LOAD 407 1.4 (DL - SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 -1.4 2 1.7
LOAD 408 1.4 (DL - SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 -1.4 2 1.7
************************************************
LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)
```

```
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 5 1.6
LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 6 1.6
LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 7 1.6
LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 8 1.6
LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 5 1.6
LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 6 1.6
LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 7 1.6
LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 8 1.6
*********************************************
LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)
```

```
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 1.0
LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 -1.0
LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 1.0
LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 -1.0
LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 1.0
LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 -1.0
LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 1.0
LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 -1.0
*********************************************
LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)
```

```
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 -0.9 5 1.6
LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 -0.9 6 1.6
LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 7 1.6
LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 8 1.6
LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 -0.9 5 1.6
LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 -0.9 6 1.6
LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 7 1.6
LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 8 1.6
************************************************
LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)
```

```
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 -0.9 11 1.0
LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 -0.9 11 -1.0
LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 12 1.0
LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 12 -1.0
LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 -0.9 11 1.0
LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 -0.9 11 -1.0
LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 12 1.0
LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 12 -1.0
**********
PERFORM ANALYSIS
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 6 STAADPRO INPUT FILE OF FOUNDATION F1B

Job Title:

Client:

Engineer:

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
5 0 -1.7 0; 6 0 -1.7 5.4; 101 0 -2 0; 102 0 -2 5.4; 114 -1 -2 -0.7;
117 -1 -2 -0.3; 118 -1 -2 0; 119 -0.5 -2 -0.7; 121 -0.5 -2 -0.3; 122 -0.5 -2 0;
123 0 -2 -0.7; 124 0 -2 -0.3; 125 0.5 -2 -0.7; 127 0.5 -2 -0.3; 128 0.5 -2 0;
129 1 -2 -0.7; 131 1 -2 -0.3; 132 1 -2 0; 136 -1 -2 6.1; 137 -1 -2 5.4;
139 -1 -2 13; 141 -0.5 -2 6.1; 142 -0.5 -2 5.4; 143 -0.5 -2 13; 145 0 -2 6.1;
146 0 -2 13; 147 0.5 -2 6.1; 148 0.5 -2 5.4; 149 0.5 -2 13; 151 1 -2 6.1;
152 1 -2 5.4; 153 1 -2 13; 158 -1 -2 0.3; 160 -1 -2 0.7; 162 -1 -2 1.03;
164 -1 -2 1.36; 166 -1 -2 2.08; 168 -1 -2 4.04; 170 -1 -2 4.37; 172 -1 -2 4.7;
174 -1 -2 5.1; 175 -0.5 -2 0.3; 176 -0.5 -2 0.7; 177 -0.5 -2 1.03;
178 -0.5 -2 1.36; 179 -0.5 -2 2.08; 180 -0.5 -2 4.04; 181 -0.5 -2 4.37;
182 -0.5 -2 4.7; 183 -0.5 -2 5.1; 184 0 -2 0.3; 185 0 -2 0.7; 186 0 -2 1.03;
187 0 -2 1.36; 188 0 -2 2.08; 189 0 -2 4.04; 190 0 -2 4.37; 191 0 -2 4.7;
192 0 -2 5.1; 193 0.5 -2 0.3; 194 0.5 -2 0.7; 195 0.5 -2 1.03; 196 0.5 -2 1.36;
197 0.5 -2 2.08; 198 0.5 -2 4.04; 199 0.5 -2 4.37; 200 0.5 -2 4.7;
201 0.5 -2 5.1; 202 1 -2 0.3; 203 1 -2 0.7; 204 1 -2 1.03; 205 1 -2 1.36;
206 1 -2 2.08; 207 1 -2 4.04; 208 1 -2 4.37; 209 1 -2 4.7; 210 1 -2 5.1;
221 -1 -2 -1.5; 222 -0.5 -2 -1.5; 223 0 -2 -1.5; 224 0.5 -2 -1.5;
225 1 -2 -1.5; 235 -1 -2 6.4; 237 -1 -2 6.8; 239 -1 -2 7.6; 241 -1 -2 8.1;
243 -1 -2 8.6; 245 -1 -2 9.1; 247 -1 -2 9.6; 249 -1 -2 10.1; 251 -1 -2 10.8;
253 -1 -2 11.2; 255 -1 -2 11.5; 257 -1 -2 11.8; 259 -1 -2 12.2;
260 -0.5 -2 6.4; 261 -0.5 -2 6.8; 262 -0.5 -2 7.6; 263 -0.5 -2 8.1;
264 -0.5 -2 8.6; 265 -0.5 -2 9.1; 266 -0.5 -2 9.6; 267 -0.5 -2 10.1;
268 -0.5 -2 10.8; 269 -0.5 -2 11.2; 270 -0.5 -2 11.5; 271 -0.5 -2 11.8;
272 -0.5 -2 12.2; 273 0 -2 6.4; 274 0 -2 6.8; 275 0 -2 7.6; 276 0 -2 8.1;
277 0 -2 8.6; 278 0 -2 9.1; 279 0 -2 9.6; 280 0 -2 10.1; 281 0 -2 10.8;
282 0 -2 11.2; 283 0 -2 11.5; 284 0 -2 11.8; 285 0 -2 12.2; 286 0.5 -2 6.4;
287 0.5 -2 6.8; 288 0.5 -2 7.6; 289 0.5 -2 8.1; 290 0.5 -2 8.6; 291 0.5 -2 9.1;
292 0.5 -2 9.6; 293 0.5 -2 10.1; 294 0.5 -2 10.8; 295 0.5 -2 11.2;
296 0.5 -2 11.5; 297 0.5 -2 11.8; 298 0.5 -2 12.2; 299 1 -2 6.4; 300 1 -2 6.8;
301 1 -2 7.6; 302 1 -2 8.1; 303 1 -2 8.6; 304 1 -2 9.1; 305 1 -2 9.6;
306 1 -2 10.1; 307 1 -2 10.8; 308 1 -2 11.2; 309 1 -2 11.5; 310 1 -2 11.8;
311 1 -2 12.2; 326 -1 -2 2.58; 328 -1 -2 3.08; 330 -1 -2 3.58;
331 -0.5 -2 2.58; 332 -0.5 -2 3.08; 333 -0.5 -2 3.58; 334 0 -2 2.58;
335 0 -2 3.08; 336 0 -2 3.58; 337 0.5 -2 2.58; 338 0.5 -2 3.08;
339 0.5 -2 3.58; 340 1 -2 2.58; 341 1 -2 3.08; 342 1 -2 3.58; 346 0 -1.7 6.1;
347 0 -1.7 11.5;
MEMBER INCIDENCES
1 101 5; 2 102 6; 240 145 346; 241 283 347;
ELEMENT INCIDENCES SHELL
16 114 117 121 119; 17 117 118 122 121; 19 119 121 124 123; 20 121 122 101 124;
22 123 124 127 125; 23 124 101 128 127; 25 125 127 131 129; 26 127 128 132 131;
33 137 136 141 142; 36 142 141 145 102; 39 102 145 147 148; 42 148 147 151 152;
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58 118 158 175 122; 59 158 160 176 175; 60 160 162 177 176; 61 162 164 178 177;
62 164 166 179 178; 64 168 170 181 180; 65 170 172 182 181; 66 172 174 183 182;
67 174 137 142 183; 68 122 175 184 101; 69 175 176 185 184; 70 176 177 186 185;
71 177 178 187 186; 72 178 179 188 187; 74 180 181 190 189; 75 181 182 191 190;
76 182 183 192 191; 77 183 142 102 192; 78 101 184 193 128; 79 184 185 194 193;
80 185 186 195 194; 81 186 187 196 195; 82 187 188 197 196; 84 189 190 199 198;
85 190 191 200 199; 86 191 192 201 200; 87 192 102 148 201; 88 128 193 202 132;
89 193 194 203 202; 90 194 195 204 203; 91 195 196 205 204; 92 196 197 206 205;
94 198 199 208 207; 95 199 200 209 208; 96 200 201 210 209; 97 201 148 152 210;
111 221 114 119 222; 113 222 119 123 223; 115 223 123 125 224;
117 224 125 129 225; 146 136 235 260 141; 147 235 237 261 260;
148 237 239 262 261; 149 239 241 263 262; 150 241 243 264 263;
151 243 245 265 264; 152 245 247 266 265; 153 247 249 267 266;
154 249 251 268 267; 155 251 253 269 268; 156 253 255 270 269;
157 255 257 271 270; 158 257 259 272 271; 159 259 139 143 272;
160 141 260 273 145; 161 260 261 274 273; 162 261 262 275 274;
163 262 263 276 275; 164 263 264 277 276; 165 264 265 278 277;
166 265 266 279 278; 167 266 267 280 279; 168 267 268 281 280;
169 268 269 282 281; 170 269 270 283 282; 171 270 271 284 283;
172 271 272 285 284; 173 272 143 146 285; 174 145 273 286 147;
175 273 274 287 286; 176 274 275 288 287; 177 275 276 289 288;
178 276 277 290 289; 179 277 278 291 290; 180 278 279 292 291;
181 279 280 293 292; 182 280 281 294 293; 183 281 282 295 294;
184 282 283 296 295; 185 283 284 297 296; 186 284 285 298 297;
187 285 146 149 298; 188 147 286 299 151; 189 286 287 300 299;
190 287 288 301 300; 191 288 289 302 301; 192 289 290 303 302;
193 290 291 304 303; 194 291 292 305 304; 195 292 293 306 305;
196 293 294 307 306; 197 294 295 308 307; 198 295 296 309 308;
199 296 297 310 309; 200 297 298 311 310; 201 298 149 153 311;
220 166 326 331 179; 221 326 328 332 331; 222 328 330 333 332;
223 330 168 180 333; 224 179 331 334 188; 225 331 332 335 334;
226 332 333 336 335; 227 333 180 189 336; 228 188 334 337 197;
229 334 335 338 337; 230 335 336 339 338; 231 336 189 198 339;
232 197 337 340 206; 233 337 338 341 340; 234 338 339 342 341;
235 339 198 207 342;
START GROUP DEFINITION
ELEMENT
_MX 16 19 22 25 59 TO 62 64 TO 66 69 TO 72 74 TO 76 79 TO 82 84 TO 86 -
89 TO 92 94 TO 96 111 113 115 117 147 TO 155 158 159 161 TO 169 172 173 175 -
176 TO 183 186 187 189 TO 197 200 201 220 TO 235
_MY 16 17 25 26 33 42 58 TO 62 64 TO 67 88 TO 92 94 TO 97 111 117 146 TO 159 -
188 TO 201 220 TO 223 232 TO 235
_SHEAR 16 17 25 26 33 42 58 TO 62 64 TO 67 70 TO 72 74 75 80 TO 82 84 85 88 -
89 TO 92 94 TO 97 111 113 115 117 146 TO 159 162 TO 168 173 176 TO 182 187 -
188 TO 201 220 TO 235
END GROUP DEFINITION
ELEMENT PROPERTY
16 17 19 20 22 23 25 26 33 36 39 42 58 TO 62 64 TO 72 74 TO 82 84 TO 92 94 -
95 TO 97 111 113 115 117 146 TO 201 220 TO 235 THICKNESS 0.5
```

Job Title:

Client:

Engineer:

```
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 240 241 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
101 102 114 117 TO 119 121 TO 125 127 TO 129 131 132 136 137 139 141 TO 143 -
145 TO 149 151 TO 153 158 160 162 164 166 168 170 172 174 TO 210 221 TO 225 -
235 237 239 241 243 245 247 249 251 253 255 257 259 TO 311 326 328 -
330 TO 342 ELASTIC MAT DIRECT Y SUBGRADE 20000 PRINT
LOAD 1 DEAD  LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
16 17 19 20 22 23 25 26 33 36 39 42 58 TO 62 64 TO 72 74 TO 82 84 TO 92 94 -
95 TO 97 111 113 115 117 146 TO 201 220 TO 235 PR GY -28
JOINT LOAD
5   FX  2.7 FY  -400.5  FZ  -1.5
5   MX  0.0 MY  -0.5    MZ  -2.1
6   FX  0.2 FY  -387.3  FZ  2.0
6   MX  2.8 MY  -0.4    MZ  1.3
346 FX  2.7 FY  -400.5  FZ  -1.5
346 MX  0.0 MY  -0.5    MZ  -2.1
347 FX  0.2 FY  -387.3  FZ  2.0
347 MX  2.8 MY  -0.4    MZ  1.3
LOAD 2 LIVE LOAD
JOINT LOAD
5   FX  0.3 FY  -50.4   FZ  0.8
5   MX  0.9 MY  -0.1    MZ  -0.3
6   FX  -0.1    FY  -51.0   FZ  -0.6
6   MX  0.0 MY  -0.1    MZ  0.2
346 FX  0.3 FY  -50.4   FZ  0.8
346 MX  0.9 MY  -0.1    MZ  -0.3
347 FX  -0.1    FY  -51.0   FZ  -0.6
347 MX  0.0 MY  -0.1    MZ  0.2
LOAD 3 SDL1 CRUSHER DYNAMIC LOAD
JOINT LOAD
5   FX  -4.3    FY  18.5    FZ  10.5
5   MX  22.3    MY  4.4 MZ  5.8
6   FX  5.0 FY  -74.8   FZ  10.0
6   MX  22.0    MY  4.3 MZ  -5.9
```

346 FX -4.3 FY 18.5 FZ 10.5
346 MX 22.3 MY 4.4 MZ 5.8
347 FX 5.0 FY -74.8 FZ 10.0
347 MX 22.0 MY 4.3 MZ -5.9
LOAD 4 SDL2 CRUSHER DYNAMIC LOAD
JOINT LOAD
5 FX 39.3 FY -24.0 FZ 1.4
5 MX 1.2 MY 0.0 MZ -55.4
6 FX 33.3 FY -19.8 FZ -1.0
6 MX -0.1 MY -0.2 MZ -46.9
346 FX 39.3 FY -24.0 FZ 1.4
346 MX 1.2 MY 0.0 MZ -55.4
347 FX 33.3 FY -19.8 FZ -1.0
347 MX -0.1 MY -0.2 MZ -46.9
LOAD 5 WIND LOAD (+X)
JOINT LOAD
5 FX 11.7 FY 2.8 FZ 0
5 MX 0.0 MY 0.0 MZ -16.4
6 FX 11.3 FY 2.6 FZ 0
6 MX 0.0 MY 0.0 MZ -15.9
346 FX 11.7 FY 2.8 FZ 0
346 MX 0.0 MY 0.0 MZ -16.4
347 FX 11.3 FY 2.6 FZ 0
347 MX 0.0 MY 0.0 MZ -15.9
LOAD 6 WIND LOAD (-X)
JOINT LOAD
5 FX -11.7 FY -3 FZ 0
5 MX 0.0 MY 0.0 MZ 16.5
6 FX -11.4 FY -2.9 FZ 0
6 MX 0.0 MY 0.0 MZ 16
346 FX -11.7 FY -3 FZ 0
346 MX 0.0 MY 0.0 MZ 16.5
347 FX -11.4 FY -2.9 FZ 0
347 MX 0.0 MY 0.0 MZ 16
LOAD 7 WIND LOAD (+Z)
JOINT LOAD
5 FX 0.1 FY 13.3 FZ 6.1
5 MX 14.6 MY 0.5 MZ -0.3
6 FX -0.1 FY -14 FZ 8.3
6 MX 15.6 MY 0.5 MZ 0.3
346 FX 0.1 FY 13.3 FZ 6.1
346 MX 14.6 MY 0.5 MZ -0.3
347 FX -0.1 FY -14 FZ 8.3
347 MX 15.6 MY 0.5 MZ 0.3
LOAD 8 WIND LOAD (-Z)
JOINT LOAD
5 FX -0.1 FY -15.1 FZ -9.2
5 MX -17.1 MY -0.7 MZ 0.3
6 FX 0.1 FY 14.5 FZ -6.6

```
6   MX  -15.9   MY  -0.6    MZ  -0.3
346 FX  -0.1    FY  -15.1   FZ  -9.2
346 MX  -17.1   MY  -0.7    MZ  0.3
347 FX  0.1 FY  14.5    FZ  -6.6
347 MX  -15.9   MY  -0.6    MZ  -0.3
LOAD 11 SEISMIC LOAD (+X)
JOINT LOAD
5   FX  12.9    FY  2.1 FZ  0
5   MX  0.0 MY  0.0 MZ  -18.3
6   FX  11.9    FY  1.7 FZ  0.1
6   MX  0.1 MY  0.0 MZ  -16.9
346 FX  12.9    FY  2.1 FZ  0
346 MX  0.0 MY  0.0 MZ  -18.3
347 FX  11.9    FY  1.7 FZ  0.1
347 MX  0.1 MY  0.0 MZ  -16.9
LOAD 12 SEISMIC LOAD (+Z)
JOINT LOAD
5   FX  -0.1    FY  18.1    FZ  4.8
5   MX  10.6    MY  0.5 MZ  -0.1
6   FX  0.1 FY  -18.2   FZ  4.7
6   MX  10.5    MY  0.5 MZ  0.1
346 FX  -0.1    FY  18.1    FZ  4.8
346 MX  10.6    MY  0.5 MZ  -0.1
347 FX  0.1 FY  -18.2   FZ  4.7
347 MX  10.5    MY  0.5 MZ  0.1
*****************LOAD COMBINATION FOR BASE PRESSURE CHECK
LOAD 301 DL+SDL1_R
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2_R
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL-SDL1_R
REPEAT LOAD
1 1.0 3 -1.0
LOAD 304 DL-SDL2_R
REPEAT LOAD
1 1.0 4 -1.0
**********************************
LOAD 305 DL + SDL1_R + LL
REPEAT LOAD
1 1.0 3 1.0 2 1.0
LOAD 306 DL + SDL2_R + LL
REPEAT LOAD
1 1.0 4 1.0 2 1.0
LOAD 307 DL - SDL1_R + LL
REPEAT LOAD
1 1.0 3 -1.0 2 1.0
LOAD 308 DL - SDL2_R + LL
```

```
REPEAT LOAD
1 1.0 4 -1.0 2 1.0
*********************************************
LOAD 309 DL + SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 310 DL + SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 311 DL + SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 312 DL + SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 313 DL + SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 314 DL + SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 315 DL + SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 316 DL + SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 317 DL - SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 5 0.75
LOAD 318 DL - SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 6 0.75
LOAD 319 DL - SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 7 0.75
LOAD 320 DL - SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 8 0.75
LOAD 321 DL - SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 5 0.75
LOAD 322 DL - SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 6 0.75
LOAD 323 DL - SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 7 0.75
LOAD 324 DL - SDL2_R + 0.75 LL + 0.75 WL(-Z)
```

```
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 8 0.75
************************************************
LOAD 325 DL + SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 326 DL + SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 327 DL + SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 328 DL + SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 329 DL + SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 330 DL + SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 331 DL + SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 332 DL + SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 333 DL - SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 0.525
LOAD 334 DL - SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 -0.525
LOAD 335 DL - SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 0.525
LOAD 336 DL - SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 -0.525
LOAD 337 DL - SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 0.525
LOAD 338 DL - SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 -0.525
LOAD 339 DL - SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 0.525
LOAD 340 DL - SDL2_R + 0.75 LL + 0.525 EL(-Z)
```

```
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 -0.525
************************************************
LOAD 341 0.6 (DL + SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 342 0.6 (DL + SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 345 0.6 (DL + SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 346 0.6 (DL + SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 349 0.6 (DL - SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 -0.6 5 1.0
LOAD 350 0.6 (DL - SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 -0.6 6 1.0
LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 7 1.0
LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 8 1.0
LOAD 353 0.6 (DL - SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 -0.6 5 1.0
LOAD 354 0.6 (DL - SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 -0.6 6 1.0
LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 7 1.0
LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)
```

```
REPEAT LOAD
1 0.6 4 -0.6 8 1.0
************************************************
LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 -0.6 11 0.7
LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 -0.6 11 -0.7
LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 12 0.7
LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 12 -0.7
LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 -0.6 11 0.7
LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 -0.6 11 -0.7
LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
```

```
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
************************************************
*Load Combination for Concrete design
************************************************
LOAD 401 1.4 (DL+SDL1_R)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2_R)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL-SDL1_R)
REPEAT LOAD
1 1.4 3 -1.4
LOAD 404 1.4 (DL-SDL2_R)
REPEAT LOAD
1 1.4 4 -1.4
************************************************
LOAD 405 1.4 (DL + SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 2 1.7
LOAD 407 1.4 (DL - SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 -1.4 2 1.7
LOAD 408 1.4 (DL - SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 -1.4 2 1.7
************************************************
LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
```

```
LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 5 1.6
LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 6 1.6
LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 7 1.6
LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 8 1.6
LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 5 1.6
LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 6 1.6
LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 7 1.6
LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 8 1.6
************************************************
LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
```

```
LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 1.0
LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 -1.0
LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 1.0
LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 -1.0
LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 1.0
LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 -1.0
LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 1.0
LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 -1.0
************************************************
LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
```

```
LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 -0.9 5 1.6
LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 -0.9 6 1.6
LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 7 1.6
LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 8 1.6
LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 -0.9 5 1.6
LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 -0.9 6 1.6
LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 7 1.6
LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 8 1.6


**********************************************
LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)
```

```
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 -0.9 11 1.0
LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 -0.9 11 -1.0
LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 12 1.0
LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 12 -1.0
LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 -0.9 11 1.0
LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 -0.9 11 -1.0
LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 12 1.0
LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 12 -1.0
**********
**********
PERFORM ANALYSIS
FINISH
```

Job Title:

Client:

Engineer:

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
5 0 -1.7 0; 6 0 -1.7 5.4; 101 0 -2 0; 102 0 -2 5.4; 114 -1 -2 -0.7;
117 -1 -2 -0.3; 118 -1 -2 0; 119 -0.5 -2 -0.7; 121 -0.5 -2 -0.3; 122 -0.5 -2 0;
123 0 -2 -0.7; 124 0 -2 -0.3; 125 0.5 -2 -0.7; 127 0.5 -2 -0.3; 128 0.5 -2 0;
129 1 -2 -0.7; 131 1 -2 -0.3; 132 1 -2 0; 136 -1 -2 6.1; 137 -1 -2 5.4;
139 -1 -2 13; 141 -0.5 -2 6.1; 142 -0.5 -2 5.4; 143 -0.5 -2 13; 145 0 -2 6.1;
146 0 -2 13; 147 0.5 -2 6.1; 148 0.5 -2 5.4; 149 0.5 -2 13; 151 1 -2 6.1;
152 1 -2 5.4; 153 1 -2 13; 158 -1 -2 0.3; 160 -1 -2 0.7; 162 -1 -2 1.03;
164 -1 -2 1.36; 166 -1 -2 2.08; 168 -1 -2 4.04; 170 -1 -2 4.37; 172 -1 -2 4.7;
174 -1 -2 5.1; 175 -0.5 -2 0.3; 176 -0.5 -2 0.7; 177 -0.5 -2 1.03;
178 -0.5 -2 1.36; 179 -0.5 -2 2.08; 180 -0.5 -2 4.04; 181 -0.5 -2 4.37;
182 -0.5 -2 4.7; 183 -0.5 -2 5.1; 184 0 -2 0.3; 185 0 -2 0.7; 186 0 -2 1.03;
187 0 -2 1.36; 188 0 -2 2.08; 189 0 -2 4.04; 190 0 -2 4.37; 191 0 -2 4.7;
192 0 -2 5.1; 193 0.5 -2 0.3; 194 0.5 -2 0.7; 195 0.5 -2 1.03; 196 0.5 -2 1.36;
197 0.5 -2 2.08; 198 0.5 -2 4.04; 199 0.5 -2 4.37; 200 0.5 -2 4.7;
201 0.5 -2 5.1; 202 1 -2 0.3; 203 1 -2 0.7; 204 1 -2 1.03; 205 1 -2 1.36;
206 1 -2 2.08; 207 1 -2 4.04; 208 1 -2 4.37; 209 1 -2 4.7; 210 1 -2 5.1;
221 -1 -2 -1.5; 222 -0.5 -2 -1.5; 223 0 -2 -1.5; 224 0.5 -2 -1.5;
225 1 -2 -1.5; 235 -1 -2 6.4; 237 -1 -2 6.8; 239 -1 -2 7.6; 241 -1 -2 8.1;
243 -1 -2 8.6; 245 -1 -2 9.1; 247 -1 -2 9.6; 249 -1 -2 10.1; 251 -1 -2 10.8;
253 -1 -2 11.2; 255 -1 -2 11.5; 257 -1 -2 11.8; 259 -1 -2 12.2;
260 -0.5 -2 6.4; 261 -0.5 -2 6.8; 262 -0.5 -2 7.6; 263 -0.5 -2 8.1;
264 -0.5 -2 8.6; 265 -0.5 -2 9.1; 266 -0.5 -2 9.6; 267 -0.5 -2 10.1;
268 -0.5 -2 10.8; 269 -0.5 -2 11.2; 270 -0.5 -2 11.5; 271 -0.5 -2 11.8;
272 -0.5 -2 12.2; 273 0 -2 6.4; 274 0 -2 6.8; 275 0 -2 7.6; 276 0 -2 8.1;
277 0 -2 8.6; 278 0 -2 9.1; 279 0 -2 9.6; 280 0 -2 10.1; 281 0 -2 10.8;
282 0 -2 11.2; 283 0 -2 11.5; 284 0 -2 11.8; 285 0 -2 12.2; 286 0.5 -2 6.4;
287 0.5 -2 6.8; 288 0.5 -2 7.6; 289 0.5 -2 8.1; 290 0.5 -2 8.6; 291 0.5 -2 9.1;
292 0.5 -2 9.6; 293 0.5 -2 10.1; 294 0.5 -2 10.8; 295 0.5 -2 11.2;
296 0.5 -2 11.5; 297 0.5 -2 11.8; 298 0.5 -2 12.2; 299 1 -2 6.4; 300 1 -2 6.8;
301 1 -2 7.6; 302 1 -2 8.1; 303 1 -2 8.6; 304 1 -2 9.1; 305 1 -2 9.6;
306 1 -2 10.1; 307 1 -2 10.8; 308 1 -2 11.2; 309 1 -2 11.5; 310 1 -2 11.8;
311 1 -2 12.2; 326 -1 -2 2.58; 328 -1 -2 3.08; 330 -1 -2 3.58;
331 -0.5 -2 2.58; 332 -0.5 -2 3.08; 333 -0.5 -2 3.58; 334 0 -2 2.58;
335 0 -2 3.08; 336 0 -2 3.58; 337 0.5 -2 2.58; 338 0.5 -2 3.08;
339 0.5 -2 3.58; 340 1 -2 2.58; 341 1 -2 3.08; 342 1 -2 3.58; 346 0 -1.7 6.1;
347 0 -1.7 11.5;
MEMBER INCIDENCES
1 101 5; 2 102 6; 240 145 346; 241 283 347;
ELEMENT INCIDENCES SHELL
16 114 117 121 119; 17 117 118 122 121; 19 119 121 124 123; 20 121 122 101 124;
22 123 124 127 125; 23 124 101 128 127; 25 125 127 131 129; 26 127 128 132 131;
33 137 136 141 142; 36 142 141 145 102; 39 102 145 147 148; 42 148 147 151 152;
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58 118 158 175 122; 59 158 160 176 175; 60 160 162 177 176; 61 162 164 178 177;
62 164 166 179 178; 64 168 170 181 180; 65 170 172 182 181; 66 172 174 183 182;
67 174 137 142 183; 68 122 175 184 101; 69 175 176 185 184; 70 176 177 186 185;
71 177 178 187 186; 72 178 179 188 187; 74 180 181 190 189; 75 181 182 191 190;
76 182 183 192 191; 77 183 142 102 192; 78 101 184 193 128; 79 184 185 194 193;
80 185 186 195 194; 81 186 187 196 195; 82 187 188 197 196; 84 189 190 199 198;
85 190 191 200 199; 86 191 192 201 200; 87 192 102 148 201; 88 128 193 202 132;
89 193 194 203 202; 90 194 195 204 203; 91 195 196 205 204; 92 196 197 206 205;
94 198 199 208 207; 95 199 200 209 208; 96 200 201 210 209; 97 201 148 152 210;
111 221 114 119 222; 113 222 119 123 223; 115 223 123 125 224;
117 224 125 129 225; 146 136 235 260 141; 147 235 237 261 260;
148 237 239 262 261; 149 239 241 263 262; 150 241 243 264 263;
151 243 245 265 264; 152 245 247 266 265; 153 247 249 267 266;
154 249 251 268 267; 155 251 253 269 268; 156 253 255 270 269;
157 255 257 271 270; 158 257 259 272 271; 159 259 139 143 272;
160 141 260 273 145; 161 260 261 274 273; 162 261 262 275 274;
163 262 263 276 275; 164 263 264 277 276; 165 264 265 278 277;
166 265 266 279 278; 167 266 267 280 279; 168 267 268 281 280;
169 268 269 282 281; 170 269 270 283 282; 171 270 271 284 283;
172 271 272 285 284; 173 272 143 146 285; 174 145 273 286 147;
175 273 274 287 286; 176 274 275 288 287; 177 275 276 289 288;
178 276 277 290 289; 179 277 278 291 290; 180 278 279 292 291;
181 279 280 293 292; 182 280 281 294 293; 183 281 282 295 294;
184 282 283 296 295; 185 283 284 297 296; 186 284 285 298 297;
187 285 146 149 298; 188 147 286 299 151; 189 286 287 300 299;
190 287 288 301 300; 191 288 289 302 301; 192 289 290 303 302;
193 290 291 304 303; 194 291 292 305 304; 195 292 293 306 305;
196 293 294 307 306; 197 294 295 308 307; 198 295 296 309 308;
199 296 297 310 309; 200 297 298 311 310; 201 298 149 153 311;
220 166 326 331 179; 221 326 328 332 331; 222 328 330 333 332;
223 330 168 180 333; 224 179 331 334 188; 225 331 332 335 334;
226 332 333 336 335; 227 333 180 189 336; 228 188 334 337 197;
229 334 335 338 337; 230 335 336 339 338; 231 336 189 198 339;
232 197 337 340 206; 233 337 338 341 340; 234 338 339 342 341;
235 339 198 207 342;
START GROUP DEFINITION
ELEMENT
_MX 16 19 22 25 59 TO 62 64 TO 66 69 TO 72 74 TO 76 79 TO 82 84 TO 86 -
89 TO 92 94 TO 96 111 113 115 117 147 TO 155 158 159 161 TO 169 172 173 175 -
176 TO 183 186 187 189 TO 197 200 201 220 TO 235
_MY 16 17 25 26 33 42 58 TO 62 64 TO 67 88 TO 92 94 TO 97 111 117 146 TO 159 -
188 TO 201 220 TO 223 232 TO 235
_SHEAR 16 17 25 26 33 42 58 TO 62 64 TO 67 70 TO 72 74 75 80 TO 82 84 85 88 -
89 TO 92 94 TO 97 111 113 115 117 146 TO 159 162 TO 168 173 176 TO 182 187 -
188 TO 201 220 TO 235
END GROUP DEFINITION
ELEMENT PROPERTY
16 17 19 20 22 23 25 26 33 36 39 42 58 TO 62 64 TO 72 74 TO 82 84 TO 92 94 -
95 TO 97 111 113 115 117 146 TO 201 220 TO 235 THICKNESS 0.5
```

Job Title:

Client:

Engineer:

```
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 240 241 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
101 102 114 117 TO 119 121 TO 125 127 TO 129 131 132 136 137 139 141 TO 143 -
145 TO 149 151 TO 153 158 160 162 164 166 168 170 172 174 TO 210 221 TO 225 -
235 237 239 241 243 245 247 249 251 253 255 257 259 TO 311 326 328 -
330 TO 342 ELASTIC MAT DIRECT Y SUBGRADE 20000 PRINT
LOAD 1 DEAD  LOAD
SELFWEIGHT Y -1
JOINT LOAD
5   FX  7.5 FY  -232.4  FZ  -6.6
5   MX  -3.2    MY  -0.8    MZ  -7.1
6   FX  5.4 FY  -201.5  FZ  1.3
6   MX  3.6 MY  -0.4    MZ  -4.1
346 FX  7.5 FY  -232.4  FZ  -6.6
346 MX  -3.2    MY  -0.8    MZ  -7.1
347 FX  5.4 FY  -201.5  FZ  1.3
347 MX  3.6 MY  -0.4    MZ  -4.1
ELEMENT LOAD
16 17 19 20 22 23 25 26 33 36 39 42 58 TO 62 64 TO 72 74 TO 82 84 TO 92 94 -
95 TO 97 111 113 115 117 146 TO 201 220 TO 235 PR GY -28
LOAD 2 LIVE LOAD
JOINT LOAD
5   FX  1.2 FY  -33.7   FZ  -0.9
5   MX  -0.3    MY  -0.3    MZ  -1.3
6   FX  0.9 FY  -28.6   FZ  -0.1
6   MX  0.7 MY  0.1 MZ  -0.8
346 FX  1.2 FY  -33.7   FZ  -0.9
346 MX  -0.3    MY  -0.3    MZ  -1.3
347 FX  0.9 FY  -28.6   FZ  -0.1
347 MX  0.7 MY  0.1 MZ  -0.8
LOAD 3 SDL1 CRUSHER DYNAMIC LOAD
JOINT LOAD
5   FX  -4.4    FY  6.0 FZ  1.2
5   MX  6.2 MY  1.5 MZ  6.0
6   FX  4.6 FY  -11.2   FZ  1.4
6   MX  6.9 MY  1.5 MZ  -5.7
```

346 FX -4.4 FY 6.0 FZ 1.2
346 MX 6.2 MY 1.5 MZ 6.0
347 FX 4.6 FY -11.2 FZ 1.4
347 MX 6.9 MY 1.5 MZ -5.7
LOAD 4 SDL2 CRUSHER DYNAMIC LOAD
JOINT LOAD
5 FX 40.6 FY -160.5 FZ -0.7
5 MX 0.3 MY -1.0 MZ -57.2
6 FX 34.4 FY -137.9 FZ 2.0
6 MX 2.3 MY 0.6 MZ -48.4
346 FX 40.6 FY -160.5 FZ -0.7
346 MX 0.3 MY -1.0 MZ -57.2
347 FX 34.4 FY -137.9 FZ 2.0
347 MX 2.3 MY 0.6 MZ -48.4
LOAD 5 WIND LOAD (+X)
JOINT LOAD
5 FX 11.6 FY -36.8 FZ -0.3
5 MX -0.2 MY -0.1 MZ -16.4
6 FX 11.2 FY -35.3 FZ 0.4
6 MX 0.3 MY 0.1 MZ -15.9
346 FX 11.6 FY -36.8 FZ -0.3
346 MX -0.2 MY -0.1 MZ -16.4
347 FX 11.2 FY -35.3 FZ 0.4
347 MX 0.3 MY 0.1 MZ -15.9
LOAD 6 WIND LOAD (-X)
JOINT LOAD
5 FX -11.8 FY 36.9 FZ 0.3
5 MX 0.2 MY 0.1 MZ 16.7
6 FX -11.4 FY 35.4 FZ -0.4
6 MX -0.3 MY -0.1 MZ 16.2
346 FX -11.8 FY 36.9 FZ 0.3
346 MX 0.2 MY 0.1 MZ 16.7
347 FX -11.4 FY 35.4 FZ -0.4
347 MX -0.3 MY -0.1 MZ 16.2
LOAD 7 WIND LOAD (+Z)
JOINT LOAD
5 FX 0.2 FY 6.1 FZ 8
5 MX 15.1 MY 0.5 MZ -0.4
6 FX -0.2 FY -5.8 FZ 8
6 MX 14.9 MY 0.0 MZ 0.4
346 FX 0.2 FY 6.1 FZ 8
346 MX 15.1 MY 0.5 MZ -0.4
347 FX -0.2 FY -5.8 FZ 8
347 MX 14.9 MY 0.0 MZ 0.4
LOAD 8 WIND LOAD (-Z)
JOINT LOAD
5 FX -0.2 FY -5.7 FZ -8.6
5 MX -15.9 MY -0.1 MZ 0.4
6 FX 0.2 FY 5.9 FZ -7.4

```
6   MX  -15.5   MY  -0.5    MZ  -0.4
346 FX  -0.2    FY  -5.7    FZ  -8.6
346 MX  -15.9   MY  -0.1    MZ  0.4
347 FX  0.2 FY  5.9 FZ  -7.4
347 MX  -15.5   MY  -0.5    MZ  -0.4
LOAD 11 SEISMIC LOAD (+X)
JOINT LOAD
5   FX  13.4    FY  -53.3   FZ  -0.5
5   MX  -0.3    MY  -0.1    MZ  -18.9
6   FX  12.4    FY  -49.4   FZ  0.6
6   MX  0.5 MY  -0.1    MZ  -17.5
346 FX  13.4    FY  -53.3   FZ  -0.5
346 MX  -0.3    MY  -0.1    MZ  -18.9
347 FX  12.4    FY  -49.4   FZ  0.6
347 MX  0.5 MY  0.1 MZ  -17.5
LOAD 12 SEISMIC LOAD (+Z)
JOINT LOAD
5   FX  0   FY  5.7 FZ  6
5   MX  12.3    MY  -0.1    MZ  -0.1
6   FX  0   FY  -5.6    FZ  6.2
6   MX  12.4    MY  -0.2    MZ  0.1
346 FX  0   FY  5.7 FZ  6
346 MX  12.3    MY  -0.1    MZ  -0.1
347 FX  0   FY  -5.6    FZ  6.2
347 MX  12.4    MY  -0.2    MZ  0.1
*****************LOAD COMBINATION FOR BASE PRESSURE CHECK
************************************************
LOAD 301 DL+SDL1_R
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2_R
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL-SDL1_R
REPEAT LOAD
1 1.0 3 -1.0
LOAD 304 DL-SDL2_R
REPEAT LOAD
1 1.0 4 -1.0
*********************************
LOAD 305 DL + SDL1_R + LL
REPEAT LOAD
1 1.0 3 1.0 2 1.0
LOAD 306 DL + SDL2_R + LL
REPEAT LOAD
1 1.0 4 1.0 2 1.0
LOAD 307 DL - SDL1_R + LL
REPEAT LOAD
1 1.0 3 -1.0 2 1.0
```

```
LOAD 308 DL - SDL2_R + LL
REPEAT LOAD
1 1.0 4 -1.0 2 1.0
*********************************************
LOAD 309 DL + SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 310 DL + SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 311 DL + SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 312 DL + SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 313 DL + SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 314 DL + SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 315 DL + SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 316 DL + SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 317 DL - SDL1_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 5 0.75
LOAD 318 DL - SDL1_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 6 0.75
LOAD 319 DL - SDL1_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 7 0.75
LOAD 320 DL - SDL1_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 8 0.75
LOAD 321 DL - SDL2_R + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 5 0.75
LOAD 322 DL - SDL2_R + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 6 0.75
LOAD 323 DL - SDL2_R + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 7 0.75
```

```
LOAD 324 DL - SDL2_R + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 8 0.75
*********************************************
LOAD 325 DL + SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 326 DL + SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 327 DL + SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 328 DL + SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 329 DL + SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 330 DL + SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 331 DL + SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 332 DL + SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 333 DL - SDL1_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 0.525
LOAD 334 DL - SDL1_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 11 -0.525
LOAD 335 DL - SDL1_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 0.525
LOAD 336 DL - SDL1_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 -1.0 2 0.75 12 -0.525
LOAD 337 DL - SDL2_R + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 0.525
LOAD 338 DL - SDL2_R + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 11 -0.525
LOAD 339 DL - SDL2_R + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 0.525
```

```
LOAD 340 DL - SDL2_R + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 -1.0 2 0.75 12 -0.525
*********************************************
LOAD 341 0.6 (DL + SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 342 0.6 (DL + SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 343 0.6 (DL + SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 344 0.6 (DL + SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 345 0.6 (DL + SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 346 0.6 (DL + SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 347 0.6 (DL + SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 348 0.6 (DL + SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 349 0.6 (DL - SDL1_R) + WL(+X)
REPEAT LOAD
1 0.6 3 -0.6 5 1.0
LOAD 350 0.6 (DL - SDL1_R) + WL(-X)
REPEAT LOAD
1 0.6 3 -0.6 6 1.0
LOAD 351 0.6 (DL - SDL1_R) + WL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 7 1.0
LOAD 352 0.6 (DL - SDL1_R) + WL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 8 1.0
LOAD 353 0.6 (DL - SDL2_R) + WL(+X)
REPEAT LOAD
1 0.6 4 -0.6 5 1.0
LOAD 354 0.6 (DL - SDL2_R) + WL(-X)
REPEAT LOAD
1 0.6 4 -0.6 6 1.0
LOAD 355 0.6 (DL - SDL2_R) + WL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 7 1.0
```

```
LOAD 356 0.6 (DL - SDL2_R) + WL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 8 1.0
************************************************
LOAD 357 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 358 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 359 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 360 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 361 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 362 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 363 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 364 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 365 0.6 (DL + SDL1_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 -0.6 11 0.7
LOAD 366 0.6 (DL + SDL1_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 -0.6 11 -0.7
LOAD 367 0.6 (DL + SDL1_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 -0.6 12 0.7
LOAD 368 0.6 (DL + SDL1_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 -0.6 12 -0.7
LOAD 369 0.6 (DL + SDL2_R) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 -0.6 11 0.7
LOAD 370 0.6 (DL + SDL2_R) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 -0.6 11 -0.7
LOAD 371 0.6 (DL + SDL2_R) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
```

```
LOAD 372 0.6 (DL + SDL2_R) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 -0.6 12 -0.7
************************************************
*Load Combination for Concrete design
************************************************
LOAD 401 1.4 (DL+SDL1_R)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2_R)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL-SDL1_R)
REPEAT LOAD
1 1.4 3 -1.4
LOAD 404 1.4 (DL-SDL2_R)
REPEAT LOAD
1 1.4 4 -1.4
************************************************
LOAD 405 1.4 (DL + SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 2 1.7
LOAD 407 1.4 (DL - SDL1_R ) + 1.7 LL
REPEAT LOAD
1 1.4 3 -1.4 2 1.7
LOAD 408 1.4 (DL - SDL2_R ) + 1.7 LL
REPEAT LOAD
1 1.4 4 -1.4 2 1.7
************************************************
LOAD 409 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 410 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 411 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 412 1.2 (DL + SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 413 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 414 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
```

```
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 415 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 416 1.2 (DL + SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 417 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 5 1.6
LOAD 418 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 6 1.6
LOAD 419 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 7 1.6
LOAD 420 1.2 (DL - SDL1_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 8 1.6
LOAD 421 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 5 1.6
LOAD 422 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 6 1.6
LOAD 423 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 7 1.6
LOAD 424 1.2 (DL - SDL2_R) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 8 1.6
************************************************
LOAD 425 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 426 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 427 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 428 1.2 (DL + SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 429 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 430 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
```

```
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 431 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 432 1.2 (DL + SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 433 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 1.0
LOAD 434 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 11 -1.0
LOAD 435 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 1.0
LOAD 436 1.2 (DL - SDL1_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 -1.2 2 0.5 12 -1.0
LOAD 437 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 1.0
LOAD 438 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 11 -1.0
LOAD 439 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 1.0
LOAD 440 1.2 (DL - SDL2_R) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 -1.2 2 0.5 12 -1.0
************************************************
LOAD 441 0.9 (DL + SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 442 0.9 (DL + SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 443 0.9 (DL + SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 444 0.9 (DL + SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 445 0.9 (DL + SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 446 0.9 (DL + SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
```

```
1 0.9 4 0.9 6 1.6
LOAD 447 0.9 (DL + SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 448 0.9 (DL + SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 449 0.9 (DL - SDL1_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 -0.9 5 1.6
LOAD 450 0.9 (DL - SDL1_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 -0.9 6 1.6
LOAD 451 0.9 (DL - SDL1_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 7 1.6
LOAD 452 0.9 (DL - SDL1_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 8 1.6
LOAD 453 0.9 (DL - SDL2_R) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 -0.9 5 1.6
LOAD 454 0.9 (DL - SDL2_R) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 -0.9 6 1.6
LOAD 455 0.9 (DL - SDL2_R) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 7 1.6
LOAD 456 0.9 (DL - SDL2_R) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 8 1.6

*********************************************
LOAD 457 0.9 (DL + SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 458 0.9 (DL + SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 459 0.9 (DL + SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 460 0.9 (DL + SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 461 0.9 (DL + SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 462 0.9 (DL + SDL2_R) + 1.0 EL(-X)
```

```
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 463 0.9 (DL + SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 464 0.9 (DL + SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
LOAD 465 0.9 (DL - SDL1_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 -0.9 11 1.0
LOAD 466 0.9 (DL - SDL1_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 -0.9 11 -1.0
LOAD 467 0.9 (DL - SDL1_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 -0.9 12 1.0
LOAD 468 0.9 (DL - SDL1_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 -0.9 12 -1.0
LOAD 469 0.9 (DL - SDL2_R) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 -0.9 11 1.0
LOAD 470 0.9 (DL - SDL2_R) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 -0.9 11 -1.0
LOAD 471 0.9 (DL - SDL2_R) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 -0.9 12 1.0
LOAD 472 0.9 (DL - SDL2_R) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 -0.9 12 -1.0
**********
PERFORM ANALYSIS
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 7 STAADPRO INPUT FILE OF FOUNDATION F4

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
9 0 1 3.3; 66 0 1 0; 101 0 -2 0; 102 0 -2 3.3; 103 1.6 -2 -1.6; 104 1.6 -2 4.9;
105 -1.6 -2 -1.6; 106 -1.6 -2 4.9; 107 1.6 -2 0; 108 -1.6 -2 0; 109 1.6 -2 3.3;
110 -1.6 -2 3.3; 111 0 -2 -1.6; 112 0 -2 4.9; 113 -1.6 -2 -0.9;
114 -0.9 -2 -0.9; 115 -0.9 -2 -1.6; 116 -1.6 -2 -0.4; 117 -0.9 -2 -0.4;
118 -0.9 -2 0; 119 -0.4 -2 -0.9; 120 -0.4 -2 -1.6; 121 -0.4 -2 -0.4;
122 -0.4 -2 0; 123 0 -2 -0.9; 124 0 -2 -0.4; 125 0.4 -2 -0.9; 126 0.4 -2 -1.6;
127 0.4 -2 -0.4; 128 0.4 -2 0; 129 0.9 -2 -0.9; 130 0.9 -2 -1.6;
131 0.9 -2 -0.4; 132 0.9 -2 0; 133 1.6 -2 -0.9; 134 1.6 -2 -0.4;
135 -1.6 -2 3.7; 136 -0.9 -2 3.7; 137 -0.9 -2 3.3; 138 -1.6 -2 4.2;
139 -0.9 -2 4.2; 140 -0.9 -2 4.9; 141 -0.4 -2 3.7; 142 -0.4 -2 3.3;
143 -0.4 -2 4.2; 144 -0.4 -2 4.9; 145 0 -2 3.7; 146 0 -2 4.2; 147 0.4 -2 3.7;
148 0.4 -2 3.3; 149 0.4 -2 4.2; 150 0.4 -2 4.9; 151 0.9 -2 3.7; 152 0.9 -2 3.3;
153 0.9 -2 4.2; 154 0.9 -2 4.9; 155 1.6 -2 3.7; 156 1.6 -2 4.2;
157 -1.6 -2 0.4; 158 -0.9 -2 0.4; 159 -1.6 -2 0.66; 160 -0.9 -2 0.66;
161 -1.6 -2 0.99; 162 -0.9 -2 0.99; 163 -1.6 -2 1.32; 164 -0.9 -2 1.32;
165 -1.6 -2 1.65; 166 -0.9 -2 1.65; 167 -1.6 -2 1.98; 168 -0.9 -2 1.98;
169 -1.6 -2 2.31; 170 -0.9 -2 2.31; 171 -1.6 -2 2.64; 172 -0.9 -2 2.64;
173 -1.6 -2 2.9; 174 -0.9 -2 2.9; 175 -0.4 -2 0.4; 176 -0.4 -2 0.66;
177 -0.4 -2 0.99; 178 -0.4 -2 1.32; 179 -0.4 -2 1.65; 180 -0.4 -2 1.98;
181 -0.4 -2 2.31; 182 -0.4 -2 2.64; 183 -0.4 -2 2.9; 184 0 -2 0.4;
185 0 -2 0.66; 186 0 -2 0.99; 187 0 -2 1.32; 188 0 -2 1.65; 189 0 -2 1.98;
190 0 -2 2.31; 191 0 -2 2.64; 192 0 -2 2.9; 193 0.4 -2 0.4; 194 0.4 -2 0.66;
195 0.4 -2 0.99; 196 0.4 -2 1.32; 197 0.4 -2 1.65; 198 0.4 -2 1.98;
199 0.4 -2 2.31; 200 0.4 -2 2.64; 201 0.4 -2 2.9; 202 0.9 -2 0.4;
203 0.9 -2 0.66; 204 0.9 -2 0.99; 205 0.9 -2 1.32; 206 0.9 -2 1.65;
207 0.9 -2 1.98; 208 0.9 -2 2.31; 209 0.9 -2 2.64; 210 0.9 -2 2.9;
211 1.6 -2 0.4; 212 1.6 -2 0.66; 213 1.6 -2 0.99; 214 1.6 -2 1.32;
215 1.6 -2 1.65; 216 1.6 -2 1.98; 217 1.6 -2 2.31; 218 1.6 -2 2.64;
219 1.6 -2 2.9;
MEMBER INCIDENCES
1 101 66; 2 102 9;
ELEMENT INCIDENCES SHELL
12 105 113 114 115; 13 113 116 117 114; 14 116 108 118 117; 15 115 114 119 120;
16 114 117 121 119; 17 117 118 122 121; 18 120 119 123 111; 19 119 121 124 123;
20 121 122 101 124; 21 111 123 125 126; 22 123 124 127 125; 23 124 101 128 127;
24 126 125 129 130; 25 125 127 131 129; 26 127 128 132 131; 27 130 129 133 103;
28 129 131 134 133; 29 131 132 107 134; 30 110 135 136 137; 31 135 138 139 136;
32 138 106 140 139; 33 137 136 141 142; 34 136 139 143 141; 35 139 140 144 143;
36 142 141 145 102; 37 141 143 146 145; 38 143 144 112 146; 39 102 145 147 148;
40 145 146 149 147; 41 146 112 150 149; 42 148 147 151 152; 43 147 149 153 151;
44 149 150 154 153; 45 152 151 155 109; 46 151 153 156 155; 47 153 154 104 156;
48 108 157 158 118; 49 157 159 160 158; 50 159 161 162 160; 51 161 163 164 162;
52 163 165 166 164; 53 165 167 168 166; 54 167 169 170 168; 55 169 171 172 170;
```

```
56 171 173 174 172; 57 173 110 137 174; 58 118 158 175 122; 59 158 160 176 175;
60 160 162 177 176; 61 162 164 178 177; 62 164 166 179 178; 63 166 168 180 179;
64 168 170 181 180; 65 170 172 182 181; 66 172 174 183 182; 67 174 137 142 183;
68 122 175 184 101; 69 175 176 185 184; 70 176 177 186 185; 71 177 178 187 186;
72 178 179 188 187; 73 179 180 189 188; 74 180 181 190 189; 75 181 182 191 190;
76 182 183 192 191; 77 183 142 102 192; 78 101 184 193 128; 79 184 185 194 193;
80 185 186 195 194; 81 186 187 196 195; 82 187 188 197 196; 83 188 189 198 197;
84 189 190 199 198; 85 190 191 200 199; 86 191 192 201 200; 87 192 102 148 201;
88 128 193 202 132; 89 193 194 203 202; 90 194 195 204 203; 91 195 196 205 204;
92 196 197 206 205; 93 197 198 207 206; 94 198 199 208 207; 95 199 200 209 208;
96 200 201 210 209; 97 201 148 152 210; 98 132 202 211 107; 99 202 203 212 211;
100 203 204 213 212; 101 204 205 214 213; 102 205 206 215 214;
103 206 207 216 215; 104 207 208 217 216; 105 208 209 218 217;
106 209 210 219 218; 107 210 152 109 219;
START GROUP DEFINITION
ELEMENT
_MX 12 13 15 16 18 19 21 22 24 25 27 28 31 32 34 35 37 38 40 41 43 44 46 47 -
49 TO 56 59 TO 66 69 TO 76 79 TO 86 89 TO 96 99 TO 106
_MY 12 TO 17 24 TO 35 42 TO 67 88 TO 107
_SHEAR 12 TO 15 18 21 24 27 TO 32 35 38 41 44 TO 57 61 TO 64 71 TO 74 -
81 TO 84 91 TO 94 98 TO 107
END GROUP DEFINITION
ELEMENT PROPERTY
12 TO 107 THICKNESS 0.5
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 PRIS YD 0.8 ZD 0.8
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
101 TO 219 ELASTIC MAT DIRECT Y SUBGRADE 20000 PRINT
LOAD 1 DEAD  LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
12 TO 107 PR GY -27
joint load
9   FX  1.6 FY  -247.3  FZ  -6.3
9   MX  0.0 MY  0.0 MZ  0.0
66  FX  -0.8    FY  -85.1   FZ  0.0
66  MX  0.0 MY  0.0 MZ  0.0
```

```
Load 2   LIVE LOAD
joint load
9   FX  0.3 FY  -349.6  FZ  -0.5
9   MX  0.0 MY  0.0 MZ  0.0
66  FX  -0.1    FY  -137.1  FZ  0.0
66  MX  0.0 MY  0.0 MZ  0.0



Load 3  SDL1- BUMKER EMPTY CHUTE EMPTY CONVEYOR STARTING
joint load
9   FX  -45.4   FY  -218.3  FZ  -8.9
9   MX  0.0 MY  0.0 MZ  0.0
66  FX  0.0 FY  -109.6  FZ  1.1
66  MX  0.0 MY  0.0 MZ  0.0



Load 4  SDL1- BUMKER FULL CHUTE FULL CONVEYOR RUNNING
joint load
9   FX  -34.2   FY  -826.9  FZ  -11.2
9   MX  0.0 MY  0.0 MZ  0.0
66  FX  -0.2    FY  -277.7  FZ  1.1
66  MX  0.0 MY  0.0 MZ  0.0



Load 5  WIND LOAD (+X)
joint load
9   FX  115.67  FY  59.37   FZ  0.93
9   MX  0.0 MY  0.0 MZ  0.0
66  FX  11.98   FY  72.1    FZ  0.01
66  MX  0.0 MY  0.0 MZ  0



Load 6  WIND LOAD (-X)
joint load
9   FX  -115.66 FY  -62.25  FZ  -0.92
9   MX  0.0 MY  0.0 MZ  0
66  FX  -11.98  FY  -72.85  FZ  -0.01
66  MX  0.0 MY  0.0 MZ  0
```

```
Load 7  WIND LOAD (+Z)
joint load
9   FX  0.39    FY  93.95   FZ  100.51
9   MX  0.0 MY  0.0 MZ  0
66  FX  0.12    FY  83.1    FZ  1.81
66  MX  0.0 MY  0.0 MZ  0



Load 8  WIND LOAD (-Z)
joint load
9   FX  0.17    FY  -78.35  FZ  -88.12
9   MX  0.0 MY  0.0 MZ  0
66  FX  -0.11   FY  -61.87  FZ  -1.79
66  MX  0.0 MY  0.0 MZ  0



Load 9  TEMPERATURE LOAD 1
joint load
9   FX  1.72    FY  -24.27  FZ  8.22
9   MX  0.0 MY  0.0 MZ  0
66  FX  2.66    FY  18.63   FZ  0.6
66  MX  0.0 MY  0.0 MZ  0



Load 10 TEMPERATURE LOAD 2
joint load
9   FX  -1.72   FY  24.27   FZ  -8.22
9   MX  0   MY  0   MZ  0
66  FX  -2.66   FY  -18.63  FZ  -0.6
66  MX  0   MY  0   MZ  0



Load 11 SEISMIC LOAD (+X)
joint load
9   FX  12.85   FY  14.41   FZ  0.19
9   MX  0.0 MY  0.0 MZ  0
66  FX  0.37    FY  7.2 FZ  0
66  MX  0.0 MY  0.0 MZ  0
```

```
Load 12 SEISMIC LOAD (+Z)
joint load
9   FX  0.23    FY  8.65    FZ  9.49
9   MX  0.0 MY  0.0 MZ  0
66  FX  0.03    FY  16.32   FZ  0.02
66  MX  0.0 MY  0.0 MZ  0
******
LOAD 301 DL+SDL1
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 9 1.0 2 1.0
LOAD 304 DL + SDL1 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 9 1.0 2 1.0
LOAD 305 DL + SDL2 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 10 1.0 2 1.0
LOAD 306 DL + SDL2 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 10 1.0 2 1.0
LOAD 307 DL + SDL1 + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 308 DL + SDL1 + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 309 DL + SDL1 + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 310 DL + SDL1 + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 311 DL + SDL2 + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 312 DL + SDL2 + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 313 DL + SDL2 + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 314 DL + SDL2 + 0.75 LL + 0.75 WL(-Z)
```

```
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 315 DL + SDL1 + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 316 DL + SDL1 + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 317 DL + SDL1 + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 318 DL + SDL1 + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 319 DL + SDL2 + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 320 DL + SDL2 + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 321 DL + SDL2 + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 322 DL + SDL2 + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 323 0.6 (DL + SDL1) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 324 0.6 (DL + SDL1) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 325 0.6 (DL + SDL1) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 326 0.6 (DL + SDL1) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 327 0.6 (DL + SDL2) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 328 0.6 (DL + SDL2) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 329 0.6 (DL + SDL2) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 330 0.6 (DL + SDL2) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
```

```
LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
****
*Load Combination for Concrete design
LOAD 401 1.4 (DL+SDL1)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 9 1.4 2 1.7
LOAD 404 1.4 (DL + SDL1 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 10 1.4 2 1.7
LOAD 405 1.4 (DL + SDL2 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 9 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 10 1.4 2 1.7
LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
```

```
LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0

LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
```

```
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 428 0.9 (DL + SDL2) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6

LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
PERFORM ANALYSIS
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023			Date	27-01-2020	11-03-2020	28-04-2022	01-11-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER'S COMMENT DATED 26-02-2020	REVISED FOR MODIFICATION OF CRUSHER STRUCTURE	AS BUILT	
Prepared By	RPK	RPK	RPK	RPK			Prepared By	RPK	RPK	RPK	RPK	
Checked	SBY	SBY	SBY	SBY			Checked	SBY	SBY	SBY	SBY	
Approved	AG	AG	AG	AG			Approved	AG	AG	AG	AG	

ATTACHMENT 8 STAADPRO INPUT FILE OF FOUNDATION F5

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 17-Jan-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
17 0 1 3.3; 74 0 1 0; 101 0 -2 0; 102 0 -2 3.3; 103 1.5 -2 -3; 104 1.5 -2 5.7;
105 -1.5 -2 -3; 106 -1.5 -2 5.7; 107 1.5 -2 0; 108 -1.5 -2 0; 109 1.5 -2 3.3;
110 -1.5 -2 3.3; 111 0 -2 -3; 112 0 -2 5.7; 113 -1.5 -2 -1.2; 114 -0.9 -2 -1.2;
115 -0.9 -2 -3; 116 -1.5 -2 -0.4; 117 -0.9 -2 -0.4; 118 -0.9 -2 0;
119 -0.4 -2 -1.2; 120 -0.4 -2 -3; 121 -0.4 -2 -0.4; 122 -0.4 -2 0;
123 0 -2 -1.2; 124 0 -2 -0.4; 125 0.4 -2 -1.2; 126 0.4 -2 -3; 127 0.4 -2 -0.4;
128 0.4 -2 0; 129 0.9 -2 -1.2; 130 0.9 -2 -3; 131 0.9 -2 -0.4; 132 0.9 -2 0;
133 1.5 -2 -1.2; 134 1.5 -2 -0.4; 135 -1.5 -2 3.7; 136 -0.9 -2 3.7;
137 -0.9 -2 3.3; 138 -1.5 -2 4.6; 139 -0.9 -2 4.6; 140 -0.9 -2 5.7;
141 -0.4 -2 3.7; 142 -0.4 -2 3.3; 143 -0.4 -2 4.6; 144 -0.4 -2 5.7;
145 0 -2 3.7; 146 0 -2 4.6; 147 0.4 -2 3.7; 148 0.4 -2 3.3; 149 0.4 -2 4.6;
150 0.4 -2 5.7; 151 0.9 -2 3.7; 152 0.9 -2 3.3; 153 0.9 -2 4.6; 154 0.9 -2 5.7;
155 1.5 -2 3.7; 156 1.5 -2 4.6; 157 -1.5 -2 0.4; 158 -0.9 -2 0.4;
159 -1.5 -2 0.66; 160 -0.9 -2 0.66; 161 -1.5 -2 0.9; 162 -0.9 -2 0.9;
163 -1.5 -2 1.32; 164 -0.9 -2 1.32; 165 -1.5 -2 1.65; 166 -0.9 -2 1.65;
167 -1.5 -2 1.98; 168 -0.9 -2 1.98; 169 -1.5 -2 2.4; 170 -0.9 -2 2.4;
171 -1.5 -2 2.64; 172 -0.9 -2 2.64; 173 -1.5 -2 2.9; 174 -0.9 -2 2.9;
175 -0.4 -2 0.4; 176 -0.4 -2 0.66; 177 -0.4 -2 0.9; 178 -0.4 -2 1.32;
179 -0.4 -2 1.65; 180 -0.4 -2 1.98; 181 -0.4 -2 2.4; 182 -0.4 -2 2.64;
183 -0.4 -2 2.9; 184 0 -2 0.4; 185 0 -2 0.66; 186 0 -2 0.9; 187 0 -2 1.32;
188 0 -2 1.65; 189 0 -2 1.98; 190 0 -2 2.4; 191 0 -2 2.64; 192 0 -2 2.9;
193 0.4 -2 0.4; 194 0.4 -2 0.66; 195 0.4 -2 0.9; 196 0.4 -2 1.32;
197 0.4 -2 1.65; 198 0.4 -2 1.98; 199 0.4 -2 2.4; 200 0.4 -2 2.64;
201 0.4 -2 2.9; 202 0.9 -2 0.4; 203 0.9 -2 0.66; 204 0.9 -2 0.9;
205 0.9 -2 1.32; 206 0.9 -2 1.65; 207 0.9 -2 1.98; 208 0.9 -2 2.4;
209 0.9 -2 2.64; 210 0.9 -2 2.9; 211 1.5 -2 0.4; 212 1.5 -2 0.66;
213 1.5 -2 0.9; 214 1.5 -2 1.32; 215 1.5 -2 1.65; 216 1.5 -2 1.98;
217 1.5 -2 2.4; 218 1.5 -2 2.64; 219 1.5 -2 2.9; 220 -1.5 -2 -1.5;
221 -0.9 -2 -1.5; 222 -0.4 -2 -1.5; 223 0 -2 -1.5; 224 0.4 -2 -1.5;
225 0.9 -2 -1.5; 226 1.5 -2 -1.5; 227 -1.5 -2 5.1; 228 -0.9 -2 5.1;
229 -0.4 -2 5.1; 230 0 -2 5.1; 231 0.4 -2 5.1; 232 0.9 -2 5.1; 233 1.5 -2 5.1;
234 -1.5 -2 -2.4; 235 -0.9 -2 -2.4; 236 -1.5 -2 -1.9; 237 -0.9 -2 -1.9;
238 -0.4 -2 -2.4; 239 -0.4 -2 -1.9; 240 0 -2 -2.4; 241 0 -2 -1.9;
242 0.4 -2 -2.4; 243 0.4 -2 -1.9; 244 0.9 -2 -2.4; 245 0.9 -2 -1.9;
246 1.5 -2 -2.4; 247 1.5 -2 -1.9; 248 -1.5 -2 -0.9; 249 -0.9 -2 -0.9;
250 -0.4 -2 -0.9; 251 0 -2 -0.9; 252 0.4 -2 -0.9; 253 0.9 -2 -0.9;
254 1.5 -2 -0.9; 255 -1.5 -2 4.2; 256 -0.9 -2 4.2; 257 -0.4 -2 4.2;
258 0 -2 4.2; 259 0.4 -2 4.2; 260 0.9 -2 4.2; 261 1.5 -2 4.2;
MEMBER INCIDENCES
1 101 74; 2 102 17;
ELEMENT INCIDENCES SHELL
14 116 108 118 117; 17 117 118 122 121; 20 121 122 101 124; 23 124 101 128 127;
26 127 128 132 131; 29 131 132 107 134; 30 110 135 136 137; 33 137 136 141 142;
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36 142 141 145 102; 39 102 145 147 148; 42 148 147 151 152; 45 152 151 155 109;
48 108 157 158 118; 49 157 159 160 158; 50 159 161 162 160; 51 161 163 164 162;
52 163 165 166 164; 53 165 167 168 166; 54 167 169 170 168; 55 169 171 172 170;
56 171 173 174 172; 57 173 110 137 174; 58 118 158 175 122; 59 158 160 176 175;
60 160 162 177 176; 61 162 164 178 177; 62 164 166 179 178; 63 166 168 180 179;
64 168 170 181 180; 65 170 172 182 181; 66 172 174 183 182; 67 174 137 142 183;
68 122 175 184 101; 69 175 176 185 184; 70 176 177 186 185; 71 177 178 187 186;
72 178 179 188 187; 73 179 180 189 188; 74 180 181 190 189; 75 181 182 191 190;
76 182 183 192 191; 77 183 142 102 192; 78 101 184 193 128; 79 184 185 194 193;
80 185 186 195 194; 81 186 187 196 195; 82 187 188 197 196; 83 188 189 198 197;
84 189 190 199 198; 85 190 191 200 199; 86 191 192 201 200; 87 192 102 148 201;
88 128 193 202 132; 89 193 194 203 202; 90 194 195 204 203; 91 195 196 205 204;
92 196 197 206 205; 93 197 198 207 206; 94 198 199 208 207; 95 199 200 209 208;
96 200 201 210 209; 97 201 148 152 210; 98 132 202 211 107; 99 202 203 212 211;
100 203 204 213 212; 101 204 205 214 213; 102 205 206 215 214;
103 206 207 216 215; 104 207 208 217 216; 105 208 209 218 217;
106 209 210 219 218; 107 210 152 109 219; 109 220 113 114 221;
111 221 114 119 222; 113 222 119 123 223; 115 223 123 125 224;
117 224 125 129 225; 119 225 129 133 226; 120 138 227 228 139;
121 227 106 140 228; 122 139 228 229 143; 123 228 140 144 229;
124 143 229 230 146; 125 229 144 112 230; 126 146 230 231 149;
127 230 112 150 231; 128 149 231 232 153; 129 231 150 154 232;
130 153 232 233 156; 131 232 154 104 233; 132 105 234 235 115;
133 234 236 237 235; 134 236 220 221 237; 135 115 235 238 120;
136 235 237 239 238; 137 237 221 222 239; 138 120 238 240 111;
139 238 239 241 240; 140 239 222 223 241; 141 111 240 242 126;
142 240 241 243 242; 143 241 223 224 243; 144 126 242 244 130;
145 242 243 245 244; 146 243 224 225 245; 147 130 244 246 103;
148 244 245 247 246; 149 245 225 226 247; 150 113 248 249 114;
151 248 116 117 249; 152 114 249 250 119; 153 249 117 121 250;
154 119 250 251 123; 155 250 121 124 251; 156 123 251 252 125;
157 251 124 127 252; 158 125 252 253 129; 159 252 127 131 253;
160 129 253 254 133; 161 253 131 134 254; 162 135 255 256 136;
163 255 138 139 256; 164 136 256 257 141; 165 256 139 143 257;
166 141 257 258 145; 167 257 143 146 258; 168 145 258 259 147;
169 258 146 149 259; 170 147 259 260 151; 171 259 149 153 260;
172 151 260 261 155; 173 260 153 156 261;
START GROUP DEFINITION
ELEMENT
_MX 49 TO 56 59 TO 66 69 TO 76 79 TO 86 89 TO 96 99 TO 106 109 111 113 115 -
117 119 TO 173
_MY 14 17 26 29 30 33 42 45 48 TO 67 88 TO 107 109 111 117 119 TO 123 128 -
129 TO 137 144 TO 153 158 TO 165 170 TO 173
_SHEAR 14 29 30 45 48 TO 57 61 TO 64 71 TO 74 81 TO 84 91 TO 94 98 TO 107 -
109 111 113 115 117 119 TO 152 154 156 158 160 TO 163 165 167 169 -
171 TO 173
END GROUP DEFINITION
ELEMENT PROPERTY
14 17 20 23 26 29 30 33 36 39 42 45 48 TO 107 109 111 113 115 117 119 TO 172 -
173 THICKNESS 0.6
```

```
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 2 PRIS YD 0.8 ZD 0.8
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
101 TO 261 ELASTIC MAT DIRECT Y SUBGRADE 20000 PRINT
LOAD 1 DEAD  LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
14 17 20 23 26 29 30 33 36 39 42 45 48 TO 107 109 111 113 115 117 119 TO 172 -
173 PR GY -27
JOINT LOAD
17 FX -0.1 FY -262.3 FZ 3
17 MX 0 MY 0 MZ 0
74 FX 0.8 FY -67 FZ -4.7
74 MX 0 MY 0 MZ 0
LOAD 2 LIVE LOAD
JOINT LOAD
17 FX -0.2 FY -340.6 FZ 2.1
17 MX 0 MY 0 MZ 0
74 FX 0.1 FY -81.5 FZ -7.4
74 MX 0 MY 0 MZ 0
LOAD 3 SDL1- BUMKER EMPTY CHUTE EMPTY CONVEYOR STARTING
JOINT LOAD
17 FX 0 FY -95 FZ -3.8
17 MX 0 MY 0 MZ 0
74 FX 0 FY -30.2 FZ -6.8
74 MX 0 MY 0 MZ 0
LOAD 4 SDL1- BUMKER FULL CHUTE FULL CONVEYOR RUNNING
JOINT LOAD
17 FX -0.4 FY -1606 FZ 15.2
17 MX 0 MY 0 MZ 0
74 FX 0.2 FY -318.1 FZ -25.6
74 MX 0 MY 0 MZ 0
LOAD 5 WIND LOAD (+X)
JOINT LOAD
17 FX 2.4 FY -276.31 FZ 3.62
17 MX 0 MY 0 MZ 0
74 FX 12.03 FY -75.75 FZ -4.36
74 MX 0 MY 0 MZ 0
```

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LOAD 6 WIND LOAD (-X)
JOINT LOAD
17 FX -2.4 FY 277.96 FZ -3.65
17 MX 0 MY 0 MZ 0
74 FX -12.03 FY 75.89 FZ 4.36
74 MX 0 MY 0 MZ 0
LOAD 7 WIND LOAD (+Z)
JOINT LOAD
17 FX -0.1 FY -293.09 FZ 58.4
17 MX 0 MY 0 MZ 0
74 FX -0.12 FY 349.95 FZ 78.69
74 MX 0 MY 0 MZ 0
LOAD 8 WIND LOAD (-Z)
JOINT LOAD
17 FX 0.1 FY 282.18 FZ -57.07
17 MX 0 MY 0 MZ 0
74 FX 0.11 FY -338.78 FZ -76.67
74 MX 0 MY 0 MZ 0
LOAD 9 TEMPERATURE LOAD 1
JOINT LOAD
17 FX -6.55 FY -8.42 FZ -36.2
17 MX 0 MY 0 MZ 0
74 FX -2.66 FY 8.91 FZ 42.37
74 MX 0 MY 0 MZ 0
LOAD 10 TEMPERATURE LOAD 2
JOINT LOAD
17 FX 6.55 FY 8.42 FZ 36.2
17 MX 0 MY 0 MZ 0
74 FX 2.66 FY -8.91 FZ -42.37
74 MX 0 MY 0 MZ 0
LOAD 11 SEISMIC LOAD (+X)
JOINT LOAD
17 FX -0.13 FY -48.31 FZ 0.49
17 MX 0 MY 0 MZ 0
74 FX 0.37 FY -8.55 FZ -0.64
74 MX 0 MY 0 MZ 0
LOAD 12 SEISMIC LOAD (+Z)
JOINT LOAD
17 FX 0 FY -72 FZ 11.54
17 MX 0 MY 0 MZ 0
74 FX -0.03 FY 97.17 FZ 17.67
74 MX 0 MY 0 MZ 0
**********
LOAD 301 DL+SDL1
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL + SDL1 + TL1 + LL
```

```
REPEAT LOAD
1 1.0 3 1.0 9 1.0 2 1.0
LOAD 304 DL + SDL1 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 9 1.0 2 1.0
LOAD 305 DL + SDL2 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 10 1.0 2 1.0
LOAD 306 DL + SDL2 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 10 1.0 2 1.0
LOAD 307 DL + SDL1 + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 308 DL + SDL1 + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 309 DL + SDL1 + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 310 DL + SDL1 + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 311 DL + SDL2 + 0.75 LL + 0.75 WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 312 DL + SDL2 + 0.75 LL + 0.75 WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 313 DL + SDL2 + 0.75 LL + 0.75 WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 314 DL + SDL2 + 0.75 LL + 0.75 WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 315 DL + SDL1 + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 316 DL + SDL1 + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 317 DL + SDL1 + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 318 DL + SDL1 + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 319 DL + SDL2 + 0.75 LL + 0.525 EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
```

```
LOAD 320 DL + SDL2 + 0.75 LL + 0.525 EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 321 DL + SDL2 + 0.75 LL + 0.525 EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 322 DL + SDL2 + 0.75 LL + 0.525 EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 323 0.6 (DL + SDL1) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 324 0.6 (DL + SDL1) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 325 0.6 (DL + SDL1) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 326 0.6 (DL + SDL1) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 327 0.6 (DL + SDL2) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 328 0.6 (DL + SDL2) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 329 0.6 (DL + SDL2) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 330 0.6 (DL + SDL2) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 3 0.6 11 -0.7
LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
REPEAT LOAD
```

```
1 0.6 4 0.6 11 -0.7
LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
****
*Load Combination for Concrete design
LOAD 401 1.4 (DL+SDL1)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 9 1.4 2 1.7
LOAD 404 1.4 (DL + SDL1 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 10 1.4 2 1.7
LOAD 405 1.4 (DL + SDL2 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 9 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 10 1.4 2 1.7
LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
```

```
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0

LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 428 0.9 (DL + SDL2) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
```

```
LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
PERFORM ANALYSIS
FINISH
```